

Spectrum PowerCC

Communication Front-End
(CFE)

Reference Guide

Preface

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Note

Please observe safety notes and warnings for your own safety.

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Document Release 05-0-01

Edition 03.2009

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History

Version	Date	Modification
05-0-01	03.2009	Updated for new functionality: Group Interrogation, Enh. Redundancy, protocol SNMP
04-2-07	Jan-2008	Section "Protocol Engineering" updated
04-2-06	Sep-2007	Updated due to review comments
04-2-05	Mar-2007	Updated for Release 4.2
04-2-04	Mar-2007	New standardized layout
04-2-03	Dec-2006	Updated as per Variance SPC400022128
04-2-02	Nov-2006	Updated as per Variance SPC400022228
04-2-01	Aug-2006	Partly updated for Release 4.2. Introducing new IMM features in chapter 'Protocol Specific Engineering'
04-1-04	Dec-2005	Screenshots replaced, related text updated
04-1-03	Apr-2005	Corrected acc. to Variances SPC400012094, SPC400012099

Preface

Safety Notes

This manual does not constitute a complete catalog of all safety measures required for operating the product in question because special operating conditions might require additional measures. For application specific safety measures refer to the respective project documentation. However, it does contain notes that must be adhered to for your own personal safety and to avoid damage to property. These notes are highlighted with a warning triangle and different keywords indicating different degrees of danger as stated in the Warning Conventions.



WARNING

WARNING indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.



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CAUTION used without safety alert symbol indicates a potentially hazardous situation which, if not avoided, may result in property damage.

NOTICE

NOTICE used without the safety alert symbol indicates a potential situation which, if not avoided, may result in an undesirable result or state.

NOTE

A note is a tip. The user does not need to follow this statement. However he will have advantages if he does.

Qualified Personnel

Only competent and authorized personnel should work with this product after becoming thoroughly familiar with all warnings, safety notices, operating instructions and maintenance procedures.

Use as described

The product must not be used for any other purposes than that described in the technical documentation. If it is used together with third-party devices and components, these must be recommended or approved by Siemens AG. The successful and safe operation of this product is dependent on adequate transportation and proper handling, storage, installation, operation and maintenance.

Purpose

The information contained within this Reference Guide should help the user to:

- Understand the structure, working mechanisms and elements of the Communications Front-End application and its interface
- Create and modify the elements of CFE as well as customize CFE according to the needs of a specific user control center installation by using the CFE user interface
- Analyze and solve problems during CFE configuration and operation (especially during commissioning)

For this purpose, this Reference Guide provides an overview of the CFE-related forms of the Engineering System as well as of the views of the Runtime Explorer. It also includes applicable screen captures and corresponding step-by-step procedures for the Process Interface “CFE” and provides overviews for every display and descriptions of the frames, fields and buttons included on each of them.

Scope

This Reference Guide describes only the user interface of the Spectrum PowerCC Communications Front-End applications.

Audience

This Reference Guide is intended to provide detailed reference and end-user functionality information about the Communications Front-End user interface for Engineers, Technicians, Dispatchers and Operators.

Prerequisites

This Reference Guide is designed and written based on the assumption that readers are already familiar with the Spectrum PowerCC product concepts, design and architecture information that is presented in higher-level overview documents. In general, readers are expected to be familiar with and have a working knowledge of their telecontrol system and its conventions. Finally, this Reference Guide assumes that readers have a working knowledge of Windows operating system, its conventions and user interface features.

Documentation Conventions

Where applicable, this Reference Guide uses quotation marks in procedure steps to indicate software field values that must be entered by users or options that must be selected.

Organization

This Reference Guide proceeds from general introductory material to detailed user interface descriptions. The user interface information and procedures included are separated between single view and tabular-based views. Additionally, the information is arranged consecutively such that it is presented in the order that replicates the popup menu option listings. The following table outlines the organization of this Reference Guide by listing chapter titles and brief content descriptions.

Volume	Contents
Tutorial	Briefly introduces the Spectrum PowerCC CFE components, data representation and configuration. It also describes the use of the instance tree and presents a short practical exercise (creating and modifying RTU data points).
User Interface	Shows the way the CFE “Runtime Explorer” can be used to check CFE status information.

Information Model Management	Provides step-by -step description of the definition of the CFE components i.e. parameterization and software setting (configurable time-outs, limits etc.) as well as CFE data entry. This includes in particular the use of the Single View editor for the creation /modification of CFE servers, COM ports, serial communication links, RTUs, RTU data points etc. Explains also the use of the table-based editors. The last part of this volume describes in detail the attributes of the individual telecontrol protocols.
Test and Diagnosis	Describes the use of the Test and Diagnosis Tool. This includes the use of the Trace functions for tracking input information from RTU to SCADA interface as well as of the Command functions in order to get information about internal states, statistics and process images (redundancy status, communication statistics, renewal check, GI status, process image, server time.

Questions?

This Reference Guide provides reference and general functionality information. It is one manual in a set of manuals that documents the Spectrum PowerCC system. Various other product modules may be mentioned or discussed briefly herein. If you have questions, either about Communications Front-End in particular or other Siemens Spectrum products in general, please contact your Siemens representative.

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Volume 1 Tutorial

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1.1 Introduction

Description / Purpose

The Tutorial Volume briefly introduces the Spectrum PowerCC Communication Front-End (CFE) components and shows how configuration data is presented in the system. Practical “hands-on” tasks convey the impression how the configuration of CFE components works in principle.

- It explores basic functionality but is not comprehensive (precedes and supplements detailed classroom training).
- It uses typical, real-world examples.

Content

The Tutorial Volume contains CFE-specific chapters focusing on primary functionality.

Intended Audience

- Engineers, Operators, Dispatchers, Data Entry / Systems Support staff
- Training, Marketing / Sales

1.2 Configuring CFE

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1.2.1 General

The Communication Front-End (CFE) is the subsystem used in Spectrum PowerCC to form the interface to the process domain.

The *CFE Reference Guide* is intended to help in setting-up and configuring the system. As a precondition the user should:

- have basic knowledge of the browser-based Engineering System and
- have already entered the SCADA data into the database.

Further, detailed knowledge of his/her telecontrol system and the characteristics of process data is highly recommended.

For details of CFE functionality refer to the document 'Communication Front-End, Technical Specification'.

1.2.2 Instances

All system data are organized in the form of a hierarchical tree, similar to a hard disk file system presented in Windows Explorer. For configuring the CFE, the following branches of the tree are of main importance:

- Network branch that contains the instance substations with further sub-instances bays, switches, measurements and the telemetered and telecontrolled digital and analog values
- The Process Interface branch (PI) that contains the instances CFE, COM ports, links, RTUs and RTU data points as well as optional device status information. The hierarchy of the process interface instances can be seen in Figure 1-2.4 and Figure 1-2.5 .

The figures hereafter include an overview of the first level of the Realtime (RT) instance as well as some typical branches of the Realtime root instance, e.g. Network branch with Substations, Process Interface with CFE and RTUs.



Note

Please note that due to the fact that instances can easily be added, removed, renamed, the instance names in these samples may differ from those given in a customized Spectrum PowerCC system.



Note

In parallel to the Network branch there may also be other branches containing tele-measured values of interest, such as

- Accounting: Telemetered integrated totals (Meter readings)
- Generation: Telemetered / telecontrolled measurements for Automatic Generation Control (AGC)

The principle of configuring this data is the same as shown hereafter for network data.

1.2.2.1 Realtime Instances (Overview)

The following diagram shows Realtime instances in the first hierarchical level available in a typical system.

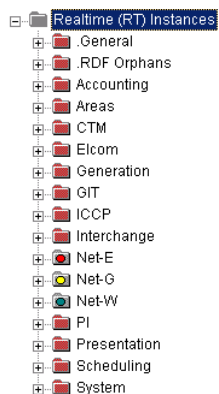


Figure 1-2.1: Overview of Realtime Instances, Root Level

1.2.2.2 Network Instances

The Network instances describe all SCADA attributes from substation name down to the data object properties.

This figure shows substations as child instances of the (electrical) network instance (extract)

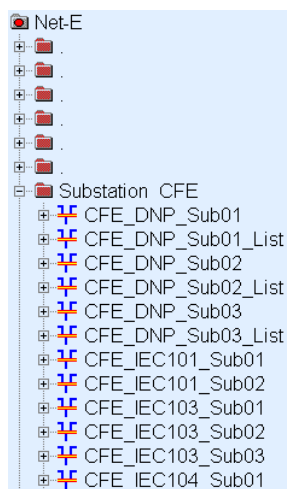


Figure 1-2.2: Substations as Child Instances of Network

This figure shows child instances of a substation (sample extract)

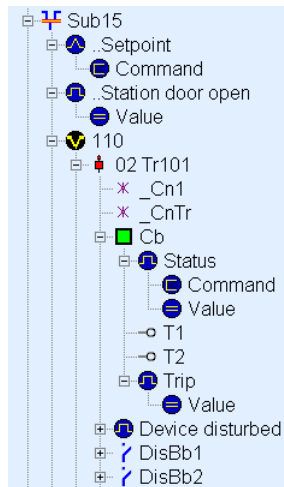


Figure 1-2.3: Sample of Child Instances of a Substation

1.2.2.3 Process Interface Instances

The Process Interface (PI) instances specify all communication related components from CFE server(s), CFE interfaces, communication links, RTU(s) down to data point properties. Common instances like characteristic curves, protocols, scan rates are also part of PI.

The following screenshot shows an extract of typical CFE (child) instances: Interface with Status Information and link to RTU.

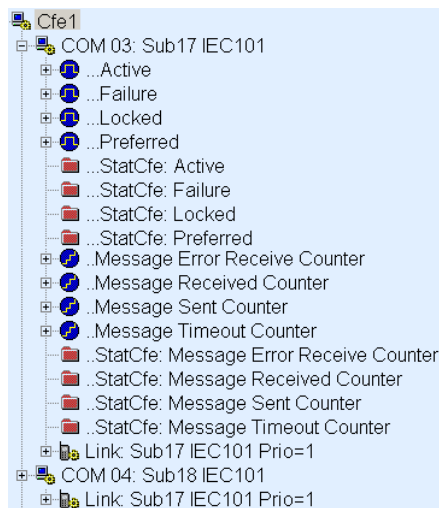


Figure 1-2.4: Typical CFE (Child) Instances

This figure shows typical RTU instances (sample extract)

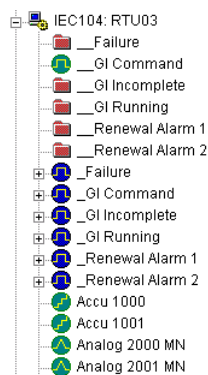


Figure 1-2.5: Typical RTU Instances (Extract)

1.2.2.4 Instances of a Remote Control Center

The following screenshot shows typical instances for configuring a Remote Control Center (RCC) including its "Logical" RTU(s), see Figure 1-2.6.

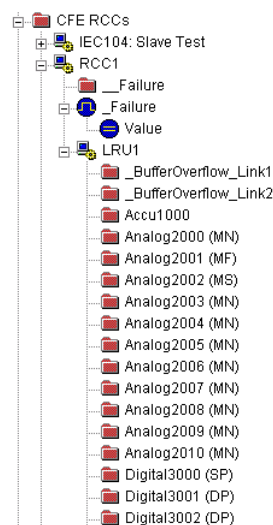


Figure 1-2.6: Instances of RCC and Logical RTU (Example)

1.2.3 IMM - Start and Job Management

The Information Model Manager (IMM) is utilized to setup a system, to add components, to edit parameters, to activate changes. Details on IMM usage can be found in the Spectrum PowerCC document *Information Model Manager - Reference Guide*, document number: SPC-RG-IMM-xx-x-xx .

To invoke IMM, you have the following options:

- from Windows **Start** button -> **Spectrum PowerCC** -> **Information Model Manager**
- from PAT (PowerCC Application Toolbar) as shown by next Figure

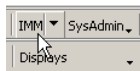


Figure 1-2.7: Starting the Information Model Manager IMM

The IMM runs up and presents a window comprising the Overview Pane in the left section and the Work Pane in the right section. The Overview Pane shows the instance tree beginning with the PSOS computer name as root node. The Work Pane shows currently existing jobs (see next figure).

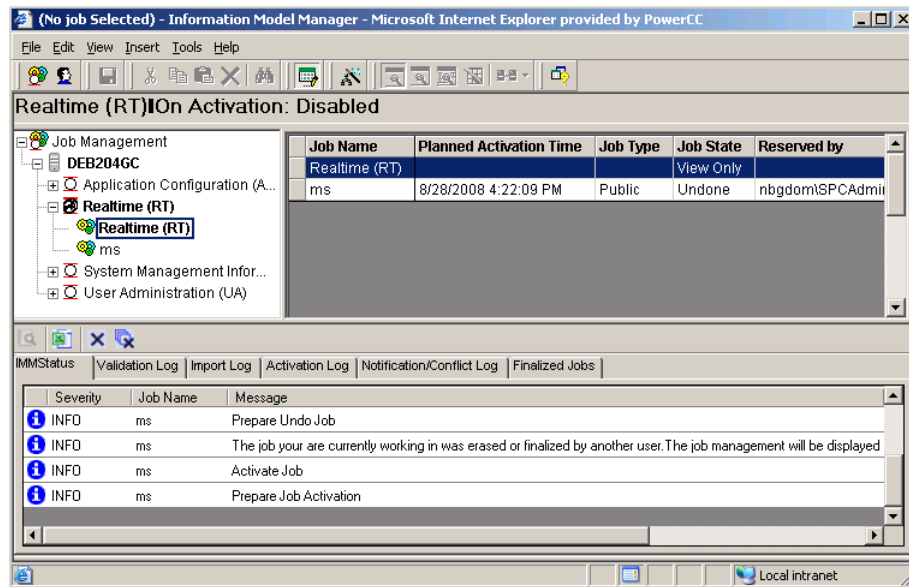


Figure 1-2.8: Information Model Manager with Realtime Context Selected

Next, either create a new job or select an existing job to continue. To start a new job, place the cursor onto Realtime (RT) DataSet in the Overview Pane, right-click and click New Job (next figure).

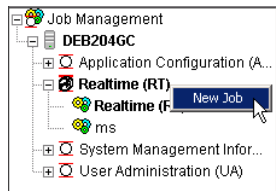


Figure 1-2.9: Creating a New Job

A dialog window appears; use this to enter a job name (required) and a job description (optional). Set the 'Planned Activation Time' to a future date, see next figure.

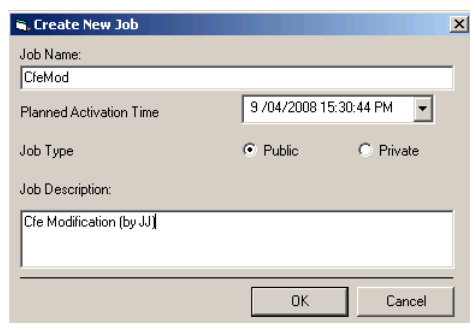


Figure 1-2.10: Entering a Job Name and (Optionally) a Job Description

After clicking OK, the instance tree is displayed in the Overview Pane where you can select the instance of interest to work with.

If you want to continue a job, right-click an existing job in the Work Pane and select Open from popup menu (see next figure).

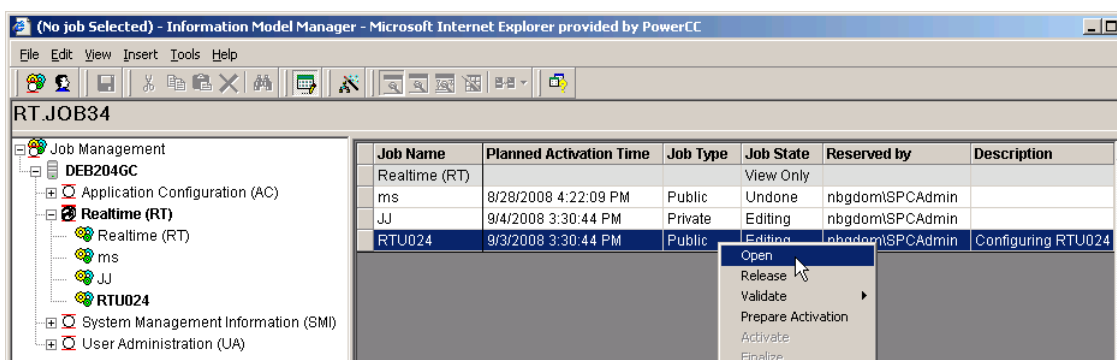


Figure 1-2.11: Resuming an Existing Job

When you select one of the instances in the Overview Pane the Single Editor in the Work Pane allows to enter or adjust the different parameter sets of this instance (next figure).

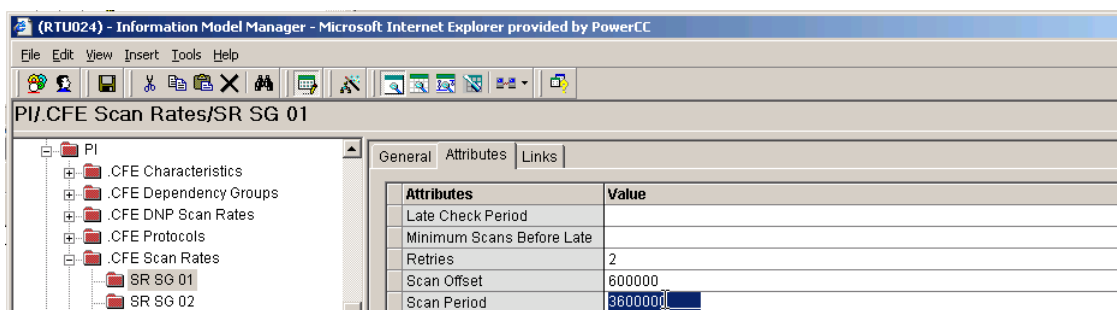


Figure 1-2.12: Overview Pane Displays the Instance Tree; in Working Area the Single Instance Editor is Opened

The parameter sets are grouped and accessible under the following tabulators:

- General
- Attributes
- Links

When you have finished your edit job you have the option either to transfer the changes to the database by the job option 'Activate' or to cancel the modification by choosing 'Erase'. If you have not completed the job and want to continue at a later date you can leave the job in edit mode and quit IMM.



Note

Instead of leaving the job in edit mode for an undetermined period of time you have also the option to export instances (including all child instances) and re-import them when ever you want to utilize them. Import / Export is a standard function of the IMM and can be accessed via the toolbar item **File**, context menu item **Import** or **Export**. For more details on Import / Export please refer to the *IMM - Reference Guide*.

1.2.4 Configuring the Process Interface

1.2.4.1 Principle

The main tasks to configure the Communication Front-End are:

1. Creating and parameterizing all necessary CFE instances such as CFE(s), COM ports, RTUs, links to RTUs as well as protocol related instances
2. Creating RTU data points as child instances of the RTU
3. Setting the protocol-specific attributes of the data points
4. Establishing links from Network (or Accounting or Generation) instances to RTU data points, see Figure 1-2.16:

Step 1 is to be performed when a new component (CFE, COM port, RTU) is added.

Steps 2 through 4 are to be performed for each data point of the RTU.



Note

It is assumed that the Network instances are already available when CFE and RTUs are being configured.

1.2.4.2 Use Case: Adding / Modifying RTU Data Points

RTU data points contain RTU specific attributes, like Object Address in monitoring and control direction as well as all CFE related processing instructions.

Each RTU data point is to be linked through the Association '(SendsTo) DiscreteValue' and/or '(ReceivesFrom) Command' to its corresponding Network (or Accounting or Generation) instance, see Figure 1-2.16

Adding a Digital RTU Data Point

To add a digital RTU data point to the CFE instance 'RTU', the instance 'Cfe Telemetered Digital' is to be created first. To do so, perform the following steps:

- 1. Select the RTU instance of interest in the Overview Pane (branch PI/CFE/CFE- RTUs). To popup the menu right-click and select **New**.

The following frame appears in the Working Pane displaying the instance types available for this area (Figure 1-2.13)

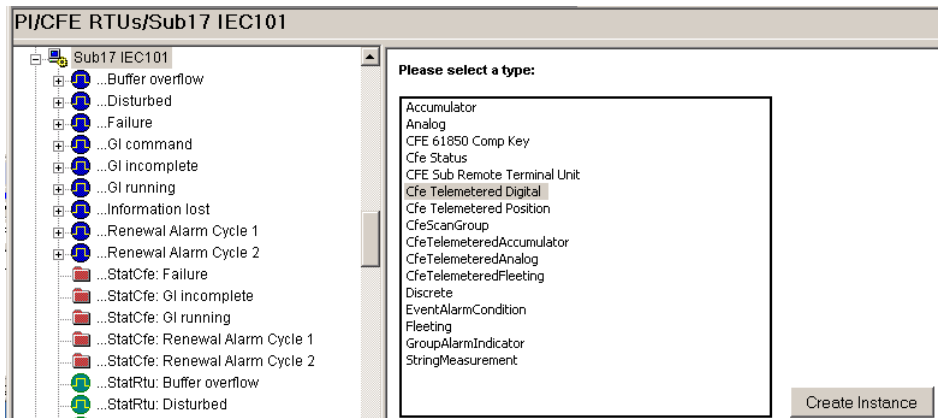


Figure 1-2.13: Selecting the Instance Type

- 2. Select *Cfe Telemetered Digital* and click **Create Instance**
- 3. The Instance Editor appears and you can enter the name and optionally a description of the new data point, see Figure 1-2.14

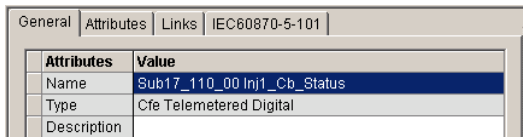


Figure 1-2.14: Naming the New Instance

- 4. Open generic tab **Attributes** to display all attributes or, alternatively, the protocol-specific tab (e.g. **IEC60870-5-101**) and edit the values (Figure 1-2.15)

Note: For protocol-specific information, such as value ranges, recommended values, interdependencies, see section "Protocol-Specific Engineering".

General Attributes Links IEC60870-5-101	
Attributes/Associations	Value
Command Qualifier	No additional definition
Command operation mode	Direct Operation
Control Direction Type	Not used
Control-TC-Address	0
Description	
Enable SOE/double transmission	False
GIRelevant	True
Intermediate State Suppression Time	0
Invert Control	False
Monitor Direction Type	Double Point
Monitor Info Conversion	NotUsed
Monitor-TC-Address	2010
Old New Check	True
Renewal check	Disabled
System Control Direction ID	Not used
System Control Extension	
(ReceivesFrom) Measurement	
Receives From (Command)	00 Inj1/Cb/Status/Command
Sends To (DiscreteValue)	00 Inj1/Cb/Status/Value

Figure 1-2.15: Editing the Attributes of the Instance Type “CfeTelemetered Digital”

- Establish the links between the measurement instances of the Network and the RTU data points, see following diagram.

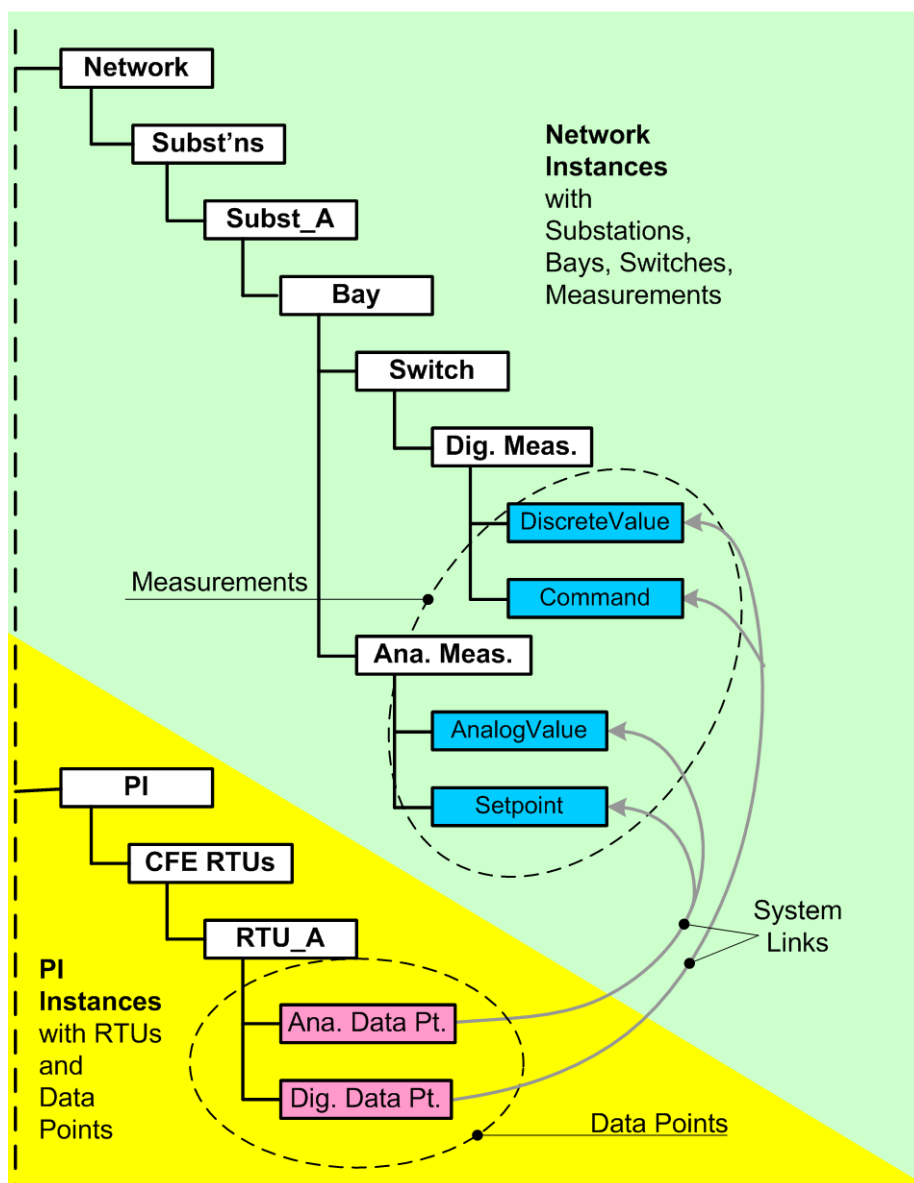


Figure 1-2.16: System Links Between RTU Data Points and Network Measurements

Open tab **Links** and enter the instance 'Network/Substation/... CB/Status/Value' into the reference field '**(SendsTo) DiscreteValue**' by drag & drop.

Hint: Please left-click the measured value from Overview Pane and hold the mouse button all the way down while dragging. Release the mouse button and "drop" it into the reference field when the cursor changes to "+", see Figure 1-2.17

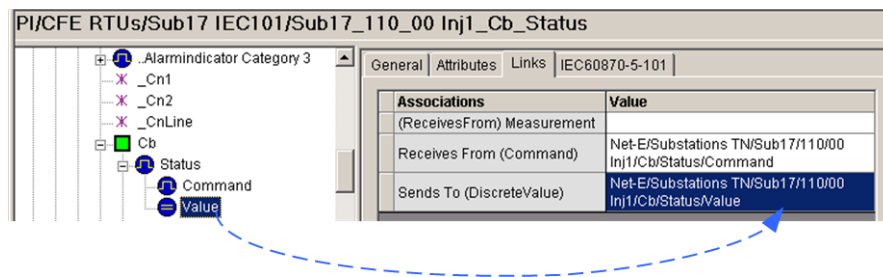


Figure 1-2.17: Creating the Link between the Network 'Discrete Value' Instance and RTU Data Point

If the Circuit Breaker (CB) is tele-controllable: also drag & drop the instance 'Network/Substation/... CB/ Status/Control' into the reference field **'(ReceivesFrom) Command'**, see Figure 1-2.18

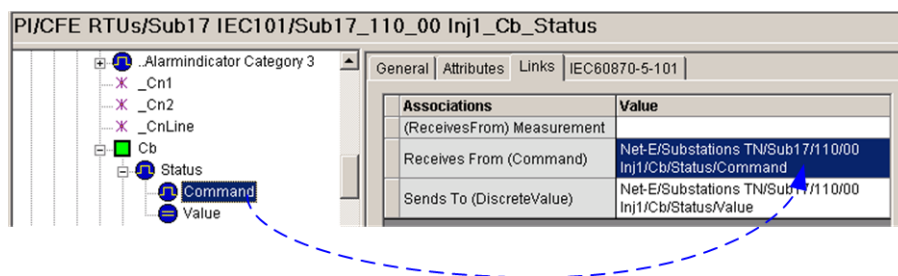


Figure 1-2.18: Linking a Network 'Command' Instance to the RTU Data Point

Duplicating Data Points

Once a data point has been created it can easily be duplicated just by copy - paste - rename. To duplicate a data point, perform the following steps:

1. Select the data point to be copied: click-right select **Copy** (Figure 1-2.19)

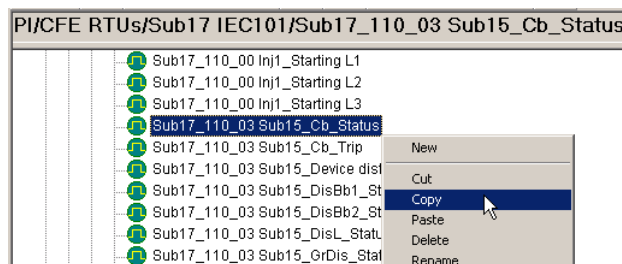


Figure 1-2.19: Copying an Instance

2. Select the parent instance under which you want to create the new instance, right-click and select **Paste** (Figure 1-2.20)

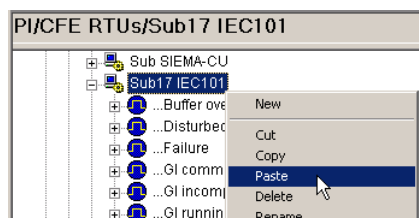


Figure 1-2.20: Pasting the Copied Instance

3. A dialog box appears: Rename the instance, click **OK** (Figure 1-2.21).

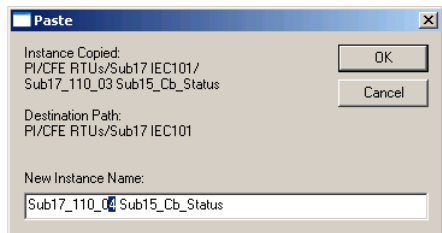


Figure 1-2.21: Renaming the Instance on Pasting

4. Check the attributes of the copied instance and adapt them if necessary (e.g. the object address in monitoring and control direction)

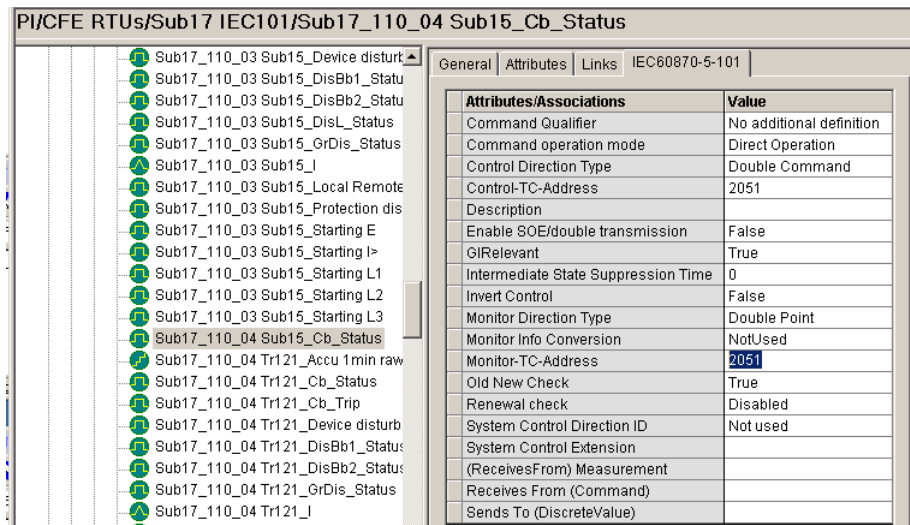


Figure 1-2.22: After Duplication: Checking and Modifying Attributes

More Information

More information and details on how to set correct attributes can be found in the Chapter "Protocol-Specific Engineering".

Volume 2 User Interface

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2.1 Introduction

The User Interface Volume introduces into CFE Status Information and describes how to use the Runtime Explorer to display this Information. The volume contains:

- Short introduction to CFE Status Information
- Using the Runtime Explorer to display CFE Status Information
- Using the Runtime Explorer to display RTU Status Information

Intended Audience

- Dispatchers / Operators / Systems Engineers
- Data Entry Technicians / Field Data Modelers
- Marketing / Sales.

2.2 CFE Presented in Runtime Explorer

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2.2.4	Runtime Explorers CFE Views	42

2.2.1 CFE /RTU Status Information - Introduction

Status data provide information about the availability of CFE components such as CFE Interface(s), link(s) to RTUs as well as for RTUs. CFE status information is derived in the CFE through periodical device monitoring routines, like Link Test. The following standard routines in CFE are applied to obtain the RTU Status Information: General Interrogation, Group Interrogation, Completeness Check, Renewal Check. Further details can also be obtained from Section 'CFE / RTU Status Information and Communication Statistics' in Chapter Engineering.

The following diagram shows an overview of CFE components and RTUs providing Status Information.

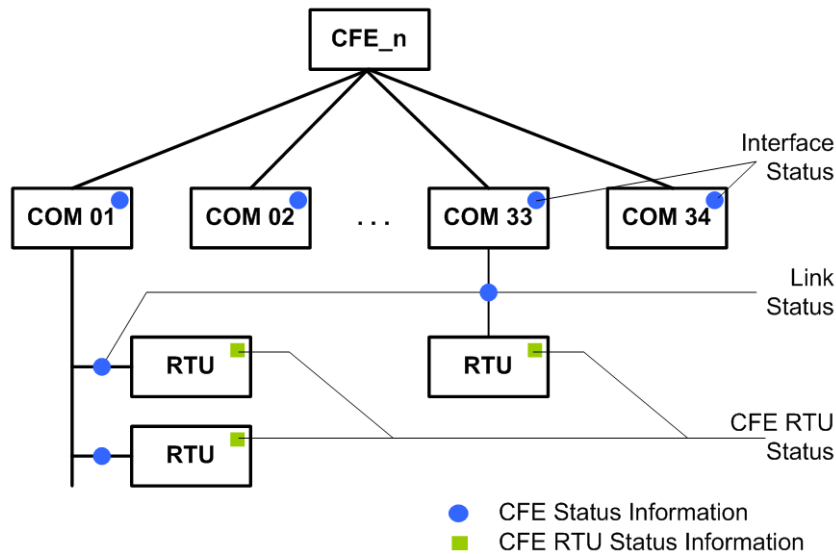


Figure 2-2.1: Overview of CFE and RTU Status Information

The Run-Time Explorer is a user interface for displaying status information including its current value of system components at a glance.

To start the Runtime Explorer, select **Tools** in the Power Application Tool Bar (PAT) and then **Runtime Explorer** in the pull-down menu.

For displaying CFE or RTU Status Information there are two "Views" available:

- Substation View, showing a comprehensive overview including RTU Status Information
- CFE View, showing selective CFE Status Information with tabs "Single", "Summary", "Failure"

You can toggle between Substation View and CFE View by means of icons in the tool bar.

2.2.2 CFE Status Information, Presented Through Runtime Explorer

If you select a particular CFE, for example CFE1, in the Overview Pane, the status of all CFE1 components are shown in the table on the right side.

The meaning of the columns is as follows:

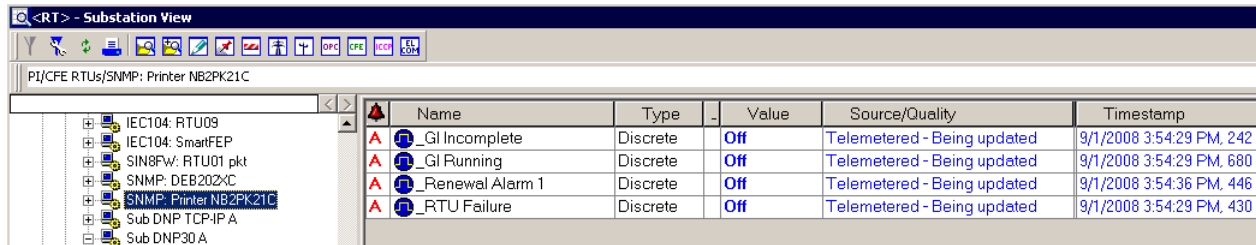
1st Column (Bell symbol):	"A" Indicates a not yet acknowledged status change
Name:	Name of the Status Info Instances
Type:	Type of the Instance (for Status Information: all are of the type <i>Digital Measurement</i>)
Value:	The current value of the Information like on, off or undefined (if not updated yet)
Source/Quality:	Indicates whether the instance is telemetered and updated
Last change:	The instant when the displayed value was most recently updated

Name	Type	Value	Source/Quality	Timestamp
Sinaut_COM05ePoll/Link03/_...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 P
Sinaut_COM05ePoll/Link03/_L...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 P
Sinaut_COM05ePoll/Link03/_...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 P
Sinaut_COM05ePoll/Link03/_...	Discrete	Undefined...	Telemetered - Not being updated	8/28/2008 2:33:03 P
TI: SNMP 2/_Active	Discrete	On	Telemetered - Being updated	8/28/2008 4:55:24 P
TI: SNMP 2/_Failure	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:24 P
TI: SNMP 2/_Locked	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:26 P
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:30 P
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP 2/Link 1 - DEB202X...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:28 PM
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:30 P
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP 2/Link 2 - NB2PK21...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP/_Active	Discrete	On	Telemetered - Being updated	8/28/2008 4:55:24 P
TI: SNMP/_Failure	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:24 P
TI: SNMP/_Locked	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP/Link 1 - DEB202XC/...	Discrete	On	Telemetered - Being updated	8/28/2008 4:55:26 P
TI: SNMP/Link 1 - DEB202XC/...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:26 P
TI: SNMP/Link 1 - DEB202XC/...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP/Link 1 - DEB202XC/...	Discrete	Off	Telemetered - Being updated	8/28/2008 4:55:23 P
TI: SNMP/Link 2 - NB2PK21C/...	Discrete	Off	Telemetered - Being updated	9/1/2008 8:39:50 AM

Figure 2-2.2: Runtime Explorer: CFE Status Information

2.2.3 RTU Status Information Presented Through Runtime Explorer

To display the RTU Status Infos select the substation instance of interest in the Overview Pane. The substation is a child instance of the instance /PI/CFE RTUs.



Name	Type	Value	Source/Quality	Timestamp
_GI Incomplete	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:29 PM, 242
_GI Running	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:29 PM, 680
_Renewal Alarm 1	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:36 PM, 446
_RTU Failure	Discrete	Off	Telemetered - Being updated	9/1/2008 3:54:29 PM, 430

Figure 2-2.3: Runtime Explorer Showing RTU Status Information

Instances are displayed in alphabetical order. To arrange Status Information items all together in top position special characters, like underscore (_) or dots (...), are often used in front of the instance name.

2.2.4 Runtime Explorers CFE Views

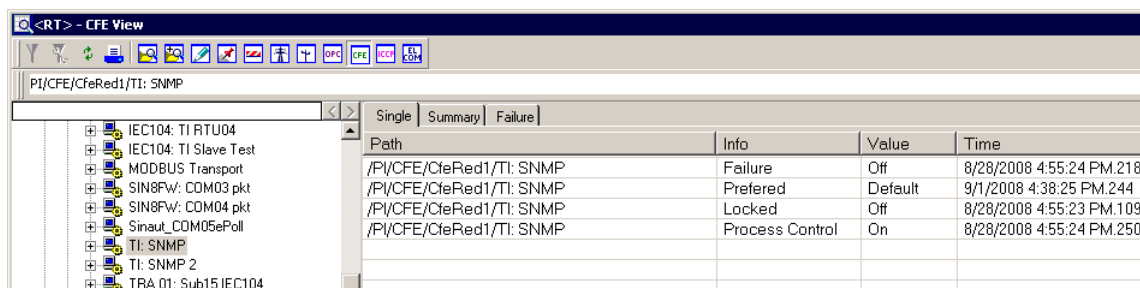
CFE Status Information can selectively be displayed in the **CFE View** of the Runtime Explorer.

The **CFE View** in the Runtime Explorer features as following:

- It displays the Status information for:
 - CFE Interfaces (Process Control, Locked, Failure, Preferred)
 - CFE Interface Links (Process Control, Locked, Failure, Preferred)
- It supports online updating
- It provides the following Views:
 - Single View
 - Summary View
 - Failure View

2.2.4.1 CFE Single View

To display the Single View select the tab **Single** from the Runtime Explorer within the CFE View.

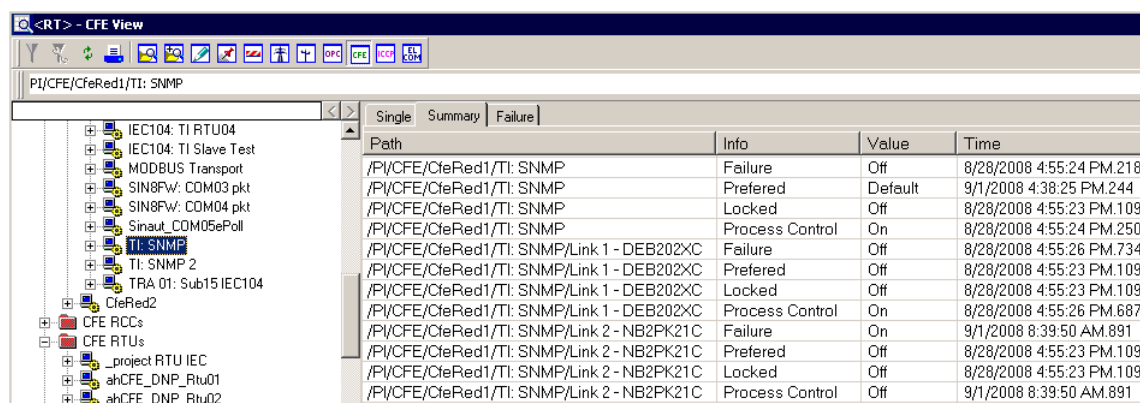


Path	Info	Value	Time
/PI/CFE/CfeRed1/TI: SNMP	Failure	Off	8/28/2008 4:55:24 PM.218
/PI/CFE/CfeRed1/TI: SNMP	Prefered	Default	9/1/2008 4:38:25 PM.244
/PI/CFE/CfeRed1/TI: SNMP	Locked	Off	8/28/2008 4:55:23 PM.109
/PI/CFE/CfeRed1/TI: SNMP	Process Control	On	8/28/2008 4:55:24 PM.250

Figure 2-2.4: Runtime Explorer: CFE Status Information, Single View

2.2.4.2 CFE Summary View

To display the Summary View select the tab **Summary** from the Runtime Explorer within the CFE View.

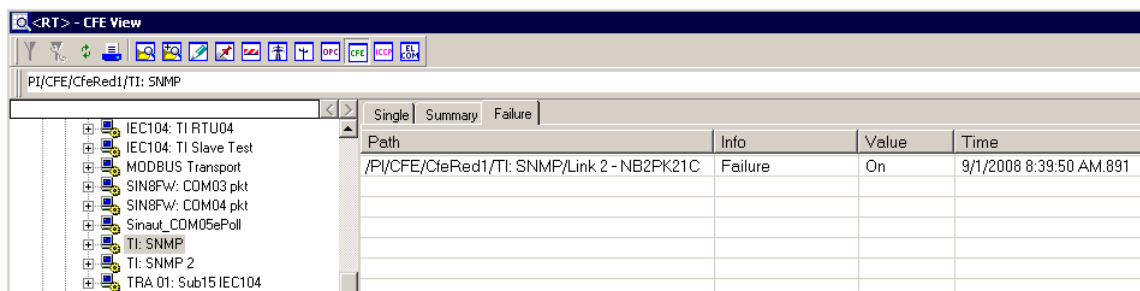


Path	Info	Value	Time
/PI/CFE/CfeRed1/TI: SNMP	Failure	Off	8/28/2008 4:55:24 PM.218
/PI/CFE/CfeRed1/TI: SNMP	Prefered	Default	9/1/2008 4:38:25 PM.244
/PI/CFE/CfeRed1/TI: SNMP	Locked	Off	8/28/2008 4:55:23 PM.109
/PI/CFE/CfeRed1/TI: SNMP	Process Control	On	8/28/2008 4:55:24 PM.250
/PI/CFE/CfeRed1/TI: SNMP/Link 1 - DEB202XC	Failure	Off	8/28/2008 4:55:26 PM.734
/PI/CFE/CfeRed1/TI: SNMP/Link 1 - DEB202XC	Prefered	Off	8/28/2008 4:55:23 PM.109
/PI/CFE/CfeRed1/TI: SNMP/Link 1 - DEB202XC	Locked	Off	8/28/2008 4:55:23 PM.109
/PI/CFE/CfeRed1/TI: SNMP/Link 1 - DEB202XC	Process Control	On	8/28/2008 4:55:26 PM.687
/PI/CFE/CfeRed1/TI: SNMP/Link 2 - NB2PK21C	Failure	On	9/1/2008 8:39:50 AM.891
/PI/CFE/CfeRed1/TI: SNMP/Link 2 - NB2PK21C	Prefered	Off	8/28/2008 4:55:23 PM.109
/PI/CFE/CfeRed1/TI: SNMP/Link 2 - NB2PK21C	Locked	Off	8/28/2008 4:55:23 PM.109
/PI/CFE/CfeRed1/TI: SNMP/Link 2 - NB2PK21C	Process Control	Off	9/1/2008 8:39:50 AM.891

Figure 2-2.5: Runtime Explorer: CFE Status Information, Summary View

2.2.4.3 CFE Failure View

To display the Failure View select the tab **Failure** from the Runtime Explorer within the CFE View.



Path	Info	Value	Time
/PI/CFE/CfeRed1/TI: SNMP/Link 2 - NB2PK21C	Failure	On	9/1/2008 8:39:50 AM.891

Figure 2-2.6: Runtime Explorer: CFE Status Information, Failure View

Volume 3 Information Model Management

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3.1 Introduction

The Engineering Volume describes procedures for customizing modules. This includes parameterization and software settings (configurable time-outs, limits, etc.) as well as data entry and information not covered in the Installation Guide.

Content

Applying Single View Editor: It is used to create or modify configuration data of the following CFE components:

- Server (CFE)
- COM port
- TCP/IP interface
- Serial communication link
- Network communication links
- RTU
- Data points of the RTU
- Characteristic Curves for analog measurements
- CFE Status information

Applying tabular-based editors:

- Communication Editor
- Table Editor

Data Import / Export

Remote Control Center

Single View Editor, Communication Editor and Table Editor are also used to create or modify configuration data of:

- Remote Control Center (RCC)
- Logical RTUs for RCC
- Data points to be forwarded to RCC(s)
- Communication links from CFE to RCC(s)
- Status information for RCC, associated interfaces and links

Intended Audience

- Service and Systems Engineers
- System Configuration Technicians
- System Support Users

3.2 Configuration Interface

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The configuration Interface IMM (Information Model Manager) presents instances and their attributes in an Explorer based user interface and allows to carry out modifications in the data model in a convenient way. Data can be displayed and edited in different types of editors, i.e.

- Single View Editor
- Tabular-Based Editors, such as
 - Communication Editor
 - Table Editor

3.2.1 Single View Editor

Creating new instances as well as modifying attributes of a particular instance can best be done using the Single View Editor within the IMM. However, to see the data of all child instances of a parent instance at a glance, the Table Editor may be preferred (see Chapter 3.2.2 *Tabular-Based Editors*)

3.2.1.1 Overview of CFE Instances

The following chart shows the hierarchical arrangement of typical CFE instances in the PI branch.

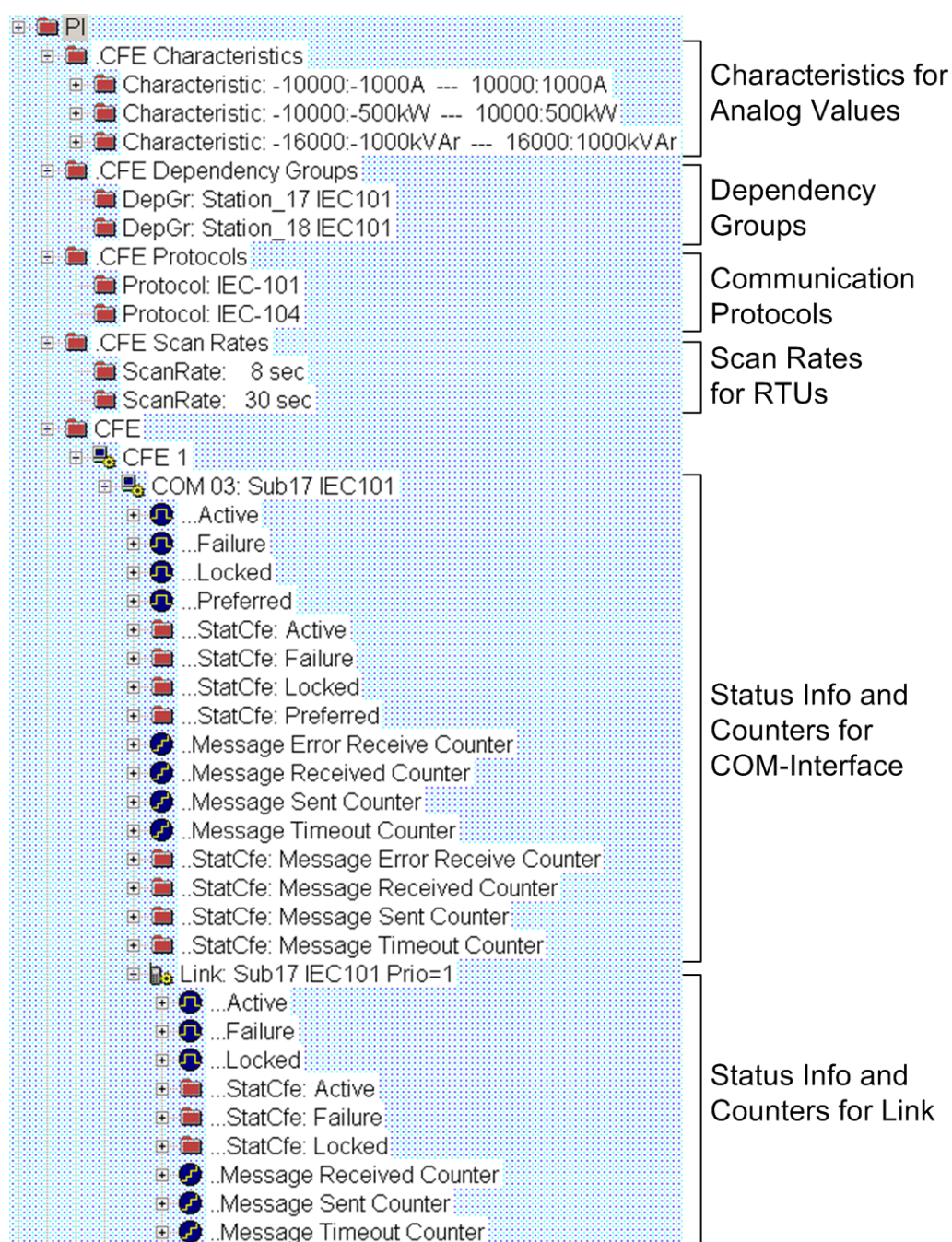


Figure 3-2.1: Overview of CFE Instances

3.2.1.2 Creating CFE Instances

The creation of instances for the components CFE Server, CFE Interface, Link to RTU, RTU, RTU Data Points, characteristic curves etc. is described in detail in the following sections.

3.2.1.2.1 Introduction

The specific characteristics and the behavior of CFE components, RTUs, communication links, RTU data points and so on must be known in PowerCC. As already shown, the PowerCC data model is made-up of instances which are organized in a hierarchical structure. These instances are the carrier of the various attributes (parameters) of the component. When a new component is to be added to the system, an instance is to be created accordingly, then the attributes are to be defined in the instance editor. Some of the instances have also links to other instances in order to use a common definition for more than one component. For example: The instance 'Characteristic Curve' may be specified just once and linked to all analog measurements having the same relationship between raw value and engineering value.

In general, the higher the rank of an instance, the less likely will be the need to create a new one. New CFE servers, for instance, are only needed if the system capacity is drastically expanded or if a non-redundant system is upgraded to a redundant one. The lower ranking instances like COM ports and RTUs, however, may more often be subject to expanding or modifying.



Note

If an instance of the desired type is already available in the system, it is easier just to duplicate an existing instance by copy / paste / rename than creating a new one. After duplicating an existing instance, you will have to check the attributes and modify them according to the specific requirements.

For some attributes it is mandatory to modify them upon copying (such as server name, COM port number or similar). Please note that different communication protocols may have different sets of attributes. For protocol-specific information on communication parameters see section "Protocol-Specific Engineering."

3.2.1.2.2 Establishing a CFE Server

To establish a new CFE server, an instance of the type *Cfe Server* is to be created and its attributes to be set. This is done performing the following steps:

1. Right-click on the parent instance 'CFE' in the Overview Pane and select **New** (Figure 3-2.2) from pop-up menu that appears (or click the "New Instance" item of the Insert menu called from the Tool bar instead).

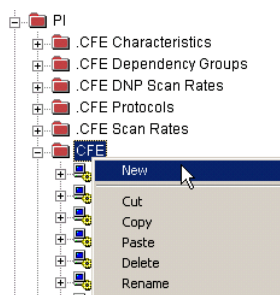


Figure 3-2.2: Creating a New CFE Server

The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area. In case of a CFE server there is only one item that can be selected.

2. Highlight *CfeServer* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new CFE unit, for example "CFE 2", "CFE 3", "CFE 4" or similar.
4. Select tab **Attributes** and fill the values (Figure 3-2.3)

Attributes	Value
Add Accumulator To Telegram Buffer	True
Add Analog To Telegram Buffer	True
Add Digital To Telegram Buffer	True
Add Fleeting To Telegram Buffer	True
Add Position To Telegram Buffer	True
Alias Name	
Area Of Responsibility Id	System
Host Name	DEB204GC
Local Name	
Message Configuration	Default
Model Resource ID	
Standby Buffer Time Span	10

Figure 3-2.3: Attributes of the Instance Type 'Cfe Server'

For more information see following table

Table 3-2.1: Attributes of the instance *CfeServer*

Attribute Label	Meaning /Comment	Default Values
Add Accumulator To Telegram Buffer	Specifying whether this type of data is to be stored in the Stand-by Buffer or not	--
Add Analog To Telegram Buffer		--
Add Digital To Telegram Buffer		--
Add Fleeting To Telegram Buffer		--
Add Position To Telegram Buffer		--
Alias Name	Optional text	--
Area Of Responsibility	System identification of an Area of Responsibility, defines the rights to access this component	Area A
Host Name	CFE server host name	--
Local Name	Human readable name of the object (optional). Only used if objects are organized in naming hierarchy	
Message Configuration	Relates the Field Resource to a message configuration for alarming	--
Standby Buffer Time Span	Time span to hold messages in the Standby Buffer. Messages older than Time Span are cleared	



Note

The attribute 'Host Name' must always be defined. It must be unambiguous in the network.

3.2.1.2.3 Establishing a Serial CFE Interface

To establish a new serial (COM) interface (a sub-component of CFE server) perform the following steps:

1. Select the relevant CFE server instance in the Overview Pane, e.g. "CFE 1" or "CFE 2". To popup the menu right-click and select **New**.

The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area.

2. Select the interface type *CfeSingleCircuitExtendedInterface* and click the button **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new interface instance, for example "COM 03", "COM 04", "COM 05", ...
4. Open tab **Links** and establish a system link to the protocol instance, e.g., 'Protocol:IEC-101' (instance type: *Cfe Protocol IEC101*).

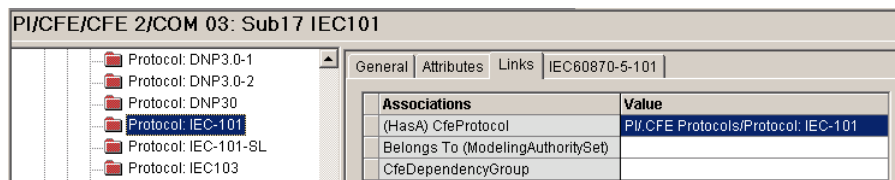


Figure 3-2.4: Linking the Protocol (Instance) to the CFE Interface (by Drag & Drop)

Note: After linking the interface instance to a protocol instance and saving the modification, in a addition to the generic 'Attribute' tab, a protocol-specific tab, showing exclusively protocol relevant attributes, will be provided.

5. Select tab **Attributes** or, which is more convenient, the protocol-specific tab and adjust the parameters. Figure 3-2.5 shows the tab 'Attributes'.

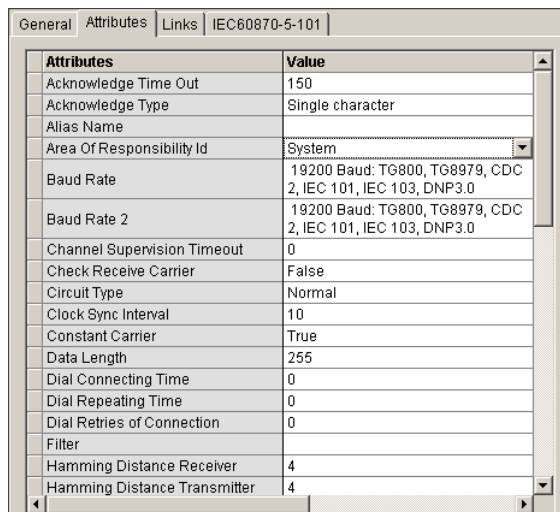


Figure 3-2.5: Attributes of the Instance Type 'CfeSingleCircuitExtendedInterface' (Extract)

Figure 3-2.6 shows the protocol-specific tab 'IEC60870-5-101' for the instance type '*CfeSingleCircuitExtendedInterface*'

General Attributes Links IEC60870-5-101	
Attributes	Value
Acknowledge Time Out	150
Acknowledge Type	Single character
Alias Name	
Area Of Responsibility Id	System
Baud Rate	19200 Baud: TG800, TG8979, CDC 2, IEC 101, IE
Constant Carrier	True
Data Length	255
Description	
Listen Mode	False
Message Configuration	Default
Polling Time	0
Postconditioning Delay	0
Preconditioning Delay	0
Serial Port	3
Transmission Mode	Balanced

Figure 3-2.6: IEC101-Specific Attributes of the Instance Type '*CfeSingleCircuitExtendedInterface*'



Note

For protocol-specific information such as value ranges, recommended values, interdependencies, see chapter "Protocol-Specific Engineering".

3.2.1.2.4 Establishing a CFE Transport Interface

Introduction

The instance type *CfeTransportInterface* is used to establish connections for TCP/IP based network protocols, like IEC6870-5-104 or DNP3.0 over TCP/IP.

To create a CFE Transport Interface perform these steps:

1. In the Overview Pane right-click the CFE server instance and select **New** from popup menu that appears.
2. The 'Type Selection Box' appears in the Work Pane showing all instance types available for this area. Select *CfeTransportInterface* and click **Create Instance and name it**.
3. Select tab **Attributes** or the protocol-specific tab and enter suitable attribute values.

General Attributes Links IEC60870-5-104	
Attributes	Value
Alias Name	
Area Of Responsibility Id	System
Cfe IP Address	157.163.53.222
Description	
Max Pending GIs	5
Message Configuration	Default

Figure 3-2.7: Attributes of the Instance Type '*CfeTransportInterface*'

NOTE: For protocol-specific information such as value ranges, recommended values, interdependencies, see chapter "Protocol-Specific Engineering"

4. Open tab **Links** and allocate a network protocol, such as 'Protocol IEC-104' to the Association (*HasA*) *Cfe Protocol* as already demonstrated by Figure 3-2.4

3.2.1.2.5 Global Protocol Parameters

Each protocol implemented in CFE has a global parameter set. These parameters are common for all links to RTUs applying this protocol. They are located within the instance tree beneath the parent instance 'CFE Protocols'.

General Attributes Links		
Attributes	Value	
Common Address Bytes	Two octets	
Communication Statistic Offset	3000	
Communication Statistic Period	60000	
Info Object Address Bytes	Three octets	
Link Address Bytes	One link address octet.	
Originator Address Bytes	No originator address	

Figure 3-2.8: Global Attributes of the Instance Type "Cfe Protocol IEC 101"



Note

If you need more than one set of global parameters for a protocol (e.g., for different types of RTUs) you may create a further version of global parameters by copy / paste / rename / modify.

The global attributes of the instance 'Protocol IEC101' are shown below.

3.2.1.2.6 Establishing Serial Communication Links to the RTUs

Introduction

There are different basic concepts to connect RTUs to CFE, see the following diagram

Type A:	Single ported RTUs with one communication path in point-to-point configuration
Type B:	Dual ported RTUs with two independent communication paths in point-to-point configuration
Type C:	Single ported RTUs with one communication path in party line configuration
Type D:	Dual ported RTUs with two independent communication paths in party line configuration

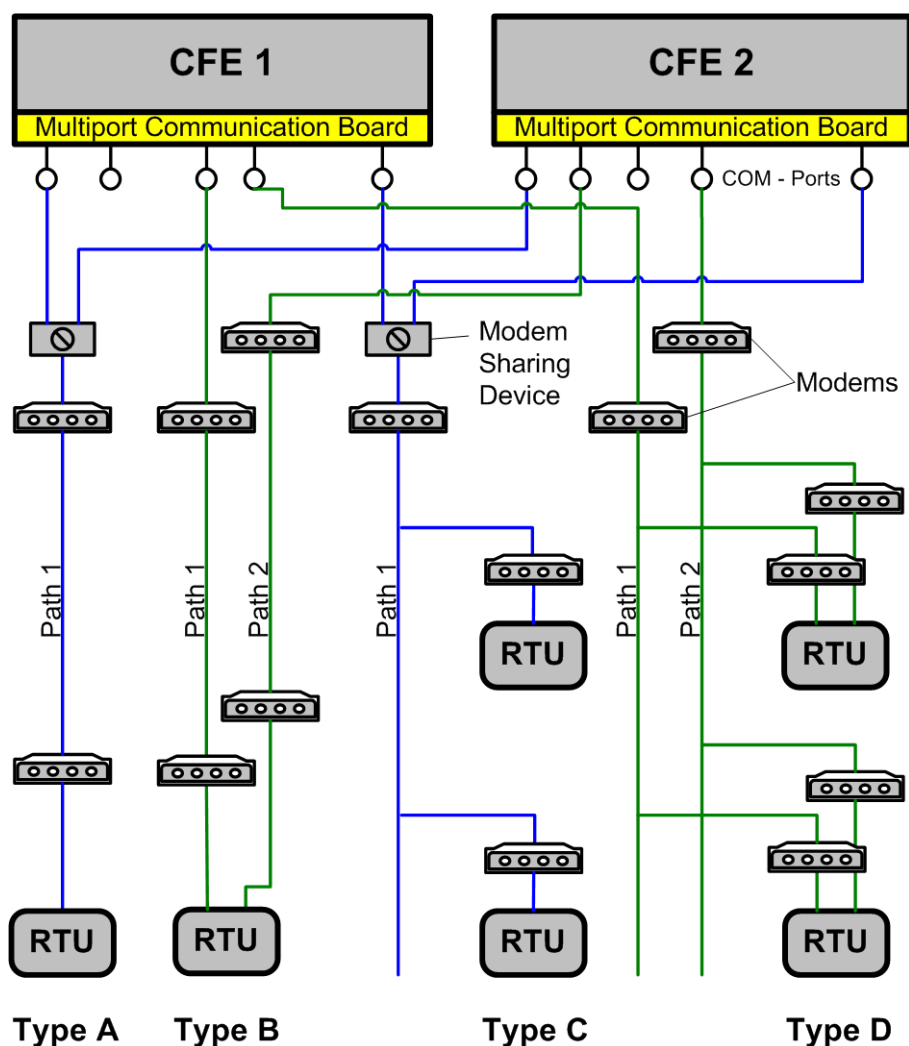


Figure 3-2.9: Basic Concepts to Connect RTUs to CFE

**Note**

In addition, for standard protocols IEC101 and IEC104, "enhanced" connection schemes can be configured, such as an independent pair of dependent links, see Chapter 3.2.4 *Enhanced Redundancy for Protocols IEC101 and IEC104*

Serial Links to RTU(s)

The principle of connecting different types of RTUs as displayed in the previous diagram is as follows:

- Single ported RTUs:

In case of CFE redundancy single ported RTUs are connected to both CFE servers. To de-couple the transmitter signals, modem sharing devices are used, one for each channel.

In non-redundant Front-End Systems each RTU occupies one serial interface at the CFE. Modem sharing devices are not needed.

- Dual ported RTUs:
 - Applicable for all protocol types: In case of CFE redundancy dual ported RTUs are connected as shown in the above Figure. One independent communication path is connected to CFE 1 the other one is connected to CFE 2
 - Applicable for IEC101 protocol: RTUs can also be connected via two pairs of dependent links, see Chapter 3.2.4 *Enhanced Redundancy for Protocols IEC101 and IEC104*

**Note**

To configure redundant CFEs the best way is to configure first CFE 1 and then duplicate CFE 1 to CFE 2. In this way all COM-ports, associated status information etc. are duplicated. After duplication the following attributes are to be checked and adapted accordingly:

- CFE Attributes: change server name
- CFE-RTU: change links of all duplicated COM ports from CFE 1 to CFE 2

Establishing a Serial Link to an RTU

To establish a new link to a CFE-RTU perform the following steps:

1. Right-click on the COM port instance of interest in the Overview Pane and select **New** from the popup menu that appears.
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area.
2. Select instance type *Cfe Interface Link* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Enter the name of the new link, for example "Link to RTU-xxx"
4. Select tab **Attributes** or the protocol specific tab and fill the values (Figure 3-2.10).

General Attributes Links IEC60870-5-101	
Attributes	Value
Alias Name	
Area Of Responsibility Id	System
Calculate Channel Delay	False
Clock synchronization interval	60
Data Link Retry	0
Data Link Timeout	5000
Dial Max Connect Time	0
General Interrogation Cycle Time	0
Link Address	200
Link Confirmation	False
Link Test Cycle Time	5000
Local Name	
Message Configuration	Default

Figure 3-2.10: Attributes of the Instance Type 'CFE Interface Link' (Extract)

**Note**

The tab **'Attributes'** contains the communication parameters for all CFE protocols in alphabetical order. For a better usability, tabs labeled with protocol names only contain subsets of protocol specific parameters.

The next figure shows an example of protocol specific communication attributes for protocol IEC60870-5-101.

General Attributes Links IEC60870-5-101	
Attributes	Value
Alias Name	
Area Of Responsibility Id	System
Clock synchronization interval	60
Description	
General Interrogation Cycle Time	0
Link Address	200
Link Test Cycle Time	5000
Message Configuration	Default
Polling Time	0
Polling Timeout	0
Priority	2
Successive Errors For RTU Fail	3

Figure 3-2.11: Protocol-Specific Attributes of the Instance Type 'Cfe Interface Link' for IEC101



Note

For protocol-specific information such as value ranges, recommended values, interdependencies, see chapter "Protocol-Specific Engineering".

3.2.1.2.7 Establishing a Dependency Group

One kind of redundant connections between CFE and RTU is of the type "dependent". In a dependent interface, you can create a group of redundant links each leading to a single-ported RTU. Such a group of links is called a "Dependency Group".

For better understanding of this matter please refer to Chapter 3.2.3.2 *Dependent Links*.

To create a Dependency Group (a child instance of '.CFE Dependency Groups') perform the following steps:

1. Right-click the instance '.CFE Dependency Groups' in the Overview Pane and select **New** from menu that pops-up.
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *CfeDependencyGroup* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new Dependency Group, for example "DepGrp RTU-xxx"
4. Select the COM-port instance to which the dependency group should belong (here: CFE 2, COM 03) and open tab **Links**. Enter the instance path of the dependency group into the reference box **CfeDependencyGroup** by drag&drop

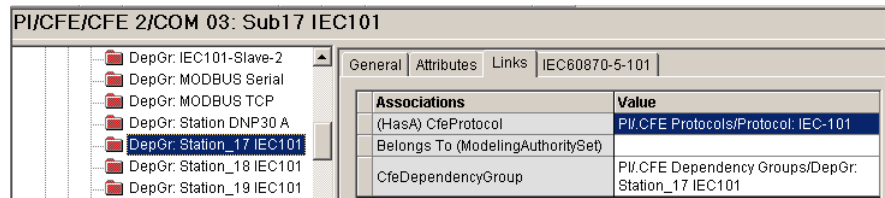


Figure 3-2.12: Establishing a System Link from Dependency Group to the Interface

3.2.1.3 Creating RTU Instances

General

The instance 'RTU' defines the device which is connected to the CFE. RTU data points are defined as child instances of the RTU.

3.2.1.3.1 Specifying an RTU

To specify an RTU - an instance of the type *Cfe Remote Terminal Unit* - perform the following steps:

1. Right-click the parent instance 'CFE RTUs' in the Overview Pane and select **New** from menu that pops-up
The 'Type Selection Box' appears in the Work Pane showing available instance types
2. Select *Cfe Remote Terminal Unit* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new RTU instance like 'RTU_nnn' or <Name of RTU location> (or whatever you consider as meaningful)
4. Select tab **Cfe Remote Terminal Unit View** and select the protocol of interest by changing the attribute 'Telecontrol Protocol Type' (Figure 3-2.13)

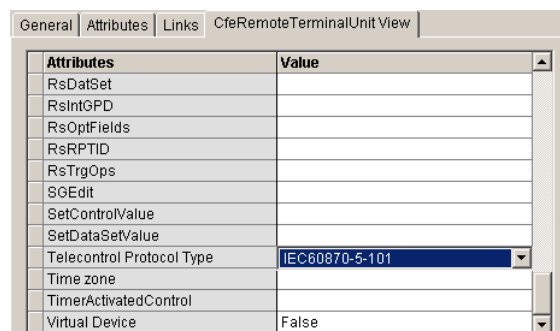


Figure 3-2.13: Attributes of the Instance Type "Cfe Remote Terminal Unit"

You may fill further attributes or, which is more convenient, use the protocol-specific tab. To get the protocol-specific tab, save and re-open the editor. Then, the tab 'Cfe Remote Terminal Unit View' is replaced with a protocol-specific tab presenting exclusively attributes required for the selected protocol (Figure 3-2.14).

General Attributes Links IEC60870-5-101	
Attributes	Value
Alias Name	
Application timeout	5000
Area Of Responsibility Id	Area A
Command Buffer Time Maximum	5000
Common Address	75
Data Point Address Structure	16 bit
Description	
Message Configuration	Default
Renewal cycle 1	0
Renewal cycle 2	0
Telecontrol Protocol Type	IEC60870-5-101

Figure 3-2.14: Protocol-Specific RTU Attributes

- Open tab **Links** and enter the path of the associated link instance (which was created in Chapter 3.2.1.2.6 *Establishing Serial Communication Links to the RTUs*) into the reference box **(ConnectsThrough) CfeInterfaceLink** of the RTU instance (using drag&drop)

PI/CFE RTUs/Sub17 IEC101													
..Message Sent Counter ..Message Timeout Counter ..StatCfe: Message Error Receive C ..StatCfe: Message Received Cour ..StatCfe: Message Sent Counter ..StatCfe: Message Timeout Count - Link: Sub17 IEC101 Prio=1 - COM 04: Sub18 IEC101 - COM 05: Sub19 IEC101	<table> <tr> <th colspan="2">General Attributes Links IEC60870-5-101</th></tr> <tr> <th>Associations</th><th>Value</th></tr> <tr> <td>(ConnectsThrough) CfeInterfaceLink</td><td>PI/CFE/CfeRed2/COM 03: Sub17 IEC101/Link: Sub17 IEC101 Prio=2</td></tr> <tr> <td></td><td>PI/CFE/CfeRed1/COM 01: Sub17 IEC101/Link: Sub17 IEC101 Prio=1</td></tr> <tr> <td>Belongs To (ModelingAuthoritySet)</td><td></td></tr> <tr> <td>Has A (CfeFiletransferDefs)</td><td></td></tr> </table>	General Attributes Links IEC60870-5-101		Associations	Value	(ConnectsThrough) CfeInterfaceLink	PI/CFE/CfeRed2/COM 03: Sub17 IEC101/Link: Sub17 IEC101 Prio=2		PI/CFE/CfeRed1/COM 01: Sub17 IEC101/Link: Sub17 IEC101 Prio=1	Belongs To (ModelingAuthoritySet)		Has A (CfeFiletransferDefs)	
General Attributes Links IEC60870-5-101													
Associations	Value												
(ConnectsThrough) CfeInterfaceLink	PI/CFE/CfeRed2/COM 03: Sub17 IEC101/Link: Sub17 IEC101 Prio=2												
	PI/CFE/CfeRed1/COM 01: Sub17 IEC101/Link: Sub17 IEC101 Prio=1												
Belongs To (ModelingAuthoritySet)													
Has A (CfeFiletransferDefs)													

Figure 3-2.15: Establishing System Link(s) from CFE Interface(s) to RTU

3.2.1.3.2 Creating a Data Concentrator

Data Concentrator within Data Model

For the definition of instances for the Data Concentrator (DACO) and the associated Sub-RTUs, you have to use specific instance types provided by the IMM (see here below).

- For Data Concentrators: the same type as for the RTUs (i.e. *Cfe Remote Terminal Unit*)
- For Sub-RTUs: a specific type of RTU (i.e. *CFE Sub Remote Terminal Unit*)

For examples of the DACO and Sub-RTU definitions in IMM, see the following section.

Examples

The following figure shows an example with an IEC60870-5-101 RTU (RTU 1) and an IEC60870-5-104 Data Concentrator (RTU 4 DACO). The Data Concentrator is linked to link 2 of the Transport Interface of CFE 1. The Sub-RTUs do not have direct links to CFE, they are child instances of the Data Concentrator.

The following figure shows the DACO setup in IMM (Example with IEC101 RTU and IEC104 DACO)

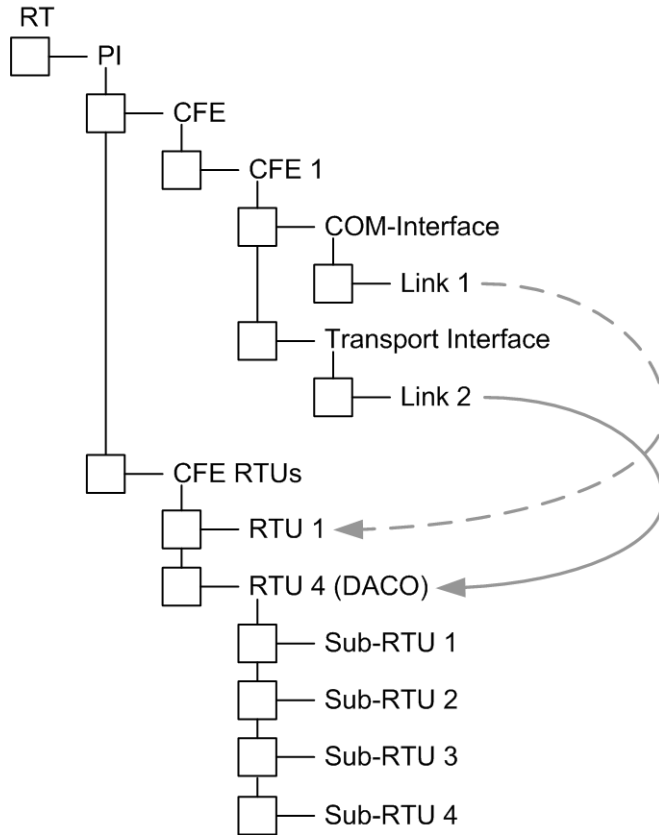


Figure 3-2.16: DACO Setup in IMM (Example of IEC101 RTU and IEC104 DACO)

The following Figure shows a similar example, but with a redundant Data Concentrator. Note that there are two CFE Transport Interfaces for the same Data Concentrator.

This figure shows the configuration of a Redundant Data Concentrator in IMM (Example)

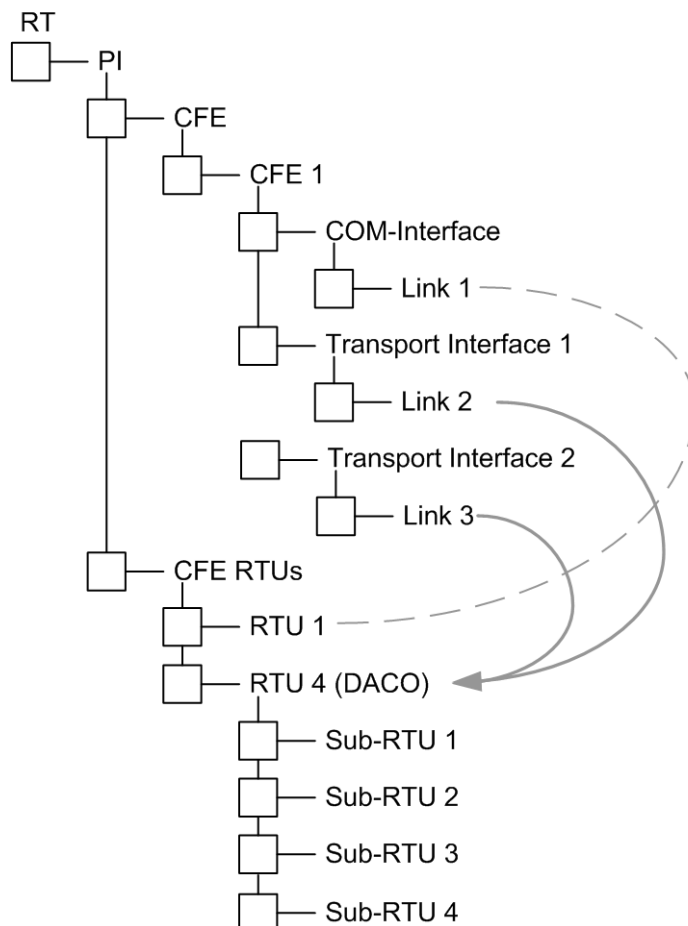


Figure 3-2.17: Configuration of a Redundant Data Concentrator in IMM (Example)

The following section describes how to:

- Create a Data Concentrator (DACO) in IMM
- Create a Sub-RTU (a child of DACO) in IMM.

Creating a Data Concentrator

Since a Data Concentrator does not differ from an ordinary RTU in the data model, it is defined in the same way. Please refer to previous Chapter 3.2.1.3.1 *Specifying an RTU*.

Creating a Data Concentrator instance in short:

1. Right-click on the instance 'CFE RTUs' in the instance tree and select **New** from appearing menu
2. Select *Cfe Remote Terminal Unit* and click **Create Instance**. The Instance Editor is opened
3. Enter the name for the new DACO-RTU into the Name row (e.g. 'CFE Data Concentrator') as well as its optional description in the Description row
4. Specify the required attributes for the newly created DACO-RTU within the **Attributes** tab. For a Data Concentrator it is mandatory to select the attribute Telecontrol Protocol Type: IEC60870-5-104

5. After saving and re-opening the editor you may take care on the following attributes in the protocol-specific tab:
 - a. Common Address: Specify the RTU identification address
 - b. Data Point Address Structure: 3 octets object address is mandatory, select '24 bit'
 - c. Renewal Cycle 1 and 2: If applicable, specify the updating check cycle [ms]

Creating a Sub-RTU

You can create a Sub-RTU instance to any RTU communicating with IEC60870-5-104 protocol (as just setup).

To create a Sub-RTU:

1. Right-click the Data Concentrator instance of interest in the instance tree and select **New** from appearing menu. As a result, the 'Type Selection Box' is opened in the Work Pane.
2. Select *Cfe Sub Remote Terminal Unit* and click **Create Instance**. As a result, the Instance Editor displays the **General** tab where you can enter name and description text for the new Sub-RTU, e.g. 'Sub-RTU 1'
3. Select the **Attribute** tab and define the attribute values for the Sub-RTU. The Sub-RTU is defined similarly as an ordinary RTU, however, consider the following restrictions for this type of RTU:
 - a. A Sub-RTU has no link to CFE
 - b. Its parent instance must be a Data Concentrator
 - c. Telecontrol Protocol Type IEC60870-5-104 to be chosen

3.2.1.3.3 Adding / Modifying RTU Data Points - General

RTU data points are instances that contain RTU specific attributes like Object Address in monitoring and control direction as well as processing instructions for CFE.

Each RTU data point is to be linked to its corresponding Network measurement instance, see also Section "Configuring the Process Interface" in the Chapter "Tutorial".

3.2.1.3.4 Adding a Digital Data Point

To add a digital data point, first create an instance of type *Cfe Telemetered Digital* by performing the following steps:

1. Right-click the RTU instance of interest in the Overview Pane and select **New** from menu that pops-up.
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *Cfe Telemetered Digital* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new data point.
4. Select the protocol-specific tab and fill the values

NOTE: The attributes required for the definition of RTU data points depend on the protocol. For protocol-specific information, such as value ranges, recommended values, interdependencies, see chapter "Protocol-Specific Engineering".

- Establish system links from Network instance to the RTU data point: Open tab **Links** and drag the Network instance 'Value' (instance type *DiscreteValue*) to the reference field (**SendsTo**) **DiscreteValue**, see next figure

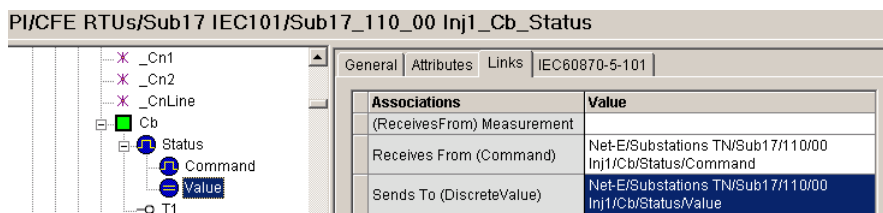


Figure 3-2.18: Linking the Network Instance 'Value' to the RTU Data Point

If the data point is tele-controllable: drag the Network instance 'Command' (Instance type *Command*) to the reference field '**ReceivesFrom (Command)**'.

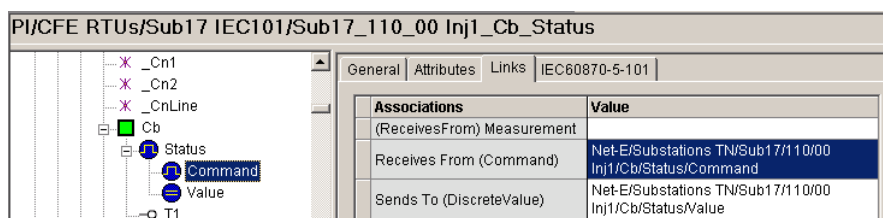


Figure 3-2.19: Linking the Network Instance 'Command' to the RTU Data Point

3.2.1.3.5 Adding A Fleeting Data Point

To add a fleeting data point perform corresponding steps as described for digital data points:

- Create an instance of type *Cfe TelemeteredFleeting* and name it
- Fill the generic tab **Attributes** or select the protocol specific tab to fill, instead
- Open tab **Links** and establish a link from corresponding Network instance to the RTU data point

3.2.1.3.6 Adding an Accumulator Data Point

To add an accumulator data point perform corresponding steps as described for digital data points:

- Create an instance of type *Cfe TelemeteredAccumulator* and name it
- Fill the generic tab **Attributes** or select the protocol specific tab to fill instead
- Open tab **Links** and establish a link from corresponding Network instance to the RTU data point



Note

The attributes for accumulator data points are a subset of type *Cfe Telemetered Digital*. For protocol-specific information, such as value ranges, recommended values, interdependencies, see chapter "Protocol Specific Engineering".

3.2.1.3.7 Adding A Step Position Data Point

To add a data point for transformer tap positions perform corresponding steps as described for other data types described before:

1. Create an instance of type *Cfe Telemetered Position* and name it
2. Fill the generic tab **Attributes** or select a protocol specific tab to fill, instead
3. Open tab **Links** and establish a link from corresponding Network instance to the RTU data point

3.2.1.3.8 Adding an Analog Value

To add an analog data point the instance *Cfe Telemetered Analog* is to be created as a child instance of the RTU instance of interest. Perform the following steps:

1. Select the 'RTU' instance of interest in the Overview Pane. To popup the menu right-click and select **New**.
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *CfeTelemeteredAnalog* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new data point
4. Select tab **Attributes** or the protocol-specific tab and fill the values
5. Depending on the type of analog value you may establish a link between the instance of the type *CfeTelemeteredAnalog* and the measured value characteristic: drag a suitable characteristic instance into the reference field of the data point (Figure 3-2.20)

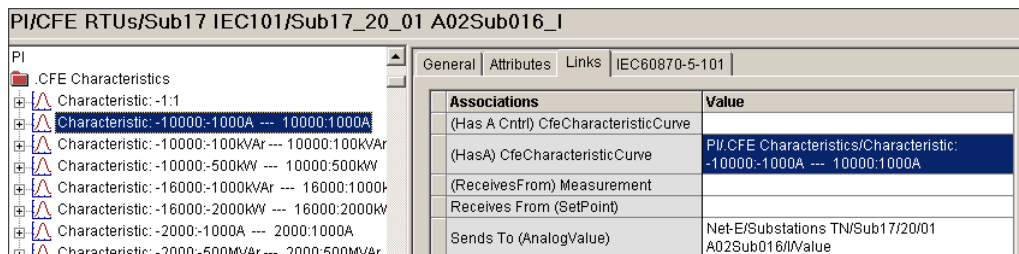


Figure 3-2.20: Linking the Characteristic Curve to the Analog RTU Data Point

For creating / modifying characteristics, refer to Chapter 3.2.1.4 *Characteristic Curves*

6. Establish a second link from Network measurement instance to the RTU data point:

If it is a measured value (only monitoring direction):

Open tab **Links** and drag the Network instance of the type *AnalogValue* into the reference field '**SendsTo (AnalogValue)**'.

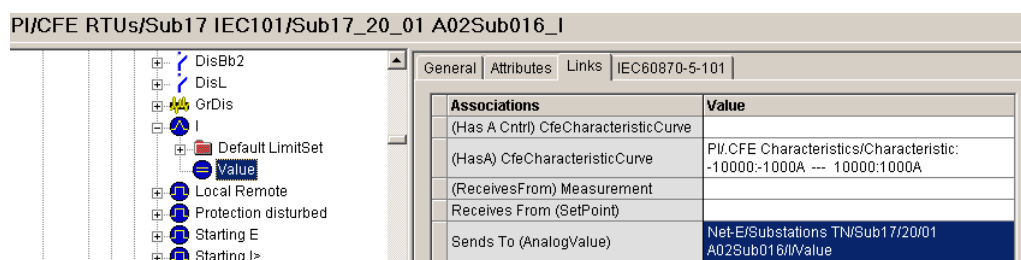


Figure 3-2.21: Linking the Network Instance 'Value' to the RTU Data Point

If it is a set-point (analog value in control direction):

Open tab **Links** and drag the Network instance of the type *Setpoint* into the reference field '**Receives-From (Setpoint)**'

3.2.1.3.9 Configuring Periodic Scans for an RTU

Scan Rate

To configure periodic scans for an RTU, an instance of the type *CfeScanRate* has to be created (as a child instance of 'CFE Scan Rates' that is type *sysCfeConfig*). To do this, perform the following steps:

1. Select the parent instance 'CFE Scan Rates' (type *sysCfeConfig*) in the Overview Pane. To popup the menu right-click and select **New**
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *CfeScanRate* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, e.g. 'Scan Rate 3 sec'
4. Select tab **Attributes** and check / adapt the values

The most important attributes of the Scan Rate instance are Scan Period and Scan Offset

Scan Period Defines the periodicity of scans in milliseconds; granularity is 1 second
Scan Offset Defines start delay related to midnight in milliseconds (implemented for a few protocols)

Scan Group

To create an instance *Scan Group* perform the following steps:

1. Select the RTU instance of interest (instance type *Cfe Remote Terminal Unit*) in the Overview Pane. To popup the menu right-click and select **New**
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select instance type *CfeScanGroup* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, like "Scangroup 1" or "Scangroup for Analogs" or similar.
4. Select tab **Attributes** and fill the values
5. Establish a system link from Scan Group to the Scan Rate instance:

Open tab **Links** and drag the 'Scan Rate' instance into the reference field *CfeScanRate* of the 'Scan Group' instance

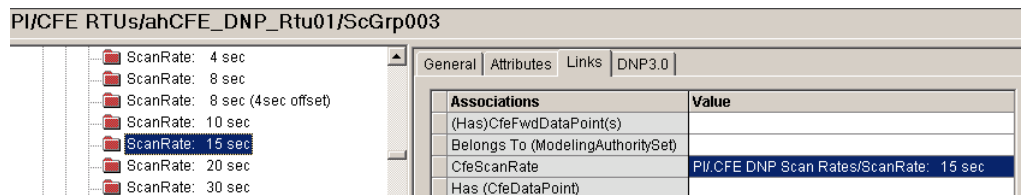


Figure 3-2.22: Link Between RTU Scan Group and Scan Rate

Scan Volume

To specify an address range within a Scan Group an instance of the type *CfeScanVolume* (as a child instance of *CfeScanGroup*) is to be created. Please perform the following steps:

1. Right-click the instance of the type *CfeScanGroup* in the Overview Pane and select **New** from menu that pops-up.
The 'Type Selection Box' appears in the Work Pane and shows all available instance type available for this area
2. Highlight *CfeScanVolume* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, like 'Scan Vol 1' or 'Integrity Scan' or similar.
4. Select tab **Attributes** and fill the values, such as Start Address, End Address, Request Mode

3.2.1.3.10 Configuring RTU Interrogation Groups for Protocols IEC101 and IEC104

For the standard protocols IEC60870-5-101 (IEC101) and IEC60870-5-104 (IEC104), in addition to the General Interrogation, up to 16 Interrogation Groups (Scan Groups) can be specified. The location of involved instances as well as the necessary system links are shown by the following generalized diagram.



Instance 'Scan Group GI' for General Interrogation and associated links to data points can be omitted if the data point attribute 'GI relevant' is used instead. Status information 'GI incomplete' and 'GI running' can be configured both, as child instance of the RTU or of an (optional) GI scan group.

In the following, all steps to establish a Scan Group including all system interlinks are demonstrated by means of an example comprising this task: For an RTU named 'RTU01', a 30 sec Scan Group for

periodical interrogation of two analog data points (DP1_analog, DP2_analog) including the status information "Group1 incomplete" is to be created.

Scan Group

To create a Scan Group (instance type *CfeScanGroup*) perform the following steps:

1. Select the RTU instance of interest in the Overview Pane. To popup the menu right-click and select **New**

The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select instance type *CfeScanGroup* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, like "Scan Group G1" or "Scan Group for 30 sec Scans" or similar
4. Select tab **Attributes** (or the protocol specific tab) and select Scan Type 'IEC101 Interrogation GroupScan' for protocol IEC101. For protocol IEC104, select 'IEC104 Interrogation GroupScan', accordingly. The other attributes can be left unchanged, see Figure 3-2.24

General Attributes Links IEC60870-5-101	
Attributes	Value
MMS Address	
Message Configuration	Default
Model Resource ID	
Monitoring Retries	0
Monitoring timeout	10000
OnChange Data	True
OnChange Quality	True
OnUpdate Data	True
ReportCtrlBlock	
Scan Group Number	-1
Scan On Event	Not used
Scan Priority	99
Scan Type	IEC101 Interrogation Group Scan

Figure 3-2.24: Attributes of the Instance Type 'CfeScanGroup'

5. Finally, establish the following system links to the Scan Group instance:
 - a. to the desired Scan Rate (Hint: for creating Scan Groups, please refer to Chapter 3.2.1.3.9 *Configuring Periodic Scans for an RTU*)
 - b. to all data points contained in this Scan Group

Open tab **Links** and drag the 'Scan Rate' instance (in this example: Scan Rate 30 sec) into the reference field *CfeScanRate* of the 'Scan Group' instance. Next, drag the desired data points into the Association "(Has)CfeDataPoint(s)" (in this example: Analog 2000, Analog 2001, Analog 2002), see Figure 3-2.25

General Attributes Links IEC60870-5-104	
Associations	Value
(Has)CfeFwdDataPoint(s)	
Belongs To (ModelingAuthoritySet)	
CfeScanRate	PI/CFE Scan Rates/SR SG 02 - 30sec
Has (CfeDataPoint)	PI/CFE RTUs/IEC104: DACO/Analog 2000 - MN
	PI/CFE RTUs/IEC104: DACO/Analog 2001 - MF
	PI/CFE RTUs/IEC104: DACO/Analog 2002 - MN

Figure 3-2.25: Links From Scan Group to Scan Rate and Different Data Points

General Interrogation and Group Interrogation

General Interrogation

There are two options to configure a General Interrogation:

1. By using the data point attribute 'GI relevant' (no Scan Group). In this case, runup and periodic GI can be set by means of the attribute 'General Interrogation Cycle Time' of the link instance as follows:
 - a. General Interrogation Cycle Time = -1: no runup GI
 - b. General Interrogation Cycle Time = 0: runup GI
 - c. General Interrogation Cycle Time = >0: runup GI and periodic GI
2. By deploying a 'GI Scan Group' instance (as a child instance of RTU) and linking all GI- relevant data points to this instance. If this option has been chosen, the behavior of the GI can be configured in a more sophisticated way, same as 'Group Interrogation', see below

Group Interrogation

The behavior of the Group Interrogation can be configured by the attributes 'Period' and 'Offset' of the Scan Rate instance as well as by the attribute 'Time Delay' of the Scan Group instance in a very flexible way, see following table and Figure 3-2.26.

Table 3-2.2: Interrogation Behavior at Different Settings for Scan Period, Scan Offset, Interrogation Delay

Period			Offset		Delay		Behavior of General- or Group Interrogation
-1	0	> 0	-1	>= 0	-1	>= 0	
X			don't care		don't care		no GI
	X		don't care		X		no GI
	X		don't care			X	1 x GI at runup + delay
		X	X				periodic GI started at runup
		X		X	X		periodic GI started at x:xx (x:xx modulo period)
		X	X			X	periodic GI started at runup and GI at runup + delay
		X		X		X	periodic GI started at x:xx and GI at runup + delay

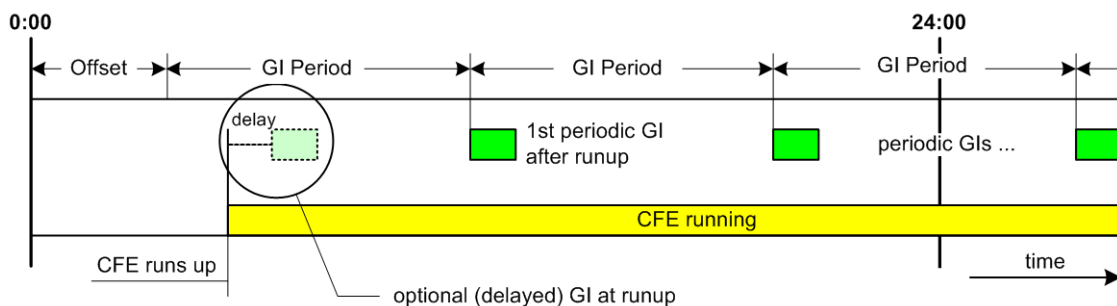


Figure 3-2.26: Periodical Group Interrogation with Offset and delayed Runup-GI

**Note**

For General Interrogation, following combinations can be considered

- No Scan Group configured: Attributes 'GI relevant' and 'General Interrogation Cycle Time' work as configured
- GI Scan Group created but not linked to a Scan Rate: Attributes 'GI relevant' and 'General Interrogation Cycle Time' work as configured; in addition, group-specific delay at runup is applied
- GI Scan Group created and linked to a Scan Rate and data points: Attributes 'GI relevant' and 'General Interrogation Cycle Time' are ignored. Period, offset and delay influence the GI behavior as indicated by above table

Scan Volume

To allocate a Scan Group to an Interrogation Group (1 to 16) an instance of the type *CfeScanVolume* (as a child instance of *CfeScanGroup*) is to be created. Please perform the following steps:

1. Right-click the instance of the type *CfeScanGroup* in the Overview Pane and select **New** from popup menu. The 'Type Selection Box' appears in the Work Pane and shows several instance types that can be chosen in this level.
2. Highlight *CfeScanVolume* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, in this example: "Scan Vol G1"
4. Select tab **Attributes** (or the protocol specific tab) and select the attribute 'Class Type' in the enumeration field, see Figure 3-2.27

Attributes	Value
Class Type	IEC101 Group Interrogation No 1
Description	

Figure 3-2.27: Attributes of the Instance Type 'CfeScanVolume'

Depending on the protocol, the following 'Class Types' can be utilized:

- a. 'IEC101 Group Interrogation No. n' (1 ... n ... 16) for protocol IEC101
- b. 'IEC104 Group Interrogation No. n' (1 ... n ... 16) for protocol IEC104
- c. 'General Interrogation' for both IEC101 and IEC104

Status Information for Scan Group(s)

The following status information is available and can be used to log or to display details of the scanning process:

- ScanGroup running
- ScanGroup incomplete

The status information 'ScanGroup running' also provides control functionality. This can be used to provide a control button for manual triggering a Group Interrogation

The creation of status information is performed in these steps:

1. Creating a digital measurement (e.g.: "G1 incomplete" or "G1 running") as a child instance of Scan Group
2. Adding a child instance of the type *DiscreteValue* to the digital measurement (e.g.: "Value")
In case of status information "G1 running", you also may create an child instance type *Command* for GI Command
3. Creating the status data point (e.g.: "G1 incomplete" or "G1 running")) as a child instance of Scan Group
4. Establishing a link between measurement value and status data point , see diagram above (Figure 3-2.23)



Note

More details on creating CFE / RTU status information are given in Chapter 3.2.1.5.2 *Creating Status Information / Statistic Counters*, later in this document

Instances involved in Group Scans are presented by extracts from Overview Tree, see next figure (Figure 3-2.28)

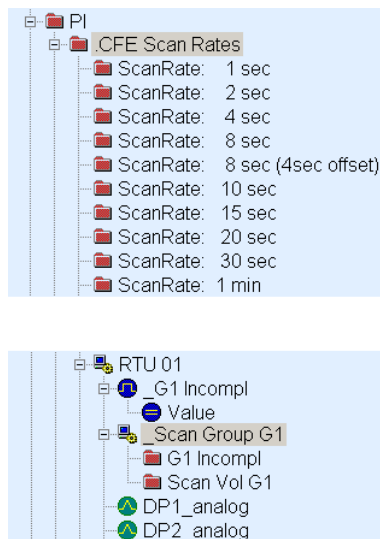


Figure 3-2.28: Overview Tree Showing Instances Involved in Group Scans (Extract)

3.2.1.4 Characteristic Curves

3.2.1.4.1 General

The characteristic curves serve to transform raw values, received from RTU, into physical values (engineering values) as required in graphical diagrams or in the archive. It depends on the protocol and the message format how the analog is encoded and what the range of valid values is.

Characteristic curves can also be used in the opposite directions, i.e. for analog setpoints. In this case the conversion is from engineering value coming from any application server (e.g. RTS) into a protocol specific raw value.

Please note that a characteristic 'curve' is not a real curve: it can be a straight line or may consist of two or more straight segments as shown hereafter in Figure 3-2.30 .

All four basic types of characteristic curves can be configured in PowerCC:

- Linear, unipolar (Figure 3-2.29)
- Non-linear unipolar (Figure 3-2.30)
- Linear, bipolar (Figure 3-2.31)
- Non-linear, bipolar

There is no need to define a characteristic curve for every single analog value in the system. One characteristic can be allocated to different analog measurements having the same transducer in the substations bay and the same type of ADC (analog to digital converter) in the RTU.

The following three diagrams show typical characteristics to convert raw values of protocol IEC60870-5-101, type ID 9 (Normalized Value) into physical value. In case of IEC101 (or IEC104) protocol the raw value can range between -2^{15} up to 2^{15} (- 32 767 ... 32 768), the physical range and unit can be any one.

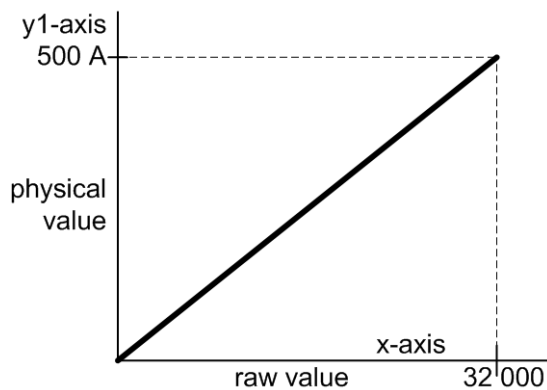


Figure 3-2.29: Linear, Unipolar Characteristic

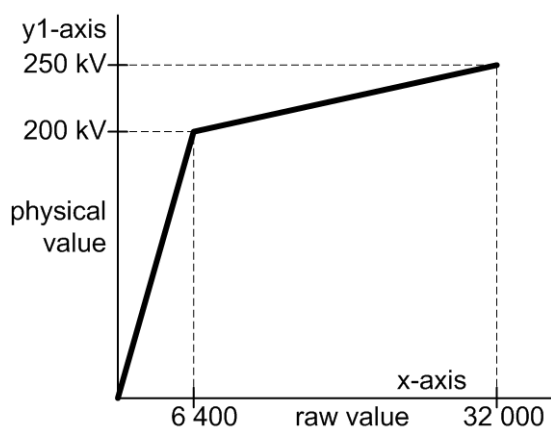


Figure 3-2.30: Non-Linear, Unipolar Characteristic

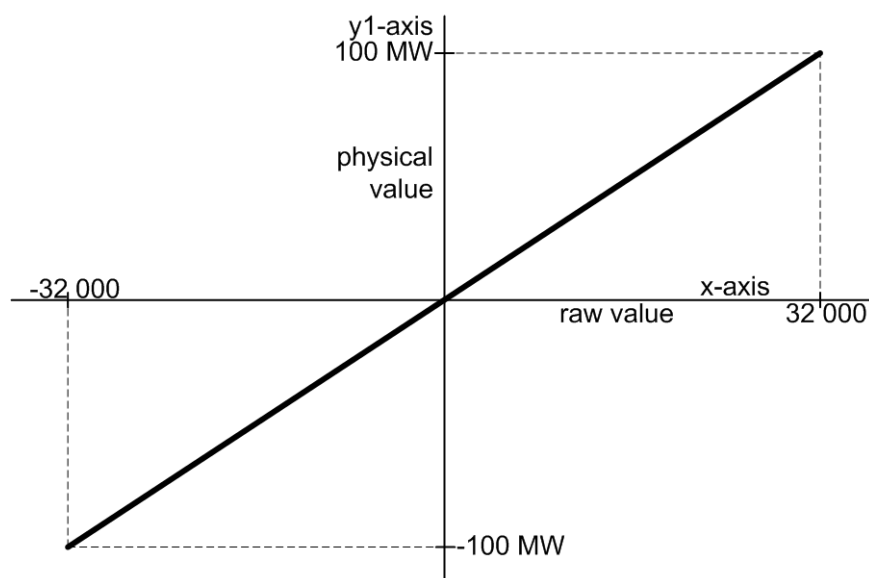


Figure 3-2.31: Linear, Bipolar Characteristic

3.2.1.4.2 Specifying a Characteristic Curve

To specify a (new) characteristic curve an instance of the type *Cfe Characteristic Curve* is to be created. Please perform the following steps:

1. Right-click the parent instance 'CFE Characteristics' (that is of type *sysCfeConfig*) in the Overview Pane and select **New**

The frame illustrated below appears in the Work Pane. Use this to select the instance type available for this area

2. Select instance type *Cfe Characteristic Curve* and click **Create Instance**
3. The Instance Editor appears. Please enter the name of the new characteristic, e.g. 'Voltage 0 - 80 - 120 kV'
4. Open tab **Attributes** and enter or adapt the x-axis and the y1-axis labels (Figure 3-2.32)

General Attributes Links	
Attributes	Value
Alias Name	
Curve Style	
Display Order	
Local Name	
Model Resource ID	
X Axis Label	Raw Value
X Axis Type	
X Axis Units	
Y Axis Type	
Y1 Axis Label	Phys. Value
Y1 Axis Units	
Y2 Axis Units	
xMultiplier	
y1Multiplier	
y2Multiplier	

Figure 3-2.32: Labeling X-Axis and Y-Axis

5. Save the new instance
6. Select the new instance in the Overview Pane and click **New** again.
7. Select the instance type *CurveSchedData* and click **Create Instance**
8. The Instance Editor appears. Please enter the name of the (first) pivot point of the curve, for example "250 kV"
9. Select tab **Attributes** and fill appropriate values into **x-Axis Data** (raw value, here: 32000) and **y1 Axis Data** (corresponding physical value, here: 250), see next figure.

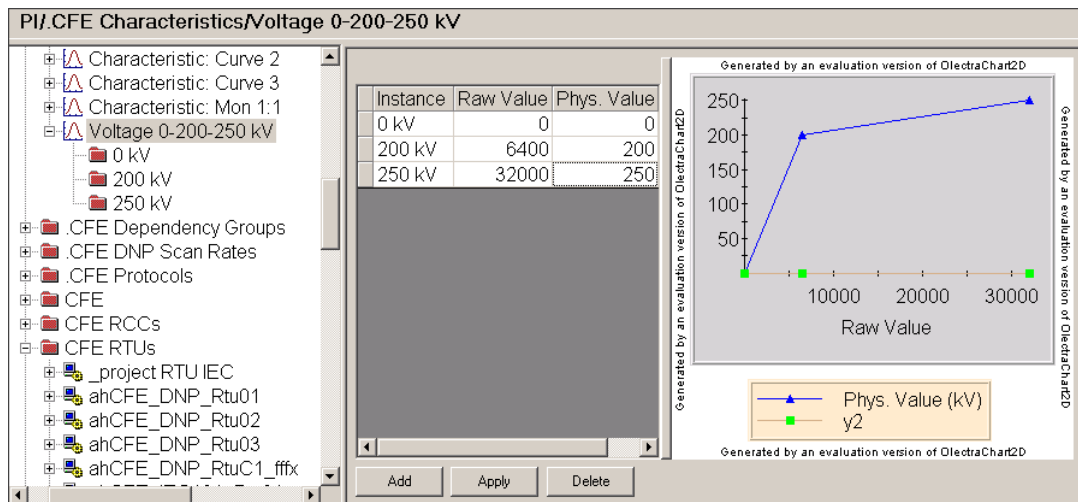


Figure 3-2.33: Defining Pivot Points of the Characteristic Curve

10. Repeat last step accordingly for every pivot point to be defined: linear characteristics need two, non-linear one more than the number of segments the characteristic curve consists of.

**Note**

Axis Data y2 through y8 are not relevant for CFE (provided for other PowerCC applications).

3.2.1.5 CFE / RTU Status Information and Communication Statistics

3.2.1.5.1 General

Status information and communication statistics is system information. Unlike process information, that is gathered from switchgear by the RTU, this type of information is generated in CFE or in RTU by software routines. Status information can be seen as optional information. There is no real need to take care on this type of data. Without any configuration work they are listed in the 'Runtime Explorer' and show their current value. However, if the customer wants to display status information or statistic counters in customized diagrams or put them into a log or into the archive, some configuration work has to be done. This is described in the following sections.

Status Information

Status information is available for RTUs and CFE components, like CFE interfaces, links to RTUs. They provide information about the availability of every component. Typical values for CFE Status information points are: FAILURE, PREFERRED, ACTIVE, LOCKED.

The following tables summarize the status information for CFE and RTU:

CFE Status Information	Interface (COM Port)	Link to RTU
FAILURE	x	x
ACTIVE (process controlling)	x	x
LOCKED	x	x
PREFERRED	only for dependent communication channels	only for independent communication channels
LISTEN MODE	x	-
CLOCKSYNC ENABLE		x

CFE RTU Status Information	Remark
RTU_FAILURE	derived in CFE
GENERAL INTERROGATION RUNNING	derived in CFE
GENERAL INTERROGATION INCOMPLETE	derived in CFE
GROUP INTERROGATION RUNNING	derived in CFE
GROUP INTERROGATION INCOMPLETE	derived in CFE
RENEWAL CHECK	derived in CFE



Note

System information that is generated and transmitted by the RTU is not considered as a type 'CFE RTU status information'. It is treated like process information. Typical RTU system information is for example: RTU DISTURBED, RTU BUFFER OVERFLOW, RTU HARDWARE FAILURE or similar.

Communication Statistics

Communication Statistics can be obtained from system counters integrated in some of the protocol software packages. Typical communication statistics are:

- Number of sent messages
- Number of received messages
- Number of received corrupted messages
- Number of scan requests

Please check the protocol of interest whether communication statistics are implemented and if so, what type of counters are available.

3.2.1.5.2 Creating Status Information / Statistic Counters

Normally, the customer need not create CFE Status Infos or statistic counters from scratch. These and all CFE status information types are pre-configured.

Should a new component be added (e.g., a new COM port) it is recommended that an existing one be copied. Start with an instance of the component that already has a full complement of CFE Status Infos defined. By making a copy, all of the CFE Status Infos, including the digital measurements, values and links, are created. On a new system, 'demo' components can be loaded and used as a starting point for making copies.

Nevertheless, in the following section the complete procedure to create RTU and CFE status information as well as communication statistics is described in detail. The general principle of creating status information and statistic counters is described in the following section.

Status Information

To display status information in system logs or in graphical diagrams the following instances types are to be created for every status information point:

1. Instance type *Discrete*
2. Child instance of *Discrete* , type *DiscreteValue* and - if applicable - *Command*

NOTE: *DiscreteValue* transports the information into logs or graphic diagrams, *Command* is used to provide controls for the status information

3. Data point instance, type *Cfe Status* (supplies the status information)

Communication Statistics

To provide Communication Statistics in diagrams the following instances are to be created:

1. Instance type *Accumulator*
2. Child instance type *AccumulatorValue*
3. Data point instance, type *CfeSystemAccumulator*

Figure 3-2.34 shows in which branch CFE status information is located as well as the links to their measurements.

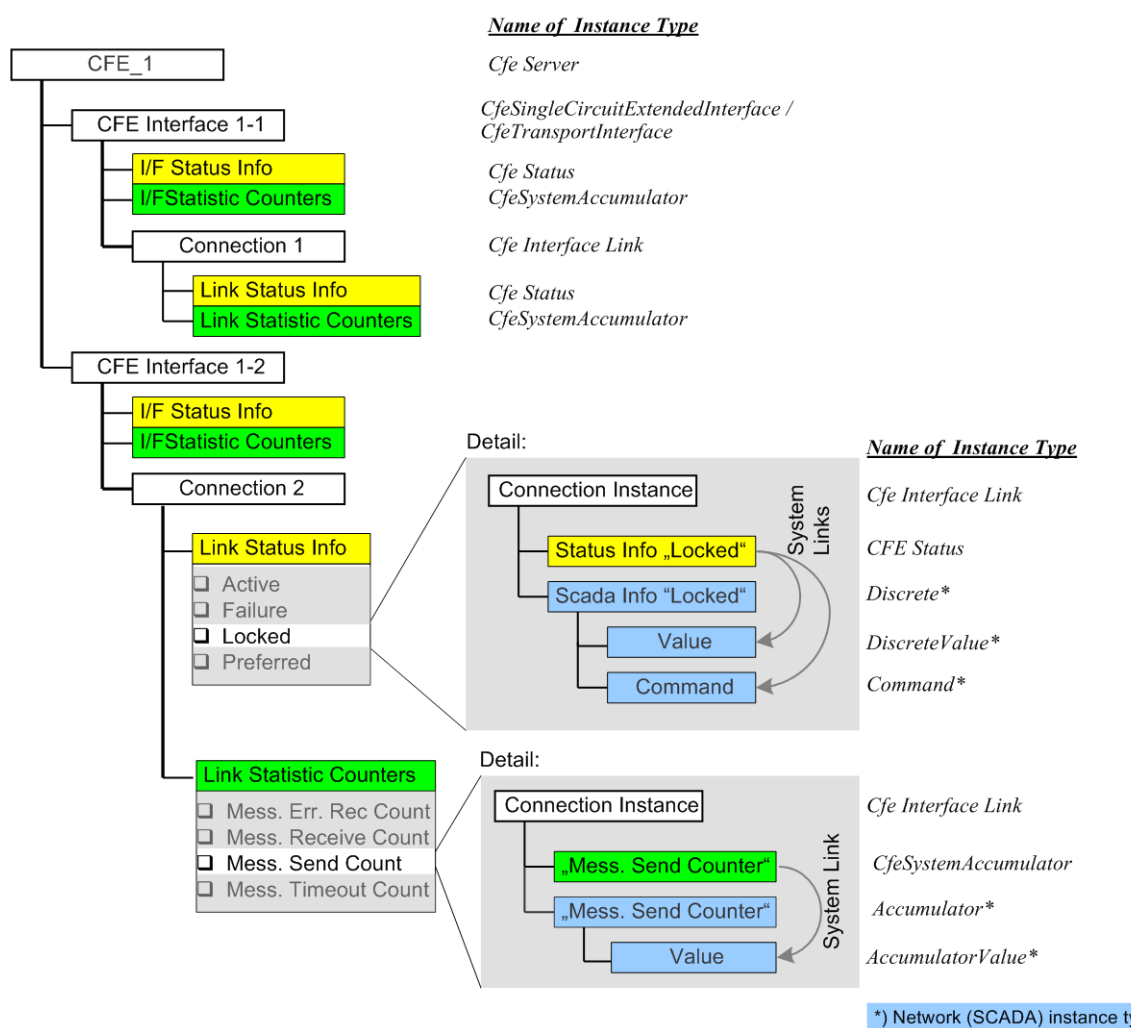


Figure 3-2.34: Principle of Configuring CFE Status Information and Communication Statistic Counters

3.2.1.5.3 Creating CFE Status Information

The creation of a CFE status information for CFE interface and / or link (connection) is performed in four steps:

1. Creating a digital measurement
2. Adding a value and / or control instance to the measurement
3. Creating a status data point
4. Establishing a link between measurement and data point

In the following example the status information 'LOCKED' is created for a serial communication port.

1. Creating a Digital Measurement

To create a digital measurement perform the following steps:

1. Select the interface or link instance of interest in the Overview Pane. To popup the menu right-click and select **New**.

The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area.
2. Select *Discrete* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new measurement instance, for example ".. Locked"
4. Select tab **Attributes** and fill the values

2. Adding a Value and / or a Control Instance to the Measurement

To add a value (instance) to the just created measurement perform the following steps:

1. Right-click the just created instance and select **New** from menu that pops-up.

The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select the instance type of interest,
 - a. *DiscreteValue* for monitoring
 - b. *Conmmand* for control functionand click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new child instance, for example "Value" or "Control", respectively
4. Select tab **Attributes** and fill the values

The procedures described above may be repeated for every CFE status information or - which is the easier way - the just created status information may be duplicated by copy / paste / rename as many times as needed.

3. Creating a Status Data Point

To create the status information data point perform the following steps:

1. Select the interface or link instance of interest in the Overview Pane. To popup the menu right-click and select **New**

The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *Cfe Status* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new CFE Status data point, for example "...Stat: Locked"
4. Select tab **Attributes** and select the appropriate Info ID, in this case: **Device locked** (Figure 3-2.35)



Figure 3-2.35: Attributes of the Instance Type 'Cfe Status' with selected Info ID 'Device locked'

4. Establishing Links Between Measurement and Data Point

Select tab **Links** (of the just created instance) and drag the instance 'Value' which was created in step 2 into the reference box **(SendsTo) MeasurementValue** and, if applicable, the instance 'Control' into the reference box 'ReceivesFrom (MeasurementControl)' (Figure 3-2.36)

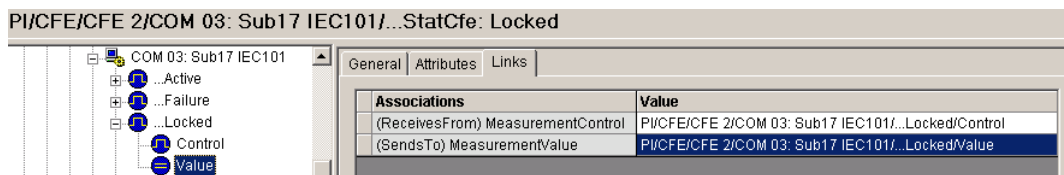


Figure 3-2.36: Linking the CFE Measurement to the CFE Status Data Point

3.2.1.5.4 Creating CFE Status Information for RTUs

Similar as the creation of the CFE status information, the RTU status information is also performed in four steps:

1. Creating a digital measurement
2. Adding a 'Value' instance to the digital measurement
3. Creating the status data point
4. Establishing a link between measurement and data point

In the following example the RTU status information 'GI Incomplete' is created.

1. Creating a Digital Measurement

To create a digital measurement for a CFE RTU status information perform the following steps:

1. Right-click the RTU instance of interest in the Overview Pane and select **New** from popup menu
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *Discrete* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the measurement, for example "_GI incomplete"
4. Select tab **Attributes** and fill the values

2. Adding a Value to the Digital Measurement

To add a value (instance) to the just created measurement perform the following steps:

1. Right-click the just created instance in the Overview Pane and select **New** from menu that pops-up.
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area

2. Select *DiscreteValue* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new child instance, for example "Value"
4. Select tab **Attributes** and fill the values

The procedures described here may be repeated for every RTU status information or - which is the easier way - the just created status information may be duplicated by copy / paste / rename as many times as needed.

3. Creating the Status Data Point

To create the status information data point perform the steps described hereafter:

1. Right-click the RTU instance of interest in the Overview Pane and select **New** from menu that pops-up.
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *Cfe Status* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, for example "...StatRTU: GI incomplete"
4. Select tab **Attributes** and choose the desired Info ID, in this example it is 'GI incomplete' (Figure 3-2.37)

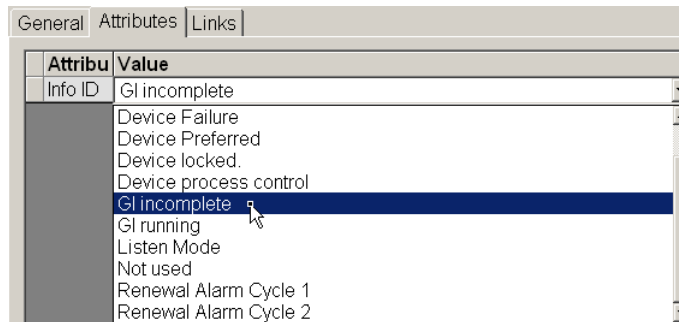


Figure 3-2.37: Attributes of the Instance Type 'Cfe Status' with Selected Info ID 'GI incomplete'

4. Establishing a Link Between Measurement and Data Point

Open tab **Links** and drag the 'Value' instance which was created in step 2 into the box (**SendsTo**) **MeasurementValue** (Figure 3-2.38)

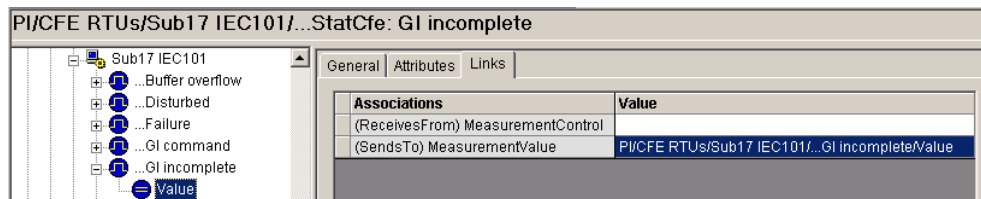


Figure 3-2.38: Linking the CFE Measurement Value to CFE Status Data Point '...GI incomplete'

3.2.1.5.5 Creating System Accumulators for Communication Errors

Communication Statistics are used to get information about the quality of communication interfaces and communications links to RTUs. For this purpose, CFE provides different communication statistic counters.

Typical counters for communication statistics are:

- Counter for sent messages
- Counter for received intact messages
- Counter for received erroneous messages
- Counter for time-outs

Similar as the creation of the CFE status information, system accumulators can be created in four steps:

1. Creating the accumulator measurement
2. Adding a value to the accumulator measurement
3. Creating the accumulator data point
4. Establishing a link between measurement and data point

In the following example the system counter 'Message Error Receive Counter' is created for a serial communication port.

1. Creating the Accumulator Measurement

To create the Accumulator Measurement for CFE interface or interface link (connection) perform the following steps:

1. Select the interface instance of interest in the Overview Pane. To popup the menu right-click and select **New**
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *Accumulator* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, in this example it is "...Message Error Receive Counter"
4. Select tab **Attributes** and fill the values

2. Adding a Value to the Accumulator Measurement

To add a Value to the just created Measurement perform the following steps:

1. Right-click the just created instance and select **New** from menu that pops-up.
The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *AccumulatorValue* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new child instance, for example 'Value'
4. Select tab **Attributes** and fill the values

The procedures described here may be repeated for other accumulators or - which is the easier way - the just created accumulator may be duplicated by copy / paste / rename as many times as needed.

3. Creating the Accumulator Data Point

To create the accumulator data point perform the steps described hereafter:

1. Select the interface or link instance of interest in the Overview Pane. To popup the menu right-click and select **New**

The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area
2. Select *CfeSystemAccumulator* and click **Create Instance**
3. The Instance Editor appears in the Work Pane. Please enter the name of the new instance, for example " ...StatCfe: Message Error Receive Counter"
4. Select tab **Attributes** and choose the **Info ID** for the desired counter type, in this example 'Message Error Receive Counter'

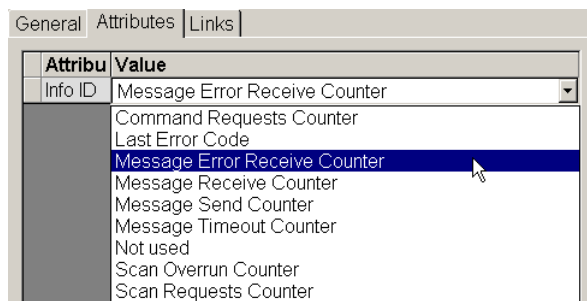


Figure 3-2.39: Attributes of the Instance Type 'CfeSystemAccumulator' with Selected Info ID

4. Establishing a Link Between Measurement and Data Point

Open tab **Links** and drag the 'Value' instance which was created in step 2 into the reference box '**MeasurementValue**' (Figure 3-2.40)



Figure 3-2.40: Linking the CFE Measurement 'Value' to the CFE Accumulator Data Point '...Error Receive Counter'

3.2.1.6 Configuring SNMP Communication

3.2.1.6.1 Introduction

It is possible to monitor network components such as printers, routers, switches, servers by means of the Simple Network Management Protocol (SNMP) along with a suitable application program for handling incoming and outgoing data, see next figure.

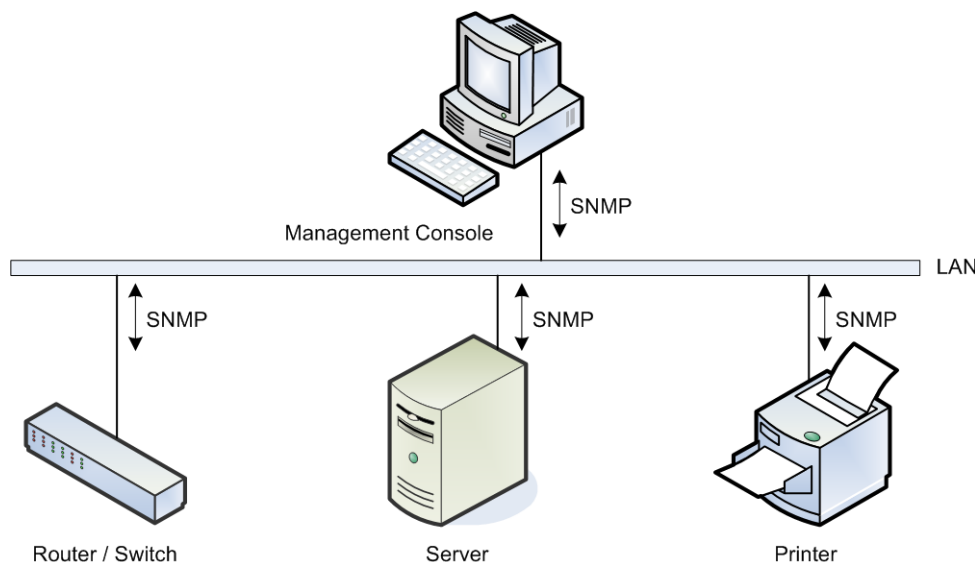


Figure 3-2.41: Monitoring Network Devices through SNMP Protocol

3.2.1.6.2 Configuring Tasks - Overview

CFE allows to monitor network devices by means of the SNMP protocol. (Controlling and setting of devices is not implemented).

Configuring the SNMP communication comprises the following basic tasks:

1. Setting-up a network interface at CFE for protocol SNMP
2. Setting-up an interface link on the transport interface for each device to be monitored by SNMP
3. Setting-up status information for interface and link(s) (optional)
4. Setting-up devices (RTUs)
5. Setting-up data points
6. Setting-up status information for device(s) (optional)
7. Setting-up corresponding "SCADA" measurement values

All instances involved in the SNMP configuration and the required system links are presented in the following generalized diagram

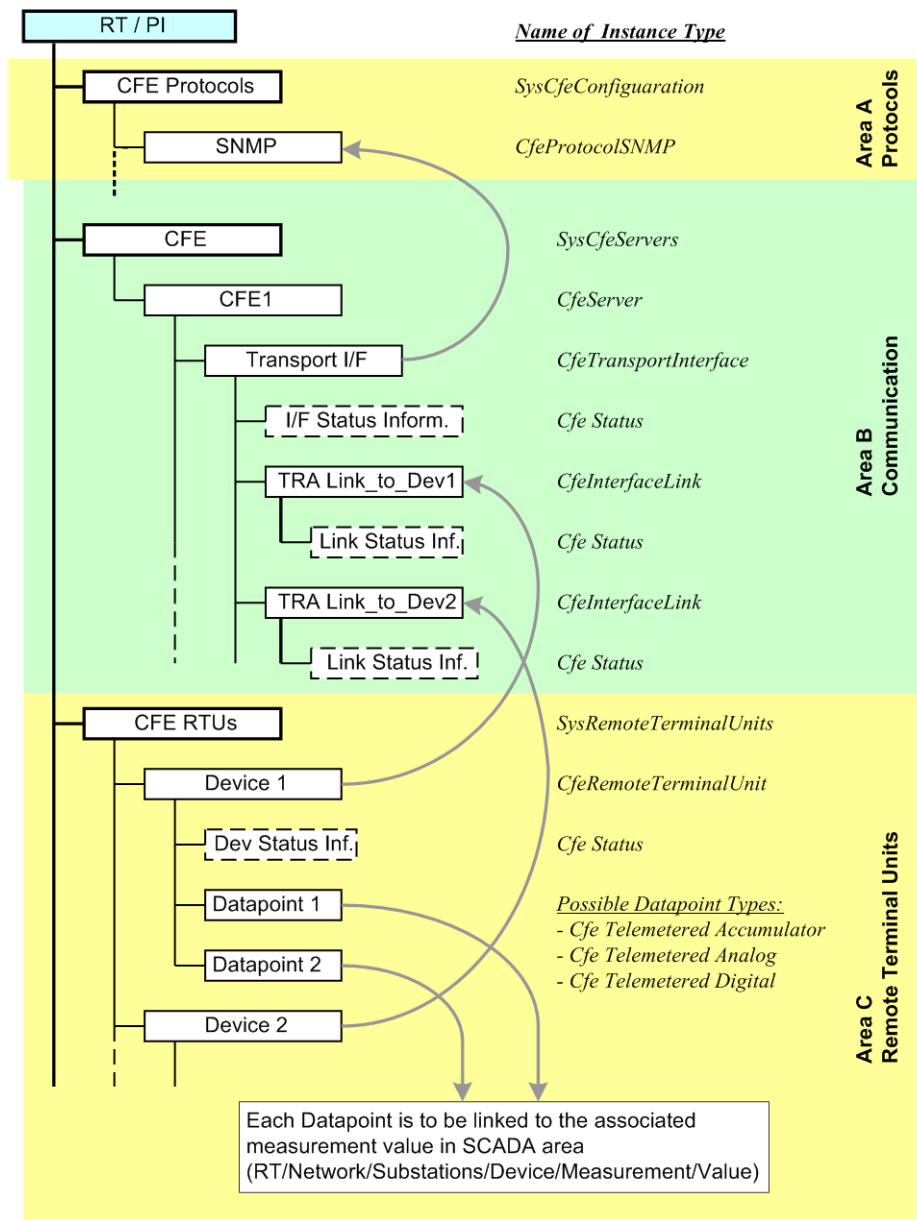


Figure 3-2.42: Instances and System Links for SNMP Communication

3.2.1.6.3 Use Case

The following use case shows all configuration steps in detail, partly illustrated by screenshots. The exemplary task consists of monitoring the number of printed pages of a network printer.

1. Setting-up the Network (Transport) Interface

To accomplish this task, perform these steps:

1. Right-click the CFE instance (type *CfeServer*), select **New** from pop-up menu, highlight instance type *CfeTransportInterface* in the 'Type Selection Box' and click **Create Instance**. Enter name and optionally, a short description, e.g., .Name: "TRA:SNMP" Description: "Transport Interface for SNMP"
2. Open tab **Attributes** and enter appropriate parameters, such as CFE IP Address, Community Name, Port Number. For details on attributes please refer to the chapter "Protocol Specific Engineering" later in this document
3. Open tab **Links** and establish a system link to the SNMP protocol instance as illustrated by the following figure.

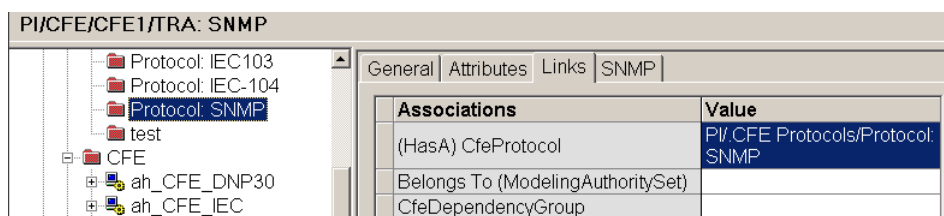


Figure 3-2.43: System Link from CFE Interface to Protocol SNMP Instance

2. Setting-up the Interface Link

To accomplish this task, perform these steps:

1. Right-click the CFE interface instance (type *CfeTransportInterface*), select **New** from pop-up menu, highlight the instance type *CfeInterfaceLink* in the 'Type Selection Box' and click **Create Instance**. Enter name and optionally, a short description, e.g., .Name: "Link: Printer01" Description: "Link to HP LaserJet 8100"
2. Open tab **Attributes** and enter appropriate parameters, such as Polling Time, IP Address, Port Number. For details on attributes please refer to the chapter 'Protocol Specific Engineering' later in this document

3. Setting-up Status Information (Optional)

The following status information can be setup for each interface and link:

- Device Failure
- Device Locked
- Device Process Control
- Device Preferred

For details on how to create status information please refer to the respective Chapter 3.2.1.5.2 *Creating Status Information / Statistic Counters*

4. Setting-up Devices

Devices, such as network printers, servers, routers and so on, are treated as ordinary RTUs. To accomplish this task, perform these steps:

1. Right-click the parent instance 'CFE RTUs' (type *SysRemoteTerminalUnits*), select **New** from pop-up menu, highlight the instance type *CfeRemoteTerminalUnit* in the 'Type Selection Box' and click **Create Instance**. Enter name and optionally, a short description, e.g., Name: "Printer01" Description: "HP LaserJet 8100 - in Room 412"
2. Open tab **Attributes** and select for the attribute 'Telecontrol Protocol Type' the enumerated item 'SNMP'. Other parameters are of minor importance and may be adapted later in the protocol specific tab. For details on attributes please refer to the chapter "Protocol Specific Engineering" later in this document.
3. Open tab **Links** and establish a system link from RTU instance to the interface link(s) created before, see next figure.

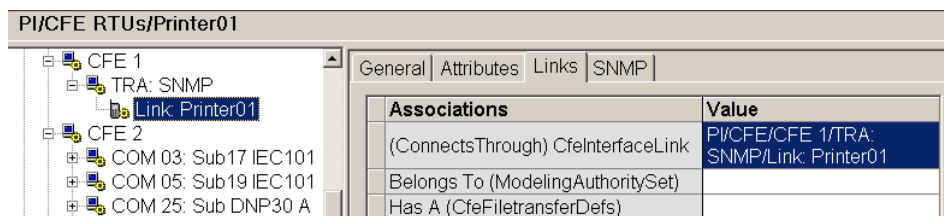


Figure 3-2.44: System Link from Device to Interface Link

5. Setting-up Data Points

For SNMP devices, the following data types can be configured:

- Accumulators (Instance type *Cfe Telemetered Accumulator*)
- Analog values (Instance type *Cfe Telemetered Analog*)
- Digital values (Instance type *Cfe Telemetered Digital*)
- Step Positions (Instance type *Cfe Telemetered Position*)

For this example (Number of printed pages), a data point of the type *Cfe Telemetered Accumulator* is to be created. To accomplish this task, perform these steps:

1. Right-click the parent instance "Printer01" (type *CfeRemoteTerminalUnit*), select **New** from pop-up menu, highlight the instance type *Cfe Telemetered Accumulator* in the 'Type Selection Box' and click **Create Instance**. Enter name and optionally, a short description, e.g., Name: "PageCount" Description: "Total Number of printed pages"
2. Open tab **Attributes** or the protocol-specific tab **SNMP** and enter the required attributes. The most important parameter is the 'Variable Address' that can be obtained from manual of the device. For details on other attributes please refer to the chapter "Protocol Specific Engineering" later in this document.
3. Open tab **Links** and establish a system link from RTU data point to the corresponding SCADA measurement value, see next figure. If the corresponding SCADA measurement value has not yet been created, you may do this now, see item 6 below.

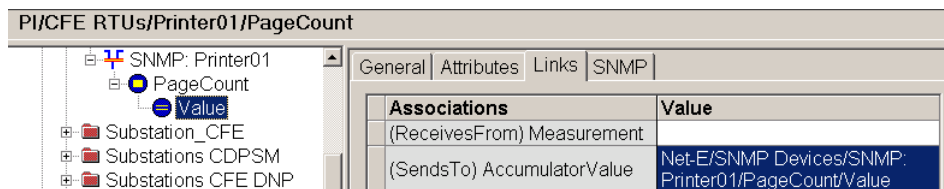


Figure 3-2.45: System Link from Data Point to SCADA Measurement Value

5. Setting-up Status Information (Optional)

The following status information can be setup for each device:

- Device Failure
- GI Incomplete
- GI Running
- Renewal Alarm Cycle 1 (or 2)

For details on how to create status information please refer to the respective Chapter 3.2.1.5.2 *Creating Status Information / Statistic Counters*

6. Setting-up SCADA Measurement Values

For SNMP devices, the following data types can be configured in SCADA area of IMM:

- Accumulators (Instance type *Accumulator*) with child instance type *AccumulatorValue*
- Analog values (Instance type *Analog*) with child instance type *AnalogValue*
- Digital values (Instance type *Discrete*) with child instance type *DiscreteValue*
- Step Position (Instance type *Discrete*) with child instance type *Discrete Value*

For this example (Number of printed pages), a corresponding SCADA accumulator value is to be created. To accomplish this task, perform these steps:

1. Right-click the parent instance type *Substation*, select **New** from pop-up menu, highlight instance type *Accumulator* in the 'Type Selection Box' and click **Create Instance**. Enter name e.g., "PageCount" and optionally, a short description
2. Open tab **Attributes** and check / adapt the attributes as appropriate.
3. Right-click the just created instance type *Accumulator*, select **New** from pop-up menu, highlight the instance type *AccumulatorValue* in the 'Type Selection Box' and click **Create Instance**. Enter name (e.g. "Value") and optionally, a short description

3.2.2 Tabular-Based Editors

Unlike the Single View Editor, the Table Editors show all child instances and their attributes at a glance as soon as the parent instance is selected in the Overview Pane. The presentation of data is in the form of a table.

Two types of table editors are available in the Information Model Manager:

1. The Communication Editor / CFE for creating automatically data points in the CFE-RTU as well as the required links between Network Substation instances and RTU data points
2. The Table Editor for all RTU related data

Table and Communication Editors can be selected by clicking buttons in the tool bar (Figure 3-2.46)

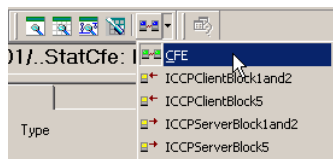


Figure 3-2.46: Selecting Different Types of Editors

3.2.2.1 CFE Communication Editor

The Communication Editor for CFE is a convenient tool to create automatically RTU data points and simultaneously the link between Network instance and RTU data point.

Use the Communication Editor as following:

1. Highlight the Network Substation of interest in the Overview Pane
2. Select Communication Editors / CFE as shown in Figure 3-2.47
3. After the substation data has been loaded into the table, select the RTU of interest, for example in the right part of the pane.
4. To reduce loading and processing time you may set the type filter to any of the available types, for example "Discrete"
5. Click on the Network instance for which you want to create an RTU data point and the necessary link(s)

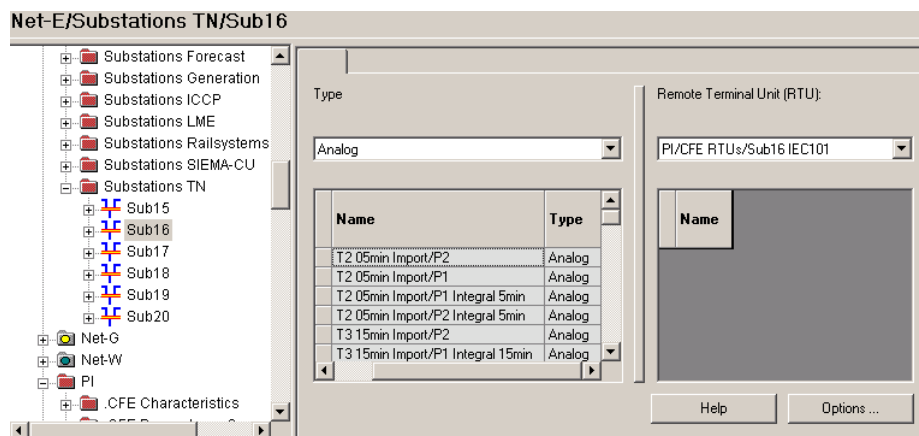


Figure 3-2.47: Communication Editor Before Links Have Been Created

After right-clicking **Create** the Communication Editor creates the following items in the RTU table for each line separately, see Figure 3-2.48.

- Instance name of the RTU data point. The name is derived from the name of the Network instance, for example:
 Network instance: 'Network/Substations TN/Sub16/20/06 Tr102/Local Remote'
 -> RTU data point name: 'Network/Substations TN/Sub16/20/06 Tr102/Local Remote'
- Link 1 (SendsValueTo) to the Network instance for the monitoring direction
- Link 2 (ReceivesControlFrom) to the Network instance for the control direction (as far as applicable)

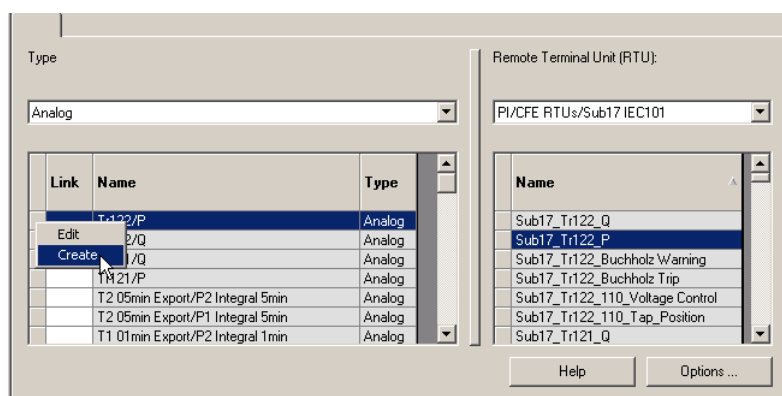


Figure 3-2.48: Communication Editor After Links Have Been Created

3.2.2.2 Table Editor

The Table Editor is mainly used to display all attributes of a particular RTU at a glance. Modification of attributes can immediately be done in the table cells.

The Table Editor can be adapted to the users requirements applying the following options:

- Setting filters
- Adjusting column width
- Hiding columns
- Sorting to each column (up/down)
- Switching text wrapping on/off.

The Table Editor for RTU data points is shown in Figure 3-2.49

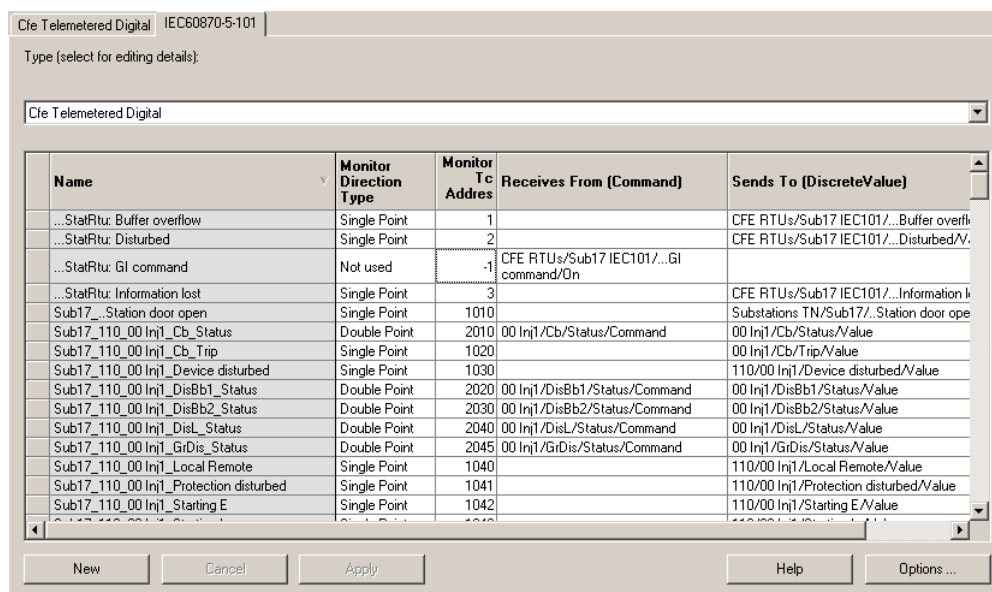


Figure 3-2.49: Table Editor Showing Data Points of the Selected RTU

In the above example the following settings have been made to customize the table:

- Filter set to a specific data type
- Text wrapping set to 'On'
- Some unused columns (or columns with identical contents) are hidden

For customizing the Table Editor these options are available:

- **Sorting:** click with left mouse button into the header field you want to sort on. Repeat the mouse click to invert the sort direction, if desired.
- **Hiding:** Click with right mouse button into the header of the table and select the columns to be hidden or made them visible again (Figure 3-2.50).
- Use the **Option** button to popup the window shown below (Figure 3-2.51). Here, further settings of the table appearance can be done:
 - Selection of data elements
 - Setting number of hierarchy path display
 - Text wrapping on/off
- To display the attributes relevant for the selected protocol use the corresponding Tab (labeled 'TG8979', 'TG800', 'CDC2' ...)

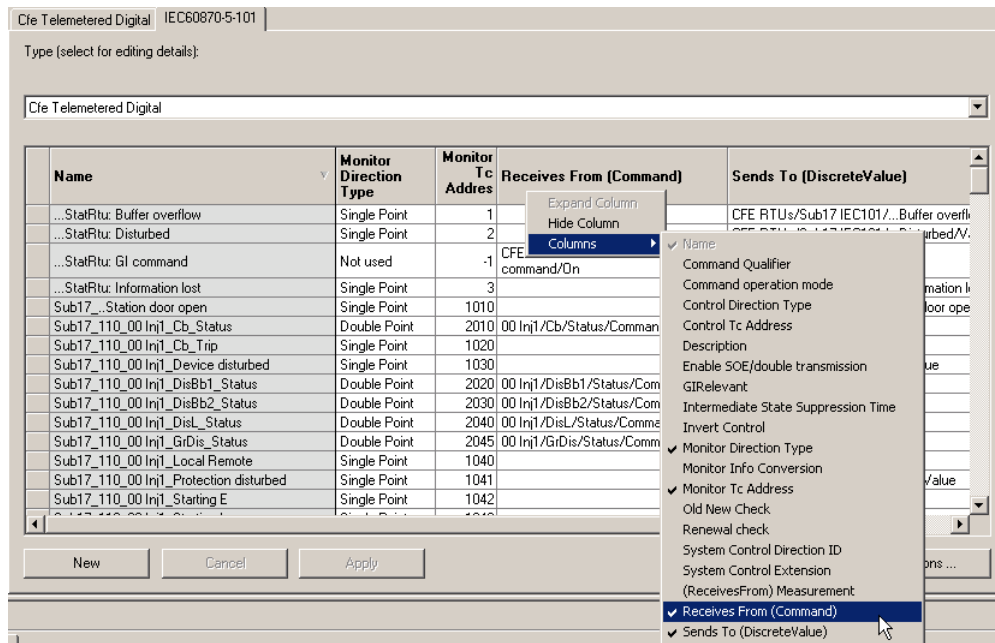


Figure 3-2.50: Hiding Columns By Right-Clicking into Header and Selecting from Menu

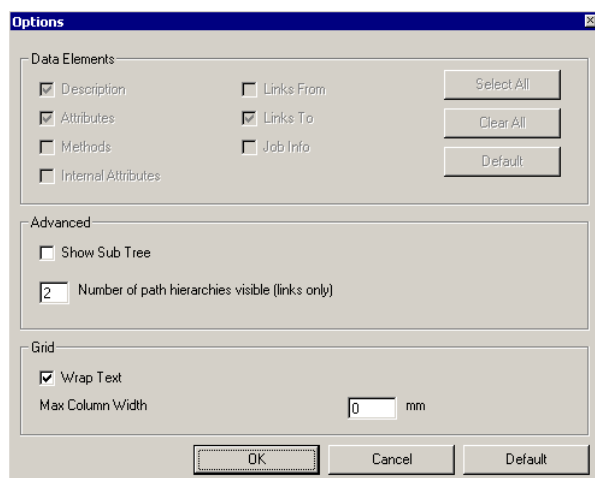


Figure 3-2.51: Window 'Options' of the Table Editor

3.2.3 Management of Redundant Links

3.2.3.1 Overview

If the Communication Front-End System is redundant, each RTU is connected twice: at CFE 1 and at CFE 2. The modem lines in control direction are de-coupled through active or passive modem sharing devices (modem splitters).

Such a Y-formed link is described in the data model as two *dependent links* (link 1, link 2) which are combined in a *Dependency Group* see section below "Dependent Links".

If the RTU(s) have two independent communication interfaces, completely redundant and independent links can be established to the RTU(s). In this case, neither modem sharing devices nor the definition of *Dependency Groups* are required, see section below, "Independent Links".

In each case, the existence of more than one link to an RTU at the Front-End System, a criterion is needed to decide whether a link becomes *active* or *standby* at system run-up and during operation.

The following settings serve as a criterion:

- In IMM: setting priority attribute to high or low
- In operation: setting the status to PREFERRED or LOCKED

Setting Priority

By setting a higher or lower priority it is defined which CFE interface and which link (to the RTU) becomes active at run-up. Interfaces and links with lower priority will operate in standby mode those with high priority become active.

Status PREFERRED

By the controllable status information PREFERRED the operator can change the preset configuration and thus invoke a change of roles between active and standby interfaces or links. The modified configuration will be retained until it is made undone manually. Even a system shutdown and run-up will not reset the configuration.

The immediate switch-over only takes place if the preference is issued to an intact standby channel. If the standby channel is out of operation when setting the preference the switch-over takes place at a later time when it is available again.



Note

Status **PREFERRED** on an interface is only applicable for dependent interfaces (connected to single ported RTUs). Status **PREFERRED** on a link is only applicable for independent interfaces (connected to dual ported RTUs).

Status LOCKED

An interface or a link can completely be deactivated by means of the status information **LOCKED**. If locking is set to an active interface or link a fail-over to the (intact) standby component takes place.



Note

Locking an active interface or link may result in a completely breakdown of the RTU communication when the standby component fails or has already failed when setting the locking.

3.2.3.2 Dependent Links

The diagram below shows dependent links (with only one communication port per RTU). The links are dependent as they use the same physical way to the RTU and thus, they must have an identical parameter setting, matching to the RTU interface.

Logical links to the same RTU interface and thus having identical communication parameters are called "Dependency Group".

This figure shows dependent links (point-to-point and party line)

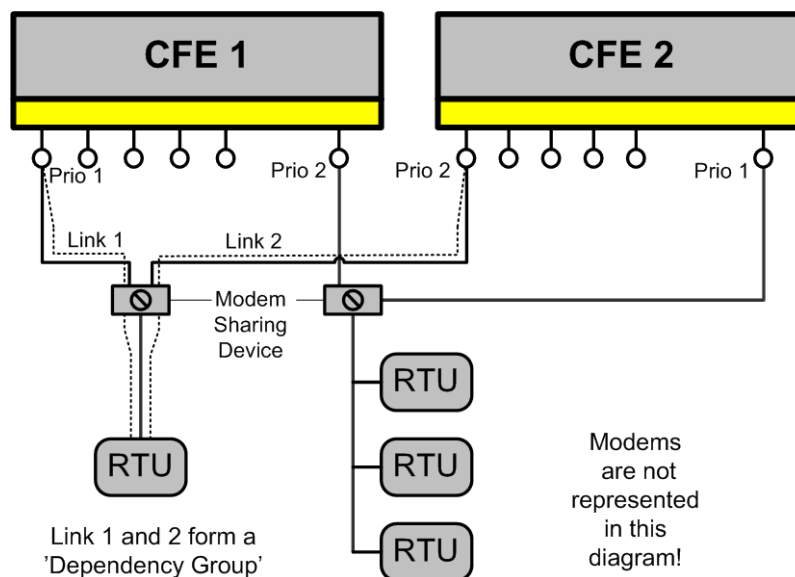


Figure 3-2.52: Dependent Links

The following rules are to be observed for configuration:

- The (two) links to an RTU must have different priority, for example link1 (at CFE 1) => priority 1, link 2 (at CFE 2) => priority 2 (or vice versa).
- For party lines, all links of an CFE interface must have the same priority

The status information PREFERRED is to be set to the CFE interface and not to the links.

3.2.3.3 Independent Links

The diagram below explains independent links (i.e. with two independent communication interfaces per RTU). The links are independent as they are physically separated and thus they could have different communication parameters. Depending on the type of RTU, also configurations with more than two links are feasible.

This figure shows independent links (point-to point and party line)

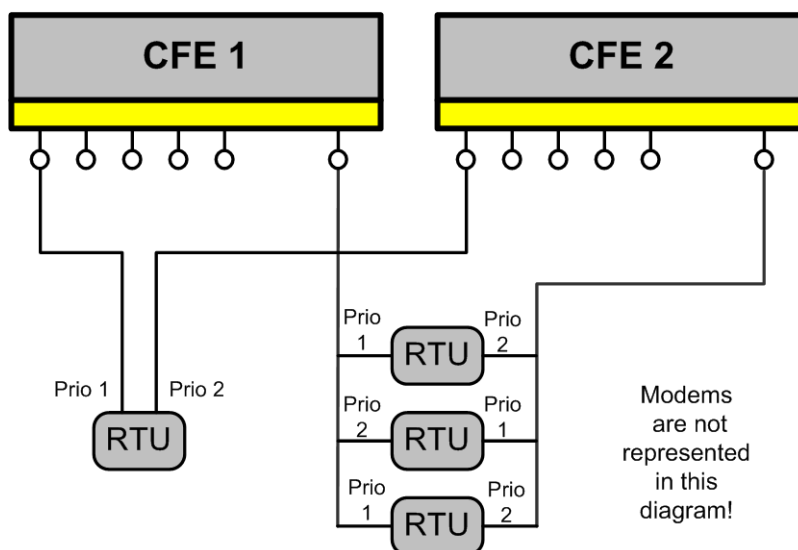


Figure 3-2.53: Independent Links

The following rules are to be observed for configuration:

- The (two) links to an RTU must have different priority, for example link1 (at CFE 1) => priority 1, link 2 (at CFE 2) => priority 2 (or vice versa)
- For party lines, the links of a CFE interface may have different priorities. However, whether the communication load can better be shared between both CFE interfaces by this method depends on the features of the communication protocol and the ability of the RTU

The status information PREFERRED is to be set to the CFE links and not to the CFE interface.

3.2.4 Enhanced Redundancy for Protocols IEC101 and IEC104

3.2.4.1 Introduction

Enhanced Redundancy provides up to four links per RTU to increase availability and flexibility of the communication system. In case of serial communication, the links are grouped in “Dependency Groups” see Figure 3-2.54

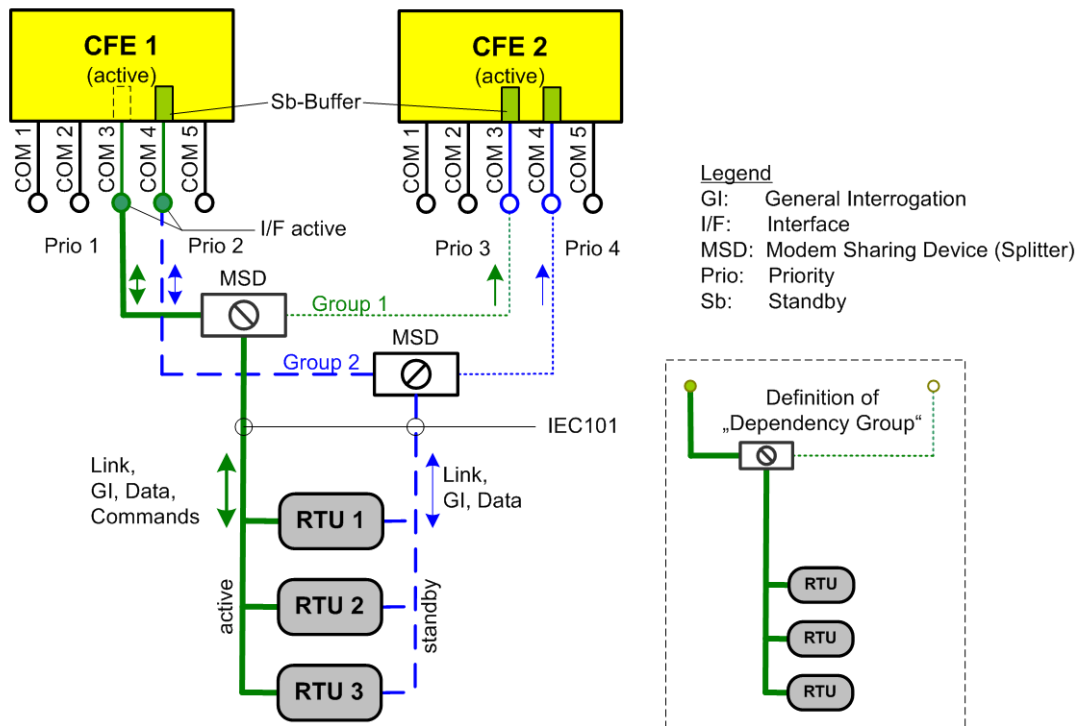


Figure 3-2.54: Enhanced IEC101 Redundancy

If communication is based on TCP/IP (i.e., protocol IEC104), the links are grouped in “Redundancy Groups”, see Figure 3-2.55

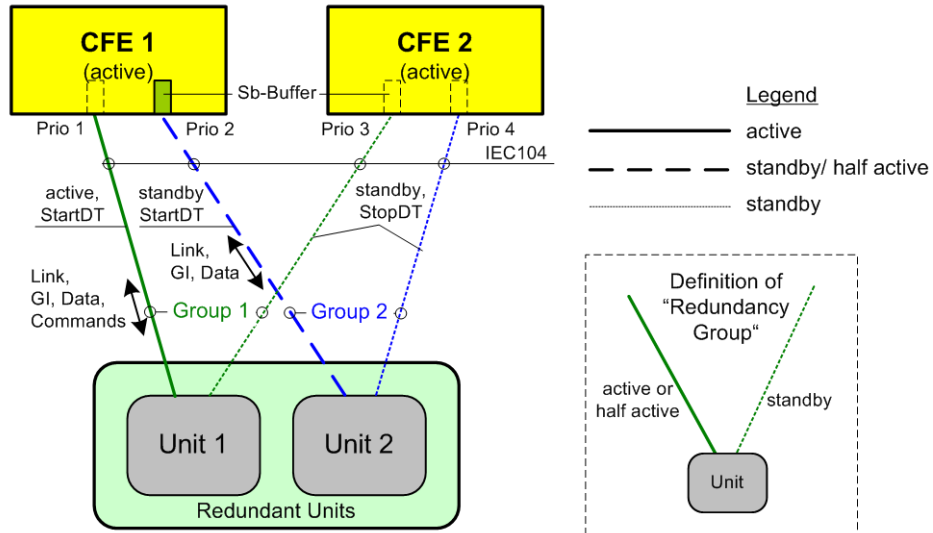


Figure 3-2.55: Enhanced IEC104 Redundancy

The following overview diagram shows all instances involved in Enhanced Redundancy and the necessary system links between them in a generalized presentation, Figure 3-2.56.

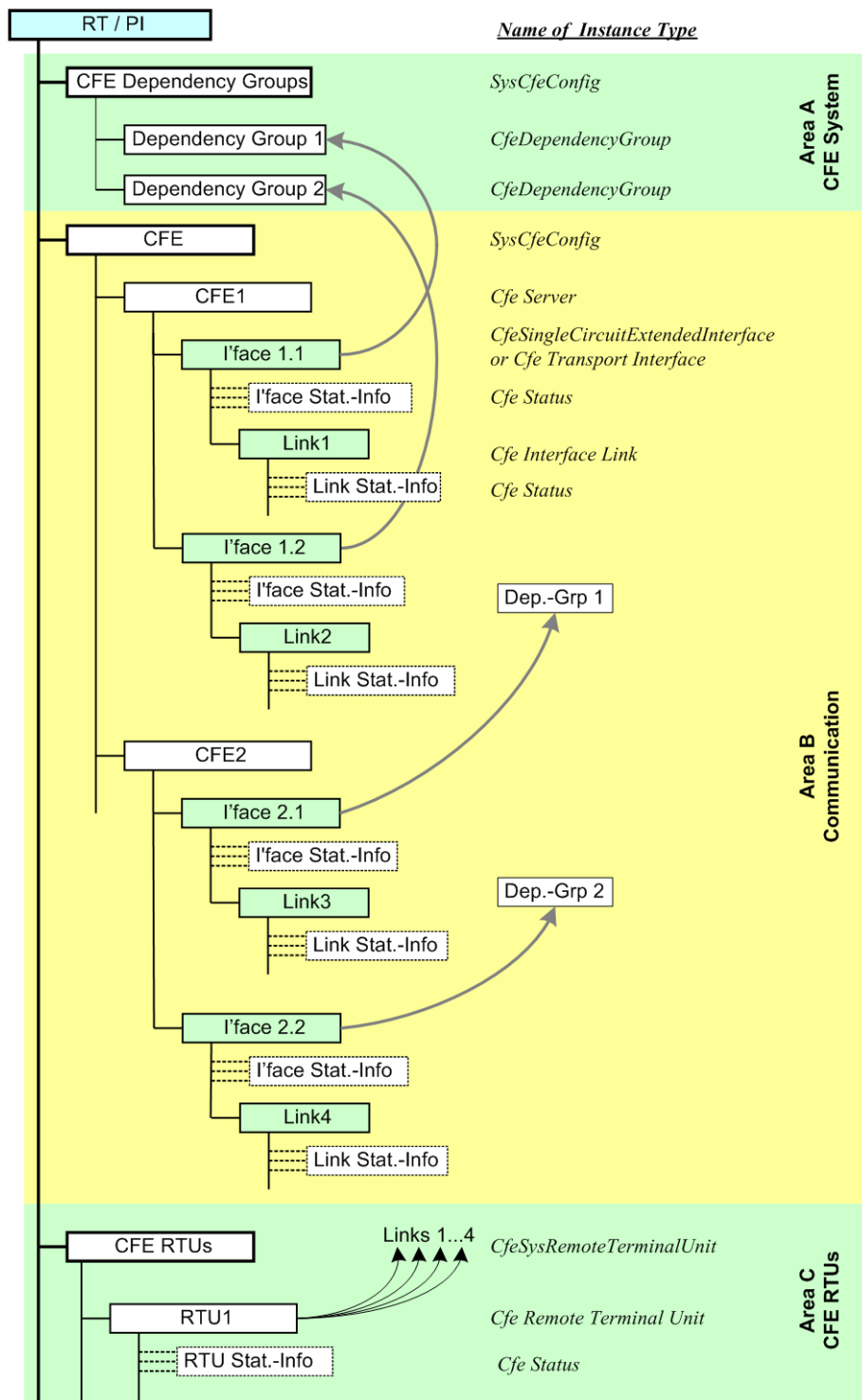


Figure 3-2.56: Instances Involved in Enhanced Redundancy

**Note**

CFE data model does not make a difference between “Dependency Groups” and “Redundancy Groups”. Instances of the type *CfeDependencyGroup* are used for both types of communication redundancy.

The only difference between IEC101 communication and IEC104 communication in CFE data model is:

1. Type of interface instance (IEC101: *CfeSingleCircuitExtendedInterface*, IEC104: *CfeTansportInterface*)
2. Status information (see Table 3-2.4)

3.2.4.2 Configuring a Redundant Communication Setup

Configuring redundant communication setups is done in these steps:

1. Creating the required Dependency Groups
2. Creating & configuring interfaces at CFE
3. Creating & configuring interface links
4. Creating connected components (RTU, Data Concentrator)
5. Adding Status Information, as appropriate, for Interface, Links, RTU

3.2.4.2.1 Creating a Dependency Group

To create a Dependency Group, perform these steps:

1. Right-click the parent instance “CFE Dependency Groups” and select **New** from popup menu
2. Select the instance type *CfeDependencyGroup* from 'Type Selection Box' in the Work Pane and click **Create Instance**
3. Enter the name of the new instance, e.g. "RTU01_DepGrp1"

3.2.4.2.2 Creating and Configuring Interfaces at CFE

To create a communication interface at CFE server, perform these steps:

1. Right-click the desired CFE server and select **New** from popup menu
2. Select the instance type according to the protocol from 'Type Selection Box'
 - a. for an IEC101 interface: *CfeSinleCircuitExtendedInterface*
 - b. for an IEC104 interface: *CfeTansportInterface*
 ... and click **Create Instance**
3. Enter the name of the new instance
4. Open the tab **Links** and allocate the interface to a protocol and one of the Dependency Groups created before, see Figure 3-2.57

General Attributes Links CfeTransportInterface View	
Associations	Value
(HasA) CfeProtocol	PI/CFE Protocols/Protocol: IEC-104
CfeDependencyGroup	PI/CFE Dependency Groups/Daco01-RedGroup 1

Figure 3-2.57: System Links from Interface to Protocol and Dependency Groups

- Open the tab **Attributes** or the protocol specific tab and attend to the attributes. Most of the attributes can be left as per default, however, some of them must be taken care about, see following table

Table 3-2.3: Important IEC101 Interface Attributes to take care about

Attribute	Remark
Baudrate	Baudrate of interface and RTU must fit together
Data Length	replace 0 with 255
Serial Port	select a free COM - port at the CFE server
Transmission Mode	must also fit to Transmission Mode of RTU

3.2.4.2.3 Creating and Configuring Interface Links

To create an interface link perform these steps:

- Right-click the CFE interface of interest and select **New** from popup menu
- Select the instance type *Cfe Interface Link* from 'Type Selection Box' in the Work Pane and click **Create Instance**
- Enter the name of the new instance
- Open the tab **Attributes** or the protocol specific tab and edit the attributes. Most of the attributes can be left as per default, however, some of them must be set individually, see table following table

Attribute	IEC101	IEC104	Remark
Link Address	1 ... n	n/a	must match to RTU link address
Priority	1 ... 4	1 ... 4	must be unique for all links of an RTU!
Port No.	n/a	2404	2404 is standard for IEC104; (other port numbers would also work at CFE)
Remote IP Address	n/a	Format: nnn.nnn.nnn.nnn	must match to RTU IP address

3.2.4.2.4 Creating Connected Components

To create a connected component (RTU or Data Concentrator) perform these steps:

- Right-click the parent instance "CFE RTUs" and select **New** from popup menu
- Select the instance type *Cfe Remote Terminal Unit* from 'Type Selection Box' in the Work Pane and click **Create Instance**
- Enter the name of the new instance
- Open the tab **Links** and allocate the RTU to all links connecting the RTU to CFE, see Figure 3-2.58

General Attributes Links IEC60870-5-104	
Associations	Value
(ConnectsThrough) CfeInterfaceLink	PI/CFE/CfeRed2/IEC104: TI DACO 2/1P Link 4
	PI/CFE/CfeRed1/IEC104: TI DACO 2/1P Link 3
	PI/CFE/CfeRed1/IEC104: TI DACO 1/1P Link 1
	PI/CFE/CfeRed2/IEC104: TI DACO 1/1P Link 2
Belongs To (ModelingAuthoritySet)	
Has A (CfeFiletransferDefs)	

Figure 3-2.58: System Links from RTU to the Interface Links

5. Open the tab **Attributes** or the protocol specific tab and attend to the following attributes as appropriate:

Attribute	Remark
Area of Responsibility	Select one Area (field must not be left empty!)
Common Address	Must match to the RTU address
Data Point Address Structure	Must match to the RTU settings
RTU Model	Only relevant for RTUs and DaCos of the type CDC, Modbus, IEC103, SMART-U-FEP. In case of SMART-U-FEP: select 'IEC104_Smart FEP'. For other RTUs and DaCos leave default 'Not used'
Local Node Unit	Only relevant for virtual RTU of SMART-U-FEP: select either 'SmartFep Arbiter RTU' or 'SmartFep Diagnosis RTU'. For other Sub-RTUs leave default 'Not used'

3.2.4.2.5 Adding Status Information for Interface, Links, RTU

Finally, you may add optional status information to interface, link and RTU. The following table summarizes status information for interfaces and links.

Table 3-2.4: Status Information Available for IEC101/IEC104 Interfaces and Links

Status Information	IEC101 Interface	IEC101 Link	IEC104 Interface	IEC104 Link
Active	Mon ³⁾	Mon	Mon	Mon
Failure	Mon	Mon	Mon	Mon
Locked	Mon/Ctr ⁴⁾	Mon/Ctr	Mon/Ctr	Mon/Ctr
Preferred ¹⁾	Mon/Ctr	Mon/Ctr	Mon/Ctr	Mon/Ctr
Listen Mode	Mon	-	-	-
Unit Active ²⁾	-	-	-	Mon
Unit Main	-	-	-	Mon/Ctr

¹⁾ *Preferred* is only useful if the links are independent

²⁾ *Unit Active* and *Unit Main* is status information for Data Concentrator of the Type “Smart FEP”

³⁾ *Mon* means: Status data point provides information for monitoring

⁴⁾ *Mon/Ctr* means: Status data point provides monitoring information and can be controlled



Note

Handling of status information is described in detail in Chapter 3.2.1.5.2 *Creating Status Information / Statistic Counters*

3.3 Configuring Slave Functionality

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Slave functionality means that CFE behaves like an RTU. This feature allows the CFE to emulate one RTU or a group of RTUs for the communication with external telecontrol masters. That way Spectrum PowerCC's Frontend is in a position to communicate to any (also non-Siemens) Remote Control Center (RCC) through standard telecontrol protocols such as IEC 60870-5-101 or IEC 60870-5-104. Configuring the necessary IMM instances is very similar as for CFE-RTU communication. This is described in the following chapter.

3.3.1 Creating and Configuring RCC Instances

The creation and parameterizing of the RCC Instances such as RCC, Logical RTU, data points to be forwarded, is described in the following sections. Please note that the examples used hereafter are based on the standard protocol IEC104Slave. Other protocols may have different attributes and settings. Detailed information for each implemented protocol can be obtained from chapter "Protocol Specific Engineering" later in this document.

3.3.1.1 Overview of RCC Instances

The following chart shows the hierarchical arrangement of RCC instances in the Realtime / Process Interface (PI) branch.

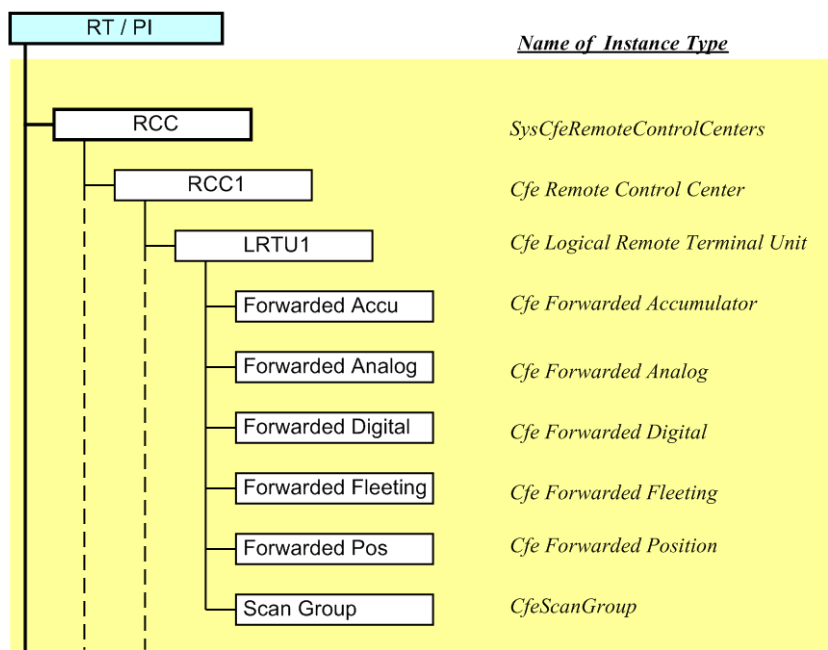


Figure 3-3.1: Hierarchical Arrangement of RCC Instances

3.3.1.2 Creating the RCC System Instance

If not yet available the RCC system instance is to be created as a child of the RT/PI instance. The instance type is named *SysCfeRemoteControlCenter*.

To create this system instance right-click the instance "PI" in the Overview Pane and select **New** from pop-up menu that appears (or click the "New Instance" item of the Insert menu called from the Tool bar instead). The 'Type Selection Box' appears in the Work Pane displaying the instance types available for this area. Select *SysCfeRemoteControlCenter* and click the **Create Instance** button.

The Instance Editor is displayed in the Work Pane. Please enter the name of the new PI component, for example "CFE RCCs" and save the new instance.

3.3.1.3 Creating and Configuring a Remote Control Center Instance

For modelling the communication behavior of the Remote Control Center in PowerCC the instance type *Cfe Remote Control Center* is provided. To create this instance right-click the parent instance of the type *SysCfeRemoteControlCenter* (named here: CFE RCCs) and select **New** from pop-up menu that appears. From 'Type Selection Box' that appears in the Work Pane select *Cfe Remote Control Center* and click the **Create Instance** button.

The Instance Editor is displayed in the Work Pane.

Tab 'General'

Tabulator 'General' of the Instance Editor is opened first. Please specify here the name of the new component, for example "RCC A". Optionally, you may enter any describing text into the row 'Description'.

Tab 'Attributes'

Next step is to open the tab *Attributes* and to specify appropriate communication parameters such as protocol to be used (IEC60870-5-101 or IEC60870-5-104), object address structure, time-outs, time formats, see Figure 3-3.2

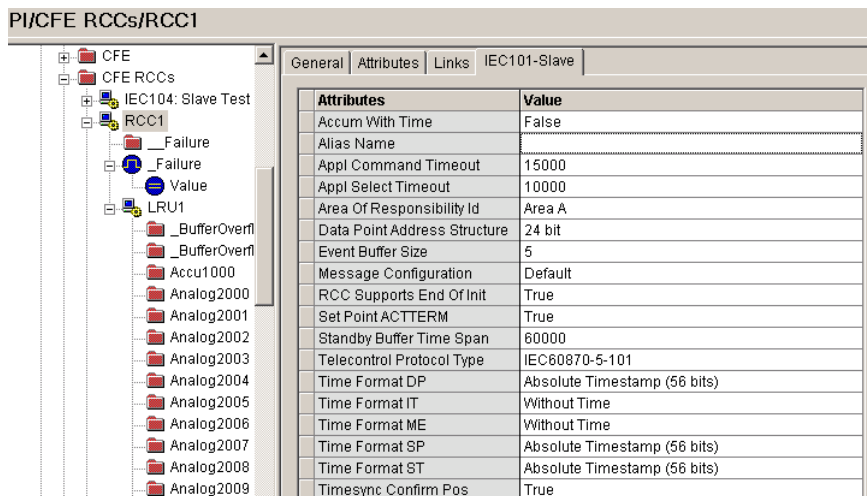


Figure 3-3.2: Communication Attributes of the Remote Control Center

Tab 'Links'

The tabulator 'Links' contains two Associations:

- (ConnectsThrough)InterfaceLink: Points to the physical link(s) that connects/connect the RCC to CFE (mandatory)
- (Overrides)InterlockingRule: Points to an interlocking rule for commands coming from RCC and to be routed to an RTU (optional)

Tab 'CfeRemote ControlCenterView'

This tab is only temporarily used and contains the same items as the tab 'Attributes'. After saving and re-opening the instance this tab is replaced with a protocol-specific tab, e.g. 'IEC104Slave'. The protocol-specific tab displays only a sub-quantity of the 'Attributes' tab.

3.3.1.4 Creating a Logical RTU

From external Control Center CFE looks like a data concentrator with connected sub-RTUs. That way it is possible to allocate forwarded data points to virtual RTUs representing real substations data originate from. Another way might be to assign spontaneous data to one Logical RTU, periodical data of high priority to another. The instance type *Cfe Logical Remote Terminal Unit* is used to specify these kind of 'logical' RTU.

To create a Logical RTU instance, right-click the just created parent instance (type *Cfe Remote Control Center*) and select **New** from pop-up menu that appears. From 'Type Selection Box' that appears in the Work Pane select *Cfe Remote Terminal Unit* and click the **Create Instance** button. The Instance Editor is displayed in the Work Pane.

Tab 'General'

Tabulator 'General' is opened first. Please enter the name of the new component, for example "RTU_A1". Filling text in the 'Description' row is optional.

Tab 'Attributes' or Protocol-Specific Tab

Next step is opening the tab *Attributes* (or the protocol-specific tab) and specifying at least the 'Common Address'. Optionally, you may define the parameters 'Area of Responsibility Id' or 'Startup Timeout' (Delay), see Figure 3-3.3

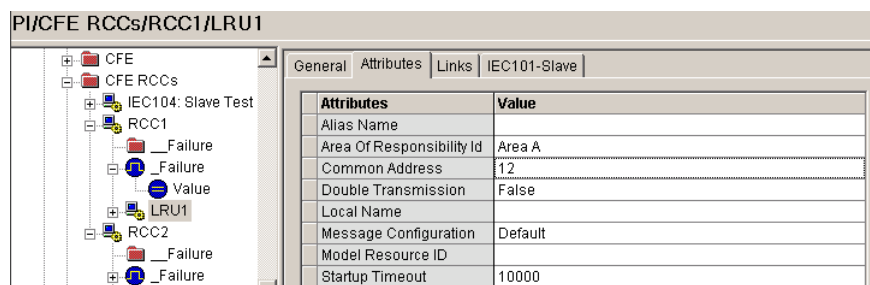


Figure 3-3.3: Communication Attributes of the Logical (RCC) RTU

Tab 'Links'

Tab 'Links' does not contain any information

3.3.1.5 Creating 'Forwarded' Data Points

Similar as a real RTU that sends data to "its" Control Center, the virtual RCC-RTU provides data for forwarding to "its" Remote Control Center. In opposite direction, controls can be defined. This enables the RCC to send control commands or setpoints via a Spectrum PowerCC Control Center to a real RTU that is connected to CFE.

The following types of data points can be configured:

- Counter values (Instance type 'Forwarded Accumulator')
- Analog values (Instance type 'Forwarded Analog')
- Digital values (Instance type 'Forwarded Digital')
- Fleeting data (Instance type 'Forwarded Fleeting')
- Step Positions (Instance type 'Forwarded Position')

To create a data point instance right-click the Logical RTU instance (type *Cfe Logical Remote Terminal Unit*) and select **New** from pop-up menu that appears. From 'Type Selection Box' that appears in the Work Pane select the desired data type for example *Cfe Forwarded Analog* and click the **Create Instance** button, see Figure 3-3.4

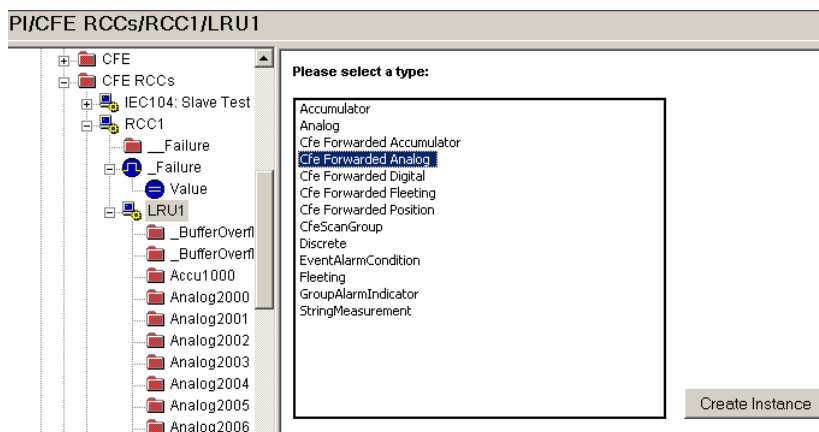


Figure 3-3.4: Type Selection Box for Selecting Data Point Instances

Then, the Instance Editor is displayed in the Work Pane.

Tab 'General'

The Instance Editor with tabulator 'General' is opened first. Please enter the name of the new component, for example "Unit1_Power". Filling text in the 'Description' row is optional.

Tab 'Attributes' or Protocol-Specific Tab

Next step is opening the tab *Attributes* (or the protocol-specific tab) and specifying the 'Monitoring Direction Type' and the 'Monitoring TC Address'. Optionally, you may change the default values of the parameters 'GI-Relevant' and 'Spontaneous Transmission' as appropriate.

Note: If 'Spontaneous Transmission' is set to 'FALSE', the forwarded data point must be assigned to a scan group (see below). Otherwise it will not be sent to the Remote Control Center at all.

The parameters 'Control Direction Type' and 'Control TC Address' are responsible for specifying a setpoint in control direction, see Figure 3-3.5

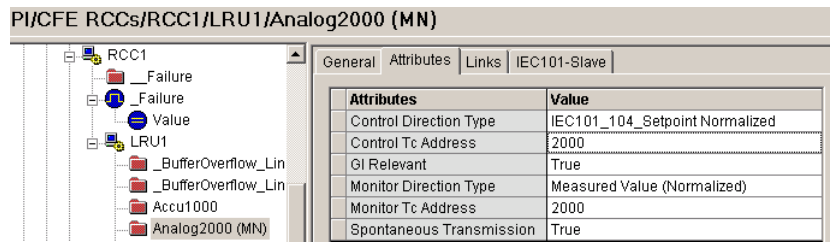


Figure 3-3.5: Attributes of the 'Forwarded Analog' Data Point

Tab 'Links'

The tabulator 'Links' contains the following associations for all data points:

(Receives From)Measurement: Points to the source of this data point in SCADA (if applicable)

(Sends Control To)Measurement: Points to the target of the data point in SCADA (if applicable)

**Note**

Unlike the system links of telemetered data points (coming from real RTUs), these links point to the instance Measurement (a child instance of *Substation*), not to the measurements child instance *Value* or *Control*

For Analog measurements the tabulator 'Links' contains two further associations:

(Has A) Cfe Characteristic Curve: Points to the characteristic curve to be applied in monitoring direction (optional)

(Has A Cntrl)Cfe Characteristic Curve: Points to the characteristic curve to be applied in control direction (optional)

The following diagram shows all possible system links for a data point of the type 'Forwarded Analog' including the links for the optional Scan Group (Figure 3-3.6).

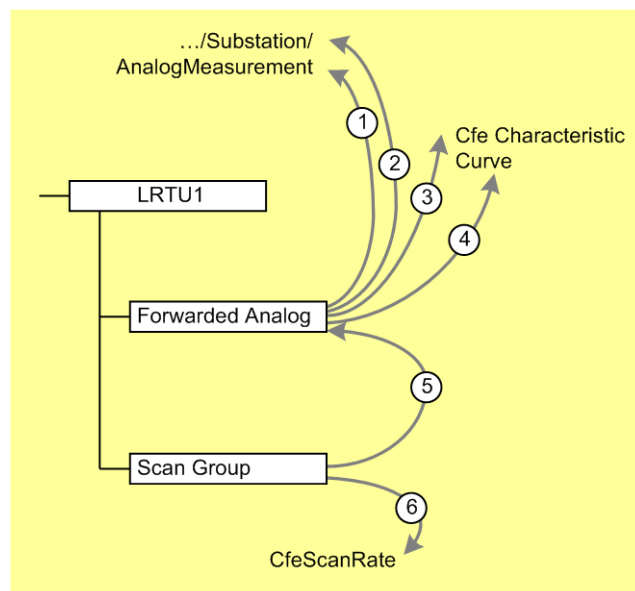


Figure 3-3.6: Possible System Links for a Data Point of the Type 'Forwarded Analog'

The following system links may be used:

1. Link to the instance type 'AnalogMeasurement' for analog data points to be sent to RCC
2. Link to the instance type 'AnalogMeasurement' for analog data received from RCC (setpoints)
3. Link to the instance type 'Cfe Characteristic Curve' for outgoing analog data
4. Link to the instance type 'Cfe Characteristic Curve' for incoming analog data (setpoints)
5. Link from instance type 'Scan Group' if the analog data point is to be sent cyclically
6. Link from Scan Group to the instance type 'CfeScanRate'

3.3.1.6 Creating a Scan Group

Scan Groups serve to transmit data from Spectrum PowerCC Control Center to the Remote Control Center cyclically. If all data is sent spontaneously, this instance is not required. If the RCC wants to receive data automatically in fixed intervals a 'Scan Group' is to be provided. To create a Scan Group instance right-click the Logical RTU instance (type *Cfe Logical Remote Terminal Unit*) and select **New** from pop-up menu that appears. From 'Type Selection Box' that appears in the Work Pane select *CfeScanGroup* and click the **Create Instance** button. The Instance Editor is displayed in the Work Pane.

Tab 'General'

Tabulator 'General' is opened first. Please enter a suitable name for the scan group, for example "ScanGroup10s".

Tab 'Attributes' or Protocol-Specific Tab

Next step is opening the tab *Attributes* (or the protocol-specific tab) and specifying the unique 'Scan Group Number' and the 'Scan Type' (select 'All_PeriodicTransm' for a cyclic scan group), see Figure 3-3.7

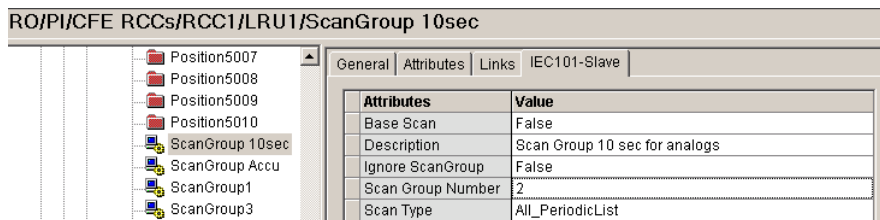


Figure 3-3.7: Scan Group Attributes

Tab 'Links'

The tabulator 'Links' contains two associations:

- (Has)CfeFwdDataPoint(s): Points to the data point(s) included in this cyclic transmission routine (mandatory for each cyclic data point)
- CfeScanRate: Points to the timer instance for this scan group (mandatory)

3.3.2 Creating and Configuring Communication Links to RCC

Establishing communication links from CFE to the external Remote Control Center is similar to those from CFE to RTUs. It is assumed that the user has already setup CFE-RTU links in IMM and as such he/she is familiar with this procedure. As such, the procedure to create and configure communication links from CFE to RCC is treated here only in brief.

These are the differences between RTU links and RCC links:

- The protocol allocated to the RCC interface is a 'Slave' type instead of a 'Master' type. In Spectrum PowerCC standard scope of supply, the IEC101-SL protocol is available for serial connections and the IEC104-SL for network connection.
- Only one link per interface can be established for an RCC (no party line communication is possible)

The following figures illustrate some typical RCC - CFE interconnections. In all these cases CFE is redundant which can be seen as the "normal" case in real installations.

1. The first two figures show serial independent connections between redundant CFE and redundant Master interface (Figure 3-3.8).

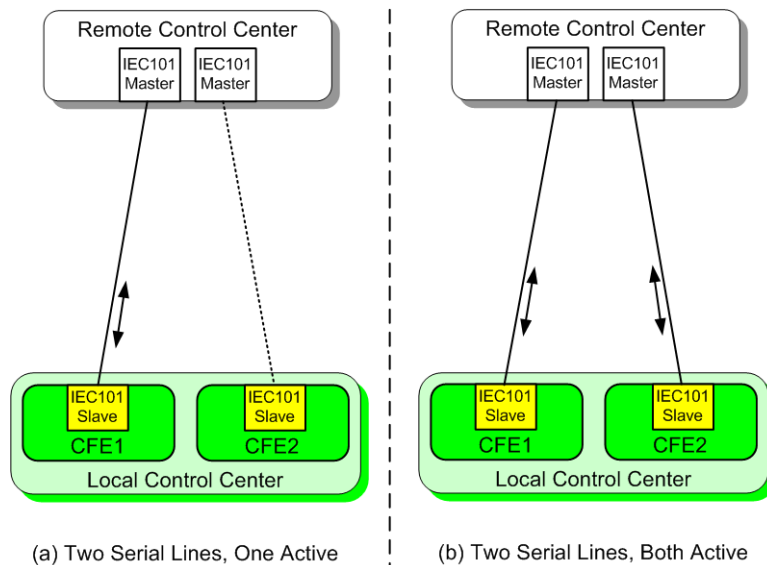


Figure 3-3.8: Master/Slave Configurations, Redundant CFE, Independent Serial Links

The following variants are possible:

- a. Redundant Master interface connected to redundant CFE, one link active, one link standby
- b. Redundant Master interface connected to redundant CFE, both links active

Note: The decision which link becomes active and whether one or both links are used simultaneously is taken by the RCC and not by CFE. There are no parameters in CFE to configure this.

2. The second group of figures illustrates different ways to connect an RCC to CFE over dependent serial links:

- a. Non-redundant Master interface connected to redundant CFE
- b. Redundant Master interface connected to redundant CFE, one link active
- c. Redundant Master interface connected to redundant CFE, both links active

Note1: In case of redundant links the RCC decides which link is to be used.

Note2: For redundant dependent links in CFE two *Dependency Groups* are to be defined.

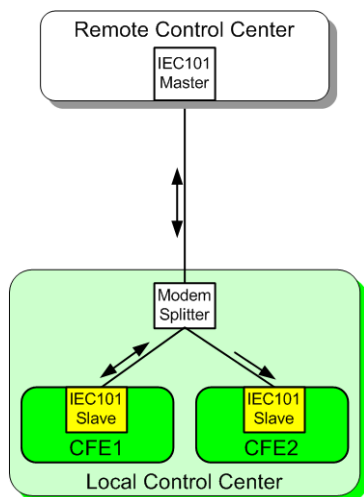


Figure 3-3.9: Master/Slave Configurations, Redundant CFE, Dependent Serial Links

3. The following two figures show basic ways to interconnect RCC and CFE using a network protocol (IEC104).
 - a. Non-redundant Master interface connected to redundant CFE
 - b. Redundant Master interface connected to redundant CFE

Note: Network links are always independent.

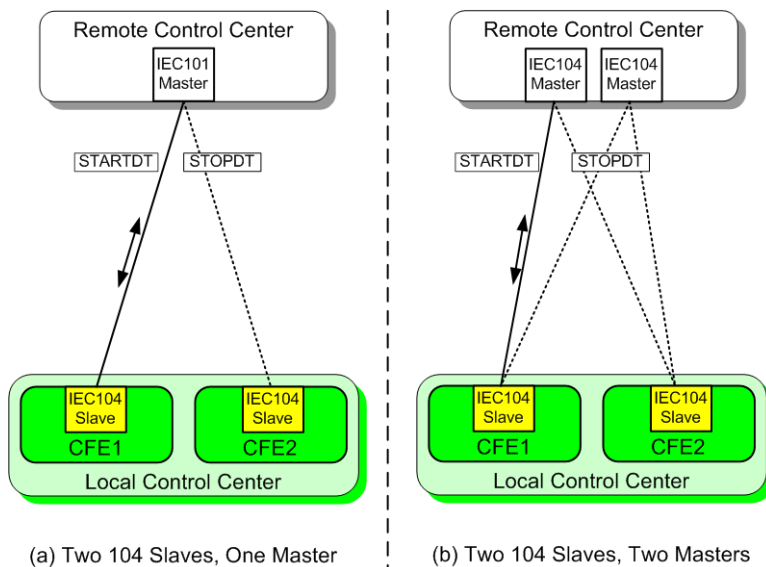


Figure 3-3.10: Master/Slave Configurations, Redundant CFE, Independent Network Links

Interfaces and communication links to RCCs are arranged in the Spectrum PowerCC data model as shown by the following diagram (Figure 3-3.11).

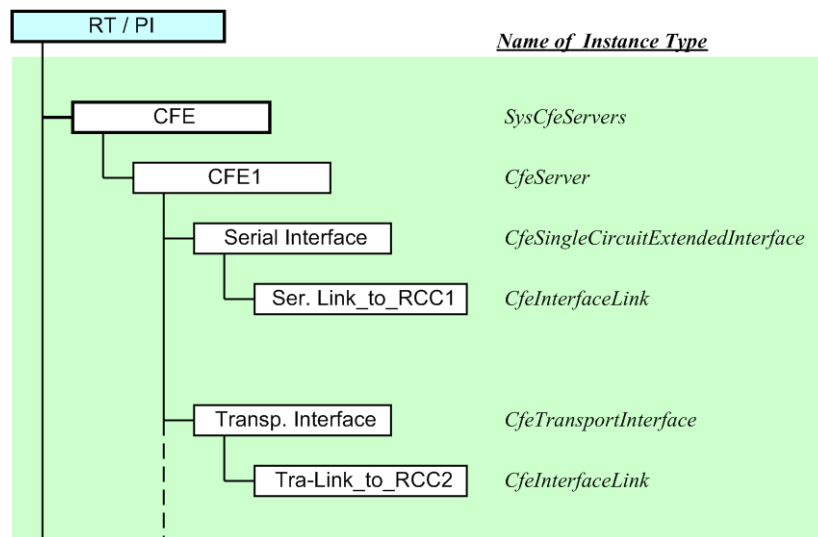


Figure 3-3.11: Hierarchical Arrangement of RCC Link Instances

3.3.2.1 Creating a Serial Link

To create a serial (IEC101) link for interconnecting an RCC to CFE follow these steps:

1. First create the interface instance type *CfeSingleCircuitExtendedInterface* as a child instance of CFE.
2. Open the tab 'Links' and establish a system link from protocol instance 'PI/CFE Protocols/IEC101-SL' to the association '(HasA)CfeProtocol', see Figure 3-3.12

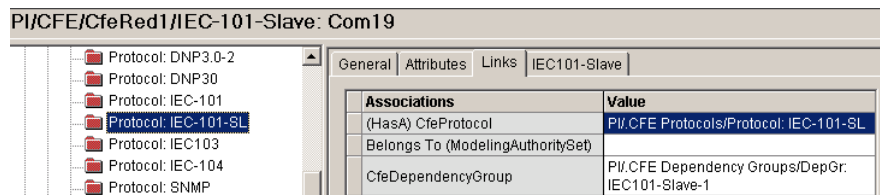


Figure 3-3.12: Establishing a System Link to the Slave Protocol Instance "IEC-101-SL"

In case of a dependent link, also assign the Association 'CfeDependencyGroup' to the instance representing the Dependency Group located in the Realtime branch 'PI/.CFE Dependency Group'

3. Finally, create the link instance type *CfeInterfaceLink* as a child instance of the interface

3.3.2.2 Creating a Network Link

To create a (TCP/IP based) network link for interconnecting an RCC to CFE follow these steps

1. First create the interface instance type *CfeTransportInterface* as a child instance of CFE
2. Open the tab 'Links' and establish a system link from protocol instance 'PI/CFE Protocols/IEC104-SL' to the association '(HasA)CfeProtocol', see Figure 3-3.13

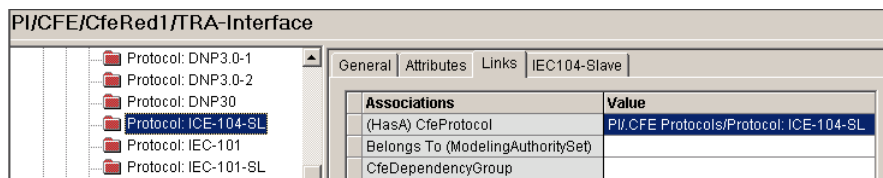


Figure 3-3.13: Establishing a System Link to the Slave Protocol Instance "IEC-104-SL"

3. Finally, create the link instance type *CfeInterfaceLink* as a child instance of the interface.



Note

Details on attributes for relevant types of interfaces and links can be obtained from chapter "Protocol Engineering" later in this document. Please also note that in case of redundant CFE this procedure is to be carried out for both CFE servers.

3.3.3 Status Information for Communication with RCCs

In order to monitor and control the communication with Remote Control Centers, appropriate status information is available in the system.

The following status information can be configured in IMM:

- Non-controllable status information:
 - Device Active (for all types of interfaces and links)
 - Device Failure (for all types of interfaces and links)
 - Buffer Overflow (for each link)
 - RCC Failure (if all links are down)
- Controllable status information:
 - Device Locked (applicable only for interfaces, not for links)
 - Device Preferred (applicable only for interfaces, not for links)

Status information for RCC and RCC-CFE communication is arranged in hierarchical order as shown by the following diagram (Figure 3-3.14).

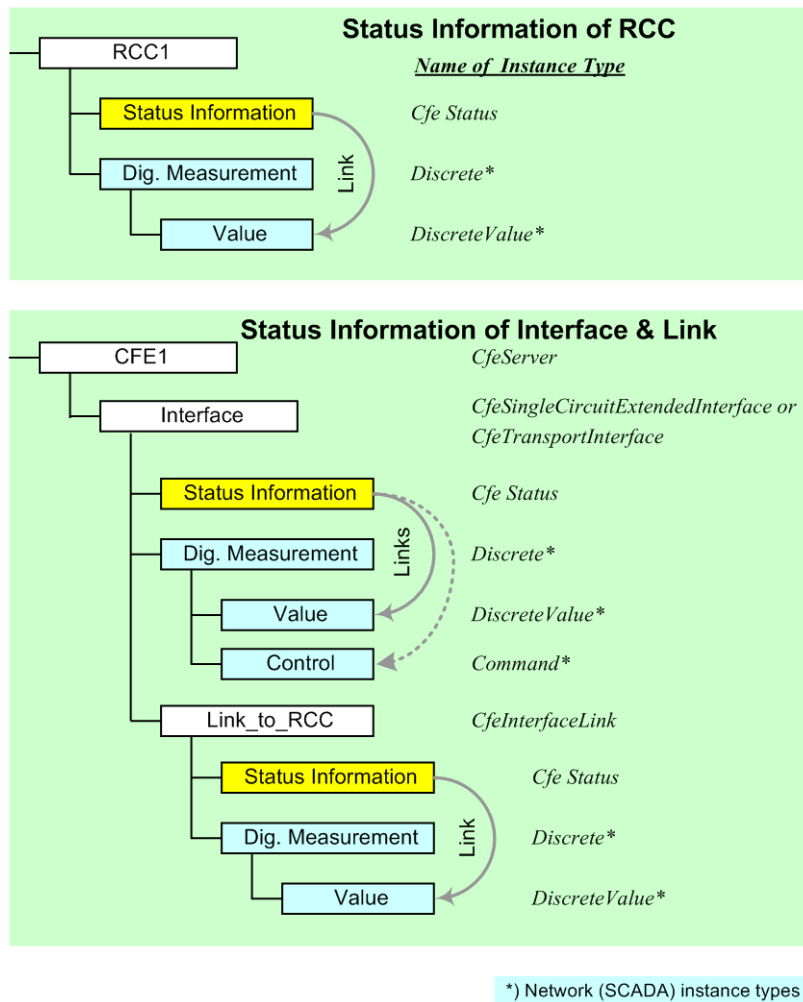


Figure 3-3.14: Arrangement of Status Information for RCC, Interface and Link

Note: System counters for communication statistics are not available for RCC connections.

Configuring status information for RCC communication is done the same way as for that of RTU communication:

1. An SCADA instance of the type *Discrete* to be created
2. A child instance of the type *DiscreteValue* is to be created
In case of a controllable status information: Also a child instance of the type *Command* is to be created
3. The desired Status Information (type *Cfe Status*) is to be created
4. Status Information and 'Value' instance are to be interlinked.

In case of a controllable status information also a link to the 'Command' instance is to be established (see above diagram)



Note

Detailed description including screenshots on creating and configuring status information can be found in Chapter 3.2.1.5.2 *Creating Status Information / Statistic Counters*.

3.4 Protocol-Specific Engineering of Master Protocols

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Introduction

In contrast to so called Peer-to-Peer protocols where all communication partners have equal rights, most of the telecontrol protocols work according to the "Master-Slave" principle. "Master Protocols" are those where CFE plays the role of a Master and the connected components (RTUs) are 'Slaves'. Protocols or protocol implementations to enable CFE to play the role of an RTU are called "Slave Protocols".

Parameters for RTU Communication

The configuration parameters for CFE interfaces, RTU communication links, the modelling of RTUs and their data points is a protocol specific task. The configuration data is laid down in 'attributes' of the respective CFE instances in the IMM. The following instance types are available to setup the different areas of communication with RTUs:

Table 3-4.1: Summary of CFE Instances Relevant for Communication with RTUs

Task	Respective Instance
Configuring global protocol settings	Cfe Protocol<protocol name>, e.g.: Cfe Protocol IEC101, Cfe Protocol DNP30, Cfe Protocol SINAUT8
Configuring the COM ports	<i>CfeSingleCircuitExtendedInterface</i>
Configuring the TCP/IP network interface:	<i>CfeTransportInterface</i>
Configuring the serial communication link to the RTUs:	<i>CfeInterfaceLink</i>
Modelling the RTU and DACO RTU:	<i>CfeRemoteTerminalUnit</i>
Setting up the scanning routine:	<i>CfeScanGroup</i> , <i>CfeScanVolume</i> , <i>CfeScanRate</i>
Specifying RTU process data points:	Digital Value: <i>CfeTelemeteredDigital</i> Analog Value: <i>CfeTelemeteredAnalog</i> Counter Value: <i>CfeTelemeteredAccumulator</i> Step Position: <i>CfeTelemeteredPosition</i> Fleeting Information: <i>CfeTelemeteredFleeting</i>
Setting up system counters:	<i>CfeSystemAccumulator</i>

Information on CFE Command Buffering

Along with PowerCC Release 4.2 the option 'Command Buffering' was introduced as a standard functionality. For better understanding, here are some details on this feature.

For all CFE protocols the Device Master module (DM) provides the option to queue RTU commands for a preset time. After the *Command Buffer Time* (t_1) has expired the affected command is canceled. In addition, the protocol modules (PM) for DNP3.0 and Sinaut 8FW, offer such a command buffer with its own "aging" monitoring timer (t_2). Please note that also the SUPY module in the SCADA package monitors each command for getting a confirmation from CFE (t_0) as well as for obtaining a reply message from RTU within a preset time, see next figure.

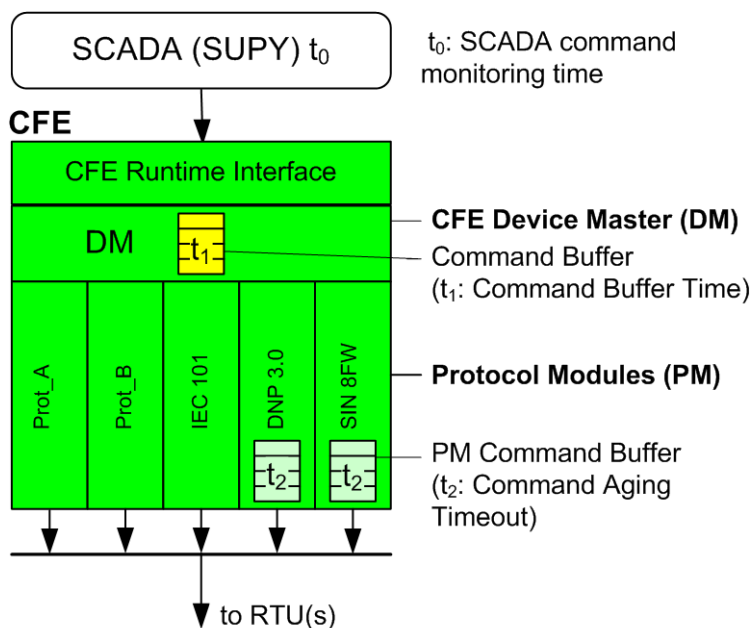


Figure 3-4.1: CFE Command Buffering

For setting correctly the different timers please follow this rule:

- t_1 and t_2 should have the same value (in case of DNP3.0 and Sinaut protocol)
- t_0 (SCADA monitoring time) should be twice as big than t_1
- for DNP3.0 the value for t_1 (and t_2) is calculated as follows:

$$t_1, t_2 > \text{duration_of_longest_scan} + \text{number_of_RTUs_on_line} * \text{command_duration}$$

In the following chapters, for each communication protocol available in CFE, the relevant instances and their attributes are presented. In the attribute tables of the instances the range of values, default values and, if applicable, recommended settings and interdependencies are indicated.



Note

In CFE editors you also will find some "System" attributes that are not CFE specific. To avoid the repetition of explanation on all places where they occur these attributes are missing in the tables hereafter. Its about the following system attributes:

- **Name** (mandatory, type String): Each instance must be named by a any (meaningful) name. All characters, such as numbers, blanks and most of the special characters are allowed to form the name. The maximum length of an instance name is limited to 32 characters.
- **Alias Name** (optional, type String): Alternative instance name
- **Area of Responsibility** (type Enumeration): Defines the access rights to the respective component. In most cases, default value is suitable

- Exchange Name (optional, type String): NERC exchange name imported from a CIM RDF file
- Description (optional, type String): any optional text describing the component
- Message Configuration (type Enumeration): Relates the Field Resource to a message configuration for alarming purposes. In most cases, default value is suitable

3.4.1 Protocol IEC 60870-5-101

Short name: IEC101

Parameterization of CFE protocol, interfaces, links and associated Remote Terminal Units (RTU) communicating with IEC101 telecontrol protocol requires the compliance of distinct rules described in the following.

3.4.1.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the protocol IEC101:

Table 3-4.2: Attributes of Instance Type *Cfe Protocol IEC101*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Common Address Bytes	Number of address bytes for the Common Address	1 or 2 octets	must suit to RTU settings
Communication Statistic Offset	Start offset related to 00:00 h	0 ... msec	As per customer requirements
Communication Statistic Period	For periodical transfer to display configured accumulators in SCADA	0 ... msec	As per customer requirements
Info Object Address Bytes	Number of address bytes for the Info Object Address	1, 2 or 3 octets	must suit to RTU settings
Link Address Bytes	Number of address bytes for the Link Address	0, 1 or 2 octets	must suit to RTU settings
Originator Address Bytes	Number of address bytes for the Originator Address	0 or 1 octet	must suit to RTU settings

3.4.1.2 CFE Interface Parameterization

IMM instance type: *CfeSingleCircuitExtendedInterface*

For configuring a CFE communication port for IEC101 protocol the following interface parameters are to be set:

Table 3-4.3: Attributes of Instance Type *CfeSingleCircuitExtendedInterface*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Acknowledge Timeout	Max time to wait for acknowledge message from RTU.	1200 b/s: 1000 msec 9600 b/s: 250 msec	Depends on baud rate
Acknowledge Type	Selection between short message and single character for link acknowledgement		As per customer requirements

Attribute Label	Meaning	Range/ Unit	Interdependencies
Area of Responsibility	AoR defines the rights to access this component	Area A Area A Gas Area A Water ... System (default)	
BaudRate	Transmission speed	1200, 1800, 2400, 4800, 9600, 19200, 38400 [bit/s]	As per customer requirements
Constant Carrier	If true modem carrier is always on (default)	False, True	
Data Length	Max length of message frames	max. 255 [byte]	As per customer requirements
Listen Mode	Mode for updating the process image (no controls)	False, True default: False	As per customer requirements
Polling Time	The interval between scans, (for RTUs in party line configuration)	0 ... n [ms]	As per customer requirements
Postconditional Delay	Delays carrier switching-off after sending messages	0 ... n [ms]	Only used in modem simplex operation
Preconditional Delay	Delays sending messages after switching-on the carrier	0 ... n [ms]	Only used in modem simplex operation
Serial Port	Number of the serial port	1 ... n	Depends on number of boards, and ports / board,
Transmission Mode	Selection between balanced (spontaneous) and unbalanced (polled) data transmission	For RTUs in party line configuration only unbalanced mode allowed	

3.4.1.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the IEC101 protocol link layer properties applied to communication lines between CFE interface and RTUs. The following attributes are to be set:

Table 3-4.4: Attributes of Instance Type *CfeInterfaceLink*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Clock synchronization interval	Time for periodical transmission of RTU synchronization messages	0 (if no synchronization needed) 60, 000 - 600, 000 [msec]	
GI Cycle Time	If periodical GI is requested	0... [msec]	As per customer requirements
Link Address	The Link Address of the device.	0 (if link address is omitted) 1 ... 255 or 1... 65535	The Link Address must match the address which has been configured in the RTU
Link Test Cycle Time	Timer for cyclically sending link test messages	0 ... n [ms]	
Polling Time	Waiting time between two data polls can be chosen individually for each link on a COM port	0 ... n [msec]	When transmission mode "unbalanced" for each link individually
Polling Time Out	Max time to wait for answer message from RTU	0 ... n [ms]	Depends on baud rate, number of stations in a party line

Attribute Label	Meaning	Range/ Unit	Interdependencies
Priority	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). 1 is highest priority	1... 8	
Successive Errors For RTU Fail	Number of successive transmission failures before a link failure is indicated	0 ... 10	

3.4.1.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters are valid for IEC101 protocol stack:

Table 3-4.5: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Application Timeout	General Interrogation monitoring time	1 ... msec	Depends on GI data volume and baudrate
Command Buffer Time Maximum	Used for command aging monitoring. Commands older than this time are deleted	0 = disabled 1... n [msec]	to be harmonized with corresponding settings in SCADA
Common Address	Common Address of the device. This must match the address which has been configured within the device itself.	1...255 / 65535	Depends on the number of common address bytes 1 or 2
Data Point Address Structure	structure of data point address as visualized on the IMM.	depends on length up to 3 octets, e.g. 8, 8-8, 8-8-8 bit structure 8-16 bit structure, 8, 16, 24 bit structure or other	Depends on the number of info object address bytes 1, 2 or 3
Renewal Cycle 1	Renewal control time in msec for the first control cycle.	0 ... n [ms]	Data points assigned to the first control cycle are subject of an updating check at the end of the cycle period.
Renewal Cycle 2	Renewal control time in msec for the second control cycle.	0 ... n [ms]	Data points assigned to the second control cycle are subject of an updating check at the end of the cycle period.
Telecontrol Protocol Type	serves to select the communication protocol for this COM port	IEC60879-5-101	

3.4.1.5 CFE Interrogation Group Parameterization

IMM instance type: *CfeScanGroup*

CFE acquires process data either by spontaneous transmission of RTU or by periodical Group Interrogations issued by CFE. To setup this functionality at least one CFE Interrogation Group has to be allocated to a "CfeRemoteTerminalUnit" instance.

Interrogation Group attributes are summarized in the following table.

Table 3-4.6: Attributes of IEC101/IEC104 *CfeScanGroup* Instance

Attribute Label	Type / Range / Default	Description / Remark
Description	String	Optional text
Delay Time	Long Integer / 0 (default) ... n [ms]	Used to delay General Interrogation at runup
Ignore Scan Group	Boolean / - False (default) - True	Used to (temporarily) disable the interrogation of this group leaving all other attributes unmodified
Scan Type	Enumeration / - All_Interrogation_General (for General Interrogation) - IEC101 Interrogation Group- Scan(for IEC101 Group Interroga- tion) - IEC104 Interrogation Group- Scan(for IEC104 Group Interroga- tion) other Scan Types are not applicable for IEC101/104	Type of Interrogation

Note: Other attributes of this instance that are not listed here are not relevant for IEC101/104 and are to be ignored.

3.4.1.6 CFE Interrogation Volume Parameterization

IMM instance type: *CfeScanVolume*

The attribute "Class Type" of the Interrogation Volume instance defines the type of Interrogation (General Interrogation / Group Interrogation 1....16)

All other attributes of the instance are not relevant for protocol type IEC101/ IEC104

Table 3-4.7: Attributes of IEC101/104 *CfeScanVolume* Instance

Attribute Label	Type / Range / Default	Description / Remark
Class Type	Enumeration / - Not used - IEC101 General Interrogation - IEC101 Group Interrogation No. 1 - IEC101 Group Interrogation No. 2 ... etc. till No.16 - IEC104 General Interrogation - IEC104 Group Interrogation No. 1 - IEC104 Group Interrogation No. 2 ... etc. till No.16	Defines the type of interrogation (General Interrogation / Group Interrogation for Group 1 through 16)

3.4.1.7 Cfe Interrogation Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic interrogation needs the specification of a period time and a starting time. This is provided by *CfeScanRate* attributes. Typically, an interrogation rate definition may be valid for more than one interrogation group. Therefore it is defined as a global parameter instance in the CFE Scanrates container instance (instance type *sysCfeConfig*).

Table 3-4.8: Attributes of IEC101/104 *CfeScanRate* Instance

Attribute Label	Type / Range / Default	Description / Remark
Scan Offset	Long Integer / [msec] -1, 0 (default) ... n (Granularity is 1 000 msec)	Offset from midnight for issuing interrogation commands -1 causes free running; interrogation starts immediately after the connection to RTU(s) is established
Scan Period	Long Integer / [msec] 1000 ... n (Granularity is 1 000 msec)	Time between start of interrogations. Consider all interrogations at one interface. It must be enough time for every scan to be answered and space for commands to be sent

Note: Other attributes of this instance that are not listed here are not relevant for IEC101/104 and are to be ignored.

3.4.1.8 Data Point Parameterization

3.4.1.8.1 RTU Data Points

For protocols IEC101 / IEC104 the following data types are distinguished:

- Digital (IMM instance type: *CfeTelemeteredDigital*)
- Analog (IMM instance type: *CfeTelemeteredAnalog*)
- Counter (IMM instance type: *CfeTelemeteredAccumulator*)
- Fleeting (IMM instance type: *CfeTelemeteredFleeting*)
- Step Position (IMM instance type: *CfeTelemeteredPosition*)

In the following table detail information on RTU data points are summarized. The column “Applied for” indicates for which data type the respective attribute is applicable (a= analog, c= counter, d= digital, f= fleeting, p= step position).

Table 3-4.9: Attributes of IEC101 / IEC104 Data Points

Attribute Label	Meaning	Applied For	Range/ Unit / Default (underlined>	Interdependencies / Remark
Command operation mode	Specification of command operation mode	d, p	- Direct - Select-Before-Operate	
Command Qualifier	Additional information to a command, for example the command duration or kind of output.	d, p	0 ... 63	

Attribute Label	Meaning	Applied For	Range/ Unit / Default (underlined>	Interdependencies / Remark
Control Direction Type	Type of command / setpoint in control direction.	a, d, p	Not used Single Command Double Command	This data point has no control. Command has only one output contact. Command has two output contacts. ON / OFF
Control Tc Address	Telecontrol address for control direction.	a, d, p	1 ... 65535	Always required if "Control Direction Type" is Single, Double or Setpoint Command.
GIRelevant	If set TRUE the data point is relevant for completeness check	a, d, f, p	False, True	Should be set "TRUE" for all spontaneously transmitted data
Intermediate State Suppression Time	Intermediate state is held back for this time	d	0 = disabled, 1 ... msec	
InvertControl	Inversion of status for command direction	d	False, True	
InvertInfo	Inverts the received information	f	False, True	Applicable for info type "Fleeting" only
Monitor Direction Type	Info type in monitor direction	a, c, d, f, p	Not used Single Point Double Point Fleeting Step Position Counter Value ...	Not used: This data point has no information in monitoring direction
Monitor Info Conversion	Inversion / conversion of information in monitoring direction	d	Not used SinglePoint invert DoublePoint invert Bit 0 DoublePoint invert Bit 0+1 DoublePoint invert Bit 1	Effect to SCADA: 1->2, 2->1 0->1, 1->0, 2->3, 3->2 0->3, 1->2, 2->1, 3->0 0->2, 1->3, 2->0, 3->1
Monitor Tc Address	Telecontrol address for monitoring direction	a, c, d, f, p	1 ... 16 776 215 (max)	For IEC101: Range depends on number of address octets used
Old New Check	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer	d, p	False, True	Should be set "TRUE" for all digital data (except fleetings)
Renewal check	Data point is subject to cyclic updating checks	a, c, d, p	Disabled Cycle 1 of renewal check enabled Cycle 2 of renewal check enabled	Should be enabled for all cyclically transmitted data
SOE_Double Transmission Enabled	Enabling processing of double-transmitted information	1, c, d, f, p	False, True	
System Control Direction ID	---	d	Not Used	Not used
System Control Extension	If "control direction type" = "System Command", this attribute provides additional information for control command processing.	d	Not Used	Not applicable

Attribute Label	Meaning	Applied For	Range/ Unit / Default (underlined>	Interdependencies / Remark
System Monitor Direction ID	If "monitor direction type" = "System Point", select a system information type from this list according to the protocol applied. (parameter only means no data point, used as protocol adjustment)	a, d, f	Not used	Not applicable
System Monitor Extension	If "monitor direction type" = "System Point", this attribute provides additional information for system info processing.	f	Not used	Not applicable
Smooth Factor	The smooth factor is used to "smooth" a noisy analog value. The formula for smoothing is: smoothed value = (current value + sf*last smoothed value) / (sf+1) where sf is the smooth factor.	a	0	
Threshold Value	To suppress analogs without significant changes to be passed-on to SCADA	a	0	
Zero Suppress Lower Border	If a value is greater than ZeroSuppressedLowerBorder and less than Zero Suppressed Upper Border it is not transmitted to the next layer.	a	0	
Zero Suppress Upper Border	See above.	a	0	

3.4.1.8.2 System Counters

IMM instance type: *CfeSystemAccumulator*

The IEC101 protocol provides system counters to obtain communication statistics. Details on errors and counters can be found in the following table.

Table 3-4.10: Attributes of Instance IEC101 *Cfe System Accumulator*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Info ID	Select the counter of interest. see chapter 2.11 in CFE-TS.	Message Send Counter Message Receive Counter Message Error Receive Counter Message Timeout Counter	Communication statistic counters are reported cyclically dependent on "Communication Statistic Period"

3.4.2 Protocol IEC 60870-5-104

Short name: IEC104

Parameterization of CFE protocol, interfaces, interface links and associated Remote Terminal Units (RTU) communicating with IEC104 transport protocol requires the compliance of distinct rules described in the following.

3.4.2.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the protocol IEC104:

Table 3-4.11: Attributes of Instance Type *Cfe Protocol IEC104*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Communication Statistic Offset	Start offset related to 00:00 h	0 ... msec	As per customer requirements
Communication Statistic Period	For periodical transfer to display configured accumulators in SCADA	0 ... msec	As per customer requirements

3.4.2.2 CFE Interface Parameterization

IMM instance type: *CfeTransportInterface*

For configuring a CFE TCP/IP interface for IEC104 protocol the following parameters are to be set:

Table 3-4.12: Attributes of Instance Type *CfeTransportInterface* for IEC104

Attribute Label	Meaning	Range/ Unit	Interdependencies
Cfe IP Address	This attribute is used for the parametrization of the CFE server IP address. The CFE IP address, the communication port and the transport protocol type "TCP" uniquely define the local Transport Endpoint.	Format: nnn.nnn.nnn.nnn	
Max. Pending GIs	Maximum number of pending General Interrogations. Only relevant to IEC104.	0 ... 65535	

3.4.2.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The interface link parameters specify the IEC104 protocol transport link layer properties applied to communication lines between CFE interface connected RTUs.

Table 3-4.13: Attributes of Instance Type *CfeInterfaceLink*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Timeout Connection Establishment	Time (t_0) the CFE waits for an response from a remote station, after the CFE have tried to establish a connection to that station.	[msec]	
Outstanding I-Format APDUs: k	Maximum number (k) of sequentially numbered I format APDUs that the remote station may have unacknowledged at a given time		
GI Cycle Time	if periodical GI is requested	[msec]	As per customer requirements
Timeout Sending Test Frame - Long Idle State	Time out (t_3) for sending test frames in case of a long idle state	[msec]	
Outstanding I-Format APDUs: w	Latest acknowledge after receiving (w) I format APDUs		
Priority	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). 1 is highest priority	1... 8	
Remote IP Address	IP address of remote communication peer. Any connection establishment requires an IP address, a defined port number (2404) and the applied transport protocol. This triplet of communication parameters defines uniquely the remote Transport Endpoint	Format: nnn.nnn.nnn.nnn	
Timeout for Ackn I- and U-Format APDUs	Time out (t_1) for acknowledges of I- and U-format APDUs	[msec]	
Timeout for Ackn - No Data Messages	Time out (t_2) for acknowledges in case of no data messages	[msec]	Note: (t_2) of station B needs to be less than (t_1) of station A

3.4.2.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters are valid for IEC104 protocol stack.

Table 3-4.14: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Application Timeout	General Interrogation monitoring time	[msec]	Depends on GI data volume and data throughput
Command Buffer Time Maximum	Used for command aging monitoring. Commands older than this time are deleted	0 = disabled1... [msec]	to be harmonized with corresponding settings in SCADA
Common Address	Common Address of the device. This must match the address which has been configured within the device itself	1 . . . 65535	Fix 2 octets
Data Point Address Structure	Structure of data point address	24 bit structure 8-16 bit structure, 16-8 bit structure, 8-8-8 bit structure or other combination.	Always 3 octets
Renewal Cycle 1	Renewal control time in msec for the first control cycle	0 ...n [ms]	Data points assigned to the first control cycle are subject of an updating check at the end of the cycle.
Renewal Cycle 2	Renewal control time in msec for the second control cycle	0 ... n [ms]	Data points assigned to the second control cycle are subject of an updating check at the end of the cycle.
Telecontrol Protocol Type	Serves to select the communication protocol for this COM port	IEC60870-5-104	

3.4.2.5 CFE Sub-RTU Parameterization

IMM instance type: *CfeSubRemoteTerminalUnit*

For RTUs connected to a Data Concentrator, the following list of Sub RTU parameters are valid:

Table 3-4.15: Attributes of Instance Type *CfeSubRemoteTerminalUnit*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Common Address	Common Address of the device. This must match the address which has been configured within the device itself.	1 . . . 65535	Always 2 octets
Data Point Address Structure	structure of data point address	24 bit structure 8-16 bit structure, 16-8 bit structure, 8-8-8 bit structure or other combination.	Always 3 octets

Attribute Label	Meaning	Range/ Unit	Interdependencies
Renewal Cycle 1	Renewal control time in msec for the first control cycle	0 ... n [ms]	Data points assigned to the first control cycle are subject of an updating check at the end of the cycle.
Renewal Cycle 2	Renewal control time in msec for the second control cycle.	0 ... n [ms]	Data points assigned to the second control cycle are subject of an updating check at the end of the cycle.
Telecontrol Protocol Type	Serves to select the communication protocol for this port	IEC60870-5-104	

3.4.2.6 Parameterization of Interrogation Groups

Interrogation Groups can be configured for type IEC104 RTUs as well as for Sub-RTUs. The attributes of the instances CfeScanGroup, CfeScanVolume and CfeScanrate are the same as for IEC101.

3.4.2.7 Data Point Parameterization

3.4.2.7.1 RTU Data Points

Same as for IEC101

3.4.2.7.2 System Counters

Not implemented

3.4.3 Protocol DNP3.0 / DNP3i

Parameterization of CFE protocol, interfaces, interface links and associated Remote Terminal Units (RTU) communicating with DNP3.0 and DNP3.0 over TCP/IP (DNP3i) telecontrol protocol requires the compliance of distinct rules described in the following.

3.4.3.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the protocols DNP3.0 / DNP3i:

Table 3-4.16: Attributes of Instance Type *Cfe Protocol DNP30*

Attribute Label	Type / Range / Default	Description / Remark
Command Aging Timeout	Long Integer / [msec] 1 000 ... 20 000	The command is removed from the Protocol-Module buffer after the given time, when not sent yet. For details on Command Aging Monitoring see also Figure 3-4.1
Communication Statistic Offset	Long Integer / [msec] -1 (default), 0 ... n	Start offset related to 00:00 h -1 means: Start immediately after CFE run-up with specified Communication Statistic Period
Communication Statistic Period	Long Integer / [msec] 0 (off, default) ... n	For periodical transfer to display configured accumulators in SCADA

3.4.3.2 CFE Serial Interface Parameterization (DNP3.0)

IMM instance type: *CfeSingleCircuitExtendedInterface*

For configuring a CFE communication port for DNP3.0 protocol the following interface parameters for dedicated line communication mode are to be set:

Table 3-4.17: Attributes of DNP3.0 *CfeSingleCircuitExtendedInterface*

Attribute Label	Type / Range / Default	Description / Remark
Baud Rate	Enumeration / [bit/s] - 300 - 600 - 1800 - 2400 - 4800 - 9600 (default) - 19200	Transmission speed for receiving and transmitting data
Channel Supervision Timeout	Long Integer / [msec] 0 (default) ... n	If a communication problem remains longer than this timeout, "Interface Failure" is reported
Check Receiver Carrier	Boolean / - False (default) - True	For Modem control. TRUE: receiver Carrier off before sending. FALSE for duplex channels TRUE for simplex channels
Constant Carrier	Boolean / - False - True (default)	For modem control. TRUE: carrier always on
Dial Connecting Time	Long Integer / [sec] 0 (default), 1 ... 60	Max. Time to establish a dial connection to an RTU. Only applicable for Port Type "Dial"
Dial Repeating Time	Long Integer / [sec] 0 (default), 1 ... 20	Time to wait before unsuccessful dialing is repeated. Only applicable for Port Type "Dial"

Attribute Label	Type / Range / Default	Description / Remark
Dial Retries Of Connection	Long Integer / 0 (default) ... n	Number of dial retries after unsuccessful dialing. Only applicable for Port Type "Dial"
Idle Time Transmit	Long Integer / [msec] 0 (default) ... 60	Minimum waiting time between transmitted message frames
Inactivity Timer	Long Integer / [sec] 0 (default) ... 180	Modem hangs up after this time of inactivity (no messages received any longer) Only applicable for Port Type "Dial"
Interbyte Timeout	Long Integer / [msec] 0 ... 16 (default)	Specifies max. gaps between message bytes
Life Cycle Time	Long Integer / [msec] 1 000 ... 60 000	Used for cyclic check of "dead" links
Listen Mode	Boolean / - False (default) - True	Mode for updating the process image (no controls)
Modem Type	String /	Name of the Modem. Exact copy of displayed name in Windows "Phone and Modem Options" / tab "Modem"
Port Type	Enumeration / - Serial - Dial	Serial: Direct communication via defined "Serial Port". Dial: Communication via configured "Phone and Modem Options" of CFE computer and "Modem Type"
Postconditioning Delay	Long Integer / [msec] 0 (default) ... 60	For modem control Time after sending the message before removing RTS.
Preconditioning Delay	Long Integer / [msec] 0 (default) ... 60	For modem control Time between setting ReadyToSend (RTS) before sending the message
Primary Address	Long Integer / 0 (default) ... 65 519	Station address of this master interface DNP3 source address
Receive Message Timeout	Long Integer / [msec] 1 ... 300 (default)	Time the communication equipment will wait to receive the first byte of the response message before declaring a timeout
Serial Port	Short Integer / 1 ... 255	Number of selected serial port. Range depends on number of boards, and ports per board

3.4.3.3 CFE Transport Interface Parameterization (DNP3i)

IMM instance type: *CfeTransportInterface*

For configuring a CFE TCP/IP interface for DNP3i protocol the following parameters are to be set:

Table 3-4.18: Attributes of DNP3.0 *CfeTransportInterface*

Attribute Label	Type / Range / Default	Description / Remark
CFE IP Address	String /	CFE server's IP address Format: nnn.nnn.nnn.nnn
Primary Address	Long Integer / 0 (default) ... 65519	Source Address Address of this server / channel
Life Cycle Time	Long Integer / [msec] 1 000 ... 60 000	Used for cyclic check of "dead" links

3.4.3.4 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the DNP3.0 protocol link layer properties applied to communication lines between CFE interface and RTUs. The following attributes are to be set:

Table 3-4.19: Attributes of DNP3.0 *CfeInterfaceLink*

Attribute Label	Type / Range / Default	Description / Remark
Calculate Channel Delay	Boolean / - False - True	If enabled channel delay is calculated prior to transmitting new time information to the RTU. DNP3.0 over serial connection True: Use objects 52 and 50 for time sync, master does the calculation. False (default): Use only object 50 variation 1 for time sync. DNP3.0 over LAN (DNP3i) True (default): Use function 24 and object 50 variation 3 for time sync, RTU does the calculation (standard) False: Use only object 50 variation 1 for time sync (only for serial communication connected via LAN).
Clock Synchronization Interval	Long Integer / [msec] 0 (default) ... 1 000 ... 60 000	Time for periodical transmission of RTU synchronization messages (0: synchronization disabled)
Data Link Retry	Short Integer / 0 (default) ... 255	Number of times the CFE repeats an unconfirmed data link message
Data Link Timeout	Long Integer / [msec] 500 ... 3 000 default: 2 000	Time interval the CFE waits for RTU confirmation of a data link message before repeating it. Only valid if Link Confirmation is enabled.
Dial Max Connect Time	Long Integer / [sec] 0 (default) ... n	Maximum time of a dialed connection. This is used to make sure that the call is eventually ended.
Link Address	Long Integer / 0 ... 65535	The Link Address must match the address which has been configured within the device itself. Range 65520 ... 65535 is used for group addressing
Link Confirmation	Boolean / - False (default) - True	If enabled, CFE requests a RTU conformation of a data link message. Only applied to SEND/CONFIRM and REQUEST/RESPONSE data link service.
Link Test Cycle Time	Long Integer / [msec] 0 (off, default) ... 20 000	Time interval for periodical link test
Port Number	Long Integer / [msec] -1 (default), 0 ... 20 000	TCP/IP port number to use when communicating with RTU. -1 to be replaced with the standard DNP port 20 000
Priority	Short Integer/ 1... 8	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). Highest priority is 1.
Remote IP Address	String / (Format: nnn.nnn.nnn.nnn)	Only for DNP3i IP Address of remote communication peer. Any connection establishment to a remote server requires an IP address, a defined port number (20000) and the applied transport protocol. That triplet of communications parameters specifies uniquely the remote transport endpoint
Successive Errors For RTU Fail	Long Integer/ 1 ... 3 (default) ... n	RTU fails after this number of consecutive errors

3.4.3.5 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following RTU parameters are valid for DNP3.0 Level 3 protocol stack

Table 3-4.20: Attributes of DNP3.0 *CfeRemoteTerminalUnit*

Attribute Label	Type / Range / Default	Description / Remark
Application Retry	Long Integer / 0 (default) ... 255	Number of times the CFE repeats an unconfirmed application message
Application Timeout	Long Integer / [msec] 500 ... 2 000 (default)	This time the CFE waits for RTU to confirm an application message before repeating it.
Command Buffer Time Maximum	Long Integer / [msec] 0 (off) ... 10 000	Used for central command aging monitoring. Commands older than this time are deleted. (To be harmonized with corresponding settings in SCADA)
Data Point Address Structure	Enumeration / - 32 bit - 8 bit : 8 bit : 8 bit : 8 bit - 16 bit : 16 bit	Selection according to the best representation of data point addressing
Enable Time Synchronization	Boolean / - False - True (default)	False: No time synchronization for this RTU at all
Phone Number	String /	Telephone number to be used in dialup access to the RTU. Only applicable for Port Type "Dial"
Renewal Cycle 1	Long Integer / [msec] 0 (default) ... n	Monitoring time for the first updating check cycle
Renewal Cycle 2	Long Integer / [msec] 0 (default) ... n	Monitoring time for the second updating check cycle
Request for Application Confirm	Boolean / - False - True (default)	Application layer message requires confirmation NOTE: Don't use it for this master application. It's not recommended through DNP (TB2000-004) Must be "False" for DNP Master applications, otherwise DNP will not work!
Telecontrol Protocol Type	Enumeration / - DNP3.0	Last table will display the selected protocol and its attributes for this instance and its children: Select DNP3.0 for this protocol
Time Zone	String / (posix syntax) Examples: - (no entry, default) - <MYT+0> - <MYT+1> - <MYT-1> - posix/Europe/Warsaw - posix/Europe/Berlin	Time Zone of the RTU - No entry: zone of own server - UTC - Constant positive offset of RTU Time to UTC, no DLST - Constant negative offset of RTU Time to UTC, no DLST - Rules of the given zone, inclusive DLST - Rules of the given zone, inclusive DLST Note: For a proper function of this "TimeZone_Calculations" environment variable needs to be set correctly to the path of "TimeZoneData". E.g.: "TZDIR"="C:\Siemens\PowerCC\Apps\CFE\TimeZoneData" "TZPATH"="C:\Siemens\PowerCC\Apps\CFE\TimeZoneData;C:\Siemens\PowerCC\Apps\CFE\TimeZoneData\posix;C:\Siemens\PowerCC\Apps\CFE\TimeZoneData\posix\Europe" (allows multiple paths and reduces the string required, e.g. without posix)

**Note**

Due to the unpredictable time needed to establish a connection, for dial-in / dial-out connections the RTU should perform the freezing of integrated totals. The CFE just reads then the “frozen” values.

3.4.3.6 CFE Scan Group Parameterization

IMM instance type: *CfeScanGroup*

CFE acquires process data either by unsolicited responses of RTU or by emitting user-defined data scans. To set up a user data scan at least one CFE Scan Group has to be allocated to a “CfeRemoteTerminalUnit” instance. Scan Group attributes are summarized in the following table:

Table 3-4.21: Attributes of DNP3.0 *CfeScanGroup*

Attribute Label	Type / Range / Default	Description / Remark
Base Scan	Boolean / - False (default) - True	Scan group describes cyclic base scan. It's continuously sent when line is free and no other commands or scans scheduled. Be careful with it's use because it can overload your communication. For fully cyclic scan no scan rate is needed
Ignore Scan Group	Boolean / - False (default) - True	Used to (temporarily) disable the complete Scan Group leaving all other attributes unmodified
Integrity Scan	Boolean / - False (default) - True	Scan group describes an integrity scan. After this scan was send a completeness check will be performed on all data points marked with Girelevant = TRUE. There can be only one Integrity Scan per RTU
Monitoring Retries	Long Integer / 0 (default)... n	Number of retries of scan control requests
Monitoring Timeout	Long Integer / [msec] 100 ... n	Time before a not answered request will be retried or indicated as failed
Scan Group Number	Long Integer / -1 (default), 0 ... n	Identification number of scan group. If used for demand scan it must be greater than 0 and unique within one <i>CfeRemoteTerminalUnit</i>
Scan on Event	Enumeration / - Not used - DNP_Initial Unsolicited Response	Start this scan group upon selected event. This scan is always started when the CFE receives an empty unsolicited message from the RTU
Scan Priority	Short Integer/ 1 ... 99 (default) ... n	Scan priority within the same activation type of scans. Internal scans have priority 10
Scan Type	Enumeration / - DNP30_Common Scan (FC01) - DNP30_Freeze (FC07) - DNP30_Freeze Reset (FC09) - DNP30_Freeze Poll (FC07 + FC 01) - DNP30_Freeze Reset Poll (FC09 + FC 01)	Type of scan. The objects or classes this scan applies to is defined in the ScanVolumes - Reads the information defined in ScanVolume - Performs a freeze without request for acknowledgement on the information defined in ScanVolume(s) - Performs a freeze on the information defined in ScanVolume(s) - Performs a freeze and reset on the information defined in ScanVolume(s) - Performs a freeze on the information defined in ScanVolume(s) and reads the frozen values

Attribute Label	Type / Range / Default	Description / Remark
	- DNP30_Freeze NoAck (FC08) - DNP30_Freeze Poll NoAck (FC08 + FC 01) - DNP30_Freeze Reset Poll NoAck (FC10 + FC 01) - DNP30_Unsolicited Enable (FC20) - DNP30_Unsolicited Disable (FC21) - DNP30_Assign to Class 0 (FC22) - DNP30_Assign to Class 1 (FC22) - DNP30_Assign to Class 2 (FC22) - DNP30_Assign to Class 3 (FC22)	- Performs a freeze and reset on the information defined in ScanVolume(s) and reads the frozen values - Performs a freeze without request for acknowledgement on the information defined in ScanVolume(s) and reads the frozen values - Performs a freeze and reset without request for acknowledgement on the information defined in ScanVolume(s) and reads the frozen values. These freeze scans are only for Counters and Analogs. "Poll" can only be used with "RequestMode" "DNP30_All data scan" - Activates / deactivates the unsolicited mode of the RTU for the objects/classes defined in SCanVolume(s) - Assigns the objects defined in ScanVolume(s) to the given class. Class 0 just cancels the assignment. In ScanVolume only assignable objects are allowed

3.4.3.7 CFE Scan Volume Parameterization

IMM instance type: *CfeScanVolume*

A scan volume instance describes the type and volume of process data to be scanned by the DNP master. Only one scan volume instance per scan group is permitted. If more than one volume is configured the sequence of the objects in a request is defined by the alphabetical order of the CfeScanVolume names.

Table 3-4.22: Attributes of DNP3.0 *CfeScanVolume*

Attribute Label	Type / Range / Default	Description / Remark
Class Type	Enumeration / - Not used - DNP30_Class0 - DNP30_Class1 - DNP30_Class2 - DNP30_Class 12 - DNP30_Class 3 - DNP30_Class 13 - DNP30_Class 23 - DNP30_Class 123 - DNP30_Class 0123	Defines the volume of data objects associated with a particular data class type or combinations of data classes. Use Object Type and not Class Type. Any selection of data class disables specified values of Object Type
End Tc Address	Long Integer / 0 ... n	End Tc Address Range Point definition. Number of points for DNP30_Count scan

Attribute Label	Type / Range / Default	Description / Remark
Object Type	Long Integer / 0100, 0101, 0102, 0200, 0201, 0202, 0203, 1000, 1002, 2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2100, 2101, 2102, 2103, 2104, 2105, 2106, 2107, 2108, 2109, 2110, 2111, 2112, 2200, 2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208, 2300, 2301, 2302, 2303, 2304, 2305, 2306, 2307, 2308, 3000, 3001, 3002, 3003, 3004, 3005, 3006, 3100, 3101, 3102, 3103, 3104, 3105, 3106, 3107, 3108, 3200, 3201, 3202, 3203, 3204, 3205, 3206, 3207, 3208, 3300, 3301, 3302, 3303, 3304, 3305, 3306, 3307, 3308, 4000, 4001, 4002, 4003, 4004, 5001, 6000, 6001, 6002, 6003, 6004, 8001	Protocol specific object type for data to be scanned in this scan volume. For DNP3.0 object/variation values type in OOVV data format OO=object group number, VV=variation number Depends upon class type. Type value is only valid for ClassType=not used.
Request Mode	Enumeration / - DNP30_All data scan - DNP30_Count scan - Range Point definition - Single Point definition	Request mode describes the scan range - All points of the given object or class. Has to be used for Scan Type "Freeze and Poll" - Amount of info objects defined in EndTcAddress. Only for events and classes! - A range of info objects starting from StartTcAddress to EndTcAddress. Only for static information! - Just one point defined by StartTcAddress
Start Tc Address	Long Integer / 0 ... n	Start Tc address for Range Point definition. Address for Single Point definition

3.4.3.8 Cfe Scan Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic scan needs the specification of a period time and a starting time. This is provided by a *CfeScanRate* attribute. Typically, a scan rate definition may be valid for more than one scan group. Therefore it is defined as a global parameter instance in the CFE DNP Scanrates container instance (instance type *sysCfeConfig*).

Table 3-4.23: Attributes of Instance DNP3.0 *CfeScanRate*

Attribute Label	Type / Range / Default	Description / Remark
Scan Offset	Long Integer / [msec] -1, 0 ... n (Granularity is 1 000 msec)	Offset from midnight for issuing scan requests -1 causes free running; scanning starts immediately after the connection to RTU(s) is established
Scan Period	Long Integer / [msec] 1000 ... n (Granularity is 1 000 msec)	Time between start of scans. Watch all cans at one interface. There must be enough time for every scan to be answered and space for commands to be sent

3.4.3.9 Data Point Parameterization

For the protocols DNP3.0 / DNP3i the following data types are distinguished:

- Digital Value (IMM instance type: *CfeTelemeteredDigital*)
- Analog Value (IMM instance type: *CfeTelemeteredAnalog*)
- Counter Value (IMM instance type: *CfeTelemeteredAccumulator*)
- CFE Status Information (IMM instance type: *CfeStatus*)
- System Counter (IMM instance type: *CfeSystemAccumulator*)

In the following tables detailed information on DNP3 RTU data points are summarized.

3.4.3.9.1 Data Type "Digital Value"

IMM Instance Type: *CfeTelemeterdDigital*

Table 3-4.24: Attributes of Instance Type *CfeTelemeterdDigital*

Attribute Label	Type / Range / Default	Description / Remark
Command Off Time	Long Integer / [msec] 0 (default) ... n	Release time for OFF-command
Command On Time	Long Integer / [msec] 0 (default) ... n	Release time for ON-command
Command Qualifier	Enumeration / - No additional definition - DNP30_Pulse On - DNP30_Pulse Off - DNP30_Latch On - DNP30_Latch Off	Additional information to a command, for example the command release duration or kind of output. Defines the DNP message the Control code: - NULL operation (0) - Pulse On (1) - Pulse Off (2) - Latch On (3) - Latch Off (4)
Command Operation Mode	Enumeration / - Direct Operation (default) - Direct Operation w/o confirmation - Select before operate	Command sequence: - (FC05) and RTU has to respond with an status - (FC06), RTU doesn't respond - (FC03 + FC 04), RTU has to respond with a status to each request
Control Direction Type	Enumeration / - Not used (default) - Single Command - Double Command - System Command	Type of digital in control direction - This data point has no control - Command has only one output contact (Obj12, Var01, Control Code - TRIP/CLOSE = NULL) - Command has two output contacts (Obj12, Var01) Only one Control Tc Address, no Control Tc Address2: (Control Code - TRIP/CLOSE = TRIP or CLOSE); Both Control Tc Addresses used: (Control Code - TRIP/CLOSE = NULL) Control Tc Address for TRIP. Control Tc Address2 for CLOSE - Has to be defined additionally in "System Control Direction ID"

Attribute Label	Type / Range / Default	Description / Remark
Control Tc Address	Long Integer / -1 (not used, default) 0 ... 65535	Telecontrol address for control direction. If protocol requires 2 addresses it's used for TRIP / OPEN / LOWER. Always required if Control Direction Type is "Single Command" or "Double Command"
Control Tc Address2	Long Integer / -1 (not used, default) 0 ... 65535	Second control telecontrol address for CLOSE / RAISE (if not same as or derived from ControlTcAddress) Only required if Control Direction Type is "Double Command"
Enable SOE/Double Transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real- time information (for archive)
GI Relevant	Boolean / - False (default) - True	True: the data point is relevant for completeness check One ScanGroup of this RTU must be marked as "IntegrityScan"
Intermediate State Suppression Time	Long Integer / [msec] 0 (off, default), 1 ... n	Intermediate state is held back for this time
Invert Control	Boolean / - False (default) - True	Inversion of status for command direction
Monitor Direction Type	Enumeration /	Type of digital in monitor direction
	- Not used (default) - Single Point - Double Point - System Point - DNP30_SingleCommand Status	- This data point has no indication - One-bit information (Obj01&02). - Two-bit information "Monitor Tc Address2" required (Obj01&02 any supported variation) - Has to be defined additionally in "System Monitor Direction ID" - Displays the binary output status read back from the RTU (Obj10, Var02)
Monitor Info Conversion	Enumeration / - Not used (default) - SinglePoint invert (1->2, 2->1) - DoublePoint invert Bit 0 (0->1, 1->0, 2->3, 3->2) - DoublePoint invert Bit 1 (0->2, 1->3, 2->0, 3->1) - DoublePoint invert Bit 0+1 (0->3, 1->2, 2->1, 3->0)	Inversion / conversion of information in monitoring direction It modifies the received information to be correctly displayed in SCADA
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... 65535	Telecontrol address for monitoring direction. If protocol requires 2 addresses it's used for TRIP / OPEN / OFF / LOW
Monitor Tc Address2	Long Integer / -1 (not used, default) 0 ... 65535	Second monitor address for ON / CLOSE / HIGH. Only required if Monitor Direction Type is "Double Point"
Old New Check	Boolean / - False (default) - True	True: The incoming value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer
Renewal Check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclical updating checks

Attribute Label	Type / Range / Default	Description / Remark
Send GI on	Enumeration / - none (default) - on appear - on disappear - on appear and disappear	Defines the trigger event for an automatic GI request. Only applied on system information generated by RTUs. Example: Buffer overflow One ScanGroup of this RTU must be marked as "IntegrityScan"
System Control Direction ID	Enumeration / - Not used (default) - DNP30_Cold Restart - DNP30_Warm Restart - DNP30_Set data to default - DNP30_ScanOnDemand - DNP30_Enable Unsolicited Mode - DNP30_Assign Class 1 - DNP30_Assign Class 2 - DNP30_Assign Class 3	If Control Direction Type is "System Command", select a system control type from this list according to the protocol applied. - Send a request with application control function code Cold Start (13) - Send a request with application control function code Warm Start (14) - Send a request with application control function code Set data to default (15) - Every ScanGroup can be started by hand. The "System Control Extension" must contain the "Scan Group Number" of "CfeScanGroup" as identification. The following controls require also a none FREEZE-"CfeScanGroup". The "System Control Extension" identifies the "CfeScanGroup" by "Scan Group Number". ON enables unsolicited mode, OFF disables unsolicited mode of the data configured in the corresponding "CfeScanGroup". ON assigns to the class, OFF assigns to class 0 (removes from class 1, 2, 3 of the data configured in the corresponding "CfeScanGroup")
System Control Extension	Long Integer / 1 ... n	"Scan Group Number" of "CfeScanGroup" as identification. If Control Direction Type is "System Command", this attribute provides additional information for control command processing
System Monitor Direction ID	Enumeration / - Not used (default) - DNP30_All station message received - DNP30_Time synchronization from master required - DNP30_Digital output points in local state - DNP30_Device trouble - DNP30_Device restart - DNP30_Function code not implemented - DNP30_Requested object(s) unknown - DNP30_Parameters not valid or out of range - DNP30_Buffer(s) overflowed - - DNP30_Requested operation already executing - DNP30_Current configuration of substation is corrupted	If Monitor Direction Type is "System Point", select a system information type from this list according to the protocol applied. (Parameter only means no data point, used as protocol adjustment) Information of Internal indication (IIN) field of application response header.
System Monitor Extension	Long Integer /	If Monitor Direction Type is "System Point", this attribute provides additional information for system info processing. Not yet needed for DNP.

3.4.3.9.2 Data Type "Analog Value"

IMM Instance Type: *CfeTelemeterdAnalog*

Table 3-4.25: Attributes of Instance Type *CfeTelemeterdAnalog*

Attribute Label	Type / Range / Default	Description / Remark
Command Off Time	Long Integer / [msec] 0 (default) ... n	Release time for OFF-command Required for Control Direction Type "DNP30_DoubleCmd Variable Count" only
Command On Time	Long Integer / [msec] 0 (default) ... n	Release time for ON-command Required for Control Direction Type "DNP30_DoubleCmd Variable Count" only
Command Qualifier	Enumeration / - DNP30_Pulse On - DNP30_Pulse Off	Additional information to a command, for example the command release duration or kind of output. Defines the DNP message the Control code: - Pulse On (1) - Pulse Off (2) Required for Control Direction Type "DNP30_DoubleCmd Variable Count" and "DNP30_DoubleCmd Variable Time" only
Command Operation Mode	Enumeration / - Direct Operation (default) - Direct Operation w/o confirmation - Select before operate	Command sequence: - (FC05) and RTU has to respond with an status - (FC06), RTU doesn't respond - (FC03 + 04), RTU has to respond with a status to each request
Control Direction Type	Enumeration / - Not used (default) - DNP30_Setpoint 16-bit - DNP30_Setpoint 32-bit - DNP30_Setpoint Short Float - DNP30_Setpoint Long Float - DNP30_DoubleCmd Variable Time - DNP30_DoubleCmd Variable Count	Type of digital in control direction. - This data point has no control - A 16 bit signed value. (Obj41, Var02) - A 32 bit signed value. (Obj41, Var01) - A short float value. (Obj41, Var03) - A long float value. (Obj41, Var04) - A digital command (Obj12, Var01) where the time field is set according to the setpoint value. (Positive sets on time and negative sets off time) - A digital command (Obj12, Var01) where the count field is set according to the setpoint value
Control Tc Address	Long Integer / -1 (not used, default), 0 ... 65535	Telecontrol address for control direction
Control Tc Address2	Long Integer / -1 (not used, default) 0 ... 65535	Second control telecontrol address for CLOSE / RAISE (if not same as or derived from ControlTcAddress) Only required if Control Direction Type is "Double Command"
Enable SOE/Double Transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real- time information (for archive)
GI Relevant	Boolean / - False (default) - True	True: the data point is relevant for completeness check One ScanGroup of this RTU must be marked as "Integrity Scan"
Monitor Direction Type	Enumeration / - Not used	Type of analog in monitor direction - This data point has no monitor information

Attribute Label	Type / Range / Default	Description / Remark
	- Measured Value Normalized (default) - DNP30_Measured Value Frozen - DNP30_Setpoint Status	- All measured values of object 30 and 32 (any supported variation) - All frozen measured values of object 31 and 33 (any supported variation) - Displays the analog output status read back from the RTU. (Obj40, any supported variation)
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... 65535	Telecontrol address for monitoring direction
Renewal Check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclical updating checks
Smooth Factor	Short Integer / 0 (default) ... n	The smooth factor is needed to "smooth" the progression of the curve of an analog value. The formula for smoothing is: smoothed value = (current value + sf*last smoothed value)/(sf+1) where sf is the smooth factor
Threshold Value	Long Integer / 0 (default) ... n	To suppress analogs without significant changes to be passed-on to SCADA
Zero Suppress Lower Border	Long Integer / 0 (default) ... n	If a value is greater than 'Zero Suppressed Lower Border' and less than Zero Suppressed Upper Border it is not transmitted to the next layer. RTUCS: ZeroDeadband
Zero Suppress Upper Border	Long Integer / 0 (default) ... n	see above (however, applicable for upper border)

3.4.3.9.3 Data Type "Counter Value"

IMM Instance Type: *CfeTelemeterdAccumulator*

Table 3-4.26: Attributes of Instance Type *CfeTelemeterdAccumulator*

Attribute Label	Type / Range / Default	Description / Remark
Enable SOE/Double Transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real- time information (for archive)
Monitor Direction Type	Enumeration / - Not used - Counter Value (default) - DNP30_Counter Value Frozen	Type of counter in monitor direction - This data point has no monitor information - All counter values of object 20 and 22 (any supported variation) - All frozen counter values of object 21 and 23 (any supported variation)
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... 65535	Telecontrol address for monitoring direction

Attribute Label	Type / Range / Default	Description / Remark
Renewal Check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclical updating checks

3.4.3.9.4 CFE Status Information

IMM instance type: *CfeStatus*

Table 3-4.27: Attributes of Instance Type *CfeStatus*

Attribute Label	Type / Range	Description / Remark
InfoID	Enumeration / - Not used - Device Failure - Device locked - Device process control - Device preferred - GI running - GI incomplete - Renewal Alarm 1 - Renewal Alarm 2	Used for CFE internal indications and commands

3.4.3.9.5 System Counters

IMM instance type: *CfeSystemAccumulator*

The DNP3 protocol provides system counters to obtain communication statistics. Details on errors and counters can be found in the following table.

Table 3-4.28: Attributes of Instance Type *Cfe System Accumulator*

Attribute Label	Type / Range / Default	Description / Remark
Info ID	Enumeration / - Last Error Code - Message Send Counter - Message Receive Counter - Message Error Receive - Counter Message Timeout - Counter Scan Overrun - Counter Scan Requests - Counter Command Requests	Last Error Code: Errors are reported spontaneously as a number Counters: Communication statistic counters are reported cyclically dependent on "Communication Statistic Period". Select the counter of interest

3.4.3.10 Additional Information on DNP3.0 Parameters

Time Parameter

The choice of meaningful timeout parameters mainly depend on the selected baud rate and the properties of the connected RTU (or IED). A DNP telegram frame allocates 10 bits per byte (1 start, 8 data and 1 stop bit). The minimum size of a DNP telegram frame is 10 bytes (characters) and the maximum size is 292 bytes.

For CFE, a DNP application frame may contain up to 9 telegram frames with max. 2048 byte application frame size.

All timing must be done so that an RTU error is indicated in an appropriate time.

Therefore the number of RTUs connected (NumberOfRtus), all timeouts and SuccesiveErrorsForRTUFail of CfeInterfaceLink have to taken in consideration!

Keep in mind that timeouts in CFE-DNP are not final (from begin of request until last received frame), they are re-triggered (from begin of request until first received frame and every received frame restarts the timer until last).

Link Layer (CfeInterfaceLink)

The **RequestLinkConfirm** tells the Substation (RTU / slave) to send a link confirmation on the request sent.

When an expected confirmation or response on link layer level is missing the CFE repeats the request a maximum number given in DataLinkRetry. Process commands are not retried!

The **DataLinkTimeout** at least has to span the transmission time of a request and the receive time of the associated confirmation or response, plus an appropriate response reaction time of the confirming device.

Reaction time of the confirming device: The selected baud rate and the type of connected device influence this time. The minimum DataLinkTimeout accepted as input is 500ms but not less than 1000ms is recommended.

This is the DataLinkTimeout_Formula:

$$\text{DataLinkTimeout [ms]} > \frac{(\text{RequestSize} + \text{ConfirmSize}) \text{ Byte} \cdot 10 \frac{\text{Bit}}{\text{Byte}}}{\text{Baudrate} \cdot \frac{\text{Bit}}{1000 \text{ ms}}} + \text{ReactionTime}$$

Figure 3-4.2: DataLinkTimeout_Formula

The ReactionTime might be typically between 1 and 2 seconds depending on the connected device.

With maximum size of 292 bytes assumed for the request telegram, and the confirmation response telegram size of 10 bytes, you get:

$$\text{DataLinkTimeout [ms]} > \frac{(292+10) \cdot 10^4}{\text{Baudrate}} + \text{ReactionTime} = \frac{3.02 \cdot 10^6}{\text{Baudrate}} + \text{ReactionTime}$$

Figure 3-4.3: DataLinkTimeout Formula with Example Values

The DataLinkTimeout is used when RequestLinkConfirm is set. Any internal link-command (LINK_RESET, LINK_TEST) used to check/start the RTU connection is also monitored with DataLinkTimeout.

If no cyclic scan is used in unsolicited mode LinkTestCycleTime assures that every connected RTU is checked regularly.

$\text{LinkTestCycleTime [msec]} > \text{Sum (DataLinkTimeout * (DataLinkRetry + 1) * SuccesiveErrorsForRTUFail) + spare ForOtherScans}$

SuccesiveErrorsForRTUFail decides when a RTU is reported as failed. An internal error counter is increased by one, when a request fails (including retries). If this counter is equal to SuccesiveErrorsForRTUFail the RTU is set to failure. Every request answered in timesets the internal error counter to zero.

LifeCycleTime of CfeSingleCircuitExtendedInterface defines the cycle for checking a faulty (not connected) RTU to get it back into operation.

LifeCycleTime [msec] shall be greater than greatest LinkTestCycleTime of the interface or use the same formula with more spare time.

Application Layer (CfeRemote TerminalUnit)

The RequestApplicationConfirm tells the Substation (RTU / slave) to send an application confirmation on the request sent.

When an expected confirmation or response on application layer level is missing the CFE repeats the request a maximum number given in ApplicationRetry. Process commands are not retried!

The ApplicationTimeout at least has to span the transmission time of a request and the receive time of the associated confirmation or response, plus an appropriate response reaction time of the confirming device. The minimum ApplicationTimeout excepted as input is 500ms but not less than 1000ms is recommended.

This is the formula for Application Layer Timeout:

$$\text{ApplicationTimeout [ms]} > \frac{(\text{RequestSize} + \text{ConfirmSize}) \text{ Byte} \cdot 10 \frac{\text{Bit}}{\text{Byte}}}{\text{Baudrate} \cdot \frac{\text{Bit}}{1000 \text{ ms}}} + \text{ReactionTime}$$

Figure 3-4.4: Formula for Application Layer Timeout

The ApplicationTimeout is used when RequestApplicationConfirm is set or when waiting for any none confirmed response from the RTU.

Internal commands / scans (CLASS_0123_SCAN for start, CLASS_SCAN on IIN, TIME_SYNC on IIN) and commands use ApplicationTimeout.

Scan Groups Defined by User (CfeScanGroup)

If RequestLinkConfirm and RequestApplicationConfirm are set they apply also to scan groups.

For all user defined Scan groups MonitoringRetry replaces ApplicationRetry and MonitoringTimeout the ApplicationTimeout. They are defined in CfeScanGroup to allow the user individual settings per scan.

The MonitoringTimeout uses the same formula like ApplicationTimeout but the ReactionTime might be different. The MonitoringTimeout shall be greater 1000ms.

Scan Rates (CfeScanRate)

The minimum ScanPeriod depends on the number of RTUs and scans performed on one interface. The sum of all scans with their timeout or answer time (take the greater one) within a given period has to be less than the period itself! The scan period shall not be less than 1000ms.

The ScanOffset shall be used only when a defined start of the scan is required (e.g. exactly at every full hour, 00:00, 01:00, 02:00,periode= 3600000 and offset=0) otherwise let the field empty or set it to "-1". Don't set ScanOffset for all scans to zero if not required, it might concentrate scans unnecessarily

Assure on one interface that all running scans get enough time to process / timeout!

It is up to the user, to apply the parameter values, if different requirements or conditions arise from the connected devices' properties, or to provide appropriate conditions if particular, obligatory parameter values have to be met. In all case watch the rules above.

Address Parameter

Explains DNP3.0 addresses:

- PrimaryAddress and LinkAddress
- ControlTcAddress for CfeDataPoints (CfeTelemeteredAnalog, CfeTelemeteredDigital)
- MonitorTcAddress for CfeDataPoints (CfeTelemeteredAccumulator, CfeTelemeteredAnalog, CfeTelemeteredDigital)

PrimaryAddress (CfeSingleCircuit ExtendedInterface)

The PrimaryAddress corresponds to the address of the master station (CFE) valid for this interface.

It allows adjusting individual address per interface (if the RTU requires different partner per connection) but is usually the same for all.

LinkAddress (CfeInterfaceLink)

The LinkAddress corresponds to the address of the substation (RTU) valid for this interface.

It allows setting individual address per link (if the RTU shall react differently on each link) but is usually the same for all links to the same RTU. For PrimaryAddress and Linkaddress the following rule is valid.

Monitoring Direction:

LinkAddress is source address

PrimaryAddress is destination address

Control Direction:

LinkAddress is destination address

PrimaryAddress is source address

ControlTcAddress (CfeDataPoint)

The ControlTcAddress corresponds to the address of the data point in control direction.

In DNP3.0 the address range starts for every data type (DNP data objects) from 0 up to maximum 65535 and has to be unique within the same data type.

CfeTelemeteredDigitals defined as double command and having separate commands for each state use this address for the first bit of this 2-bit information (trip, lower, down, off).

ControlTcAddress2 (CfeTelemeteredDigital)

Only CfeTelemeteredDigitals defined as double command and having separate commands for each state (only double commands with control code [6, 7] as NULL = CommandQualifier as "No additional definition") use this address for the second bit of its 2-bit information (close, raise, up, on).

All other data points don't need the second control address!

The ControlTcAddress2 corresponds to the second address of the data point in control direction.

The address range starts from 0 up to maximum 65535 and has to be unique within the same data type.



Note

Please check the RTU configuration for correct use of ControlTcAddress2!

MonitorTcAddress (CfeDataPoint)

The MonitorTcAddress corresponds to the address of the data point in monitor direction.

In DNP3.0 the address range starts for every data type (DNP data objects) from 0 up to maximum 65535 and has to be unique within the same data type.

CfeTelemeteredDigitals defined as double point use this address for the first bit of this 2-bit information (tripped, off, open).

MonitorTcAddress2 (CfeTelemeteredDigital)

Only CfeTelemeteredDigitals defined as double point use this address for the second bit of its 2-bit information (closed, on).

All other data points don't need the second control address!

The MonitorTcAddress2 corresponds to the second address of the data point in monitor direction.

The address range starts from 0 up to maximum 65535 and has to be unique within the same data type.

Cold/ Warm Restart of RTU

When the operator issues a cold or warm restart command to the RTU the following happens:

- CFE sends a restart request command to the RTU
- Optionally, the RTU sends a service message back asking for a recovery period. In this case, the CFE waits for the requested time before it resumes the communication with the RTU

During the RTU recovering period the CFE has the following behavior:

- Scan activity is stopped for the affected RTU
- Commands from operator are not passed-on to this RTU

The following events result in resuming the communication with the RTU before the recovery time is over:

- A fail-over between redundant links / interfaces occurs
- The link to the RTU or the corresponding interface is locked / unlocked
- An unsolicited message with INN "restart bit" set is received from RTU

As soon as the communication is resumed a class 1/ 2/ 3/ 0 scan is issued to the RTU.



Note

There will be no communication interruption after RTU restart if the CFE does not receive any recovery time request, or one of less than 20 milliseconds, or any error was detected.

3.4.3.11 TAPI Configuration for Dial-in / Dial-out

It is required that TAPI properties are set for the modem being used. To get the TAPI configuration, go to "Settings -> Control Panel -> Phone and Modem Options". In the tab "Modem" there should be a list of TAPI modems. If the list is still empty you will need to add your modem.

Under "Modem - Properties" the telegraphic speed is to be set (same as set for "CfeSingleCircuitExtendedInterface"). Uncheck the box of the option "Wait for dial tone before dialing".

3.4.4 Protocol CDC Type II

Short name: CDC2

Parameterization of CFE interfaces, interface links and associated Remote Terminal Units (RTU) communicating with CDC (Type) 2 telecontrol protocol requires the compliance of distinct rules described in the following.

3.4.4.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the protocol CDC2:

Table 3-4.29: Attributes of Instance Type *Cfe Protocol CDC2*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Communication Statistic Offset	Start offset related to 00:00 h	0 ... msec	As per customer requirements
Communication Statistic Period	For periodical transfer to display configured accumulators in SCADA	0 ... msec	As per customer requirements
SOE Dump Block Size	Size of SOE Dump Blocks		
Scan 1 SOE Block Size	Size of Scan SOE Blocks	0 ...	RTU settings
Scan 2 SOE Block Size			
Scan 3 SOE Block Size			
Scan 3X SOE Block Size			

3.4.4.2 CFE Interface Parameterization

IMM instance type: *CfeSingleCircuitExtendedInterface*

For configuring a CFE communication port for CDC2 protocol the following interface parameters for dedicated line communication mode are to be set:

Table 3-4.30: Attributes of Instance Type *CfeSingleCircuitExtendedInterface*

Attribute Label	Meaning	Range/ Unit	Interdependencies
BaudRate	Transmission speed	200, 300, 600, 1200, 1800, 2400, 4800, 9600, 19200, [bit/s]	as per customer requirements
Check Receive Carrier	If True, receive carrier must be off before send	False, True	Must be False, if Constant Carrier is True
Clock synchronization interval	Time for periodical transmission of RTU synchronization message.		as per customer requirements
Constant Carrier	True means send carrier always on.	False, True	Must be False, if Check Receive Carrier is True
Interbyte Timeout	Time period permitted between received bytes before a timeout error is declared.	0 . . . (16) [msec]	
Listen Mode	Mode for updating the process image (no controls)	False, True	

Attribute Label	Meaning	Range/ Unit	Interdependencies
Outage base	Number of scans to detect RTU failure.	1 . . . x (4)	
Parity	The type of parity used on the circuit	Even, Odd, None	
Postconditioning Delay	Time after the message is transmitted that the communication will wait to drop the carrier .	0 . . . xyz (0) [msec] 0 means no delay	
Preconditioning Delay	The Time the communication equipment will wait after asserting the carrier before beginning to transmit data.	0 . . . xyz (30) [msec] 0 means no delay	
Receive Message Timeout	Time that the communication will wait to receive the first byte of the message response, before declaring a timeout.	0 . . . xyz (1000) [msec]	
Serial Port	Number of the serial port	3 . . . 50	Depends on number of boards, and ports / board,
Squelch	Squelch time in milliseconds.	. . . (3) [msec]	

3.4.4.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the CDC2 protocol link layer properties applied to communication lines between CFE and RTUs.

Table 3-4.31: Attributes of Instance Type *CfeInterfaceLink*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Link Address	The Link Address of the device. This must match the address which has been configured within the device itself.	0, (if link address is omitted, 1 . . . 255 / 65535	Depends on the number of the link address bytes 0/1/2 0 only for RTUs in balanced mode permitted.
Outage Limit	Number of communication failures within 'Outage Base Scan' before RTU communication is declared to be failed.	1 . . .	Must be less than Outage Base Scan
Priority	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). 1 is highest priority	1 . . . 8	

3.4.4.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters are valid for CDC2 protocol stack:

Table 3-4.32: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Meaning	Range/ Unit	Interdependencies
RTU Model	RTU Model.	Model 550 Model 560 Model 565	
Command Buffer Time Maximum	Used for command aging monitoring. Commands older than this time are deleted	0 = disabled1... msec	to be harmonized with corresponding settings in SCADA
Data Point Address Structure	structure of data point address	16bit: [10bit]: 6bit	Always 4 octets
Enable time synchronization	RTU is individually enabled / disabled for time synchronization. The time is sent as broadcast, if all RTUs on a line are enabled.	False, True	
Listen Timeout	Timeout for Listen Mode	[...]	Required if Listen Mode = True
Renewal Cycle 1	Renewal control time in msec for the first control cycle.		Data points assigned to the first control cycle are subject of an updating check at the end of the cycle.
Renewal Cycle 2	Renewal control time in msec for the second control cycle.		Data points assigned to the second control cycle are subject of an updating check at the end of the cycle.
Telecontrol Protocol Type	serves to select the communication protocol for this COM port	CDC2	None

3.4.4.5 CFE Scan Group Parameterization

IMM instance type: *CfeScanGroup*

CFE acquires process data either by unsolicited responses of RTU or by emitting user-defined data scans. To set up a user data scan at least one CFE Scan Group has to be allocated to a “CfeRemoteTerminalUnit” instance. Scan Group attributes are summarized in the following table:

Table 3-4.33: Attributes of Instance Type *CfeScanGroup*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Ignore ScanGroup	Used to (temporarily) disable the complete Scan Group leaving all other attributes unmodified	False, True default: False	
Scan Group Number	Identification of a Scan Group.		This number must be unique over all Scan Group items associated to a Cfe Terminal unit.

Attribute Label	Meaning	Range/ Unit	Interdependencies
ScanType	Type of scan	CDC2_All_Data Force CDC2_Analog and Position Force CDC2_Digital Force CDC2_SOE Force CDC2_SOE Log	None

3.4.4.6 CFE Scan Volume Parameterization

IMM instance type: *CfeScanVolume*

A scan volume instance describes the type and volume of process data to be scanned by the CDC2 master. Only one scan volume instance per scan group is permitted.

Table 3-4.34: Attributes of Instance Type *CfeScanVolume*

Attribute Label	Meaning	Range/ Unit	Interdependencies
End Tc Address	End Tc Address for range Point definition.		Within the defined address range, greater than Start Tc Address (only if Range Point definition selected).
Start Tc Address	Start Tc Address for range Point definition, address for Single Point definition		Within the defined address range.
Request Mode	Request Mode describes the type of scan	CDC2_ . . . Range Point Definition Single Point Definition	

3.4.4.7 Cfe Scan Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic scan needs the specification of a period time and a starting time. This is provided by an *CfeScanRate* instance. Typically, a scan rate definition may be valid for more than one scan group.

Table 3-4.35: Attributes of Instance Type *CfeScanRate*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Late Check Period	Period over which the late scan check is performed.	same as TG 800	
Minimum Scans Before Late	Minimum number of completed scans below which a late scan is counted	same as TG 800	
Retries	Number of retries to be issued for scans at this rate.	same as TG 800	
ScanOffset	Offset from midnight for issuing scan in msec	same as TG 800	
Scan Period	Time between start of scans in msec	same as TG 800	

3.4.4.8 Data Point Parameterization

3.4.4.8.1 RTU Data Points

For protocol CDC2 the following data types are distinguished:

- Digital (IMM instance type: *CfeTelemeteredDigital*)
- Analog (IMM instance type: *CfeTelemeteredAnalog*)
- Fleeting (IMM instance type: *CfeTelemeteredFleeting*)
- Counter (IMM instance type: *CfeTelemeteredAccumulator*)
- Step Position (IMM instance type: *CfeTelemeteredPosition*)

In the following table detail information on RTU data points are summarized. The column “Applied for” indicates for which data type the respective attribute is applicable (a= analog, c= counter, d=digital, f= fleeting p= step position).

Table 3-4.36: Attributes of CDC2 Data Points

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Control Direction Type	Type of command / set-point in control direction.	a, d, p	Not used Single Command Double Command Setpoint ...	This data point has no control. Command has only one output contact. Command has two output contacts. ON / OFF
Control Tc Address	Telecontrol address for control direction.	a, d, p	0 ...	Always required if “Control Direction Type” is Single, Double or Setpoint Command.
Control Tc Address2	2nd Telecontrol address for control direction.	d, p	0 ...	
Enable SOE/ Double transmission	Enabling SOE / Double Transmission processing. Some protocols allow transmitting data points twice: first without time stamp, then with time stamp	d	False, True	TRUE: Information received from the RTU with real time is treated as SOE and not evaluated by SCADA but stored in HIS (if configured). Information received from the RTU without real time is evaluated by SCADA. FALSE: All data is evaluated by SCADA.
GIRelevant	If set TRUE the data point is relevant for completeness check	a, d, p	False, True	Should be set “TRUE” for all spontaneously transmitted data
Intermediate State Suppression Time	Intermediate state is held back for this time	d	0 = disabled, 1 ... msec	
InvertControl	Inversion of status for command direction	d, p	False, True	
Monitor Direction Type	Info type in monitor direction	a, d, f, p	Not used Single Point Double Point Measured Value Step Position Counter Value ...	Data point has no information in monitoring direction.

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Monitor Info Conversion	Inversion / conversion of information in monitoring direction	d	Not used SinglePoint invert DoublePoint invert Bit 0 DoublePoint invert Bit 0+1 DoublePoint invert Bit 1	Effect to SCADA: 1->2, 2->1 0->1, 1->0, 2->3, 3->2 0->3, 1->2, 2->1, 3->0 0->2, 1->3, 2->0, 3->1
Monitor-TC-Address	Telecontrol address in monitoring direction	a, c, d, f, p	-1 (not used) 0 ... n	
Monitor-TC-Address 2	Second telecontrol address in monitoring direction	d	- 1 (not used) 0 ... n	
Monitor Soe Address	Address 1 for SOE data points	d	- 1 (not used) 0 ... n	
Monitor Soe Address 2	Address 2 for SOE data points	d	- 1 (not used) 0 ... n	
Monitor Tc Address	Telecontrol address for monitoring direction	a, c, d, p	0 ...	
Old New Check	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer	d, p	False, True	Should be set "TRUE" for all digital data (except fleetings)
Output time count	Multiple of the base contact closure time defined in the RTU.	d	0	
Renewal check	Data point is subject to cyclic updating checks	a, c, d, p	Disabled Cycle 1 of renewal check enabled Cycle 2 of renewal check enabled	Should be set "TRUE" for cyclically transmitted data
Smooth Factor	The smooth factor is used to "smooth" a noisy analog value. The formula for smoothing is: smoothed value = (current value +sf*last smoothed value)/(sf+1) where sf is the smooth factor.	a	0	
System Control Direction ID	If "monitor direction type" ="System Point", select a system information type from this list according to the protocol applied. (parameter only means no data point, used as protocol adjustment)	d	Not Used	not applicable

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
-System Control Extension				not applicable for CDC2
-System Control Extension				
-System Monitor Direction ID				
Threshold Value	To suppress analogs without significant changes to be passed-on to SCADA	a	0	
Zero Suppress Lower Border	If a value is greater than ZeroSuppressedLower-Border and less than Zero Suppressed Upper Border it is not transmitted to the next layer.	a	0	
Zero Suppress Upper Border	See above.	a	0	

3.4.4.8.2 System Counters

IMM instance type: *CfeSystemAccumulator*

The CDC2 protocol provides system counters to obtain communication statistics. Details on errors and counters can be found in the following table.

Table 3-4.37: Attributes of Instance Type *Cfe System Accumulator*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Info ID	Select the counter of interest see chapter 2.11 in CFE-TS.	Last Error Code	Errors are reported spontaneously as a number with following ID: 1Error Data Length 2Failure Interface 3Failure Device 4Invalid Message 5Error Msg Checksum 13Error OutputBuf Overflow 16Error Control 17Error SW Logic
		Message Send Counter Message Receive Counter Message Error Receive Counter Message Timeout Counter Scan Overrun Counter Scan Requests Counter Command Requests Counter	Communication statistic counters are reported cyclically dependent on "Communication Statistic Period"

3.4.5 Protocol TG800

Parameterization of CFE protocol, interfaces, interface links and associated Remote Terminal Units (RTU) communicating with TG800 telecontrol protocol requires the compliance of distinct rules described in the following.

3.4.5.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the protocol TG800:

Table 3-4.38: Attributes of Instance Type *Cfe Protocol TG800*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Communication Statistic Offset	Start offset related to 00:00 h	0 ... msec	As per customer requirements
Communication Statistic Period	For periodical transfer to display configured accumulators in SCADA	0 ... msec	As per customer requirements

3.4.5.2 CFE Interface Parameterization

IMM instance type: *CfeSingleCircuitExtendedInterface*

For configuring a CFE communication port for TG800 protocol the following interface parameters for dedicated line communication mode are to be set:

Table 3-4.39: Attributes of *CfeSingleCircuitExtendedInterface*

Attribute Label	Meaning	Range/ Unit	Interdependencies
BaudRate	Transmission speed	300, 600, 1200, 1800, 2400, 4800, 9600, 19200, 38400 [bit/s]	As per customer requirements
Circuit Type	Type of circuit	Backup, Command, Normal, Spontaneous	Only relevant if more than one Cfe-Communication circuits are defined on a CfeMultiple Circuit Interface object.
Clock Synchronization Interval	Time for periodical transmission of RTU synchronization message.	0 (if no synchronization needed). 60 ... 600 [msec]	As per customer requirements
Constant Carrier	True means Carrier always on.	False, True	
Inactivity Timer	Time period after that the circuit is considered inactive before the modem hangs up.	1 ... [msec]	Only relevant if Modem type defined.
Interbyte Timeout	Time period permitted between received bytes before a timeout error is declared.	1 ... 16 [msec]	
Modem Type	Name of the modem command set defined in MODEM_COMMANS. If not empty this is a dial up line.	[TXT]	

Attribute Label	Meaning	Range/ Unit	Interdependencies
Parity	The type of parity used on the circuit	Even, None, Odd	
Postconditioning Delay	Time after the message is transmitted that the communication will wait to drop the carrier.	[msec]	
Preconditioning Delay	The Time the communication equipment will wait after asserting the carrier before beginning to transmit data.	20 [msec]	
Receive Message Timeout	Time that the communication will wait to receive the first byte of the message response, before declaring a timeout.	1000 [msec]	
Serial Port	Number of the serial port	3 . . . 50	Depends on number of boards, and ports / board
Successive Errors For circuit Failure	RTU fails after this number of consecutive errors.		

3.4.5.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the TG800 protocol link layer properties applied to communication lines between CFE and RTUs. The following attributes are to be set:

Table 3-4.40: Attributes of *CfeInterfaceLink*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Link Address	The Link Address of the device.	0, (if link address is omitted, 1 . . . 255 always 1 octet link address	The Link Address must match the address which has been configured within the device itself.
Priority	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). 1 is highest priority	1 ... 8	
Sucessive Errors for RTU Failure	Each circuit fails after the consecutive number of fails.	1 . . . n	

3.4.5.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters are valid for TG800 protocol stack:

Table 3-4.41: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Command Buffer Time Maximum	Used for command aging monitoring. Commands older than this time are deleted	0 = disabled 1... msec	to be harmonized with corresponding settings in SCADA
Common Address	Common Address of the device.	1 . . . 1020	The Common Address must match the address which has been configured within the device itself.
Data Point Address Structure	structure of data point address	[6bit]:1bit:[2bit]:3bit:4bit:[1bit]:15bit	Always 4 octets.
Enable time synchronization	RTU is individually enabled / disabled for time synchronization. The time is sent as broadcast, if all RTUs on a line are enabled.	False, True	
Phone Number	Telephone number to be used in dial up access to RTU.	False, True	True only if dial up access to RTU.
Renewal Cycle 1	Renewal control time in msec for the first control cycle.		Data points assigned to the first control cycle are subject of an updating check at the end of the cycle.
Renewal Cycle 2	Renewal control time in msec for the second control cycle.		Data points assigned to the second control cycle are subject of an updating check at the end of the cycle.
Telecontrol Protocol Type	serves to select the communication protocol for this COM port	TG800	None

3.4.5.5 CFE Scan Group Parameterization

IMM instance type: *CfeScanGroup*

CFE acquires process data either by unsolicited responses of RTU or by emitting user-defined data scans. To set up a user data scan at least one CFE Scan Group has to be allocated to a "CfeRemoteTerminalUnit" instance. Scan Group attributes are summarized in the following table:

Table 3-4.42: Attributes of Instance Type *CfeScanGroup*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Ignore ScanGroup	Used to (temporarily) disable the complete Scan Group leaving all other attributes unmodified	False, True default: False	

Attribute Label	Meaning	Range/ Unit	Interdependencies
ScanType	Type of scan	TG800_Accum Freeze Read TG800_Accum Freeze Reset Read TG800_Accum Line Freeze Read TG800_Accum Read TG800_All Data Force TG800_Analog and Position Force TG800_Digital Force TG800_General Change	

3.4.5.6 Cfe Scan Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic scan needs the specification of a period time and a starting time. This is provided by an *CfeScanRate* object. Typically, a scan rate definition may be valid for more than one scan group. A set of scan rates is predefined in the container instance "CFE DNP Scan Rates".

Table 3-4.43: Attributes of Instance Type *CfeScanRate*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Late Check Period	Period over which the late scan check is performed.	[msec]	
Minimum Scans Before Late	Minimum number of completed scans before a late scan is initiated		
Retries	Number of retries to be issued for scans at this rate.		
ScanOffset	Offset from midnight for issuing scan in msec	500 ... 60000 [msec] none or -1 : no offset	
Scan Period	Time between start of scans in msec	1 ... [msec]	

3.4.5.7 Data Point Parameterization

3.4.5.7.1 RTU Data Points

For protocol TG800 the following data types are distinguished:

- Digital (IMM instance type: *CfeTelemeteredDigital*)
- Analog (IMM instance type: *CfeTelemeteredAnalog*)
- Counter (IMM instance type: *CfeTelemeteredAccumulator*)
- Step Position (IMM instance type: *CfeTelemeteredPosition*)

In the following table detail information on RTU data points are summarized. The column “Applied for” indicates for which data type the respective attribute is applicable (a= analog, c= counter, d=digital, p= step position).

Table 3-4.44: Attributes of TG800 Data Points

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Command PNB	String to be sent to the RTU instead of a control address	a, d, p		
Control Direction Type	Type of command / setpoint in control direction.	a, d, p	Not used Single Command Double Command	Not used: This data point has no control. Command has only one output contact. Command has two output contacts. ON / OFF
Control Tc Address	Telecontrol address for control direction.	a, d, p	0 ... 65535	Always required if “Control Direction Type” is Single, Double or Setpoint Command.
Enable SOE/ Double transmission	Enabling SOE / Double Transmission processing. Some protocols allow transmitting data points twice: first without time stamp, then with time stamp	d	False, True	TRUE: Information received from the RTU with real time is treated as SOE and not evaluated by SCADA but stored in HIS (if configured). Information received from the RTU without real time is evaluated by SCADA. FALSE: All data is evaluated by SCADA.
GIRelevant	If set TRUE the data point is relevant for completeness check	a, d, p	False, True	Should be set “TRUE” for all spontaneously transmitted data
Intermediate State Suppression Time	Intermediate state is held back for this time	d	0 = disabled, 1 ... msec	
InvertControl	Inversion of status for command direction	d, p	False, True	
Monitor Direction Type	Info type in monitor direction	a, c, d, p	Not used Single Point Double Point Measured Value Step Position Counter Value ...	Not used: This data point has no information in monitoring direction.
Monitor Info Conversion	Inversion / conversion of information in monitoring direction	d	Not used SinglePoint invert DoublePoint invert Bit 0 DoublePoint invert Bit 0+1 DoublePoint invert Bit 1	Effect to SCADA: 1->2, 2->1 0->1, 1->0, 2->3, 3->2 0->3, 1->2, 2->1, 3->0 0->2, 1->3, 2->0, 3->1
Monitor Tc Address	Telecontrol address for monitoring direction	a, c, d, p	0 ...	
Monitor Soe Adress2	Second SoE address for ON/CLOSE in monitoring direction	d		
Old New Check	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer	d, p	False, True	Should be set “TRUE” for all digital data (except fleetings)

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Renewal check	Data point is subject to cyclic updating checks	a, c, d, p	Disabled Cycle 1 of renewal check enabled Cycle 2 of renewal check enabled	Should be set "TRUE" for all cyclically transmitted data
Smooth Factor	The smooth factor is used to "smooth" a noisy analog value. The formula for smoothing is: $\text{smoothed value} = (\text{current value} + \text{sf} * \text{last smoothed value}) / (\text{sf} + 1)$ where sf is the smooth factor.	a	0	
System Control Direction ID	If "monitor direction type" = "System Point", select a system information type from this list according to the protocol applied. (parameter only means no data point, used as protocol adjustment)	d	Not used	not applicable
System Control Extension	If "control direction type" = "System Command", this attribute provides additional information for control command processing.	d	Not used	not applicable
System Monitor Direction ID	If "monitor direction type" = "System Point", select a system information type from this list according to the protocol applied. (parameter only means no data point, used as protocol adjustment)	d	Not used	not applicable
System Monitor Extension	If "monitor direction type" = "System Point", this attribute provides additional information for system info processing.	d	Not used	not applicable
Threshold Value	To suppress analogs without significant changes to be passed-on to SCADA	a	0	
Time delay	Minimum time delay in milliseconds the RTUCS introduces between two successive SBO commands to the same RTU. (e.g. for protocols TG800, TG8979, CDC2)	d		

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Zero Suppress Lower Border	If a value is greater than ZeroSuppressedLower-Border and less than Zero Suppressed Upper Border it is not transmitted to the next layer.	a	0	
Zero Suppress Upper Border	See above.	a	0	

3.4.5.7.2 System Counters

Not implemented.

3.4.6 Protocol TG8979

Parameterization of CFE interfaces, interface links and associated Remote Terminal Units (RTU) communicating with TG8979 telecontrol protocol requires the compliance of distinct rules described in the following.

3.4.6.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the protocol TG8979:

Table 3-4.45: Attributes of Instance Type *Cfe Protocol TG8979*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Communication Statistic Offset	Start offset related to 00:00 h	0 ... msec	As per customer requirements
Communication Statistic Period	For periodical transfer to display configured accumulators in SCADA	0 ... msec	As per customer requirements

3.4.6.2 CFE Interface Parameterization

IMM instance type: *CfeSingleCircuitExtendedInterface*

For configuring a CFE communication port for TG8979 protocol the following interface parameters for dedicated line communication mode are to be set:

Table 3-4.46: Attributes of Instance Type *CfeSingleCircuitExtendedInterface*

Attribute Label	Meaning	Range/ Unit	Interdependencies
BaudRate	Transmission speed	same as TG800	As per customer requirements
Circuit Type	Type of circuit	same as TG800	Only relevant if more than one Cfe-Communication circuits are defined on a CfeMultiple circuit interface.
Clock synchronization interval	Time for periodical transmission of RTU synchronization message.	same as TG800	As per customer requirements
Constant Carrier	True means carrier always on.	same as TG800	
Inactivity Timer	Time period before the circuit is considered inactive before the modem hangs up.	same as TG800	Only relevant if Modem type defined.
Interbyte Timeout	Time period permitted between received bytes before a timeout error is declared.	same as TG800	
Life Cycle Time	Cycle time, when a faulty RTU / Link is tested to get it back in operation.	same as TG800	
Listen Mode	Mode for updating the process image (no controls)	False, True	
Modem Type	Name of the modem command set defined in MODEM_COMMANS. If not empty this is a dial up line.	same as TG800	
Parity	The type of parity used on the circuit	same as TG800	

Attribute Label	Meaning	Range/ Unit	Interdependencies
Postconditioning Delay	Time after the message is transmitted that the communication will wait to drop the carrier.	same as TG800	
Preconditioning Delay	The Time the communication equipment will wait after asserting the carrier before beginning to transmit data.	same as TG800	
Receive Message Timeout	Time that the communication will wait to receive the first byte of the message response, before declaring a timeout.	same as TG800	
Serial Port	Number of the serial port	3. . . 50	Depends on number of boards, and ports / board,
Successive Errors For circuit Failure	RTU fails after this number of consecutive errors.	same as TG800	

3.4.6.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the TG8979 protocol link layer properties applied to communication lines between CFE and RTUs. The following attributes are to be set:

Table 3-4.47: Attributes of Instance Type *CfeInterfaceLink*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Link Address	The Link Address of the device. This must match the address which has been configured within the device itself.	same as TG800	
Priority	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). 1 is highest priority	same as TG800	
Successive Errors for RTU Failure	Each circuit fails after the consecutive number of fails.	same as TG800	

3.4.6.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters are valid for TG8979 protocol stack:

Table 3-4.48: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Command Buffer Time Maximum	Used for command aging monitoring. Commands older than this time are deleted	0 = disabled 1... msec	to be harmonized with corresponding settings in SCADA
Data Point Address Structure	structure of data point address	12bit	Dependant on data type e.g. analog input 12bit, accumulator 8 bit

Attribute Label	Meaning	Range/ Unit	Interdependencies
Enable time synchronization	RTU is individually enabled / disabled for time synchronization. The time is sent as broadcast, if all RTUs on a line are enabled.	same as TG800	
Listen Timeout	Timeout for Listen Mode	[...]	Required if Listen Mode = True
Phone Number	Telephone number to be used in dial up access to RTU.	False, True	True only if dial up access to RTU.
Renewal Cycle 1	Renewal control time in msec for the first control cycle.		Data points assigned to the first control cycle are subject of an updating check at the end of the cycle.
Renewal Cycle 2	Renewal control time in msec for the second control cycle.		Data points assigned to the second control cycle are subject of an updating check at the end of the cycle.
Telecontrol Protocol Type	serves to select the communication protocol for this COM port	TG8979	

3.4.6.5 CFE Scan Group Parameterization

IMM instance type: *CfeScanGroup*

CFE acquires process data either by unsolicited responses of RTU or by emitting user-defined data scans. To set up a user data scan at least one CFE Scan Group has to be allocated to a “CfeRemoteTerminalUnit” instance. Scan Group attributes are summarized in the following table:

Table 3-4.49: Attributes of Instance Type *CfeScanGroup*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Ignore ScanGroup	Used to (temporarily) disable the complete Scan Group leaving all other attributes unmodified	False, True default: False	
Scan Group Number	Identification of a Scan Group.		
ScanType	Type of scan	TG8979_accum Line Freeze Read TG8979_accum Read TG8979_All_Data Force TG8979_Analog and Position TG8979_Digital Force TG8979_General Change . . . TG8979_SOE Change TG8979_SOE Force	

3.4.6.6 CFE Scan Volume Parameterization

IMM instance type: *CfeScanVolume*

A scan volume instance describes the type and volume of process data to be scanned by the TG8979 master. This feature is normally not used.

Table 3-4.50: Attributes of Instance Type *CfeScanVolume*

Attribute Label	Meaning	Range/ Unit	Interdependencies
End Tc Address	End Tc Address for range Point definition.		Within the defined address range, greater than Start Tc Address (only if Range Point definition selected).
Start Tc Address	Start Tc Address for range Point definition, address for Single Point definition		Within the defined address range
Request Mode	Request Mode describes the type of scan	Range Point Definition Single Point Definition	None

3.4.6.7 Cfe Scan Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic scan needs the specification of a period time and a starting time. This is provided by an *CfeScanRate* object. Typically a scan rate definition may be valid for more than one scan group.

Table 3-4.51: Attributes of Instance Type *CfeScanRate*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Late Check Period	Period over which the late scan check is performed.	same as TG 800	
Minimum Scans Before Late	Minimum number of completed scans below which a late scan is counted	same as TG 800	
Retries	Number of retries to be issued for scans at this rate.	same as TG 800	
ScanOffset	Offset from midnight for issuing scan in msec	same as TG 800	
Scan Period	Time between start of scans in msec	same as TG 800	

3.4.6.8 Data Point Parameterization

3.4.6.8.1 RTU Data Points

For protocol TG8979 the following data types are distinguished:

- Digital (IMM instance type: *CfeTelemeteredDigital*)
- Analog (IMM instance type: *CfeTelemeteredAnalog*)
- Fleeting (IMM instance type: *CfeTelemeteredFleeting*)
- Counter (IMM instance type: *CfeTelemeteredAccumulator*)
- Step Position (IMM instance type: *CfeTelemeteredPosition*)

In the following table detail information on RTU data points are summarized. The column “Applied for” indicates for which data type the respective attribute is applicable (a= analog, c= counter, d=digital, f= fleeting p= step position).

Table 3-4.52: Attributes of TG8979 Data Points

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
ADC Group Number		a		
Control Direction Type	Type of command / set-point in control direction.	a, d, p	Not used Single Command Double Command Setpoint ...	Not used: This data point has no control. Command has only one output contact. Command has two output contacts.
Control Tc Address	Telecontrol address for control direction.	a, d, p	0 ...	Always required if “Control Direction Type” is Single, Double or Setpoint Command.
GIRelevant	If set TRUE the data point is relevant for completeness check	a, d, p	False, True	Should be set “TRUE” for all spontaneously transmitted data
Intermediate State Suppression Time	Intermediate state is held back for this time	d	0 = disabled, 1 ... msec	
InvertControl	Inversion of status for command direction	d, p	False, True	
Monitary Change	RTU has Change Memory capability on this point	d	False, True	
Monitor Direction Type	Info type in monitor direction	a, c, d, p	Not used Single Point Double Point Measured Value Step Position Counter Value ...	Not used: Data point has no information in monitoring direction.
Monitor Info Conversion	Inversion / conversion of information in monitoring direction	d	Not used SinglePoint invert DoublePoint invert Bit 0 DoublePoint invert Bit 0+1 DoublePoint invert Bit 1	Effect to SCADA: 1->2, 2->1 0->1, 1->0, 2->3, 3->2 0->3, 1->2, 2->1, 3->0 0->2, 1->3, 2->0, 3->1
Monitor Tc Address	Telecontrol address for monitoring direction	a, c, d, p	0 ...	
Old New Check	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer	d, p	False, True	Should be set “TRUE” for all digital data (except fleetings)
On Duration	Monitoring time for tap changer	p	0 ...	
Output time count	Multiple of the base contact closure time defined in the RTU.	d	0	
Renewal check	Data point is subject to cyclic updating checks	a, c, d, p	Disabled Cycle 1 of renewal check enabled Cycle 2 of renewal check enabled	Should be set “TRUE” for all cyclically transmitted data

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Smooth Factor	The smooth factor is used to "smooth" a noisy analog value. The formula for smoothing is: smoothed value = (current value +sf*last smoothed value) / (sf+1) where sf is the smooth factor.	a	0	
System Control Direction ID	If "monitor direction type" ="System Point", select a system information type from this list according to the protocol applied. (parameter only means no data point, used as protocol adjustment)	d	Not Used	not applicable
System Control Extension	If "control direction type" ="System Command", this attribute provides additional information for control command processing.	d	Not Used	not applicable for TG8979
System Monitor Direction ID	If "monitor direction type" ="System Point", select a system information type from this list according to the protocol applied. (parameter only means no data point, used as protocol adjustment)	d	Not Used	not applicable for TG8979
System Monitor Extension	If "monitor direction type" ="System Point", this attribute provides additional information for system info processing.		Not Used	not applicable for TG8979
Threshold Value	To suppress analogs without significant changes to be passed-on to SCADA	a	0	
Time delay	Minimum time delay in milliseconds the RTUCS introduces between two successive SBO commands to the same RTU.(e.g. for protocols TG800, TG8979, CDC2)	d		
Zero Suppress Lower Border	If a value is greater than ZeroSuppressedLowerBorder and less than Zero Suppressed Upper Border it is not transmitted to the next layer.	a	0	
Zero Suppress Upper Border	See above.	a	0	

3.4.6.8.2 System Counters

IMM instance type: *CfeSystemAccumulator*

The TG8979 protocol provides system counters to obtain communication statistics. Details on errors and counters can be found in the following table.

Table 3-4.53: Attributes of Instance Type *Cfe System Accumulator*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Info ID	Select the counter of interest see chapter 2.11 in CFE-TS.	Last Error Code	Errors are reported spontaneously as a number with following ID: 1Error Data Length 2Failure Interface 3Failure Device 4Invalid Message 5Error Msg Checksum 13Error OutputBuf Overflow 16Error Control 17Error SW Logic
		Message Send Counter Message Receive Counter Message Error Receive Counter Message Timeout Counter Scan Overrun Counter Scan Requests Counter Command Requests Counter	Communication statistic counters are reported cyclically dependent on "Communication Statistic Period"

3.4.7 Protocol TG065 / TD065

Parameterization of CFE protocol, interfaces, interface links and associated Remote Terminal Units (RTU) communicating with TG065 / TD065 telecontrol protocol requires the compliance of distinct rules described in the following.

3.4.7.1 Common Protocol Parameters

Table 3-4.54: Attributes of Instance Types *Cfe Protocol TG065 / Cfe Protocol TD065*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Communication Statistic Offset	Start offset related to 00:00 h	0 ... msec	As per customer requirements
Communication Statistic Period	For periodical transfer to display configured accumulators in SCADA	0 ... msec	As per customer requirements

3.4.7.2 CFE Interface Parameterization

IMM instance type: *CfeSingleCircuitExtendedInterface*

For configuring a CFE communication port for TG065/ TD065 protocol the following interface parameters for dedicated line communication mode are to be set:

Table 3-4.55: Attributes of Instance Type *CfeSingleCircuitExtendedInterface*

Attribute Label	Meaning	Range/ Unit	Interdependencies
BaudRate	Transmission speed	300, 600, 1200, 1800, 2400, 4800, 9600, 19200, 38400 [bit/s]	As per customer requirements
Circuit Type	Type of circuit	Backup, Command, Normal, Spontaneous	Only relevant if more than one CfeCommunication circuits are defined on a CfeMultiple Circuit Interface object.
Constant Carrier	True means carrier always on.	False, True	
Inactivity Timer	Time period after that the circuit is considered inactive before the modem hangs up.	1 ... [msec]	Only relevant if Modem type defined.
Interbyte Timeout	Time period permitted between received bytes before a timeout error is declared.	1 ... 16 [msec]	
Modem Type	Name of the modem command set defined in MODEM_COMMANS. If not empty this is a dial up line.	[TXT]	
Parity	The type of parity used on the circuit	Even, None, Odd	
Postconditioning Delay	Time after the message is transmitted that the communication will wait to drop the carrier.	[msec]	
Preconditioning Delay	The Time the communication equipment will wait after asserting the carrier before beginning to transmit data.	20 [msec]	

Attribute Label	Meaning	Range/ Unit	Interdependencies
Receive Message Timeout	Time that the communication will wait to receive the first byte of the message response, before declaring a timeout.	1000 [msec]	
Serial Port	Number of the serial port	3 . . . 50	Depends on number of boards, and ports / board,
Successive Errors For Circuit Failure	RTU fails after this number of consecutive errors.		

3.4.7.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The interface link parameters specify the TG065 / TD065 protocol link layer properties applied to communication lines between CFE and RTUs.

Table 3-4.56: Attributes of Instance Type *CfeInterfaceLink*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Link Address	The Link Address of the device.	0, (if link address is omitted, 1 . . . 255 always 1 octet link address	The Link Address must match the address which has been configured within the device itself.
Priority	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). 1 is highest priority	1 . . . 8	
Successive Errors For RTU Failure	Each circuit fails after the consecutive number of fails.	1 . . .	

3.4.7.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters are valid for TG065 / TD065 protocol stack:

Table 3-4.57: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Common Address	Common Address of the device.	1 . . . 1020	The Common Address must match the address which has been configured within the device itself.
Command Buffer Time Maximum	Used for command aging monitoring. Commands older than this time are deleted	0 = disabled 1... msec	to be harmonized with corresponding settings in SCADA
Data Point Address Structure	Structure of data point address	[6bit]:1bit:[2bit]:3 bit:4bit:[1bit]:15bit	Always 4 octets.
Phone Number	Telephone number to be used in dial up access to RTU.	False, True	True only if dial up access to RTU.

Attribute Label	Meaning	Range/ Unit	Interdependencies
Renewal Cycle 1	Renewal control time in msec for the first control cycle.		Data points assigned to the first control cycle are subject of an updating check at the end of the cycle.
Renewal Cycle 2	Renewal control time in msec for the second control cycle.		Data points assigned to the second control cycle are subject of an updating check at the end of the cycle.
Telecontrol Protocol Type	serves to select the communication protocol for this COM port	Tx065	None

3.4.7.5 CFE Scan Group parameterization

IMM instance type: *CfeScanGroup*

CFE acquires process data either by unsolicited responses of RTU or by emitting of user-defined data scans. To set up a user data scan at least one CFE Scan Group has been linked to CfeRemoteTerminal object.

Table 3-4.58: Attributes of Instance Type *CfeScanGroup*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Ignore Scan Group	Used to (temporarily) disable the complete Scan Group leaving all other attributes unmodified	False, True default: False	
Scan Type	Type of Scan	Tx065_Accum Freeze Read Tx065_Accum Freeze Reset Read Tx065_Accum Line Freeze Read Tx065_Accum Read Tx065_All Data Force Tx065_Analog and Position Force Tx065_Digital Force Tx065_General Change	

3.4.7.6 Cfe Scan Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic scan needs the specification of a period time and a starting time. This is provided by an CfeScanRate instance. Typically, a scan rate definition may be valid for more than one scan group.

Table 3-4.59: Attributes of Instance Type *CfeScanRate*

Attribute Label	Meaning	Range/ Unit	Interdependencies
Late Check Period	Period over which the late scan check is performed.	[msec]	
Minimum Scans Before Late	Minimum number of completed scans before a late scan is initiated		
Retries	Number of retries to be issued for scans at this rate.		

Attribute Label	Meaning	Range/ Unit	Interdependencies
Scan Offset	Offset from midnight for issuing scan in msec	500 ... 60000 [msec] none or -1 : no offset	
Scan Period	Time between start of scans in msec	1 ... [msec]	

3.4.7.7 Data Point Parameterization

3.4.7.7.1 RTU Data Points

For the protocols TG065 /TD065 the following data types are distinguished:

- Digital (IMM instance type: *CfeTelemeteredDigital*)
- Analog (IMM instance type: *CfeTelemeteredAnalog*)
- Counter (IMM instance type: *CfeTelemeteredAccumulator*)
- Step Position (IMM instance type: *CfeTelemeteredPosition*)

In the following table detail information on RTU data points are summarized. The column "Applied for" indicates for which data type the respective attribute is applicable (a= analog, c= counter, d= digital, p= step position)

Table 3-4.60: Attributes of TG065 /TD065 Data Points

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Control Direction Type	Type of command / set-point in control direction.	a, d, p	Not used Single Command Double Command Setpoint ...	Not used: This data point has no control. Command has only one output contact. Command has two output contacts.
Control Tc Address	Telecontrol address for control direction.	a, d, p	0 ...	Always required if "Control Direction Type" is Single, Double or Setpoint Command.
GIRelevant	If set TRUE the data point is relevant for completeness check	a, d, p	False, True	Should be set "TRUE" for all spontaneously transmitted data
Intermediate State Suppression Time	Intermediate state is held back for this time	d	0 = disabled, 1 ... msec	
Invert Control	Inversion of status for command direction	d, p	False, True	
Monitory Change	RTU has Change Memory capability on this point	d	False, True	
Monitor Direction Type	Info type in monitor direction	a, c, d, p	Not used Single Point Double Point Measured Value Step Position Counter Value ...	Not used: Data point has no information in monitoring direction.

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Monitor Info Conversion	Inversion / conversion of information in monitoring direction	d	Not used SinglePoint invert DoublePoint invert Bit 0 DoublePoint invert Bit 0+1 DoublePoint invert Bit 1	Effect to SCADA: 1->2, 2->1 0->1, 1->0, 2->3, 3->2 0->3, 1->2, 2->1, 3->0 0->2, 1->3, 2->0, 3->1
Monitor Tc Address	Telecontrol address for monitoring direction	a, c, d, p	0 ...	
Old New Check	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer	d, p	False, True	Should be set "TRUE" for all digital data (except fleetings)
Renewal check	Data point is subject to cyclic updating checks	a, c, d, p	Disabled Cycle 1 of renewal check enabled Cycle 2 of renewal check enabled	Should be set "TRUE" for all cyclically transmitted data
Smooth Factor	The smooth factor is used to "smooth" a noisy analog value. The formula for smoothing is: smoothed value = (current value +sf*last smoothed value) / (sf+1) where sf is the smooth factor.	a	0	
System Control Direction ID	If "monitor direction type" ="System Point", select a system information type from this list according to the protocol applied. (parameter only means no data point, used as protocol adjustment)	d	Not Used	not applicable
System Monitor Extension	If "monitor direction type" ="System Point", this attribute provides additional information for system info processing.		Not Used	not applicable for TG8979
Threshold Value	To suppress analogs without significant changes to be passed-on to SCADA	a	0	
Time delay	Minimum time delay in milliseconds the RTUCS introduces between two successive SBO commands to the same RTU (e.g. for protocols TG800, TG8979, CDC2)	d		

Attribute Label	Meaning	Applied for	Range/ Unit	Interdependencies
Zero Suppress Lower Border	If a value is greater than ZeroSuppressedLower-Border and less than Zero Suppressed Upper Border it is not transmitted to the next layer.	a	0	
Zero Suppress Upper Border	See above.	a	0	

3.4.7.7.2 System Counters

Not implemented.

3.4.8 Protocol Sinaut 8FW (DPDM)

Parameterization of CFE interfaces, links and associated Remote Terminal Units (RTU) communicating with Sinaut 8FW telecontrol protocol requires the compliance of distinct rules described in the following.

3.4.8.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the protocol Sinaut 8FW:

Table 3-4.61: Attributes of Instance Type *Cfe Protocol Sinaut8FW*

Attribute Label	Type / Range / Default	Description / Remark
Command Aging Timeout	Long Integer / [msec] 1 000 ... 20 000 default: 5 000	The command is removed from Device Master command buffer after the given time, when not sent yet. For details on Command Aging Monitoring see Figure 3-4.1
Communication Statistic Offset	Long Integer / [msec] -1 (default) 0 ... n	Start offset related to 00:00 h -1 means: Start immediately after CFE run-up with specified Communication Statistic Period
Communication Statistic Period	Long Integer / [msec] -1 (off, default) 0 ... n	For periodical transfer of configured statistic counters to SCADA
Operate Commport HighOhmic	Boolean / - False (default) - True	For switching the output of the COM port into high-ohmic state. Only for type BCOM interface card!
Pulse Modulation Type	Enumeration / - Pulse Duration Modulation (PDM)	For selecting modulation type. Must fit to RTU settings
Station Address Receive	Enumeration / - with station address (7 bit, default) - with extended station address (8 bit) - without station address	RTU address in monitoring direction. Must fit to RTU settings
Station Address Transmit	Enumeration / - with station address (7 bit, default) - with extended station address (8 bit) - without station address	RTU address in command direction. Must fit to RTU settings

3.4.8.2 CFE Interface Parameterization

IMM instance type: *CfeSingleCircuitExtendedInterface*

For configuring a CFE communication port for Sinaut 8FW /DPDM protocol the following interface parameters for dedicated line communication mode are to be set

Table 3-4.62: Attributes of Instance Type *CfeSingleCircuitExtendedInterface*

Attribute Label	Type / Range / Default	Description / Remark
Baud Rate	Enumeration / [bit/s] - 50 - 150 - 200 (default) - 300 - 600	Transmission speed, RTU receiving direction (DPDM). As per customer requirements and RTU settings

Attribute Label	Type / Range / Default	Description / Remark
Baud Rate2	Enumeration / [bit/s] - 50 - 150 - 200 (default) - 300 - 600	Transmission speed, RTU transmitting direction (DPDM). As per customer requirements and RTU settings
Channel Supervision Timeout	Long Integer / [msec] 0 (default) ... n	If the received signal stays longer in mark polarity than this timeout, "Interface Failure" is reported. (10 000 msec is recommended)
Check Receiver Carrier	Boolean / - False - True (default)	For Modem control. TRUE: receiver Carrier off before sending. FALSE for duplex channels, TRUE for simplex channels
Clock Sync Interval	Long Integer/ [msec] 0 (off) 60 000 ... 600 000 (default)	Specifies the RTU clock synchronization interval. Also used for correct time formation. 0 means: no real-time message is used
Constant Carrier	Boolean / - False (default) - True	For modem control. TRUE sets carrier always on
Filter	Long Integer / [%] 0 (off, default) ... 25	Suppression of parasitic signal spikes within the message. For DPDM: percentage of short pulse.
Hamming Distance Receiver	Enumeration / - 2 - 4 (default) - 6	Error detection [d] for received messages
Hamming Distance Transmitter	Enumeration / - 2 - 4 (default) - 6	Error protection [d] for transmitted messages
Idle Time Receive	Long Integer / [msec] 10 ... 500 default: 200	Minimum waiting time between received messages. Value (standard: 3 * Long pulse) must fit to the selected Baud Rate.
Idle Time Transmit	Long Integer / [msec] 0 (default) ... 500	Minimum waiting time between transmitted messages. Value (standard: 3 * Long pulse) must fit to the selected Baud Rate
Invert Receive Signal	Boolean / - False (default) - True	If 'True', incoming signal polarity is inverted
Invert Transmit Signal	Boolean / - False (default) - True	If 'True', outgoing signal polarity is inverted
Life Cycle Time	Long Integer / [msec] 1 000 ... 60 000 (default)	Checks RTU and link to RTU after failure to recognize recovery
Listen Mode	Boolean / - False (default) - True	Mode for updating the process image only (no controls)
LongShortPulse Ratio Receiver	Float / 2 ... 5 default: 2.2	Ratio of long to short character for receiver.
LongShortPulse Ratio Transmitter	Float / 2 ... 5 default: 2.2	Ration of long to short character for transmitter.
Maximum Bit Distortion	Long Integer / [%] 0 ... 50 default: 40	Maximum allowed bit distortion for receiver.

Attribute Label	Type / Range / Default	Description / Remark
Polling Retry	Short Integer / 0 ... 10 default: 2	Number of polling retries for one station.
Polling Time	Long Integer / [msec] 1 000 ... 10 000 default: 5 000	The interval between scans, (for RTUs in party line configuration). Only applied in "Balanced"-mode.
Polling Timeout	Long Integer / [msec] 0 ... 60	Max time to wait for an answer from a station.
Postconditioning Delay	Long Integer / [msec] 0 (default) ... 60	Delay after message transmission before RTS is switched off. Depends on modem characteristics
Preconditioning Delay	Long Integer / [msec] 0 (default) ... 60	Delay before message transmission after RTS is switched on. Depends on modem characteristics
Serial Port	Short Integer / 3 ... 34	Number of selected serial port. Range depends on number of boards, and ports per board
Transmission Mode	Boolean / - Unbalanced (default) - Balanced	The Transmission Mode can be balanced or unbalanced. Party line connection requires "Unbalanced" mode

3.4.8.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The interface link parameters specify the Sinaut 8FW protocol link layer properties applied to communication lines between CFE and RTUs.

Table 3-4.63: Attributes of Instance Type *CfeInterfaceLink*

Attribute Label	Type / Range / Default	Description / Remark
Calculate Channel Delay	Boolean / - False (default) - True	If enabled, causes the protocol stack to calculate channel delay time prior to writing a new time to the RTU
Data Link Retry	Short Integer / 0 (default) ... 255	Number of retries of the test command
Data Link Timeout	Long Integer / [msec] 0 ... 2 000 (default)	Time for CFE to wait for RTU answer on the test command, before repeating it
General Interrogation Cycle Time	Long Integer / [msec] 0 ... n (Granularity is 1 000 msec)	Cycle time for periodically repeated GI commands
Link Test Cycle Time	Long Integer / [msec] 0 ... n (Granularity is 1 000 msec)	Time interval for periodically repeated test commands
Priority	Short Integer / 1 (default) ... 8	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over). 1 is highest priority
Successive Errors For RTU Fail	Long Integer / 0 ... 3 (default) ... n	RTU fails after this number of consecutive errors

3.4.8.4 CFE RemoteTerminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters are valid for Sinaut 8FW protocol stack:

Table 3-4.64: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Type / Range / Default	Description / Remark
Application Timeout	Long Integer / [msec] 500 ... 2 000 (default)	Time for CFE to wait for RTU to confirm an application message before repeating it
Command Buffer Time Maximum	Long Integer / [msec] 0 (default) ... 10 000	Used for central command aging monitoring. Commands older than this time are deleted. To be harmonized with corresponding settings in SCADA
Common Address	Long Integer / 0 (virtual RTU) 1 ... 127 (7 address bits) 1 ... 255 (8 address bits)	Station address of the RTU. Virtual RTU sends commands to all RTUs on the party line. This must match the address which has been configured within the device itself. Depends on the number of common address bits 0, 7 or 8 (extended address range).
Data Point Address Structure	Enumeration / - 32 bit - 8bit : 16bit : 8bit	Selection is made according to the best representation of the data point address
Enable Time Synchronization	Boolean / - False (default) - True	RTU is individually enabled / disabled for time synchronization. The time is sent as broadcast, if all RTUs on a line are enabled.
MSC Acknowledge	Short Integer / 0 ... 3 (default) ... 25	MSC acknowledge after n received messages
MSC Retry	Short Integer / 0 ... 3 (default) ... 10	Max. retries to get the missing messages before communication restarts with GI command
MSC run-up Acknowledge	Short Integer / 0 ... 3 (default) ... 5	MSC acknowledge after n received messages (run-up / overflow), starting with first.
MSC Timeout	Long Integer / [msec] 0 ... 60 000 default: 10 000	Timeout for recalling missing messages before retry or communication restart
MSC at GI	Boolean / - False (default) - True	The message sequence code is increased through GI response messages
MSC is active	Boolean / - False - True (default)	True: MSC processing is activated
MSC max Gap	Short Integer / 0 ... 2 ... 25	After reaching this gap in the MSC, communication is restarted with GI.
Renewal Cycle 1	Long Integer / [msec] 0 (default) ... n	Timeout for the first control cycle. Data points assigned to the control cycle are subject of an updating check at the end of the cycle.
Renewal Cycle 2	Long Integer / [msec] 0 (default) ... n	Timeout for the second control cycle. Data points assigned to the control cycle are subject of an updating check at the end of the cycle.
Telecontrol Protocol Type	Enumeration / - SINAUT 8FW	Last table displays the selected protocol and its attributes for this instance and its children: Select SINAUT 8FW for this protocol

3.4.8.5 CFE Scan Group Parameterization

IMM instance type: *CfeScanGroup*

CFE acquires process data either by unsolicited responses of RTU or by emitting user-defined data scans. To set up a user data scan at least one CFE Scan Group has to be allocated to a “CfeRemoteTerminalUnit” instance. Scan Group attributes are summarized in the following table:

Table 3-4.65: Attributes of Instance Type *CfeScanGroup*

Attribute Label	Type / Range / Default	Description / Remark
Base Scan	Boolean / - False (default) - True	Scan group describes cyclic base scan. It's continuously sent when line is free and no other commands or scans scheduled. Be careful with its use because it can overload your communication. For continuous scan no base scan rate to be specified
Ignore Scan Group	Boolean / - False (default) - True	Used to (temporarily) disable the complete Scan Group leaving all other attributes unmodified
Integrity Scan	Boolean / - False (default) - True	True (recommended): This scan group replaces the internal GI command and can be initiated by a ScanOnDemand command. The period will be the valid 'Scan Period' and 'ScanOffset' (@CfeScanRate) of this scan group. If not supplied it uses 'General Interrogation Cycle Time' (@CfeInterfaceLink) and no offset. There should be only one Integrity Scan per RTU and it must have a Class Type = SIN8_Poll List1 (GI) @CfeScanVolume. False: Overlapping Scans without integrity check are allowed (for special applications only)
Monitoring Retries	Long Integer / 0 (default) ... 5	Number of retries of scan control requests
Monitoring Timeout	Long Integer / [msec] 100 ... n	Monitoring period for scan control request. Recommended value: 1 000 msec
Scan Group Number	Long Integer / -1 or 0 (not used, -1 default) 1 ... n	Identification number of scan group. If used for demand scan it must be greater than 0 and unique within one <i>CfeRemoteTerminalUnit</i>
ScanType	Enumeration / - SIN8_Common Scan - SIN8_Freeze - SIN8_Group Scan	Type of scan - None polling commands (e.g. test) - For sending a counter freeze (SIN8_SwitchingCommand) - For polling requests For relation see 'Class Type' of <i>CfeScanVolume</i>

3.4.8.6 CFE Scan Volume Parameterization

IMM instance type: *CfeScanVolume*

A scan volume instance describes the type and volume of process data to be scanned by the CFE. Only one scan volume instance per scan group is permitted.

Table 3-4.66: Attributes of Instance Type *CfeScanVolume*

Attribute Label	Type / Range / Default	Description / Remark
Class Type	Enumeration / - SIN8_Poll List1 (GI) - SIN8_Poll List2 (Counter) - SIN8_Poll List3 - SIN8_Poll List4 - SIN8_Poll Quick Check - SIN8_Poll Quick Scan - SIN8_SetDate (CH18) - SIN8_Switching Command (CD1) - SIN8_Sync (CH16) - SIN8_Test Command (CH1) - SIN8_Switching Command (CD1)	Defines the volume of data objects associated with a Scan Type Scan Type = SIN8_Group Scan - GI interrogation - Counter interrogation - Interrogation of list 3 information - Interrogation of list 4 information - Quick Check, only in balanced mode (polling) - Quick Scan, only in balanced mode (polling) Note: If any scan of the same Class Type is started before the first is terminated the running scan will be aborted. This can be avoided by proper settings of the attributes 'Scan Period' and 'Scan Offset' (@ <i>CfeScanRate</i>) Scan Type = SIN8_Common Scan - Set date of RTU - Normal switching command which needs to be sent periodically - Time synchronization - Test command Scan Type = SIN8_Freeze - Normal switching command used as freeze command
Start Tc Address	Long Integer / 1 ... n	Tc Address for switching commands only. Rules of CfeTeleme-teredDigital - "Control Tc Address" apply. No structured entry possible.

3.4.8.7 Cfe Scan Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic scan needs the specification of a period time and a starting time. This is provided by an *CfeScanRate* instance. Typically, a scan rate definition may be valid for more than one scan group.

Table 3-4.67: Attributes of Instance Type *CfeScanRate*

Attribute Label	Type / Range / Default	Description / Remark
Scan Offset	Long Integer / [msec] -1 ... 0 ... n (Granularity: 1 000 msec)	Offset from midnight for issuing scans -1 = free running, scans start immediately when the connection to RTU is available. 0 = no offset to midnight
Scan Period	Long Integer / [msec] 1 000 ... n (Granularity: 1 000 msec)	Time between start of scans. Note: Consider all cans at an interface! There must be enough time for every scan to be answered and space for commands to be sent

3.4.8.8 Data Point Parameterization

For protocol Sinaut 8FW the following data types are distinguished:

- Digital Value (IMM instance type: *CfeTelemeteredDigital*)
- Analog Value (IMM instance type: *CfeTelemeteredAnalog*)
- CounterValue (IMM instance type: *CfeTelemeteredAccumulator*)
- Step Position (IMM instance type: *CfeTelemeteredPosition*)
- Fleeting Information (IMM instance type: *CfeTelemeteredFleeting*)
- CFE Status Information (IMM instance type: *CfeStatus*)
- System Counters (IMM instance type: *CfeSystemAccumulator*)

In the tables hereafter detail information on RTU data points for each data type are summarized.

3.4.8.8.1 Data Type "Digital Value"

IMM Instance Type: *CfeTelemeterdDigital*

Table 3-4.68: Attributes of Instance Type *CfeTelemeterdDigital*

Attribute Label	Type / Range / Default	Description / Remark
Command Qualifier	Enumeration / - No additional definition - SIN8_Long duration pulse (Isolator) - SIN8_Persistent output - SIN8_Short duration pulse (CB)	Specifies the command release duration.
Command Operation Mode	Enumeration / - Direct Operation - Check-back-before-operate - SIN8_Continuous Operation	Specification of command operation mode
Control Direction Type	Enumeration / - Not used - Single Command - Double Command - System Command	Type of command in control direction. - This data point has no control - Command has only one output contact - Command has two output contacts (on/off) - Needs a "System Control Direction ID"
Control Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for control direction. Always required if "Control Direction Type" is Single or Double Command
Enable SOE/Double Transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real-time information for archiving
GI Relevant	Boolean / - False (default) - True	Includes data point in Completeness Check at GI.
Intermediate State Suppression Time	Long Integer / [msec] 0 (off, default) ... 1 ... n	Intermediate state is held back for this time
Invert Control	Boolean / - False (default) - True	Inversion of status for command direction

Attribute Label	Type / Range / Default	Description / Remark
Monitor Direction Type	Enumeration / - Not used - Single Point - Double Point - System Point	Info type in monitor direction - No information in monitoring direction. - One bit info - Two bit info (on / off) - Needs a "System Monitor Direction ID"
Monitor Info Conversion	Enumeration / - Not used (default) - Single Point invert (1->2, 2->1) - Double Point invert Bit 0 (0->1, 1->0, 2->3, 3->2) - Double Point invert Bit 1 (0->2, 1->3, 2->0, 3->1) - Double Point invert Bit 0&1 (0->3, 1->2, 2->1, 3->0)	Inversion / conversion of information in monitoring direction. It modifies the received information to be correctly displayed in SCADA
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for monitoring direction Always required if "Monitor Direction Type" is Single or Double Point
Old New Check	Boolean / - False (default) - True	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer
Renewal check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclic updating checks
Send GI on	Enumeration / - none (default) - on appear - on disappear - on appear and disappear	Defines the trigger event for an automatic GI request. Only applied on system information generated by RTUs. Example: Buffer overflow
System Control Direction ID	Enumeration / - Not used (default) - SIN8_Group Interrogation (CH6) - SIN8_Interposing Test Command (CH22) - SIN8_MSCKack (CH2) parameter only - SIN8_POLL_ZFBIT (CH24) - SIN8_ResetCommand (CH4) - SIN8_ScanOnDemand - SIN8_StartupAck (CH3) - SIN8_Synchronization (CH16-18) parameter only - SIN8_Test Command (CH1)	If "Monitor Direction Type" is "System Point", select a system information type from this list according to the protocol applied. ("parameter only" means no data point, used as protocol adjustment) - Needs a valid Control Tc-Address - For interposing interface check - Needs a valid Control Tc-Address - Requests the error bits / information - Resets the RTU (set "System Control Extension") - Demand Scan (set "System Control Extension") - For sending a start-up Acknowledge - Needs a valid Control-Tc-Address - For sending a test message
System Control Extension	Long Integer / SIN8_ScanOnDemand: Scan Group Number" of "CfeScanGroup" as identification: 1 – n SIN8_ResetCommand(CH4): 0: RL=000(FWP) 2:RL=010(MC210)	If "Control Direction Type" is "System Command", this attribute provides additional information for control messages

Attribute Label	Type / Range / Default	Description / Remark
System Monitor Direction ID	Enumeration / - Not used (default) - SIN8_AckCBO (MH4) parameter only - SIN8_AckStartUp (MH10) parameter only - SIN8_AckSync (MH7-9) parameter only - SIN8_ErrorPolling2 (MH15) parameter only - SIN8_Error_ZBBIT (MH16) - SIN8_Error_ZBBIT_FWA (MH17) - SIN8_Error_ZFBIT (MH13) - SIN8_InfoLost by Master - SIN8_LastStop (MH14) parameter only - SIN8_OverflowBit - SIN8_RTU started - SIN8_SingleErrorRecord (MH2) parameter only - SIN8_TestRecord (MH1) parameter only	If "Monitor Direction Type" is "System Point", select a system information type from this list according to the protocol applied. ("parameter only" means no data point, used as protocol adjustment; they need a valid Monitor Tc-Address)
System Monitor Extension	Long Integer / SIN8_Error_ZFBIT (MH13) SIN8_Error_ZBBIT (MH16) SIN8_Error_ZBBIT_FWA (MH17) = Bit Position 0 ... 31	If "Monitor Direction Type" = "System Point", this attribute provides additional information for system info processing. Location of the information within the 32-bit pattern

3.4.8.8.2 Data Type "Analog Value"

IMM Instance Type: *CfeTelemeterdAnalog*

Table 3-4.69: Attributes of Instance Type *CfeTelemeterdAnalog*

Attribute Label	Type / Range / Default	Description / Remark
Control Direction Type	Enumeration / - Not used - SIN8_Setpoint Scaled 8-bit	Type of setpoint in control direction
Control Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for control direction. Always required if "Control Direction Type" is "Setpoint"
Enable SOE/Double Transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real- time information for archiving
G I Relevant	Boolean / - False (default) - True	Includes data point in Completeness Check at GI
Monitor Direction Type	Enumeration / - Not used - SIN8_Measured Value 11 bit - SIN8_Measured Value 8 bit	Info type in monitor direction - No information in monitoring direction. - 11 bit resolution - 8 bit resolution

Attribute Label	Type / Range / Default	Description / Remark
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for monitoring direction Always required if "Monitor Direction Type" is a Measured Value
Old New Check	Boolean / - False (default) - True	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer
Renewal check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclic updating checks
Smooth Factor	Short Integer / 0 (off, default) ... n	The smooth factor can be used to "smooth" the progression of the curve of an analog value. The formula for smoothing is: smoothed value = (current value +sf*last smoothed value)/(sf+1) where sf is the smooth factor
Threshold Value	Long Integer / 0 (off, default) ... n	To suppress analogs without significant changes to be passed-on to SCADA
Zero Suppress Lower Border	Long Integer / 0 (off, default) ... n	If a value is greater than ZeroSuppressedLowerBorder and less than Zero Suppressed Upper Border it is not transmitted to the next layer.
Zero Suppress Upper Border	Long Integer / 0 (off, default) ... n	See above, however, applicable for upper border

3.4.8.8.3 Data Type "Counter Value"

IMM Instance Type: *CfeTelemeterdAccumulator*

Table 3-4.70: Attributes of Instance Type *CfeTelemeterdAccumulator*

Attribute Label	Type / Range / Default	Description / Remark
Enable SOE/Double Transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real-time information for archiving
Monitor Direction Type	Enumeration / - Not used - SIN8_Counter Value BCD - SIN8_Counter Value Binary	Info type in monitor direction - No information in monitoring direction. - Counter value is BCD encoded - Counter value is binary encoded
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for monitoring direction Always required if "Monitor Direction Type" is Single or Double Point
Renewal check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclic updating checks

3.4.8.8.4 Data Type "Step Position"

IMM Instance Type: *CfeTelemeterdPosition*

Table 3-4.71: Attributes of Instance Type *CfeTelemeterdPosition*

Attribute Label	Type / Range / Default	Description / Remark
Command Qualifier	Enumeration / - No additional definition - SIN8_Long duration pulse (Isolator) - SIN8_Persistent output - SIN8_Short duration pulse (CB)	Specifies the command release duration.
Command operation mode	Enumeration / - Direct Operation - Check-back-before-operate - SIN8_Continuous Operation	Specification of command operation mode
Control Direction Type	Enumeration / - Not used - Regulation Step Command	Type of command in control direction.
Control Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for control direction. Always required if "Control Direction Type" is Single or Double Command
Enable SOE/Double transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real- time information for archiving
GIRelevant	Boolean / - False (default) - True	Includes data point in Completeness Check at GI.
InvertControl	Boolean / - False (default) - True	Inversion of status for command direction
Monitor Direction Type	Enumeration / - Not used - SIN8_Step Position BCD - SIN8_Step Position Binary	Info type in monitor direction - No information in monitoring direction - Position is BCD encoded - Position is binary encoded
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for monitoring direction Always required if "Monitor Direction Type" is Single or Double Point
Old New Check	Boolean / - False (default) - True	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer
Renewal check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclic updating checks

3.4.8.8.5 Data Type "Fleeting Information"

IMM Instance Type: *CfeTelemeterdFleeting*

Table 3-4.72: Attributes of Instance Type *CfeTelemeterdFleeting*

Attribute Label	Type / Range / Default	Description / Remark
Enable SOE/Double Transmission	Boolean / - False (default) - True	Enabling double transmission of this object. TRUE: same data point is received twice from RTU: 1st message w/o time information (for SCADA) 2nd message with real- time information for archiving
GI Relevant	Boolean / - False (default) - True	Includes data point in Completeness Check at GI.
Monitor Direction Type	Enumeration / - Not used - Fleeting Point - System Fleeting	Info type in monitor direction - No information in monitoring direction. - One bit info - System info, needs a "System Monitor Direction ID"
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for monitoring direction Always required if "Monitor Direction Type" is Single or Double Point
Send GI on	Enumeration / - none (default) - on appear - on disappear - on appear and disappear	Defines the trigger event for an automatic GI request. Only applied on system information generated by RTUs. Example: Buffer overflow
System Monitor Direction ID	Enumeration / - SIN8_Error_ZBBIT (MH16) - SIN8_Error_ZBBIT_FWA (MH17) - SIN8_Error_ZFBIT (MH13) - SIN8_InfoLost by Master - SIN8_OverflowBit - SIN8_RTU started	If "Monitor Direction Type" is "System Point", select a system information type from this list according to the protocol applied.
System Monitor Extension	Long Integer / - SIN8_Error_ZFBIT (MH13) - SIN8_Error_ZBBIT (MH16) - SIN8_Error_ZBBIT_FWA (MH17) = Bit Position 0 ... 31	If "Monitor Direction Type" is "System Point", this attribute provides additional information for system info processing. Location of the information within the 32-bit pattern

3.4.8.8 CFE Status Information

IMM instance type: *CfeStatus*

Table 3-4.73: Attributes of Instance Type *CfeStatus*

Attribute Label	Type / Range	Description / Remark
InfoID	Enumeration / - Device Failure - Device locked - Device process control - Device preferred - GI running - GI incomplete - Renewal Alarm 1 - Renewal Alarm 2	Used for CFE internal indications and commands

3.4.8.8.7 System Counters

IMM instance type: *CfeSystemAccumulator*

The Sinaut protocol provides system counters to obtain communication statistics. Details on errors and counters can be found in the following table.

Table 3-4.74: Attributes of Instance Type *CfeSystemAccumulator*

Attribute Label	Type / Range / Default	Description / Remark
Info ID	Enumeration / - Message Send Counter - Message Receive Counter - Message Error Receive Counter - Message Timeout Counter	Counter(s) of interest to be selected. Communication statistic counters are reported cyclically, dependent on "Communication Statistic Period"

3.4.8.9 Sinaut 8FW Addressing Scheme

The data model of Spectrum PowerCC uses the type "Long Integer" (4 bytes) for addressing the information objects. It is structured, bytes have different meaning:

Subsystem			Byte Address						Information Number			
31	...	24	23	...	16	15	...	8	7	...	0	bit
Byte 3			Byte 2			Byte 1			Byte 0			

Allocation of the Address Bytes:

Byte 0: "Information Number" is for digital and fleeting information in the range from 0 through 7 and for other information 0 (full bytes used for this information).

Byte 1 & 2: "Byte address", range 0 ... 1023

Byte 3: "Subsystem" address, range 0 ... 7.

The data input can be made easier by using a structured input defined through the attribute "DataPointAddressStructure" (of the instance type *CfeRemoteTerminalUnit*) The following structures can be used: 8-16-8 Byte3-Byte2/1- Byte0

3.4.9 Protocol IEC 60870-5-103

Short name: IEC103

Parameterization of CFE protocol, interfaces, interface links and associated devices communicating with IEC 60870-5-103 protocol requires the compliance of distinct rules described in the following.

3.4.9.1 Common Protocol Parameters

The following parameter set is common for all connected devices communicating with CFE using the protocol IEC103:

Table 3-4.75: Attributes of Instance Type *CfeProtocolIEC103*

Attribute Label	Type / Range / Default	Description / Remark
Name, Description	String	Name, description of the instance (presently no "real" CFE specific attributes required)

3.4.9.2 CFE Interface Parameterization

IMM instance type: *CfeSingleCircuitExtendedInterface*

These parameters specify the IEC103 interface properties applied to the serial communication ports of the CFE:

Table 3-4.76: Attributes of Instance Type *CfeSingleCircuitExtendedInterface*

Attribute Label	Type / Range / Default	Description / Remark
Baud Rate	Enumeration / [bit/s] - 1 200 - 2 400 - 4 800 - 9 600 (default) - 19 200 - 38 400	Transmission speed for receiving and transmitting data
Clock Sync Interval	Long Integer/ [msec] 0 (off) ... 600 000 (default) ... 86 400 000)	Defines the time sync interval for all RTUs at this interface
Life Cycle Time	Long Integer/ [msec] 5 000 ... 300 000	Used for cyclic check of "dead" links
Serial Port	Short Integer/ 1...255	Number of selected serial port. Range depends on number of boards and ports per board

3.4.9.3 CFE Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the IEC103 link properties applied to communication links between CFE interface and connected devices:

Table 3-4.77: Attributes of Instance Type *CfeInterfaceLink*

Attribute Label	Type / Range / Default	Description / Remark
Link Address	Long Integer/ 1...254	not used, set as Common Address in RTU
Clock Synchronisation Interval	Long Integer/ [msec] 0 (off, default) 86 400 000	Clock synchronization period for sending synchronization messages to the connected device(s)
General Interrogation Cycle Time	Long Integer/ [msec] 0 (off, default) 86 400 000	Time interval for the general interrogation (GI)
Priority	Short Integer/ 1 ... 8	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic fail-over) 1 is highest priority
Successive Errors For RTU Fail	Long Integer/ 0 ... 250 default: 3	After n successive errors the RTU status is set to 'Failure'

3.4.9.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following list of RTU parameters is applicable for IEC103 protocol stack.

Table 3-4.78: Attributes of Instance Type *CfeRemoteTerminalUnit*

Attribute Label	Type / Range / Default	Description / Remark
Common Address	Long Integer/ 1... 254	Station address, also used for link address
Data Point Address Structure	Enumeration/ - 8:8 - 8:8:8 - 8:8:8:8	For the best representation of your data point address use a byte-representation
Renewal Cycle1	Long Integer/ [msec] 0 ... 2 ³²	Update check 1 for cyclically transmitted data
Renewal Cycle2	Long Integer/ [msec] 0 ... 2 ³²	Update check 2 for cyclically transmitted data
RTU Model	Enumeration/ - Not used - IEC103_(default)	Specifies the RTU model. Standard is IEC103 communication
Telecontrol Protocol Type	Enumeration/ - IEC60870-5-103	Last table displays the selected protocol and its attributes for this instance and its children: Select "IEC60870-5-103" for this protocol

3.4.9.5 Parameterization of Data Scans

Scan Groups, Scan Volumes, Scan Rates are not used for IEC103

3.4.9.6 Data Point Parameterization

For protocol IEC103 the following data types are distinguished:

- Digital Value (IMM instance type: *CfeTelemeteredDigital*)
- Analog Value (IMM instance type: *CfeTelemeteredAnalog*)
- Counter Value (IMM instance type: *CfeTelemeteredAccumulator*)
- Fleeting Information (IMM instance type: *CfeTelemeteredFleeting*)
- CFE Status Information (IMM instance type: *CfeStatus*)

In the tables hereafter detailed information on RTU process data points for each data type are summarized.

3.4.9.6.1 Data Type "Digital Value"

IMM instance type: *CfeTelemeterdDigital*

Table 3-4.79: Attributes of Instance Type *CfeTelemeterdDigital*

Attribute Label	Type / Range / Default	Description / Remark
ControlDirection Type	Enumeration / - Not used (default) - Single Command - Double Command - System Command	Type of command in control direction. - This data point has no control - Command has only one output contact - Command has two output contacts (on/off) - Needs a "System Control Direction ID"
ControlTc Address	Long Integer/ -1 (not used, default) 0 ... 2^{32}	Object Address in control direction
GIRelevant	Boolean / - False (default) - True	Includes data point in Completeness Check at GI
Intermediate State Suppression Time	Long Integer / [msec] 0 (off, default) 1 ... n	Inverts the monitored signal
Invert Control	Boolean/ - False (default) - True	Inversion of status for command direction
Monitor Direction Type	Enumeration / - Not used - IEC103_Single Point - IEC103_Double Point - IEC103_Single Point w. Rel.Time - IEC103_Double Point w. Rel.Time	Data type specified for monitor direction

Attribute Label	Type / Range / Default	Description / Remark
Monitor Info Conversion	Enumeration/ - Not used (default) - SinglePoint invert (1->2, 2->1) - DoublePoint invert Bit 0 (0->1, 1->0, 2->3, 3->2) - DoublePoint invert Bit 1 (0->2, 1->3, 2->0, 3->1) - DoublePoint invert Bit 0+1 (0->3, 1->2, 2->1, 3->0)	Inversion / conversion of information in monitoring direction. It modifies the received information to be correctly displayed in SCADA
Monitor Tc-Address	Long Integer/ -1 (not used, default) 0 ... 2^{32}	Object Address in monitor direction. Structure see Chapter 3.4.9.7 <i>IEC103 Addressing Scheme</i> Address Parameter. Position is required
Old New Check	Boolean/ - False (default) - True	If set, the actually transmitted value is compared to the image value. If they are identical, the value is not written into the image and not transmitted to the next layer.
Renewal Check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Update check for cyclically transmitted data
System Control Direction ID	Enumeration / - Not used (default) - IEC103_Freeze Command SC	If "Control direction type" is "System Command": select a system control type from this list according to the protocol applied

3.4.9.6.2 Data Type "Analog Value"

IMM instance type: *CfeTelemeteredAnalog*

Table 3-4.80: Attributes of Instance Type *CfeTelemeteredAnalog*

Attribute Label	Type / Range / Default	Description / Remark
GI Relevant	Boolean / - False (default) - True	Includes data point in Completeness Check at GI
Monitor Direction Type	Enumeration / - Not used - IEC103_Measured Value I - IEC103_Measured Value w Rel.Time - IEC103_Measured Value II	Data type specified for monitoring direction
Monitor Tc-Address	Long Integer / -1 (not used, default), 0 ... 2^{32}	Object Address in monitor direction. Structure see Chapter 3.4.9.7 <i>IEC103 Addressing Scheme</i> Position is required if more than one value is in a message
Renewal Check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Update check for cyclically transmitted data
Smooth Factor	Short Integer / 0 (default) ... n	The function is needed to "smooth" the progression of the curve of an analog value. The formula for smoothing is: smoothed value = (current value + sf*last_smoothed_value)/(sf+1) where sf is the smooth factor

Attribute Label	Type / Range / Default	Description / Remark
Threshold Value	Long Integer / 0 (default) ... n	To prevent analogs without significant changes to be passed-on to SCADA
Zero Suppress Lower Boarder	Long Integer / 0 (default) ... n	If a value is greater than 'Zero Suppressed Lower Border' and less than Zero Suppressed Upper Border it is not transmitted to the next layer. RTUCS: ZeroDeadband
Zero Suppress Upper Boarder	Long Integer / 0 (default) ... n	see above (however, applicable for upper border)

3.4.9.6.3 Data Type "Counter Value"

IMM instance type: *CfeTelemeteredAccumulator*

Table 3-4.81: Attributes of Instance Type *CfeTelemeteredAccumulator*

Attribute Label	Type / Range / Default	Description / Remark
Monitor Tc-Address	Long Integer / -1 (not used, default) 0 ... 2^{32}	Object Address in monitor direction. Structure see Chapter 3.4.9.7 <i>IEC103 Addressing Scheme</i>
Monitor Direction Type	Enumeration / - Counter Value (default)	Data type specified for monitoring direction
Renewal Check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	Data point is subject to cyclic updating checks

3.4.9.6.4 Data Type "Fleeting Information"

IMM instance type: *CfeTelemeteredFleeting*

Table 3-4.82: Attributes of Instance Type *CfeTelemeteredFleeting*

Attribute Label	Type / Range /Default	Description / Remark
Monitor Tc-Address	Long Integer / -1 (not used, default) 0 ... 2^{32}	Object Address in monitoring direction. Structure see Chapter 3.4.9.7 <i>IEC103 Addressing Scheme</i> . Position not required
Monitor Direction Type	Enumeration / - Not used - IEC103_Fleeting Point - IEC103_Fleeting Point w. Rel.Time	Data type specified for monitor direction
Invert Monitor Info	Boolean / - False (default) - True	Inverting of information in monitoring direction. If set 'True', it modifies the received information to be correctly displayed in SCADA
GI Relevant	Boolean / - False (default) - True	Includes data point in Completeness Check at GI

3.4.9.6.5 CFE Status Information

The following table shows the complete parameters required for IEC 103 RTU – Cfe Status instances. The instance 'CfeStatus' for Cfe – devices remains unchanged.

Table 3-4.83: Attributes of Instance Type *CfeStatus*

Attribute Label	Type / Range	Description / Remark
Infold	Enumeration / - Device Failure - Device locked - Device process control - Device preferred - GI running - GI incomplete - Renewal Alarm 1 - Renewal Alarm 2	Used for CFE internal indications and commands

3.4.9.7 IEC103 Addressing Scheme

The data model of Spectrum PowerCC uses the type "Long Integer" (4 bytes) for addressing the information objects. It is structured, every byte has an individual meaning:

SICAM Infotyp			Position/ Index			Function Type			Information Number			
31	...	24	23	...	16	15	...	8	7	...	0	bit
Byte 3			Byte 2			Byte 1			Byte 0			

Allocation of the Address Bytes:

Byte 0:	"Information Number" (second octet of the Information Object Identifier in the message) is required; range 0...255
Byte 1:	"Function Type" (first octet of the Information Object Identifier in the message) is required; range 0...255
Byte 2:	"Position/ Index" is only required when one object defined by byte 0/1 contains more than one information (e.g. 2 or more measure values). It defines the position of the value within the object; range 1...254 (0 is internally always converted to 1)
Byte 3:	"SICAM Infotype" is optional, see protection configuration. The input can be made easier by using a structured input defined through the attribute "DataPointAddressStructure" (of the instance type <i>CfeRemoteTerminalUnit</i>) The following structures can be used: 8-8 Byte1-Byte0 8-8-8 Byte2-Byte1-Byte0 8-8-8-8 Byte3-Byte2-Byte1-Byte0

3.4.10 Protocol MODBUS

3.4.10.1 Common Protocol Parameters

The following parameter set is common for all RTUs communicating with CFE using the MODBUS protocol

Table 3-4.84: Attributes of Instance Type *Cfe ProtocolMODBUS*

Attribute Label	Type / Range / Default	Description / Remark
Command Aging Timeout	Long Integer / [msec] 1 000 ... 20 000	The command is removed from the Protocol-Module buffer after the given time, when not sent yet. For details on Command Aging Monitoring see also Figure 3-4.1
Communication Statistic Offset	Long Integer / [msec] -1 (default) 0 ... n	Start offset related to 00:00 h -1 means: Start immediately after CFE run-up with set Communication Statistic Period
Communication Statistic Period	Long Integer / [msec] 0 (off, default) ... n	For periodical transfer to display configured accumulators in SCADA
Family Transmission Mode	Enumeration / - Serial Character (default) - Serial Binary	Specifies MODBUS Family Transmission Mode for serial transmission
TcAddress Offset Coils	Long Integer / -65535 ... 0 ... 65535 default: 0	Address offset for Coils FC01 and FC05 and FC15 ¹⁾
TcAddress Offset Input Status	Long Integer / -65535 ... 0 ... 65535 default: 0	Address offset for Input Status FC02 ¹⁾
TcAddress Offset Holding Register	Long Integer / -65535 ... 0 ... 65535 default: 0	Address offset for Holding Register FC03, FC06 and FC16 ¹⁾
TcAddress Offset Input Register	Long Integer / -65535 ... 0 ... 65535 default: 0	Address offset for Input Register FC04 ¹⁾

¹⁾ FC: Function Code

3.4.10.2 CFE Transport Interface Parameterization

IMM instance type: *CfeTransportInterface*

For configuring a CFE TCP/IP interface for MODBUS protocol the following parameters are to be set:

Table 3-4.85: Attributes of MODBUS *CfeTransportInterface*

Attribute Label	Type / Range / Default	Description / Remark
Life Cycle Time	Long Integer / [msec] 1000 ... n	Used for cyclic check of "dead" links

3.4.10.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the MODBUS protocol link layer properties applied to communication lines between CFE interface and RTUs. The following attributes are to be set:

Table 3-4.86: Attributes of MODBUS *CfeInterfaceLink*

Attribute Label	Type / Range / Default	Description / Remark
Link Address	Long Integer / 1 ... 247	The Address of the device
Priority	Short Integer / 1 ... 8	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic failover). 1 is highest priority
Remote IP Address	String / (Format: nnn.nnn.nnn.nnn)	IP Address of remote communication peer. Any connection establishment to a remote server requires an IP address, a defined port number (502) and the applied transport protocol. That triplet of communications parameters specifies uniquely the remote Transport Endpoint.
Successive Errors For RTU Fail	Long Integer / 1 ... n (default: 3)	RTU fails after this number of consecutive errors

3.4.10.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following RTU parameters are valid for MODBUS protocol stack

Table 3-4.87: Attributes of MODBUS *CfeRemoteTerminalUnit*

Attribute Label	Type / Range / Default	Description / Remark
Application Retry	Long Integer / 0 (default)... 255	Number of times the CFE repeats an unconfirmed application message
Application Timeout	Long Integer / [msec] 500 ... 2000 (default)	This time the CFE waits for RTU to confirm an application message before repeating it
Data Point Address Structure	Enumeration / - 16 bit - 32 bit	For the best representation of data point address use "16 bit".
Renewal Cycle 1	Long Integer / [msec] 0 (default) ... n	Renewal check time for the first check cycle
Renewal Cycle 2	Long Integer / [msec] 0 (default) ... n	Renewal check time in for the second check cycle
RTU Model	Enumeration / - Not used (default) - MODBUS_M84 - MODBUS_M484 - MODBUS_M184_OR_M384 - MODBUS_M584 - MODBUS_M884 - MODBUS_M984	Specifies the RTU model. This will determine the maximum amount of data that can be requested per FC. NOTE: the smallest RTU model will be used on a circuit that supports multidrop RTUs. In case "Not used" is selected "MODBUS_M84" is used Specifies the RTU model. This will determine the maximum amount of data that can be requested per function. NOTE: the smallest RTU model will be used on a circuit that supports multidrop RTUs.

Attribute Label	Type / Range / Default	Description / Remark
Telecontrol Protocol Type	Enumeration / - MODBUS	Last table will display the selected protocol and its attributes for this instance and its children: Select MODBUS for this protocol

3.4.10.5 Cfe Scan Rate Parameterization

IMM instance type: *CfeScanRate*

Any periodic scan needs the specification of a period time and a starting time. This is provided by a *CfeScanRate* attribute. Typically, a scan rate definition may be valid for more than one scan group.

Table 3-4.88: Attributes of Instance MODBUS *CfeScanRate*

Attribute Label	Type / Range / Default	Description / Remark
ScanOffset	Long Integer / [msec] -1, 0 ... n (Granularity is 1000 ms)	Offset from midnight for issuing scan -1 = free running, scans start immediately when the connection to RTU is available
Scan Period	Long Integer / [msec] 1000 ... n (Granularity is 1000 ms)	Time between start of scans. Watching all scans at an interface. There must be enough time for every scan to be answered and space for commands to be sent

3.4.10.6 Data Point Parameterization

For the MODBUS protocol the following data types are distinguished:

- Digital Value (IMM instance type: *CfeTelemeteredDigital*)
- Analog Value (IMM instance type: *CfeTelemeteredAnalog*)
- Accumulator (IMM instance type: *CfeTelemeteredAccumulator*)
- CFE Status Information (IMM instance type: *CfeStatus*)
- System Counters (IMM instance type: *CfeSystemAccumulator*)

In the tables hereafter detailed information on data points for each data type are summarized.

3.4.10.6.1 Data Type "Digital Value"

IMM Instance type: *CfeTelemeteredDigital*

Table 3-4.89: Attributes of Instances for MODBUS *CfeTelemeteredDigital*

Attribute Label	Type / Range / Default	Description / Remark
Command Direction Type	Enumeration / - Not used (default) - Single Command - Double Command - System Command	Type of digital in control direction. - This digital has no control - Single Command is sent with FC05 - Double Command is sent with FC15 ²⁾ - Needs a "System Control Direction ID"
Control-Tc-Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for control direction. The real address transmitted (0 ... 65535) depends on "TcAddress Offset Coils" of the instance "CfeProtocolMod-Bus". It must be unique within an RTU

Attribute Label	Type / Range / Default	Description / Remark
Intermediate State Suppression Time	Long Integer / [msec] 0 (off, default), 1 ... n	Intermediate state is held back for this time
InvertControl	Boolean / - False (default) - True	Inversion of status for command direction
Monitor Direction Type	Enumeration / - Not used - Single Point (default) - Double Point - MODBUS_Input Status Single - MODBUS_Input Status Double	Type of digital in monitor direction. Single Points and Double Points are received with FC01. MODBUS_Input Status Single and MODBUS_Input Status Double are received with FC02. The telecontrol addresses within one FC have to be unique.
Monitor Info Conversion	Enumeration / - Not used (default) - SinglePoint invert (1->2, 2->1) - DoublePoint invert Bit 0 (0->1, 1->0, 2->3, 3->2) - DoublePoint invert Bit 1 (0->2, 1->3, 2->0, 3->1) - DoublePoint invert Bit 0+1 (0->3, 1->2, 2->1, 3->0)	Inversion / conversion of information in monitoring direction. It modifies the received information for correct display in SCADA.
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... 65535	Telecontrol address for monitoring direction. The real address received (0 ... 65535) depends on the configured "TcAddress Offset ..." of the instance "CfeProtocolModBus". It must be unique within an RTU.
Monitor Tc Address2	see above	Second monitor address for Doublepoint digitals. If not entered (-1) "Monitor Tc Address" + 1 is used for the second point.. Same rules apply as for "Monitor Tc Address".
Old New Check	Boolean / - False (default) - True	If set, the actually transmitted value is compared to the image value, if they are equal, the value is not written to the image and not transmitted to the next layer.
Renewal check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	If enabled, data point is subject to cyclic updating checks.
System Control Direction ID	Enumeration / - Not used (default) - MODBUS_ScanOnDemand	If "Control direction type" is "System Command", select a system control type from this list according to the protocol applied.
System Control Extension	Long Integer / 1 ... n	If "System Control Direction ID" is "MODBUS_ScanOnDemand" you have to set here the "Scan Group Number" of the Scan Group.

2) Double Commands use "Control TcAddress +1" as second point

3.4.10.6.2 Data Type "Analog Value"

IMM Instance type: *CfeTelemeteredAnalog*

Table 3-4.90: Attributes of Instances for MODBUS *CfeTelemeteredAnalog*

Attribute Label	Type / Range / Default	Description / Remark
Command Direction Type	Enumeration / - Not used (default) - MODBUS_Single Register 16 Int - MODBUS_Single Register 32 Int - MODBUS_Single Register Short Float	Type of analog setpoint in control direction. A signed 16 bit integer setpoint sent with FC06 (occupies 1 Tc address) A signed 32 bit integer setpoint sent with FC16 (occupies 2 Tc addresses) A signed 32 bit short float setpoint sent with FC16 (occupies 2 Tc addresses)
Control-Tc-Address	Long Integer / -1 (not used, default) 0 ... n	Telecontrol address for control direction. The real address transmitted (0 ... 65535) depends on "TcAddress Offset Coils" of the instance "CfeProtocolModBus". It must be unique within an RTU
Info Structure	Enumeration / - Default - 32Bit HiWoHiBy-HiWoLoBy-LoWoHiBy-LoWoLoBy - 32Bit LoWoHiBy-LoWoLoBy-HiWoHiBy-HiWoLoBy	MODBUS defines only 16bit (word) values. For MODBUS it allows to set the structure rule for not defined 32bit (double word) values. Hi = High Lo = Low Wo = Word By = Byte This attribute applies to control and monitor direction
Monitor Direction Type	Enumeration / - Not used - MODBUS_Holding Register 16 Int signed - MODBUS_Holding Register 32 Int signed - MODBUS_Holding Register Short Float - MODBUS_Input Register 16 Int signed - MODBUS_Input Register 32 Int signed - MODBUS_Input Register Short Float	Type of digital in monitor direction A signed 16 bit integer value received with FC03 (occupies 1 Tc address). A signed 32 bit integer value received with FC03 (occupies 2 Tc addresses). A signed 32 bit short float value received with FC03 (occupies 2 Tc addresses). A signed 16 bit integer value received with FC04 (occupies 1 Tc address). A signed 32 bit integer value received with FC04 (occupies 2 Tc addresses). A signed 32 bit short float value received with FC04 (occupies 2 Tc addresses).
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... 65535	Telecontrol address for monitoring direction. The real address received (0 ... 65535) depends on the configured "TcAddress Offset ..." of the instance "CfeProtocolModBus". "Monitor Tc Address" + 1 is used for the second address. The telecontrol addresses within one FC have to be unique within an RTU
Renewal check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	If enabled, data point is subject to cyclic updating checks

Attribute Label	Type / Range / Default	Description / Remark
Smooth Factor	Short Integer / 0 (default) ... n	The smooth factor is needed to "smooth" the progression of the curve of an analog value. The formula for smoothing is: smoothed value = (current value +sf*last smoothed value)/(sf+1) where sf is the smooth factor.
Threshold Value	Long Integer / 0 (default) ... n	To suppress analogs without significant changes to be passed-on to SCADA
Zero Suppress Lower Border	Long Integer / 0 (default) ... n	If a value is greater than 'Zero Suppressed Lower Border' and less than Zero Suppressed Upper Border it is not transmitted to the next layer. RTUCS: ZeroDeadband
Zero Suppress Upper Border	Long Integer / 0 (default) ... n	see above (however, applicable for upper border)

3.4.10.6.3 Data Type "Counter Value"

IMM instance type: *CfeTelemeteredAccumulator*

Table 3-4.91: Attributes of Instances for MODBUS *CfeTelemeteredAccumulator*

Attribute Label	Type / Range / Default	Description / Remark
Info Structure	Enumeration / - Default - 32Bit HiWoHiBy-HiWoLoBy-LoWoHiBy-LoWoLoBy - 32Bit LoWoHiBy-LoWoLoBy-HiWoHiBy- - HiWoLoBy	MODBUS defines only 16bit (word) values. For MODBUS it allows to set the structure rule for not defined 32bit (double word) values. Hi = High Lo = Low Wo = Word By = Byte This attribute applies to control and monitor direction
Monitor Direction Type	Enumeration / - Not used - MODBUS_Holding Register 16 Int signed - MODBUS_Holding Register 32 Int signed - MODBUS_Holding Register 32 Int signed - MODBUS_Input Register 32 Int signed	Type of information in monitor direction A signed 16 bit integer value received with FC03 (occupies 1 Tc address) A signed 32 bit integer value received with FC03 (occupies 2 Tc addresses) A signed 16 bit integer value received with FC04 (occupies 1 Tc address) A signed 32 bit integer value received with FC04 (occupies 2 Tc addresses)
Monitor Tc Address	Long Integer / -1 (not used, default) 0 ... 65535	Telecontrol address for monitoring direction. The real address received (0 ... 65535) depends on the configured "TcAddress Offset ..." of the instance "CfeProtocolModBus". "Monitor Tc Address" + 1 is used for the second address. It must be unique within an RTU
Renewal check	Enumeration / - Disabled (default) - Cycle 1 of renewal check enabled - Cycle 2 of renewal check enabled	If enabled, data point is subject to cyclic updating checks

3.4.10.6.4 CFE Status Information

IMM instance type: *CfeStatus*

Table 3-4.92: Attributes of Instance Type *CfeStatus*

Attribute Label	Type / Range	Description / Remark
InfoID	Enumeration / - Not used - Device Failure - Device locked - Device process control - Device preferred - Renewal Alarm 1 - Renewal Alarm 2	Used for CFE internal indications and commands

3.4.10.6.5 System Counters

IMM instance type: *CfeSystemAccumulator*

The MODBUS protocol provides system counters to obtain communication statistics. Details on errors and counters can be found in the following table.

Table 3-4.93: Attributes of Instance Type *CfeSystemAccumulator*

Attribute Label	Type / Range / Default	Description / Remark
Info ID	Enumeration / - Last Error Code - Message Send Counter - Message Receive Counter - Message Error Receive - Counter Message Timeout - Counter Scan Overrun - Counter Scan Requests - Counter Command Requests	Last Error Code: Errors are reported spontaneously as a number Counters: Communication statistic counters are reported cyclically dependent on "Communication Statistic Period". Select the counter of interest

3.4.10.7 Additional Information on MODBUS Parameters

Address Parameters

The following attributes deal with MODBUS addresses:

- "TcAddress Offset Coils", "TcAddress Offset Input Status", "TcAddress Offset Holding Register" and "TcAddress Offset Input Register" of instance *CfeProtocolModBus*
- "Control Tc Address" for *CfeDataPoints* (*CfeTelemeteredAnalog*, *CfeTelemeteredDigital*)
- "Monitor Tc Address" for *CfeDataPoints* (*CfeTelemeteredAccumulator*, *CfeTelemeteredAnalog*, *CfeTelemeteredDigital*)
- "Start Tc Address" for *CfeScanVolume*
- "End Tc Address" for *CfeScanVolume*

The "TcAddress Offset ..." specifies an offset which is added to the "TcAddress" and the result is the real address in the message. The offset allows the user to configure the data point address in its device even if

it differs from the message address. The offset applies to "Control Tc Address", "Monitor Tc Address", "Start Tc Address" and "End Tc Address".

Formula: "address in message" = "configured Tc Address" + "TcAddress Offset ..."

The relation between the attribute TcAddressOffset (in instance CfeProtocolMosBus) and Function Code is presented in the following table

TcAddress Offset	Function Code (FC)	Description / Remark
Coils	01	
Coils	05, 15	
Input Status	02	
Holding Register	03	
Holding Register	06, 16	
Input Register	04	



Note

The address of function codes in one row of the table above have to be unique!!

Some of the Control and Monitor Direction Types need 2 message addresses. Only "Monitor Tc Address2" of CfeTelemeteredDigital allows configuring a second TcAddress. In all other cases are "Control Tc Address" + 1 or "Monitor Tc Address" + 1 used. These addresses can't be used double either.

Example

CfeProtocolModBus: "TcAddress Offset Holding Register" = -40001

CfeTelemeteredAnalog: "Monitor Direction Type" = MODBUS_Holding Register 32 Int signed

CfeTelemeteredAnalog: "Monitor Tc Address" = 40001

CfeScanVolume: "Class type" = MODBUS_Holding_Register

CfeScanVolume: "Start Tc Address" = 40001

Address in request and response message: $40001 + (-40001) = 0$

As MODBUS_Holding Register 32 Int signed needs 2 addresses, 40002 can't be used again for any other "Monitor Tc Address" if "Monitor Direction Type" = Holding_Register ... (FC03) is used.

RTU Model

The attribute "RTU model" (in instance CfeRemoteTerminalUnit) determines the maximum amount of data that can be requested / sent per FC.

FC01 R CS	Read Coil Status
FC02 R IS	Read Input Status
FC03 R HR	Read Holding Register
FC04 R IR	Read Input Register
FC05 W SC	Write (force) Single Coil

FC06 W SR	Write (force) Single Register
FC15 W MC	Write (force) Multiple Coil
FC16 W MR	Write (force) Multiple Register

RTU Model	FC01 R CS	FC02 R IS	FC03 R HR	FC04 R IR	FC05 W SC FC06 W SR	FC15 W MC FC16 W MR
MODBUS_M84	64	64	32	4	1	64
MODBUS_M484	512	512	125	32	1	800
MODBUS_M184_OR_M384	600	800	100	100	1	800
MODBUS_M584	2000	2000	125	125	1	800
MODBUS_M884	2000	2000	125	125	1	800
MODBUS_M984	2000	2000	125	125	1	800

**Note**

The smallest RTU model is used on a circuit that supports multidrop RTUs. If the configured request exceeds the limit the requests will be split into several small ones. A better control by the user is possible by configuring several pairs of ScanGroups with same "Scan Type" and a ScanVolume with different addressing.

Time Parameters

The choice of meaningful timeout parameters mainly depend on the selected baud rate and the properties of the connected RTU (or IED). All timings must be done that way that an RTU error is indicated within an acceptable time period. Therefore the number of RTUs connected (NumberOfRtus), all timeouts, number of configured periodic ScanGroups and SuccessiveErrorsForRTUFail of CfeInterfaceLink have to be taken in consideration!

Scan Rates (CfeScanGroup / CfeScanRate)

The minimum ScanPeriod depends on the number of RTUs and scans performed on an interface. The sum of all scans with their timeout or answer time (take the greater one) within a given period has to be less than the period itself!

The scan period shall not be less than 1 000 ms. The ScanOffset shall be used only when a defined start of the scan is required (e.g. exactly at every full hour, 00:00, 01:00, 02:00,periode= 3 600 000 and offset=0) otherwise set it to "-1". Don't set ScanOffset for all scans to zero if not required, it might concentrate scans unnecessarily.

Ensure on every interface that all running scans get enough time to be processed. It is up to the user to adjust the parameter values, if different requirements or conditions arise from the connected devices' properties. In all cases, the rules above must be followed.

3.4.11 Protocol SNMP



Note

Please note that attributes presented by the IMM Instance Editor but not listed in the tables hereafter are not relevant for this protocol and as such they can be ignored.

3.4.11.1 Common Protocol Parameters

The following parameter set is common for all devices communicating with CFE using the SNMP protocol

Table 3-4.94: Attributes of Instance Type *CfeProtocol/SNMP*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the Instance, e.g. "Protocol SNMP"
Communication Statistic Offset	Long Integer / [msec] -1 (default) 0 ... n	Not implemented for SNMP
Communication Statistic Period	Long Integer / [msec] 0 (off, default) ... n	Not implemented for SNMP

3.4.11.2 CFE Transport Interface Parameterization

IMM instance type: *CfeTransportInterface*

For configuring a CFE TCP/IP interface for SNMP protocol the following parameters are to be set:

Table 3-4.95: Attributes of SNMP *CfeTransportInterface*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the instance interface
CfeIPAddress	String	Host IP Address
CommunityName	String	Name of the host SNMP community
Port Number	Long Integer / 0 ... n	Number of the local TCP/IP port for SNMP protocol. Default value is 162.

3.4.11.3 CFE Interface Link Parameterization

IMM instance type: *CfeInterfaceLink*

The (interface) link parameters specify the SNMP protocol link layer properties applied to communication links between CFE interface and connected devices. The following attributes are to be set:

Table 3-4.96: Attributes of SNMP *CfeInterfaceLink*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the instance
Polling Time	Long Integer / [ms] 0 ... n	Defines the time period the device is polled. Recommended value is 60 000 ms

Attribute Label	Type / Range / Default	Description / Remark
Port Number	Long Integer / 0 ... n	SNMP TCP/IP Port used to communicate with the connected device
Priority	Short Integer / 1 ... 8	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for run-up and during automatic failover). 1 is highest priority
Remote IP Address	String / (Format: nnn.nnn.nnn.nnn)	IP Address of remote communication peer. Any connection establishment to a remote server requires an IP address, a defined port number and the applied transport protocol. That triplet of communications parameters specifies uniquely the remote Transport Endpoint.

3.4.11.4 CFE Remote Terminal Unit Parameterization

IMM instance type: *CfeRemoteTerminalUnit*

The following "RTU" parameters are valid for SNMP protocol stack

Table 3-4.97: Attributes of SNMP *CfeRemoteTerminalUnit*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the instance
Community Name	String	Name of the device SNMP community
Request Trigger	Enumeration / - Not Used (default) - No Request - Send General Request	Specifies how to handle incoming "Trap messages"
Telecontrol Protocol Type	Enumeration / ... - SNMP ...	Instance shows a list of available protocols. Select SNMP for this protocol



Note

The instance types *CfeScangroup*, *CfeScanVolume*, *CfeScanRate* are not in use for protocol SNMP

3.4.11.5 Data Point Parameterization

For the SNMP protocol the following data types can be configured:

- Digital Value (IMM instance type: *CfeTelemeteredDigital*)
- Analog Value (IMM instance type: *CfeTelemeteredAnalog*)
- Accumulator (IMM instance type: *CfeTelemeteredAccumulator*)

In the tables hereafter detailed information on data points for each data type are summarized.

3.4.11.5.1 Data Type "Digital Value"

IMM Instance type: *CfeTelemeteredDigital*

Table 3-4.98: Attributes of Instances for SNMP *CfeTelemeteredDigital*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the instance
Monitor Direction Type	Enumeration / - Not used - Single Point - Double Point	Type of data point. Select either "Single Point" or "Double Point"
VariableAddress	String	SNMP OID (Object Identifier), e.g. 1.3.6.1.2.1.2.2.1.7.1

3.4.11.5.2 Data Type "Analog Value"

IMM Instance type: *CfeTelemeteredAnalog*

Table 3-4.99: Attributes of Instances for SNMP *CfeTelemeteredAnalog*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the instance
Monitor Direction Type	Enumeration / - Not used - Measured Value (Normalized)	Type of data point. Select "Measured Value"
VariableAddress	String	SNMP OID (Object Identifier), e.g. 1.3.6.1.2.1.25.2.3.1.5.2

3.4.11.5.3 Data Type "Counter Value"

IMM instance type: *CfeTelemeteredAccumulator*

Table 3-4.100: Attributes of Instances for SNMP *CfeTelemeteredAccumulator*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the instance
Monitor Direction Type	Enumeration / - Not used - Counter Value	Type of data point. Select "Counter Value"
VariableAddress	String	SNMP OID (Object Identifier), e.g. 1.3.6.1.2.1.1.3.0

3.4.11.5.4 Data Type "Step Position"

IMM instance type: *Cfe Telemetered Position*

Table 3-4.101: Attributes of Instances for SNMP *Cfe Telemetered Position*

Attribute Label	Type / Range / Default	Description / Remark
Name	String	Name of the instance
Monitor Direction Type	Enumeration / - Not used - Step Position	Type of data point. Select "Step Position"
VariableAddress	String	SNMP OID (Object Identifier), e.g. 1.3.6.1.2.1.2.2.1.7.1

3.4.11.5.5 CFE Status Information

Status information can optionally be configured for being processed in SCADA like regular process data.

IMM instance type: *CfeStatus*

Cfe Status Information for Interface and Link

Table 3-4.102: Attributes of Instance Type *CfeStatus* for Interfaces & Links

Attribute Label	Type / Range	Description / Remark
Infold	Enumeration / - Not used - Device Failure - Device locked - Device process control - Device preferred	Each Infold can optionally be configured as status information for being processed in SCADA

Cfe Status Information for Connected Devices (RTU)

Table 3-4.103: Attributes of Instance Type *CfeStatus* for Devices

Attribute Label	Type / Range	Description / Remark
Infold	Enumeration / - Not used - Device Failure - GI incomplete - GI running - Renewal Alarm Cycle 1 - Renewal Alarm Cycle 2	see above

3.5 Protocol-Specific Engineering of Slave Protocols

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Slave protocols are "inverted" (standard) telecontrol protocols. For the communication aspect, CFE is a 'Slave' partner for an external 'Master' component. This way data exchange from/to a Remote Control Center (RCC) can easily be established: CFE behaves just like one or several RTU(s) connected to the RCC. CFE 'Slave' functionality allows to forward any (in IMM accordingly configured) data point from Spectrum PowerCC Realtime data base to an RCC as well as to route commands received from RCC to RTUs associated with CFE.

3.5.1 Slave Protocols IEC60870-5-101-SL and IEC60870-5-104-SL, Overview

Short names: IEC101-SL, IEC104-SL

Parameterization of CFE protocol, interfaces, communication links and associated Remote Control Centers (RCC) communicating via the protocol IEC60870-5-101 or protocol IEC60870-5-104 requires the compliance of distinct rules described in the following.

For configuring Master/Slave communication by means of the Information Model Manager (IMM) the user has to know what IMM instances are involved, what their attributes mean and how the related instances are arranged in the hierarchical ordered instance tree. The following table gives an overview of these instances.

Table 3-5.1: Instance Types Specifying the CFE - RCC Communication

Instance Type	Specifies ...
<i>Cfe Protocol IEC101 Slave</i>	General Protocol IEC101-SL parameters
<i>Cfe Protocol IEC104 Slave</i>	General Protocol IEC104-SL parameters
<i>CfeSingleCircuitInterfaceExtended</i>	Serial (RS232) interface parameters
<i>CfeTransportInterface</i>	Network (TCP/IP) interface parameters
<i>CfeInterfaceLink</i>	Link parameters (both serial and network)
<i>Cfe Remote Control Center</i>	General Remote Control Center parameters
<i>Cfe Logical Remote Terminal Unit</i>	Logical RTU parameters
<i>Cfe Forwarded Accumulator</i>	Parameters for data points to be sent to the RCC such as: Accumulator values, Analog values, Digital values, Fleeting data, Transformer step positions
<i>Cfe Forwarded Analog</i>	
<i>Cfe Forwarded Digital</i>	
<i>Cfe Forwarded Fleeting</i>	
<i>Cfe Forwarded Positions</i>	
<i>CfeScanGroup</i>	RTU scan groups controlling the periodical data transmission to RCC
<i>CfeScanRate</i>	Timing of periodical data transmission

The following diagram gives an overview of all IMM instances involved in CFE - RCC communication as well as their hierarchical position (Figure 3-5.1).

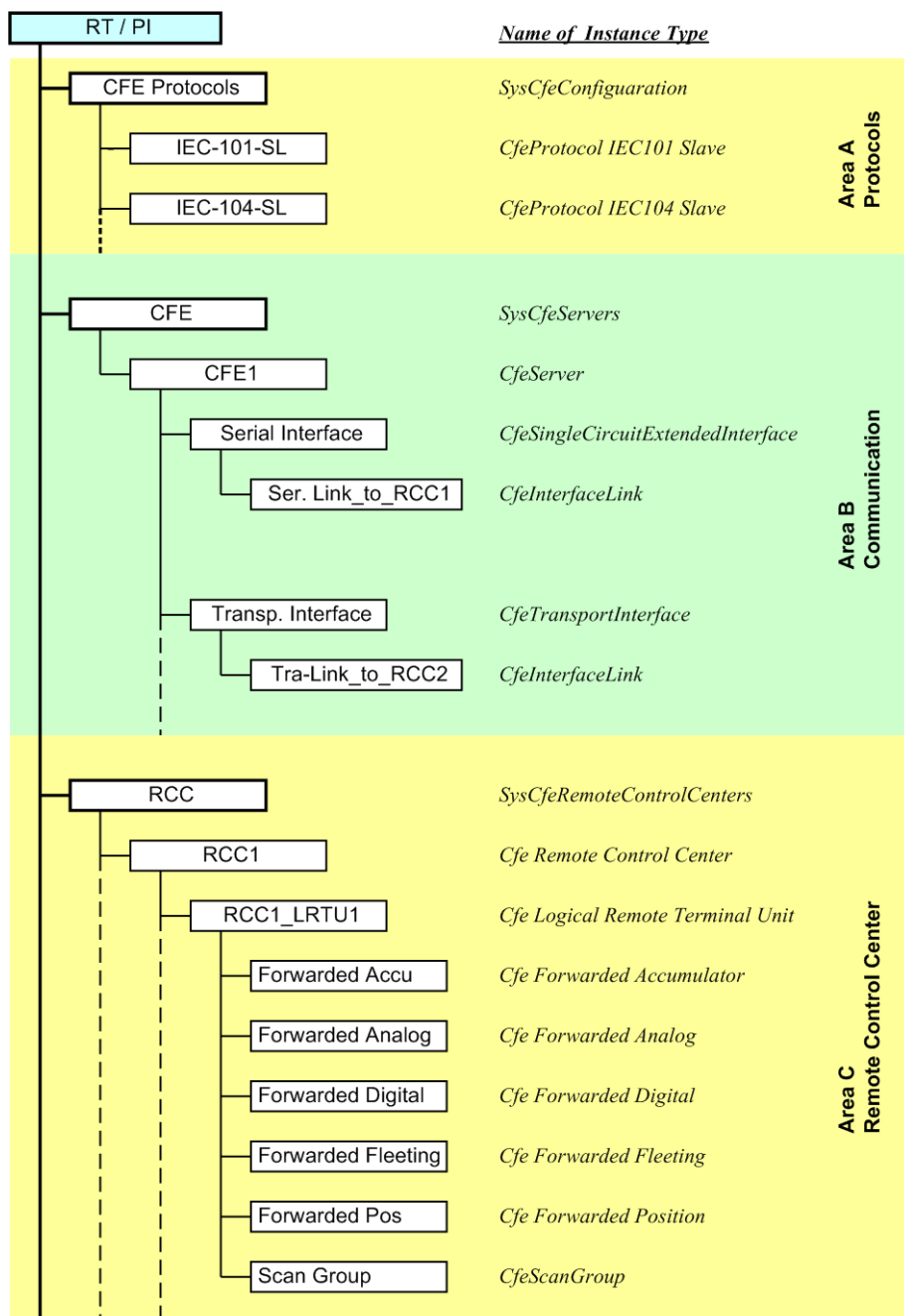


Figure 3-5.1: IMM Instances Involved in CFE - RCC Communication

**Note**

Default values should be applied for all attributes displayed in IMM editors that are not indicated in the following tables.

3.5.2 Specific Parameters for RCC Communication Over Serial Links

3.5.2.1 Common Protocol Parameters

The following parameter set is common for all RCCs communicating with CFE using the protocol IEC101SL :

Table 3-5.2: Attributes of Instance Type *Cfe Protocol IEC101 Slave*

Attribute Label	Type/ Range	Default Value	Description/ Remark
Common Address Bytes	Enumeration/ 1 or 2 octets	2 octetts	Number of address bytes for the Common Address
Communication Statistic Offset	Long Integer/ 0 ... msec	0	Start offset related to 00:00 h
Communication Statistic Period	Long Integer/ 0 ... msec	0	For periodical transfer to display configured accumulators in SCADA
Info Object Address Bytes	Enumeration/ 1, 2 or 3 octets	3 octets	Number of address bytes for the Info Object Address
Link Address Bytes	Enumeration/ 0, 1 or 2 octets	2 octets	Number of address bytes for the Link Address
Originator Address Bytes	Enumeration/ 0 or 1 octet	0 octets	Number of address bytes for the Originator Address
Strict Class Reply	Boolean/ False, True	False	If False: Response to class 2 scan may also contain class 1 data

3.5.2.2 Parameterization the Serial Interface for IEC101-SL

IMM instance type: *CfeSingleCircuitInterfaceExtended*

The interface parameters specify the communication properties of serial communication ports (COM ports) provided for connecting RCCs.

Table 3-5.3: Attributes of Instance Type *CfeSingleCircuitInterfaceExtended*

Attribute Label	Type/ Range	Default Value	Description/ Remark
Acknowledge Type	Enumeration/	Single Char	To select acknowledgement type: Short message or single character
Baud Rate	Enumeration/	-	Transmission speed
Data Length	Short Integer		Max length of message frames
Serial Port	Short Integer/ 1 ... 64	-	Number of the COM port
Transmission Mode	Enumeration/	unbalanced	To select between balanced (polling) and unbalanced (spontaneous) transmission

3.5.2.3 Parameterizing the Serial Link for IEC101-SL

IMM Instance Type: *CfeInterfaceLink*

The interface link parameters specify the IEC101-SL protocol serial link layer properties applied to communication links established between CFE and RCCs.

Table 3-5.4: Attributes of Instance Type *CfeInterfaceLink* for IEC101-SL

Attribute Label	Type/ Range	Default Value	Description/ Remark
Link Address	Long Integer/ 0 (if link address is omitted) 1 ...255 or 1...65535		The link address of the device
Link Test Cycle Time	Long Integer/ 0 ... 2^{32} [msec]	0	Cycle time for periodical link test frames to the RCC
Priority	Short Integer/ 1 ... n	0	Priority of a link. This attribute will be used by CFE Line Level Redundancy Management to decide which link has to be activated (relevant for runup and during automatic failover and only for dependent links)
Successive Errors For RTU Fail	Long Integer 0 ... 10	3	Number of successive transmission failures before a link failure is indicated

3.5.3 Specific Parameters for RCC Communication Over Network Links

3.5.3.1 Parameterizing the Transport Interface for IEC104-SL

IMM instance type: *CfeTransportInterface*

The transport interface parameters specify IEC104-SL interface properties applied to network ports for connecting RCCs.

Table 3-5.5: Attributes of Instance Type *CfeTransportInterface* for IEC104-SL

Attribute Label	Type/ Range	Default Value	Description/ Remark
Cfe IP Address	String/	-	IP address, format nnn.nnn.nnn.nnn
Max Pending GIs	Long Integer/	-	not relevant for Slave protocols
Primary Address	Long Integer/	0	not relevant

3.5.3.2 Parameterizing the Transport Link for IEC104-SL

The transport link parameters specify the IEC104-SL protocol transport link layer properties applied to communication links established between CFE and RCCs.

Table 3-5.6: Attributes of Instance Type *CfeInterfaceLink* for IEC104-SL

Attribute	Type/ Range	Default Value	Description/ Remark
Outstanding I-Format APDUs: k	Short Integer/ 1 ... 2^{16}	12	Stop sending I-frames after k unacknowledged frames
Outstanding I-Format APDUs: w	Short Integer/ 1 ... 2^{16}	8	Acknowledgement after w I-format APDUs latest
Port Number	Long Integer/ 1 ... 2^{32}	-1	Network port for TCP transport layer. IEC104 normally uses port no. 2404
Priority	Short Integer/ 1 ... n	0	Priority not relevant for IEC104-SL, the RCC decides with StartDT/StopDT and works only in independent mode
Remote IP Address	String/		Not relevant for IEC104-SL
Timeout Sending Test Frame - Long Idle State	Long Integer/ 0 ... 2^{32} [msec]	20,000	Send test frame after t_3 milliseconds without I-frames
Timeout for Ackn - No Data Message	Long Integer/ 0 ... 2^{32} [msec]	10,000	Send acknowledgement latest after t_2 milliseconds
Timeout for Ackn I and U Format APDU	Long Integer/ 0 ... 2^{32} [msec]	15,000	Acknowledges must arrive within t_1 milliseconds

3.5.4 Common Parameters for each Type of RCC Communication

3.5.4.1 Parameterizing the Remote Control Center

IMM instance type: *Cfe Remote Control Center*

This instance specifies general communication properties of the Remote Control Center in the PowerCC data model. The attributes are common for all logical RTUs of an RCC.

Table 3-5.7: Attributes of the Instance Type *Cfe Remote Control Center*

Attribute	Type/ Range	Default Value	Description/ Remark
Accum With Time	Boolean/	False	Time stamp for accumulator values even in requested mode
Appl Command Time-Out	Long Integer/ 0 ... 2^{32} ms	15,000	Timeout for the responds to a command
Appl Select TimeOut	Long Integer/ 0 ... 2^{32} ms	10,000	Timeout for the execute command after a select command
Data Point Address Structure	Enumeration/	-	Address structure of data points
Event Buffer Size	Short Integer/ 1 ... 2^{16}	2048	Maximum number of entries in the event buffers of data types

Attribute	Type/ Range	Default Value	Description/ Remark
RCC Supports End Of Init	Boolean/	True	Specifies whether the RCC reacts on 'End-of-Init' message by GI request. If 'False' CFE sends GI data after runup without GI request as a 'Background GI'
Set Point ACTTERM	Boolean/	True	Specifies whether a setpoint command is terminated by an 'Activation-Termination' message
Standby Buffer Time Span	Long Integer/ 0 ... 2^{32} [msec]	60,000	Specifies how long information is held in the Standby Buffer
Telecontrol Protocol Type	Enumeration/	IEC60870-5-104	Telecontrol protocol type
Time Format DP	Enumeration/ Without Time, Relative Timestamp (24 bits), Absolute Timestamp (56 bits)	Without Time	Time format for double point information
Time Format IT			Time format for integrated totals
Time Format ME			Time format for analog values
Time Format SP			Time format for single point information
Time Format ST			Time format for step position information
Timesync Confirm Pos	Boolean/ False, True	True	Specifies whether a time synchronization message from RCC is positively confirmed (although PowerCC cannot be synchronized via this way)

3.5.4.2 Parameterizing the Logical RTU(s) of an RCC

IMM instance type: *Cfe Logical Remote Terminal*

For an external Control Center, CFE can emulate logical RTUs as many as appropriate. For the RCC, CFE behaves like a Data Concentrator with connected Sub-RTUs. The following set of parameters is to be setup for each Logical RTU of an RCC:

Table 3-5.8: Attributes of Instance Type *Cfe Logical Remote Terminal*

Attribute	Type/ Range	Default Value	Description/ Remark
Common Address	Long Integer/ 1 ... 2^8 , 1 ... 2^{16} , 1 ... 2^{24}	0	Common Address of ASDU Range of Common Address depends on how many address bytes are used
Double Transmission	Boolean/	False	Not supported
StartupTimeout	Long Integer/ 0 ... 2^{32} [msec]	0	To avoid the transmission of still invalid data after CFE runup, start of data transmission can be delayed by this attribute

3.5.4.3 CFE Scan Group Parameterization

IMM instance type: *CfeScanGroup*

For Scan Groups the following list of parameters are applicable:

Table 3-5.9: Attributes of Instance Type *CfeScanGroup*

Attribute	Type/ Range	Default Value	Description/ Remark
BaseScan	Boolean/	False	The data points of this group are transmitted in the background (normally applied for analogs)
ScanGroupID	Short Integer/ 0 ... 2^{16}	-1	Scan Group identifier, used as follow: All_InterrogationGeneral: Range 1-16 (groups) All_InterrogationAccum: Range 1-4 (groups), 5 (all counters) All_PeriodicList: Range 0-3 with and 0-3 without BaseScan
ScanType	Enumeration/	0	Scan type options, e.g.: - All_InterrogationGeneral (GI) - All_InterrogationAccum (Counter values) - All_PeriodicList (for cyclic data transmission) ...

3.5.4.4 Parameterizing the Forwarding of Data Points

3.5.4.4.1 Data Type "Digital Values"

IMM instance type: *Cfe Forwarded Digital*

For digital values to be sent to the RCC the following parameters are applicable:

Table 3-5.10: Attributes of Instance Type *Cfe Forwarded Digital*

Attribute	Type/ Range	Default Value	Description/ Remark
Control Direction Type	Enumeration/ not used, Single command, double command, ...	Not used	When used: single or double command in control direction enabled
Control Tc Address	Long Integer/ 0 ... 2^{24}	-1	Control direction, telecontrol address (-1 = not used)
GI Relevant	Boolean/	True	If 'True' data point is sent on GI request
Invert Control	Boolean/	False	If True: command is inverted (ON gets OFF and vice versa)
Invert Info	Boolean/	False	If True: information is inverted (appearing gets disappearing and vice versa)
Monitor Direction Type	Enumeration/ none, Single Point, Double Point, ...	Single Point	Data type specified for monitor direction. Either 'Single Point' or 'Double Point' to be selected
Monitor Tc Address	Long Integer/ 0 ... 2^{24}	-1	Monitor telecontrol address. (-1 = not used)

Attribute	Type/ Range	Default Value	Description/ Remark
Spontaneous Transmission	Boolean	True	Specifies whether this data point is spontaneously transmitted (after change)

3.5.4.4.2 Data Type "Analog Values"

IMM instance type: *Cfe Forwarded Analog*

For analog values to be sent to the RCC the following parameters are applicable:

Table 3-5.11: Attributes of Instance Type *Cfe Forwarded Analog*

Attribute	Type/ Range	Default Value	Description/ Remark
Control Direction Type	Enumeration/ not used, IEC101_104_Setpoint Float IEC101_104_Setpoint Normalized IEC101_104_Setpoint Scaled	Not used	When used: setpoint command in control direction enabled
Control Tc Address	Long Integer/ 0 ... 2^{24}	-1	Control direction, telecontrol address. (-1 = not used)
GI Relevant	Boolean/	True	If 'True' data point is sent on GI request
Monitor Direction Type	Enumeration/ not used, IEC101_104_Measured Value Float IEC101_104_Measured Value Scaled Measured Value Normalized	Measured Value (Normalized)	Data type specified for monitoring direction.
Monitor Tc Address	Long Integer/ 0 ... 2^{24}	-1	Monitor direction, telecontrol address. (-1 = not used)
Spontaneous Transmission	Boolean/	True	Specifies whether this value is spontaneously transmitted (after change)

3.5.4.4.3 Data Type "Accumulator Values"

IMM instance type: *Cfe Forwarded Accumulator*

For counter values to be sent to the RCC the following parameters are applicable:

Table 3-5.12: Attributes of Instance Type *Cfe Forwarded Accumulator*

Attribute	Type/ Range	Default Value	Description/ Remark
Control Tc Address	Long Integer/	-1	not used
Monitor Direction Type	Enumeration/	Counter Value	Data type specified for monitor direction. 'Counter Value' to be selected
Monitor Tc Address	Long Integer/ 0 ... 2^{24}	-1	Monitor telecontrol address. (-1 = not used)
Spontaneous Transmission	Boolean/ False, True	True	Specifies whether counter values are spontaneously transmitted (when they arrive from origin RTU)

3.5.4.4.4 Data Type "Transformer Step Positions"

IMM instance type: *Cfe Forwarded Position*

For transformer step positions to be sent to the RCC the following parameters are applicable:

Table 3-5.13: Attributes of Instance Type *Cfe Forwarded Position*

Attribute	Type/ Range	Default Value	Description/ Remark
Control Direction Type	Enumeration/ Not used, Regulation Step Command	Not used	When used: Regulation command Higher/Lower enabled
Control Tc Address	Long Integer/ 0 ... 2^{24}	-1	Control telecontrol address. (-1 = not used)
GI Relevant	Boolean/	True	If 'True' data point is sent on GI request
Invert Control	Boolean/	False	If True: Regulation command is inverted (higher gets lower and vice versa)
Monitor Direction Type	Enumeration/ Not used, Step Position	Step Position	Data type specified for monitor direction. 'Step Position' to be selected
Monitor Tc Address	Long Integer/ 0 ... 2^{24}	-1	Monitor telecontrol address. (-1 = not used)
Spontaneous Transmission	Boolean/	True	Specifies whether this data point is spontaneously transmitted (after change)

3.5.4.4.5 Data Type "Fleeting Data"

IMM instance type: *Cfe Forwarded Fleeting*

For fleeting information (events) to be sent to the RCC the following parameters are applicable:

Table 3-5.14: Attributes of Instance Type *Cfe Forwarded Fleeting*

Attribute Label	Type/ Range	Default Value	Description/ Remark
Control Tc Address	Long Integer/ 0 ... 2^{24}	-1	not used
GI Relevant	Boolean/	True	If 'True' data point is sent on GI request
Invert Info	Boolean/	False	If True: information is inverted (appearing > disappearing)
Monitor Direction Type	Enumeration/	Fleeting Point	Data type specified for monitor direction. 'Fleeting Point' to be selected
Monitor Tc Address	Long Integer/ 0 ... 2^{24}	-1	Monitor telecontrol address. (-1 = not used)
Spontaneous Transmission	Boolean/	True	Specifies whether this data point is spontaneously transmitted (after change)

3.5.4.4.6 CFE Status Information

IMM instance type: *CfeStatus*

Table 3-5.15: Attributes of Instance Type *CfeStatus*

Attribute Label	Type / Range	Description / Remark
InfoID	Enumeration / - Buffer Overflow - Device Failure - Device locked - Device process control - Device preferred	Used for CFE internal indications and commands

3.6 SMART-U-FEP

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3.6.1 Introduction

SMART-U-FEP is a PC-based data concentrator with embedded protocol converter. The equipment is able to independently communicate with associated Sub-RTUs. It preprocesses incoming data from Sub-RTUs and it has the ability to route data points from one RTU to another RTU. If the SMART-U-FEP setup is redundant, monitoring of components and failover actions are carried out by its own software. Communication with external hosts is carried out through the TCP/IP based protocol IEC60870-5-104 (short: IEC104).

SMART-U-FEP System Setup - Overview

The Smart Front-End-Processor (SMART-U-FEP) consists of the following components:

- Main Unit
- Backup Unit (optional)
- ARBITER (optional)

SMART-U-FEP can be operated in a redundant or non-redundant mode.

If the equipment is redundant, Main Unit and Backup Unit are equal devices. Both are able to play the active or the "half-active" role when communicating with CFE. The ARBITER device is only utilized if SMART-U-FEP is redundant, i.e., the Backup Unit exists. The ARBITER provides two RTUs that are serially connected to both Main Unit and Backup Unit.

Main Unit and Backup Unit may be interfaced to the redundant CFE via redundant Ethernet hubs as shown by the next figure (Figure 3-6.1)

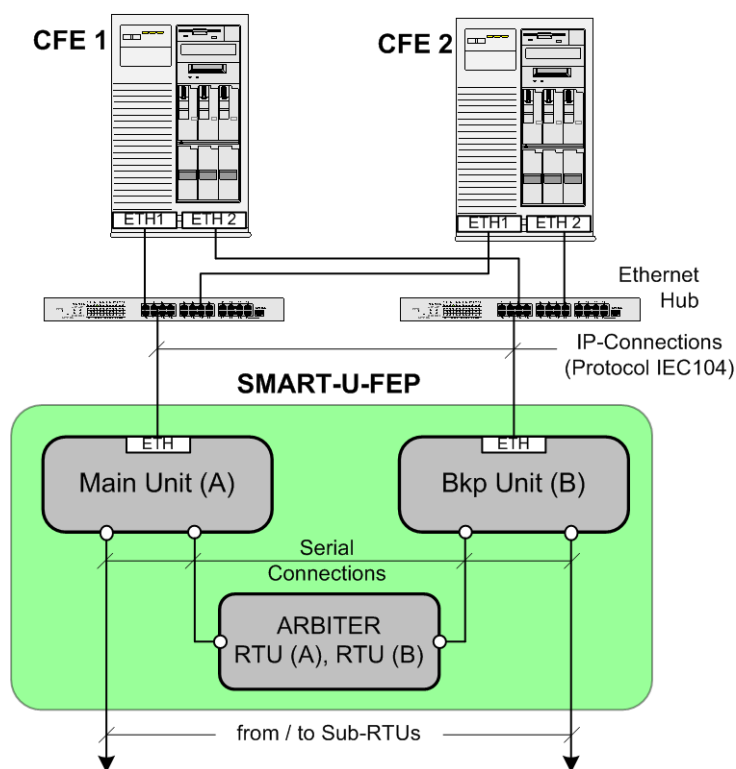


Figure 3-6.1: SMART-U-FEP System Setup - Overview

Typically, SMART-U-FEP is interfaced to Sub-RTUs of different vendors and types. For communication with SMART-U-FEP, these RTUs may use serial standard or proprietary protocols, such as IEC101, Granit, RTP80, MKT, UTK and some others. Also type IEC104 Sub-RTUs may be connected to SMART-U-FEP (through the Ethernet interface). Communication with CFE is always done through the standard protocol IEC104.

SMART-U-FEP System Setup - Details

The next figure shows a typified CFE - SMART-U-FEP setup in details. This configuration is also used as an exemplary setup referred to in this document.

- RTU1 and RTU2: type IEC101 RTUs , operated in party line mode, redundant channels
- RTU3: Dual ported RTU, type RTP80 (may consist of two single ported RTUs connected in parallel to the process), point-to-point configuration
- RTU4: Single ported type Granit RTU, point-to-point configuration, non-redundant channel

Sub-RTUs are mapped in both units into standardized data areas (RTU1 ... RTU4). From these data areas, SMART-U-FEP delivers updated RTU information to CFE. which has no direct access to any of the Sub-RTUs.

In addition to the mapped RTUs, SMART-U-FEP provides the following virtual RTUs:

- UNI RTU: Provides diagnosis information derived from communication channels
- TECHNO RTU: Provides diagnosis information from SMART-U-FEP for a local graphical interface, used by system administrators and service staff
- SERVER RTU: Provides detailed information about IP-connections to external host (CFE), used by system administrators and service staff

If the SMART-U-FEP is redundant, the ARBITER Unit is a mandatory part of the assembly. This equipment decides which unit (Main or Backup) becomes active and which one "half active". The ARBITER also provides two virtual RTUs that are connected through serial channels with Main- and Backup-Unit. Further, a "MONITOR" parameter set is to be configured for both, Main Unit and Backup Unit. The MONITOR provides settings about the use of the "Telemon.exe" - program. It can be deployed either as a "TELEMON" telecontrol server or as a "TECHNO" remote client, but nothing of its information is of interest for SCADA applications.

3.6.2 Configuring SMART-U-FEP (dBase Files)

3.6.2.1 Two Types of Configuration Data

All configuration data is stored and maintained in the central database of Spectrum PowerCC (on PSOS). For SMART-U-FEP, two types of configuration data have to be considered:

1. Configuration data for SMART-U-FEP: This type of data is used to configure SMART-U-FEP, i.e., to bring it in a position to communicate with its Sub-RTUs, preprocess and deliver data in a standardized form to CFE. Configuration data for SMART-U-FEP is always made up in dBase format. The dBase files, automatically generated by the Config-Converter, are copied by Windows File Copy function into dedicated folders of SMART-U-FEP. After resetting the SMART-U-FEP new configuration data becomes effective.
2. Configuration data for CFE: This data is needed to configure CFE in order to communicate with SMART-U-FEP and process data stemming from Sub-RTUs. It has to evaluate the IEC104 frames coming from SMART-U-FEP, identify the delivered information and pass it on to SCADA. In opposite direction, commands issued by the operator or by SCADA programs, appropriate IEC104 messages have to be prepared and sent to SMART-U-FEP. Optional CFE status information on communication issues with SMART-U-FEP , including Sub-RTUs, can be configured here. Configuration data is maintained in the DOR database and transferred to CFE applying SMMF (Shared Memory Mapped File) mechanism. Modifications of configuration data become effective as soon as the IMM job has been activated.

The complete scope of data management for both, SMART-U-FEP and CFE is presented by the next figure (Figure 3-6.3)

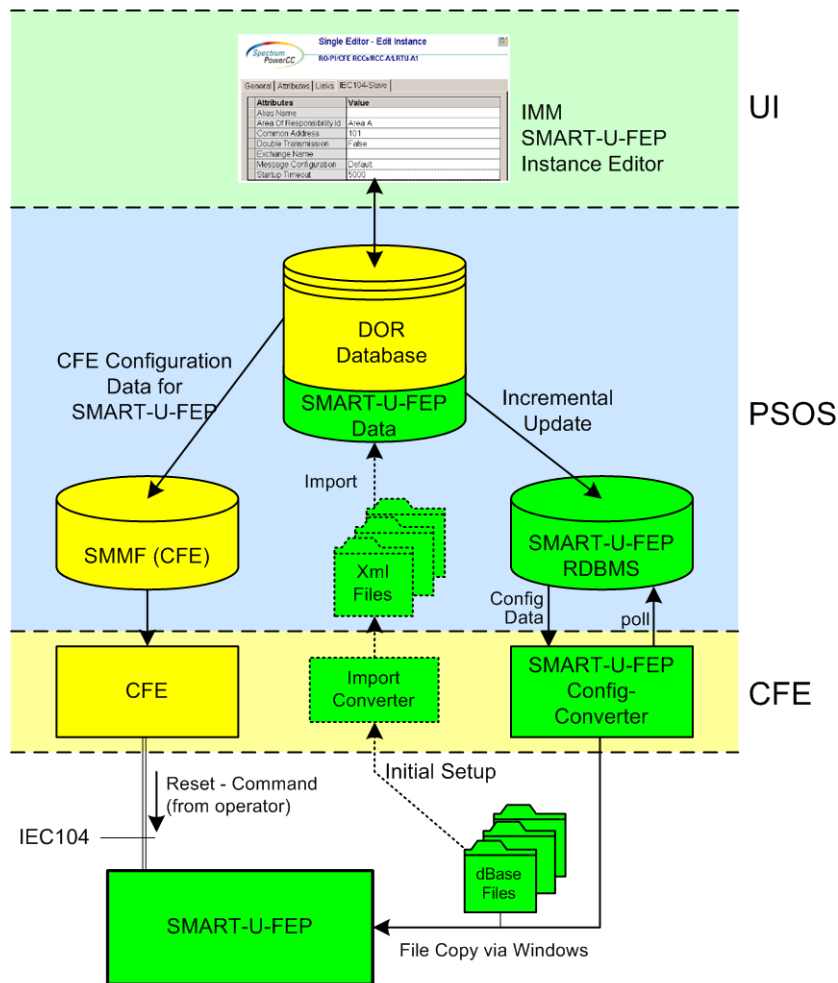


Figure 3-6.3: Configuring SMART-U-FEP by dBase Files and CFE by SMMF

3.6.2.2 SMART-U-FEP IMM Data Model - Overview

The Information Model Manager (IMM) presents configuration data in a concise and well structured arrangement. All SMART-U-FEP related data can be found in a dedicated "folder". It is a "Container Instance" (i.e., an instance without attributes) of the type "sysSmartFEPConfig".

This instance has two sub-containers of the type

SmartFEPGeneral	Contains some global parameter instances for protocols and characteristics for analog values and accumulators
SmartFEP	This is a container instance for a number of further instances, see next definition list

The instance type "SmartFEP" comprises these containers and major instances:

Main Unit Specials	Settings for main components ARBITER, SERVER, TECHNO, UNI
Backup Unit Specials	Settings for backup components ARBITER, SERVER, TECHNO, UNI
Connections	Parameters of communication devices and channels
Retransmit Data	Used for optional routing of data points from one RTU to another RTU

RTU

Settings for RTUs, protocols and data points

All major instances that are used to configure SMART-U-FEP are presented in the following simplified overview diagram (Figure 3-6.4)

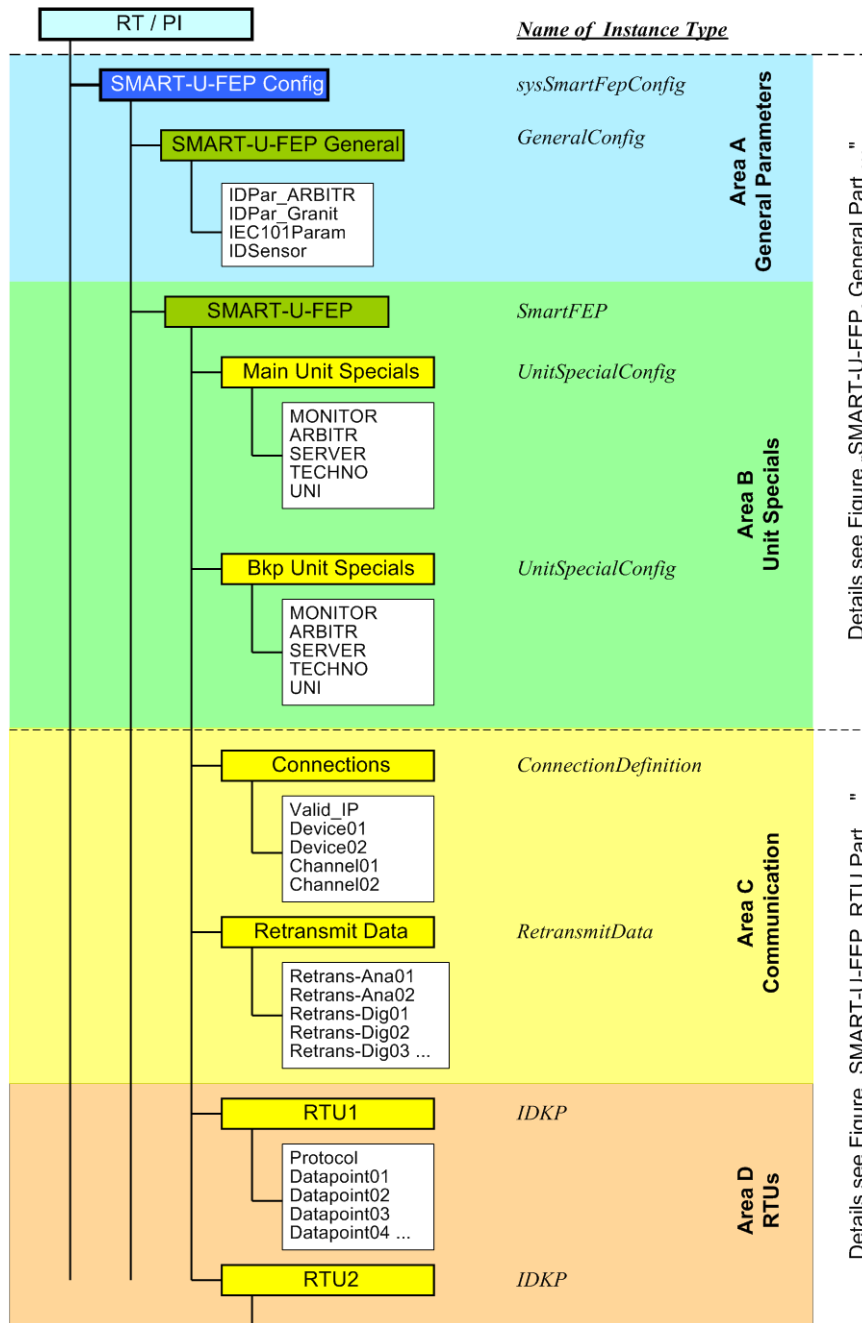


Figure 3-6.4: SMART-U-FEP: Configuration Data - Overview

Further details as well as required system links between instances are shown by the following two diagrams

- SMART-U-FEP, Details of "General Part" (Figure 3-6.5)
- SMART-U-FEP, Details of "Connections" and "RTUs" (Figure 3-6.6)

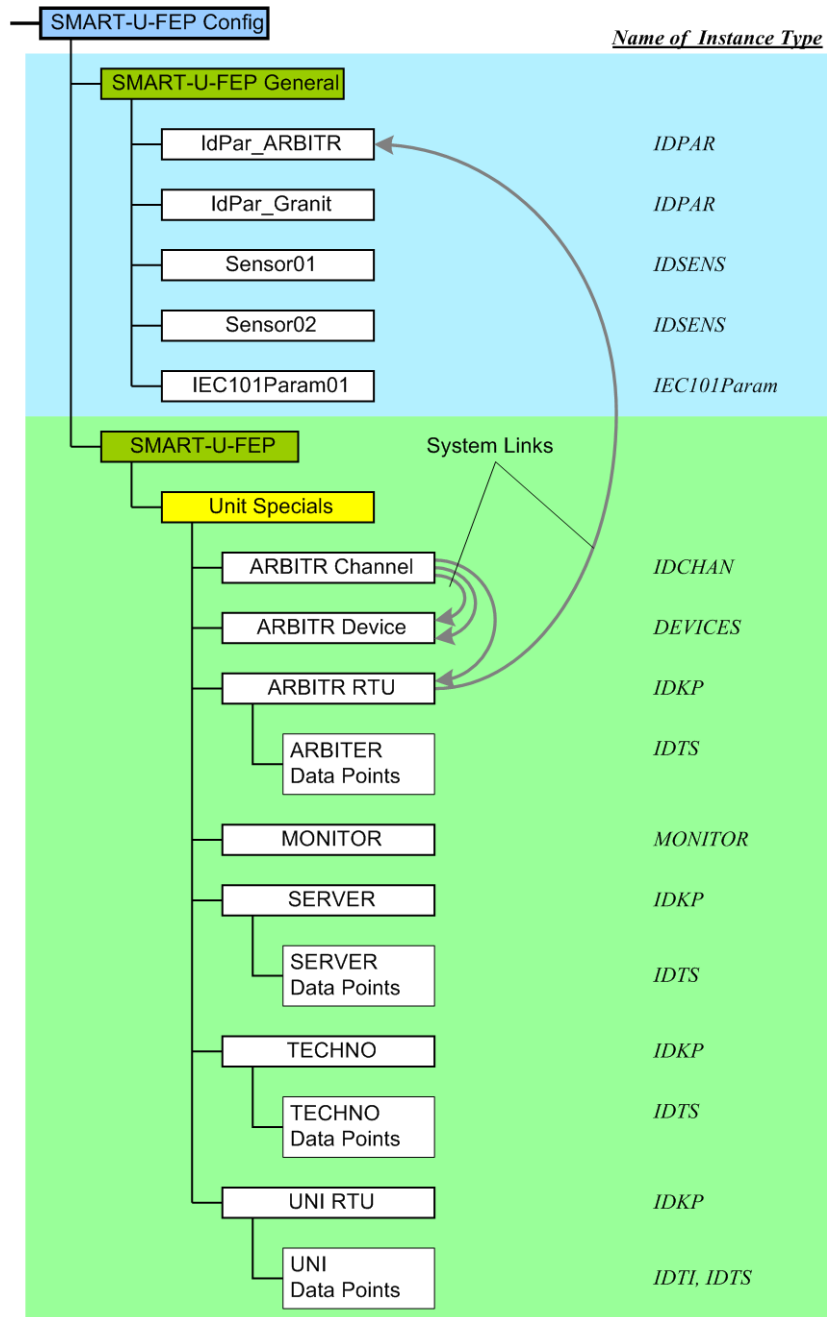


Figure 3-6.5: SMART-U-FEP, Details of "General Part"

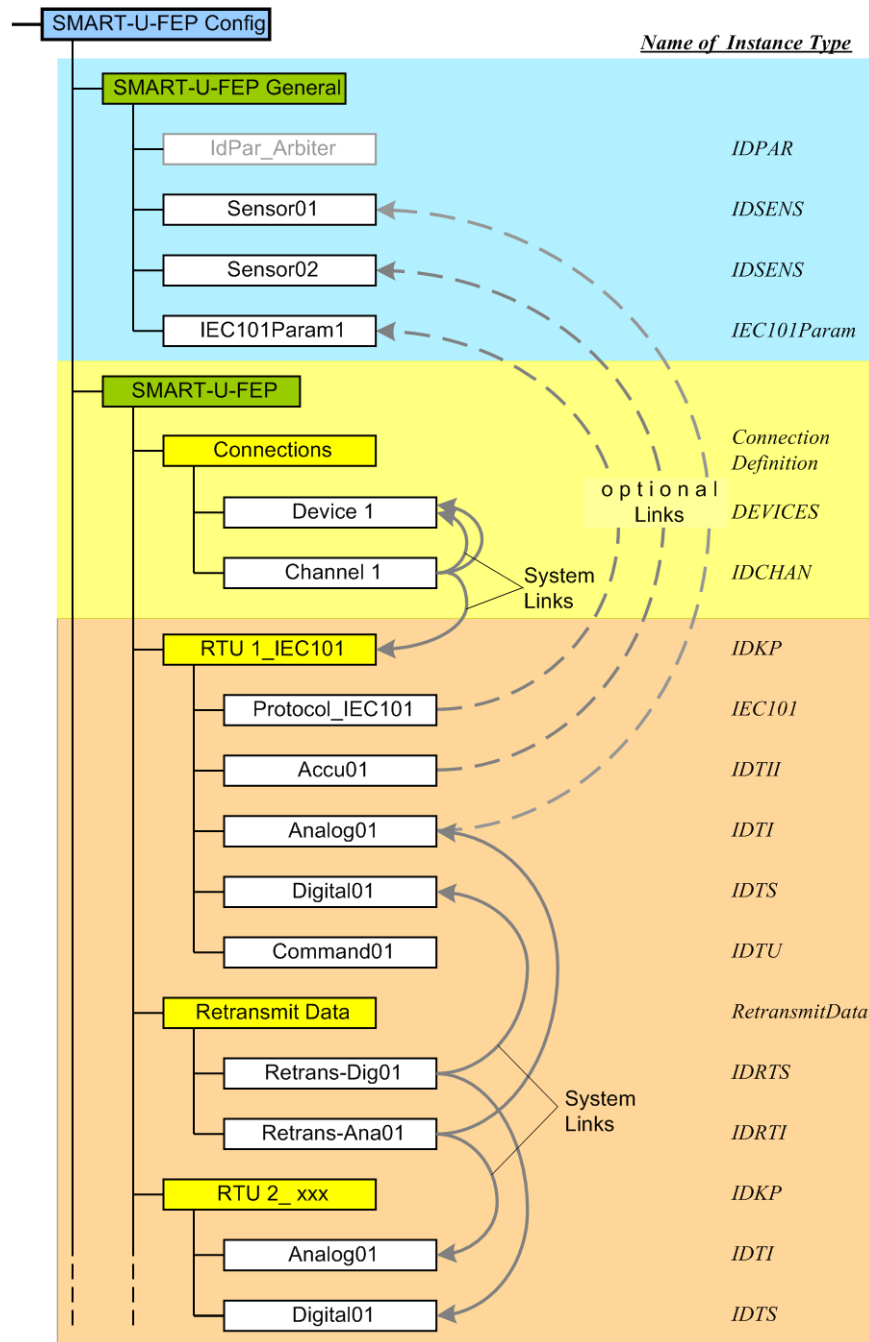


Figure 3-6.6: SMART-U-FEP, Details of "Connections" and "RTUs"

3.6.2.3 SMART-U-FEP IMM Instances, Overview

In the following section, all instances required to configure a SMART-U-FEP are summarized. This might be helpful for starting from scratch as well as for extending or modifying an existing data model.



Note

Each instance features a specific instance type. In addition, each instance has got a name that is unique within child instances of the same parent instance. The name of the instance can freely be given and changed by the user, whereas the type is fixed. To identify an instance, even at deviant name, also the instance type is indicated in the following tables. Instance type can be obtained from IMM Instance Editor, tab "General" when selecting the instance name in the Overview Tree.

3.6.2.3.1 First Hierarchical Level - Main Container for SMART-U-FEP

The first hierarchical level is built by an instance of the type *sysSmartFEPConfig*. All configuration data for SMART-U-FEP are concentrated in this container instance. In the following tables, container instances are presented in bold.

Parent Instance Type	Parent Instance Name	SMART-U-FEP Instance Type	Name as per Sample Setup
sysProcessInterface	PI	sysSmartFEPConfig	SMART-U-FEP Config

3.6.2.3.2 Instances of the Second Hierarchical Level

In the second hierarchical level there are only two containers:

Parent Instance Type	Parent Instance Name	Available Child Instance Types	Name as per Sample Setup
sysSmartFEPConfig	SMART-U-FEP Config	SmartFEPGeneral	SMART-U-FEP General
		SmartFEP	SMART-U-FEP

3.6.2.3.3 Instances of the Third Hierarchical Level

In the third hierarchical level there are the following instances:

Parent Instance Type	Parent Instance Name	Available Child Instance Types	Instance Name as per Sample Setup	Links	Attribute Details see ...
SmartFEPGeneral	SMART-U-FEP General	IDPAR	IDPar_ARBITER	-	Table 3-6.1
		IEC101Param	IEC101 Parameter	-	Table 3-6.2
		IDSENS	Sensor	-	Table 3-6.3

Parent Instance Type	Parent Instance Name	Available Child Instance Types	Instance Name as per Sample Setup	Links	Attribute Details see ...
SmartFEP	SMART-U-FEP	UnitSpecials	Main Unit Specials	-	-
			Backup Unit Specials	-	-
		ConnectionDefinitions	Connections	-	-
		IDKP	RTU1 ... RTU4	yes	Table 3-6.5
		RetransmitData	Retransmit Data	-	-

3.6.2.3.4 Instances of the Forth Hierarchical Level

Parent Instance Type	Name of the Parent Instance	Available Child Instance types	Instance Name as per Sample Setup	Links	Attribute Details see ...
UnitSpecials	Main Unit Specials /	MONITOR	MONITOR	-	Table 3-6.4
	Backup Unit Specials	IDKP	ARBITER RTU	yes	Table 3-6.5
			SERVER RTU	yes	
			TECHNO RTU	yes	
			UNI RTU	yes	
ConnectionDefinition	Connections	DEVICES	Device ...	-	Table 3-6.7
		KIO_UTM	-	yes	Table 3-6.8
		IDCHAN	Channel ...	yes	Table 3-6.9
		VALIDIP	IP-Connection ...	-	Table 3-6.6
IDKP	RTU1 ... RTU4	GRANITAFB	Prot_GRANIT	yes	Table 3-6.10
		IEC101	Prot_IEC101	yes	Table 3-6.12
		IEC104	-	yes	Table 3-6.11
		IDMKT	-	yes	Table 3-6.13
		KIO	-	yes	Table 3-6.14
		RPT80	Prot_RPT80	yes	Table 3-6.15
		TM800	-	yes	Table 3-6.16
		UTM7INP	-	yes	Table 3-6.17
IDKP	RTU1 ... RTU4 ARBITER RTU, UNI RTU	IDTII	Accu	yes	Table 3-6.18
		IDTI	Analog	yes	Table 3-6.19
		IDTS	Digital	-	Table 3-6.20
		IDTU	Command	-	Table 3-6.21
RetransmitData	Retransmit Data	IDRTI	Retrans_Analog	yes	Table 3-6.22
		IDRTS	Retrans_Digital	yes	

3.6.2.4 SMART-U-FEP Attribute Tables

In the following section all SMART-U-FEP instance tables are itemized. You will find SMART-U-FEP related Attributes, their data type and meaning.

Additional information and more details about SMART-U-FEP parameters can be obtained from document "CONFIGURATION AND SOFTWARE OF CSRS "SMART-U-FEP" issued by RTSoft.

Table 3-6.1: Attributes of Instance Type IDPAR

Additional Parameters of Granit Protocol Note: The parameters of this table, starting from P0 to P6, refer only to the Granit-Micro modification. All the other parameters are used by all modifications of the GRANIT protocol.		
Attribute (Label)	Type	Meaning / Remark
NAME	String	Optional name
P0	Long Integer	0 – the KP number has the size of 1 byte 1 – the KP number has the size of 2 bytes
P1		0 – the opening flag of the frame has the size of 1 byte 1 – the opening flag of the frame has the size of 2 bytes
P2		0 – frames with no time marker 1 – frames with a time marker
P3		0 – 8 bit TIs are used 1 – 12 bit TIs are used
P4		0 – the frame is not broken into multiple packets 1 – the frame may be broken into multiple packets (used only for 12 bit TIs)
P5		0 – the frame does not contain any fields with information about the transmission wait time 1 – the frame contains fields with information about the transmission wait time
P6		0 – 64 TSs in the group 1 – 32 TSs in the group
P7		Operation mode: 0 - master 1 - slave
P8		0 - the Granit protocol driver sends requests and information packets without waiting for a response/receipt 1 – the Granit protocol driver waits 3 seconds for KP response to the request or packet receipt, and is silent on the line
P9		0 – the Granit protocol driver continually retransmits TIs and TSs, in a cycle 1- the Granit protocol driver retransmits TIs and TSs only after it receives a request from a connected device. When the request is expected, data is transmitted sporadically
T0		The operation mode of the backup channel 0 - data reception and transmission 1 – complete silence 2 – only meanders are exchanged
T1		The length of the interframe gap is the tenth parts of the previous packet 0 – the interframe gap is constant and equal to 3 bytes N – the length of the interframe gap
T2		0 – TS numbering in the byte starting with the most significant bit 1 – TS numbering in the byte starting with the least significant bit
T3		Supporting the frames of the additional set "KVM-micro" (only for the GRANIT_EKAT modification) 0 - disabled 1- enabled

Table 3-6.2: Attributes of Instance Type IEC101Param

Additional Parameters of Protocol IEC101		
Attribute (Label)	Type	Meaning / Remark
DIR	Boolean	Direction Bit (True: Bit set)
INTERROG	Long Integer	Priority of data transmission at the General Interrogation Request
NOT_100	Boolean	Attribute for ASDU100 sending a (General Interrogation Request) False: ASDU100 is sent when the channel is initialized, any further sendings are defined by the KP configuration parameters. True: ASDU100 is not sent either during channel initialization, or in any other event, irrespective of the other configuration parameters
SPORAD_TI	Long Integer	Priority of the transmission of changed TI values
SPORAD_TS	Long Integer	Priority of the transmission of changed TS values
TI_SEQ	Enumeration	Value 0: In response to a station interrogation TI values are transmitted by ASDUs with SQ = 0; Value 1: In response to a station interrogation TI values are transmitted by ASDUs with SQ = 1.
TS_SEQ	Enumeration	Value 0: In response to a station interrogation TS values are transmitted by ASDUs with SQ = 0; Value 1: In response to a station interrogation TS values are transmitted by ASDUs with SQ = 1.
WAIT_70	Boolean	Attribute for KP initialization end wait. True: initialization end from the KP (ASDU70) is waited for, then the General Interrogation Request is sent False: General Interrogation Request is sent after connection is set up (after the initialization procedure is completed at the channel level)

Table 3-6.3: Attributes of Instance Type IDSSENS

Characteristic for Analog Values, Accumulators		
Attribute (Label)	Type	Meaning / Remark
A	Long Integer	Calibration factor A
B	Long Integer	Calibration factor B
NAME_SENS	String	Sensor Name (optional)
TIP_TX	Long Integer	The type of sensor property (0, 1, 2, 3) defines the formula of converting the code value of TI or TII to a physical value.

Table 3-6.4: Attributes of Instance Type MONITOR

FEP Unit Special, Type "MONITOR" and "TECHNO"		
Attribute (Label)	Type	Meaning / Remark
dBaseDIRECTORY	String	Location of the dBase files
TECHNO_IPADDRESS	String	For server type "TECHNO": IP address
TECHNO_PORT	Long Integer	For server type "TECHNO": IP port, preferably 2404

FEP Unit Special, Type "MONITOR" and "TECHNO"		
Attribute (Label)	Type	Meaning / Remark
TELEMON_FEP_NUMBER	Long Integer	For server type "MONITOR": The number of TM-RSD (range of values: 0 – 255). It is used for addressing data in the IEC-870-5-104 protocol. Each TM-SRD within this system must have a unique FEP_NUMBER
TELEMON_NOOF-CON	Long Integer	For server type "MONITOR": Number of connections
TELEMON_NOT_WIN	Boolean	For server type "MONITOR": True: disable graphical display False: enable graphical display
TELEMON_OIKTIME-OUT	Long Integer	For server type "MONITOR": Timeout (milliseconds) for the dropout of applied information from RTIS, after which the own retransmission SMART-U-FEP activates for signals with the enabled feature P_OIK in the retransmission tables IDRTS and IDRTI.
TELEMON_PORT	Long Integer	For server type "MONITOR": IP port, preferably 2404
TELEMON_P_WDOG	Boolean	For server type "MONITOR": The Identifier shows whether the hardware WatchDog of SRD SMART-U-FEP is allowed. The response time is 20 seconds. True: Watchdog is enabled False: Watchdog is disabled

Table 3-6.5: Attributes of Instance Type IDKP

KP / Remote Terminal Unit (RTU, Sub-RTU)		
Attribute (Label)	Type	Meaning / Remark
KP_RQST	Long Integer	Interval (milliseconds) to define KP poll
KP_TYPE	Enumeration	KP type (ARBITER, SERVER, TECHNO, UNI ...)
LOC_KP	Long Integer	Local (physical) number of a KP. It is used when trunk connection of multiple KPs in the same direction is arranged. This number is installed on a KP and transmitted in the telecontrol information frame to identify this KP
L_GR_TI	Long Integer	Number of TIs transmitted in the same group
L_GR_TII	Long Integer	Number of TIIs transmitted in the same group
L_GR_TS	Long Integer	Number of TSs transmitted in the same group
L_GR_TU	Long Integer	Number of TUs transmitted in the same group
NAME_KP	String	The dispatcher name of a KP
N_FILE	Long Integer	Overall number of files on a KP
N_GR_TS	Long Integer	Number of TS groups on a KP
N_GR_TU	Long Integer	Number of TU groups on a KP
N_TI	Long Integer	Overall number of TIs on a KP
N_TII	Long Integer	Overall number of TIIs on a KP
P_OTKL	Boolean	The feature shows whether the current record is not used, if it is 'True' this record is not used.
P_TR_TU	Boolean	The feature shows whether the TU command address has to be translated
RES_KP	Long Integer	The number of the backup logical KP. If there is no backup KP, the field is 0. It is used to display data over the backup channel in the Telecontrol Operator Workstation.

KP / Remote Terminal Unit (RTU, Sub-RTU)		
Attribute (Label)	Type	Meaning / Remark
RES_LOC_KP	Long Integer	Local (physical) number of a KP on the backup channel (optional field). It is filled, if the local address of a KP on the backup channel is different from the address on the primary channel. If both addresses are the same, nulls may be inserted in this field.
SYS_KP	Long Integer	The system (logical) numbers of a KP – irrespective of the protocol type, all KPs in the complex are numbered in the user-defined order. The system number may assume values from 1 to Nmax, where Nmax is the highest number defined during the system generation. The system number of a KP is used when multiple parameters of the system (TI, TS, TII, TU) are described. The system number of a KP is used when an ASDU address is formed in IEC 870-5-104.
TII_RQST	Long Integer	Interval (milliseconds) to periodically poll a TII, if it is 0, the TII is not periodically polled
TIMEOUT	Long Integer	Interval (milliseconds) to define KP dropout
TI_RQST	Long Integer	Interval (milliseconds) to periodically poll a TI, if it is 0, the TI is not periodically polled
TM_TYPE	Enum	TC protocol (COMPAS, GRANIT, IEC101, IEC104, KIO, KIT, MKT, TM800, UTK, UTM ...)
TS_RQST	Long Integer	Interval (milliseconds) to periodically poll a TS, if it is 0, the TS is not periodically polled
TU_TIMEOUT	Long Integer	Interval (milliseconds) required to wait for receipts, responses to polls and TU commands

Table 3-6.6: Attributes of Instance Type VALID IP

IEC104 Communication Links to CFE		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional text
P_OTKL	Boolean	True: disable (block) this record
VALID_IP	String	The IP address of the remote LAN computer (CFE) allowed connecting to this semi-complex TM SRD from the CSRS "SMART-U-FEP"

Table 3-6.7: Attributes of Instance Type DEVICES

Serial Communication Interfaces (COM, ISA)		
Attribute (Label)	Type	Meaning / Remark
BAUDRATE	Enum	Transmission speed over the serial communication channel (Baud) for SOCKET_104: 0 (the value is ignored); for DS18S20 (Thermometer): 9600
CHAN_TYPE	Enum	Channel type. Possible values: Duplex, Half-duplex, Simplex. For devices supported as of now, the value of the field is Duplex
DATA	Long Integer	Number of data bits
DEV_NAME	Enum	Logical name of the device defining the type. Options: COM, FILE_EMUL, FILE_OUT, ADAPTER_ISA, ASYNC_ISA, SOCKET_104, DS18S20.
EMULFILE	String	For FILE_EMUL, FILE_OUT devices – the name of the emulation file, e.g., Mkt3_1.bin
IPADDRESS	String	For SOCKET_104 devices - the IP address string, for example "169.254.203.3"
IPPORT	Long Integer	For SOCKET_104 devices - the IP port. Typical for IEC104 is port # 2404
N_DEV	Long Integer	Device Number
PARITY	Enum	The parity field may be assigned to one of the three values: E (even parity check), O (odd parity check) or N (no check)
PH_CHAN	Long Integer	The physical number of the channel in the adapter or the number of the COM port. For files and SOCKET_104 it is 0
P_ASYNC	Boolean	The attribute shows whether the device is an asynchronous type. For the time being, developers use it. By default P_ASYNC is 'False'
P_INV_INP	Boolean	The attribute shows whether the signal has to be inverted at reception (connection polarity change) default: False
P_INV_OUT	Boolean	The attribute shows whether the signal has to be inverted at transmission (connection polarity change) default: False
P_REF_INP	Boolean	The attribute shows whether the signal has to be reversed at reception (changing the sequence of bits in bytes to reverse). True: Always True for ADAPTER_ISA False: For other types
P_REV_OUT	Boolean	The attribute shows whether the signal has to be reversed at transmission (changing the sequence of bits in bytes to reverse). True: Always True for ADAPTER_ISA False: For other types
STOPBIT	Enum	For the stop field (the number of stop bits) the allowed values are 1 or 2 For DS18S20 (Thermometer) stop=1;

Table 3-6.8: Attributes of Instance Type KIO_UTM

Configuration Table for Data Exchange Using the RPT-PC Protocol		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional field for comment or description
N_UTM	Long Integer	The number of the RPT-PC direction

Table 3-6.9: Attributes of Instance Type IDCHAN

RTU Communication Channel		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional field for comment or description
N_CHAN	Long Integer	Number of the logical data channel
P_ONE_OUT	Boolean	This attribute shows whether retransmission is active only in one primary channel of the pair : PRIMARY-BACKUP. (optional field, the default value is 'False')
TIMEOUT	Long Integer	Interval (milliseconds) to define channel timeout
TM_TYPE	Enum	"Tele Monitor" type, i.e., Name of TC protocol. (Implemented: RPT80, UTM7, MKT3, TM800A, 512, GRANIT, TK113, SM01)

Table 3-6.10: Attributes of Instance Type GRANITAFB

Protocol Type Granit		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Additional data description field (optional)
INP_OUT	Enum	Direction feature: 1: Received data 0: Retransmitted data
N_AFB	Long Integer	Value of the least significant byte in AFB number

Table 3-6.11: Attributes of Instance Type IEC101

Protocol Type IEC101		
Attribute (Label)	Type	Meaning / Remark
ADDR_TI	Long Integer	Starting address of the used range of TI addresses in the maintained equipment (reception/transmission in the communication channel)
ADDR_TII	Long Integer	Starting address of the used range of TII addresses in the maintained equipment (reception/transmission in the communication channel)
ADDR_TS	Long Integer	Starting address of the used range of TS addresses in the maintained equipment (reception/transmission in the communication channel)
ADDR_TU	Long Integer	Starting address of the used range of TU addresses in the maintained equipment (reception/transmission in the communication channel)
COMMENT	String	Additional data description field (optional)
LNK_ADDR	Long Integer	Value of the channel address, link address (IEC-870-5-101 standard)
LNK_SIZE	Enum	Length (bytes) of the channel address: 1: 1 byte 2: 2 bytes
L_ADDR_DUI	Enum	Length (bytes) of the ASDU shared address: 1: 1 byte 2: 2 bytes

Protocol Type IEC101		
Attribute (Label)	Type	Meaning / Remark
L_IOI	Enum	Length (bytes) of the Information Object Address (IOI): 1: 1 byte 2: 2 bytes 3: 3 bytes
L_SECTION	Long Integer	Information section length (maximum is 255)
L_TI	Long Integer	Maximum number of TIs used in the maintained equipment
L_TII	Long Integer	Maximum number of TIIs used in the maintained equipment
L_TS	Long Integer	Maximum number of TSs used in the maintained equipment
L_TU	Long Integer	Maximum number of TUs used in the maintained equipment
P_ORIGIN	Enum	Length (bytes) of the transmission reason: 0: 1 byte 1: 2 bytes
P_SECTION	Boolean	Attribute specifies whether or not information sections are used: False: not used True: used
TIME_PROC	Long Integer	Interval (milliseconds) – equipment response time
TIME_SYNC	Long Integer	Interval (milliseconds) that defines the timing intervals. If 0, no timing is used

Table 3-6.12: Attributes of Instance Type IEC104

Protocol Type IEC104		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional text
EX_ADDR		Not used in current implementation
NOT_100	Boolean	The attribute that disables sending ASDU100 (General Interrogation Request) False: (default value) send ASDU100 at the period specified in the configuration for this system KP True: do not send ASDU 100, regardless of the other configuration parameters
N_IS_TIME	Long Integer	Not used in current implementation
P_EX_ADDR	Boolean	Not used in current implementation
STD_ADDR	Boolean	False: Retransmit address is equal to end-to-end address True: Retransmit address is equal 0x1000 + TS end-to-end address, 0x2000 + TI end-to-end address, 0x3000 + TII end-to-end address, 0x4000 + TU end-to-end address
TIME_SYNC	Boolean	Period (seconds) of clock synchronization command. Value 0: no clock synchronization command
TOTAL_SEND	Boolean	Transmitting the entire volume of data set described for this system KP. The entire volume of data set is transmitted in cycle with the period specified in the KP_RQST field in the ID_KP.dbf table. False: Default value, do not transmit the entire volume of data True: Transmit the entire volume of data

Table 3-6.13: Attributes of Instance Type IDMKT

Protocol Type MKT		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional field for comment or description
INVERT	Boolean	The Identifier shows whether data (TI or TS group) is inverted
N_MKT	Long Integer	The sequence number of TI or TS group in the frame of the MKT1 or MKT2 protocol. Numbering starts from 1
REVERS	Boolean	The attribute shows whether data (TI or TS group) is reversed (changing the sequence order of bit in a byte): False: Value of receiving byte (TS or TI group) is not reversed, True: Value of receiving byte (TS or TI group) is reversed. Usage only in the MKT-1 protocol

Table 3-6.14: Attributes of Instance Type KIO

Protocol Type KIO. The instance defines forwarding for the RPT-PC protocol		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional field for comment or description
INP_OUT	Long Integer	Direction feature: 1: Received data 0: Retransmitted data
N_KIO	String	The hexadecimal address of TI, or of TS group in the RPT-PC protocol channel

Table 3-6.15: Attributes of Instance Type RPT80

Protocol Type RTP80		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional field for comment or description
INP_OUT	Long Integer	Direction feature: 1: Received data 0: Retransmitted data
N_RPT	Long Integer	Address accompanying TI or TS group in the AIST protocol channel (RPT80-RPT80).

Table 3-6.16: Attributes of Instance Type TM800

Protocol Type TM800		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional field for comment or description

Protocol Type TM800		
Attribute (Label)	Type	Meaning / Remark
TI_FILTER	Long Integer	Identifier for switching on the TI filter mode. Switching mode of checking of the feasible changing. The changes up to $\pm 25\%$ are possible. That is to say that if TI value changes are no more 25% then they transmitted in the in the SmartFep kernel. In the case of exceeding the TI threshold, the change of the TI value will not be transmitted in the SF kernel: 0: mode off 1: mode on
TS_FILTER	Long Integer	Identifier for switching on the TS filter mode. The mode of majority voting "two of three" is switching on. The last three received TS are analyzed and in the case of changing TS it is sent in SmartFEP kernel: The values are the next if two last three TS are 0 – 0, if they are 1 – 1: 0: mode off 1: mode on
VOID_BITS	Long Integer	Identifier of indication the switching on the special receiving mode. It is used, when new sub-loop starts with new byte. Not used bits of the last byte of the previous sub-loop are ignored, and interruption at marker receiving is not raised: 0: mode off 1: mode on

Table 3-6.17: Attributes of Instance Type UTM7INP

Protocol Type UTM7INP		
Attribute (Label)	Type	Meaning / Remark
COMMENT	String	Optional field for comment or description
N_UTM7	Long Integer	The sequence number of a TI or TS group in the UTM7 frame. Numbered from 1 to 7. The numbers are assigned at the UTM7 device

Table 3-6.18: Attributes of Instance Type IDTII

Accumulator Data Points		
Attribute (Label)	Type	Meaning / Remark
EX_ADDR	Long Integer	Not used in current implementation, value set to "0"
IEC104Address	Long Integer	IP address of this data point when transmitted via IEC104 link to CFE
Name	String	Optional field for comment or description
N_CH	Long Integer	The sequence number of a TII within a group, it may assume values from 1 to L_GR_TII, where L_GR_TII is a IDKP table value corresponding to this KP
N_GR	Long Integer	the sequence number of the TII group, it may assume values from 1 to N_TII/L_GR_TII, where N_TII, L_GR_TII are IDKP table values corresponding to this KP
P_EX_ADDR	Boolean	Not used in current implementation, value set to 'False'
P_OTKL	Boolean	The attribute shows whether the current record is not used, if it is 'True', this record is not used

Table 3-6.19: Attributes of Instance Type IDTI

Analog Data Points		
Attribute (Label)	Type	Meaning / Remark
APERT	Long Integer	The aperture value in quanta. The new value of the parameter will be sent to transmission if in comparison with the previous value the value changed by the value that is greater than the aperture (APERT value). Aperture filtering is enabled by the P_APERT key
EX_ADDR	Long Integer	Not used in current implementation, value set to "0"
HIGHLIMIT	Long Integer	The higher limit value that triggers data transmission even if the aperture is not exceeded
IEC104Address	Long Integer	Object address of this data point when transmitted via IEC104 link to CFE
LOWLIMIT	Long Integer	The lower limit value that triggers data transmission even if the aperture is not exceeded
NAME	String	Optional field for comment or description
N_CH	Long Integer	The sequence number of a TI within a group, it may assume values from 1 to L_GR_TI, where L_GR_TI is a IDKP table value corresponding to this KP
N_GR	Long Integer	the sequence number of the TI group, it may assume values from 1 to N_TI/L_GR_TI, where N_TI, L_GR_TI are IDKP table values corresponding to this KP
P_APERT	Boolean	Attribute indicates whether aperture filtering is on. Default is 'False'
P_EX_ADDR	Boolean	Not used in current implementation, value set to 'False'
P_FLOAT	Boolean	Sets the data output format. If P_FLOAT is 'True', data will be output to RTIS, and for internal retransmission in the short float format. If P_FLOAT is 'False', data will be output in codes without conversion. P_FLOAT = 'True' has priority over P_SCALE = 'True'
P_HIGH	Boolean	Attribute indicates whether the higher limit is used
P_LOW	Boolean	Attribute indicates whether the lower limit is used
P_OTKL	Boolean	Attribute indicates whether the current record is not used, if it is 'True', this record is not used
P_OUT	Boolean	An additional feature defining whether this signal is a received one (P_OUT = False) or a retransmitted one (P_OUT = True). The P_OUT value is used to organize signal filtering when viewing in the telecontrol operator workstation windows is performed
P_SCALE	Boolean	Sets the data output format. If P_SCALE = 'True', data will be outputted to RTIS and for internal retransmission as an integer value that has been scaled using a corresponding formula from the IDSENS table (SVA format). If P_SCALE = 'False', data will be outputted in codes without conversion. (Also see note below).
P_SMOOTH	Boolean	The feature shows whether evening-out (averaging) is used
SMOOTH	Long Integer	Integer parameter that shows on which of the previous values evening out (averaging) of this TI should be based

Table 3-6.20: Attributes of Instance Type IDTS

Digital Data Points		
Attribute (Label)	Type	Meaning / Remark
EX_ADDR	Long Integer	Not used, must be 0
IEC104Address	Long Integer	Object address of this data point when transmitted via IEC104 link to CFE
Name	String	Optional field for comment or description
N_CH	Long Integer	Sequence number of a TS within a group
N_GR	Long Integer	Sequence number of a TS group
P_EX_ADDR	Boolean	Not used in current implementation, value set to 'False'
P_INV	Boolean	Attribute indicates whether this signal is inverted
P_OTKL	Boolean	Attribute indicates whether the current record is not used, if it is 'True', this record is not used
P_OUT	Boolean	An additional feature defining whether this signal is a received one (P_OUT= 'False') or a retransmitted one (P_OUT= 'True'). The P_OUT value is used to organize signal filtering when viewing in the telecontrol operator workstation windows is performed

Table 3-6.21: Attributes of Instance Type IDTU

Digital Commands		
Attribute (Label)	Type	Meaning / Remark
EX_ADDR	Long Integer	Not used in current implementation, value set to "0"
IEC104Address	Long Integer	IP address of this data point when transmitted via IEC104 link to CFE
NAME	STRING	Optional field for comment or description
N_CH	Long Integer	Sequence number of a TU within a group
N_GR	Long Integer	Group number in the TU command
N_GR_TS	Long Integer	Not used. Must be equal to 0
N_TU_TS	Long Integer	Not used. Must be equal to 0
P_DIRECT	Boolean	Not used. Must be set to 'False'
P_DOUBLE	Boolean	Not used. Must be set to 'False'
P_EMUL	Boolean	Not used. Must be set to 'False'
P_EX_ADDR	Boolean	Not used. Must be set to 'False'
P_LONG	Boolean	Not used. Must be set to 'False'
P_OTKL	Boolean	Attribute indicates whether the current record is not used, if it is 'True', this record is not used
P_PERSTST	Boolean	Not used. Must be set to 'False'
P_SHORT	Boolean	Not used. Must be set to 'False'

Table 3-6.22: Attributes of Instance Types IDRTI, IDRTS

Retransmitted Data Points (Analog, Digital)		
Attribute (Label)	Type	Meaning / Remark
NAME	String	Optional field for comment or description
P_OIK	Boolean	The feature is enabled (True) for the signals that can be retransmitted both from RTIS and via an internal retransmission mechanism. Retransmission from RTIS has higher priority. If a channel with RTIS drops out, or data non-arrival timeout was triggered (the OIKTIMEOUT field in the MONITOR.DBF field), internal retransmission is activated. If 'True' for signals transmitted from RTIS, scaling is not enabled, but for internal retransmission it is on.
P_OTKL	Boolean	Attribute indicates whether the current record is not used, if it is 'True', this record is not used

3.6.2.5 SMART-U-FEP Parameter Table

The following table indicates allowed parameters for the implemented types of devices and protocols

Table 3-6.23: Allowed Parameters

Device / Protocol	KP_TYPE	TM_TYPE	N_TI ¹	L_GR_TI	N_GR_TS	L_GR_TS ¹	Comment
Power system frequency meters	NORMAL	FREQ	1	1	0	1	Frequency meter IVCh-1
	VER1	FREQ	1	1	0	1	Frequency meter IVCh-1\Sch
	VER2	FREQ	10	10	0	1	Frequency meter IVCh-1/SP/TV
	NORMAL	FREQCD	1		0	1	Frequency meter (Vitebsk)
Precise time clock	NORMAL	PCHZ4	7	7	0	0	Time correction by radio signals
Thermometer DS18S20	NORMAL	DS18S	1	1	0	0	
PLC DL-405	NORMAL	DL405	125	125	8	8	
RPT80 (AIST) (RPT-RPT)	NORMAL	RPT80	496 ²	496	496 ²	8	Reception and retransmission
	RETR		248 ³	248	248 ³	8	Only retransmission
	USOTM		496 ²	496	496 ²	8	USOTM – RPT modification
RPT-PC	NORMAL	KIO	< ... > ⁴	< ... >	< ... >	8	RostovEnerg
	P_NPST		< ... >	< ... >	< ... >	8	RostovEnerg (Board)
	VER1		< ... >	< ... >	< ... >	8	Lipetsk RDU (Board)
	VER2		< ... >	< ... >	< ... >	8	Ivanovo RDU (Board)
	VER3		< ... >	< ... >	< ... >	8	Omsk RDU (Board)
KP UTM-7	NORMAL	UTM7	7	7	7	4	
P TM512	NORMAL	TM512	64	8	64	8	Reception
	RETR		64	8	64	8	Retransmission
P TM800A	NORMAL	TM800A	128	8	64	8	Reception
	RETR		128	8	64	8	Retransmission
SPIDER ABB	NORMAL	RP570	250	250	10	16	SPIDER ABB protocol

Device / Protocol	KP_TYPE	TM_TYPE	N_TI ¹	L_GR_TI	N_GR_TS	L_GR_TS ¹	Comment
KP GRANIT, devices over GRANIT protocol	NORMAL	GRANIT	256	16	16	64	Classic GRANIT (Zhitomir), GRANIT – Systel, USD05 (SYSTEL)
	RETR		256	32	16	64	Retransmission Classic GRANIT
	EKAT		256	32	16	64	Ekaterinburg-version Granit (bound to AFB)
	RETR_AFB		256	32	16	64	Retransmission bound to AFB
	MICRO		256	32	16	32	Reception Granit-MICRO protocol (Granit expanded)
	RETR_MICRO		256	32	16	32	Retransmission Granit-MICRO protocol
	MAP		256	16	16	64	Retransmission of TI and TS to BOARD
	P_ASI		256	16	0	0	Retransmission of TI to BOARD (KTsOTI SYSTEL)
KP TK113	NORMAL	TK113	120	6	16	16	Sometimes called TK-112
KP KONUS	NORMAL	KONUS	128	8	16	14	KP KONUS
Devices operating over INTERLINK protocol	HDLC	INTERLINK	< ... >	< ... >	1	< ... >	Reception Synchronous option of INTERLINK protocol
	RETR_HDLC		< ... >	< ... >	1	< ... >	Transmission Synchronous option of INTERLINK protocol
	NORMAL		< ... >	< ... >	1	< ... >	Reception Asynchronous option of INTERLINK protocol
	RETR		< ... >	< ... >	1	< ... >	Transmission Asynchronous option of INTERLINK protocol
KP UTK-1	NORMAL	UTK1	1	1	1	4	Reception
	RETR		1	1	1	4	Retransmission
KP MKT-1A	NORMAL	MKT1	5	5	5	7	Reception
KP MKT-1B	NORMAL	MKT1	10	10	10	7	Reception
KP MKT-2A	NORMAL	MKT2	10	10	10	8	Reception
	RETR		10	10	10	8	Retransmission
KP MKT-2B	NORMAL	MKT2	20	20	20	8	Reception
	RETR		20	20	20	8	Retransmission
KP MKT-2V	NORMAL	MKT2	30	30	30	8	Reception
	RETR		30	30	30	8	Retransmission
KP MKT-3	NORMAL	MKT3	64	8	64	8	Reception
Device “KIT”	NORMAL	KIT	32	8	10	8	“KIT” type device (Khabarovsk RDU
KOMPAS 1.1	NORMAL	COMPAS1	< ... >	8	< ... >	8	
KP TC120	NORMAL	TC120	< ... >	4	< ... >	8	Reception
UMA -1U	ARBITR	SM01	16	16	1	38	Reservation Device “SMART-U-FEP”

Device / Protocol	KP_TYPE	TM_TYPE	N_TI ¹	L_GR_TI	N_GR_TS	L_GR_TS ¹	Comment
KP IEC101	MASTER	IEC101	4095	4095	1	4095	Reception or transmission Unbalanced mode
	SLAVE		4095	4095	1	4095	Reception or transmission Unbalanced mode
	BALANCE		4095	4095	1	4095	Reception or transmission Balanced mode
	RETR		4095	4095	1	4095	Transmission Unbalanced/ balanced mode
KP IEC104	NORMAL	IEC104	4095	4095	1	4095	Reception or transmission SMART-FEP acts as a client (MASTER)
Dataset	UNI	n/a	**	16	***	8	Status and operation history of UTM and TC channels
	TECHNO		16	4	0	0	Information used in Telecontrol Operator's Workstation
	SERVER		30	3	10	1	Data on connected network clients

Legend to Table:

- 1) maximum tolerable values of the parameters, the others are fixed
- 2) the combined number of TIs and TS groups at reception or at transmission must not be in excess of 248. Protocol restriction
- 3) RETR type KP differs from MASTER, SLAVE and BALANCE types KP by prohibition on sending the general interrogation request. In other words, there is a restriction on sending 100 type ASDU with transmission reason 6 and P/N = 0 bit in this KP type implementation (positive acknowledgement)
- 4) (notation "<...>") setting of actual values
- ** $\max(\text{SYS_KP}, \text{RES_KP}, \text{N_CHAN}) \cdot 16$ – maximum value for KP numbers (primary and backup) and numbers of channels multiplied by the number of TIs in the group for a UNI type KP, i.e., by 16
- *** $\max(\text{SYS_KP}, \text{RES_KP}, \text{N_CHAN})$ – maximum value for KP numbers (primary and backup) and channel numbers

3.6.2.6 Using IMM to Configure SMART-U-FEP (Examples)

3.6.2.6.1 General Approach

In Spectrum PowerCC data model, all parameters that are required to specify a system component are defined by instances.

In general, an instance contains the following three areas, selectable through tabulators (tabs):

1. **General:** Name, type, description
2. **Attributes:** A list of attributes (parameters)
3. **Links:** Reference to another instance(s), optional

Many instance types can also have child instances. This way it is easily possible to set up hierarchical structures. In the Overview Pane of the IMM, the instances look like a file tree presented in Windows Explorer, see Figure 3-6.7.

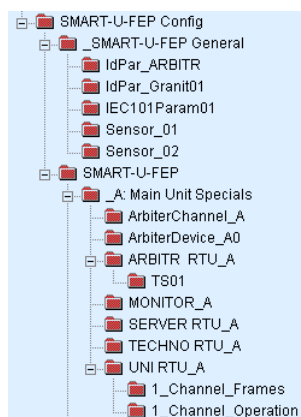


Figure 3-6.7: Hierarchical Structure of Configuration Data in IMM

To support the idea of structuring the configuration data, also instances without attributes and links are deployed, they are called "Container Instances".

Basically, if a new component is to be added to the system, the following actions are to be taken:

1. Creating a new instance
2. Editing the Attributes
3. Establishing necessary system links, as far as required

If the instance has child instances, this procedure is to be repeated for every child instance.



Note

Once an instance has been created, further instances of the same type can also be created by copy / paste / rename which is more convenient in most cases.

All three basis actions are demonstrated by the following example: .

An RTU communication channel is to be added to the SMART-U-FEP configuration. The following settings are used in this example:

General	Name: "Channel 4"
Attributes	TIMEOUT: 2 000 ms, TM_TYPE : RPT80
Links	Links to instances "Device 4A" and "Device 4B"

Creating Instances

To create an instance, right-click on the appropriate parent instance (in this example: '_Connections') in the Overview Pane and select **New** from pop-up menu that appears. Alternatively, you may highlight the parent instance (by point and left-click) and enter <CTR-I>. In both cases, the 'Type Selection Box' appears in the Work Pane and displays the (child) instance types available for the selected parent instance, see Figure 3-6.8

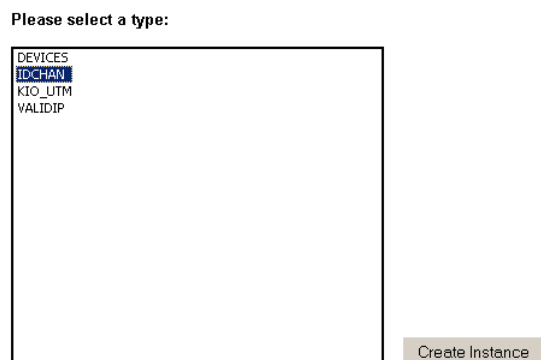


Figure 3-6.8: Creating an Instance

Select one of the instance types from the box (in this example: IDCHAN) and click **Create Instance**. The Instance Editor appears in the Work Pane. Please enter the name of the new component, for this example: "Channel 4"

Editing Attributes

Open tab **Attributes** and modify the attributes as indicated above

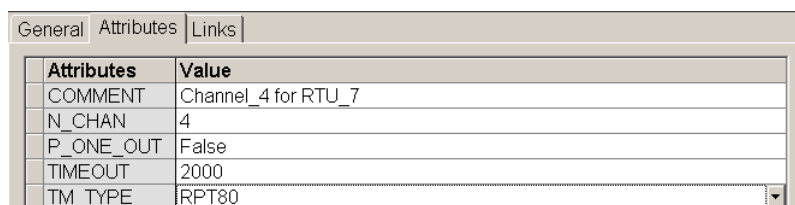


Figure 3-6.9: Attributes of the Instance "Channel"

Establishing System Links

Open tab **Links** and establish the required system links by "Drag & Drop" as follows: Select the desired instance in the Overview Pane and drag it to the reference field in the Instance Editor. As soon as the cursor reaches the target field it changes to a + sign and you can release the right mouse button: The reference instance "drops" into the target field, see Figure 3-6.10.

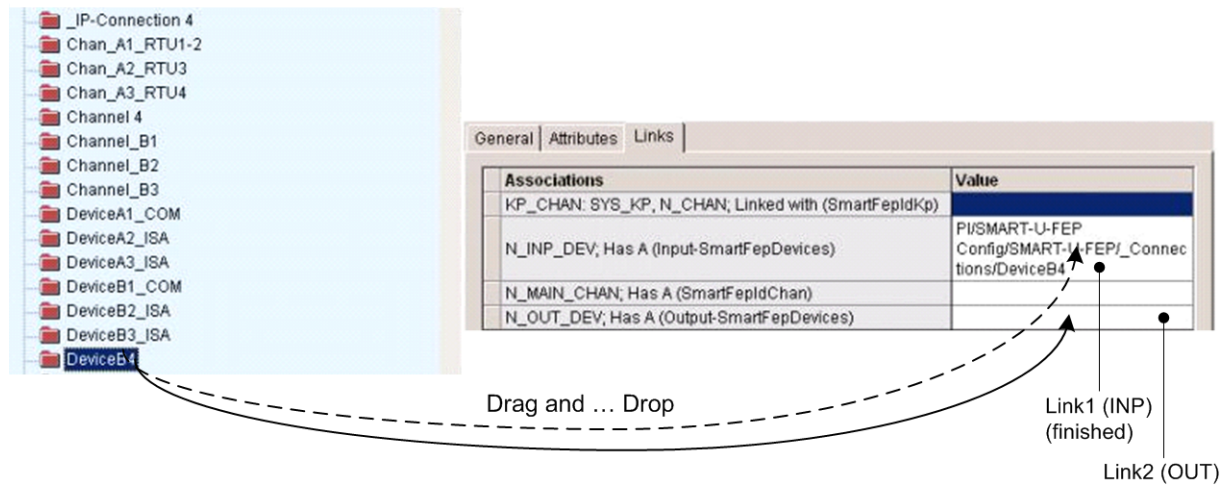


Figure 3-6.10: Establishing System Links

3.6.2.6.2 Typical SMART-U-FEP Configuration Tasks

In the following section, some typical configuration tasks are demonstrated by means of suitable examples. The samples are based on the hardware setup shown by Figure 3-6.2. This layout is reflected in IMM through the following instance tree in the Overview Pane:

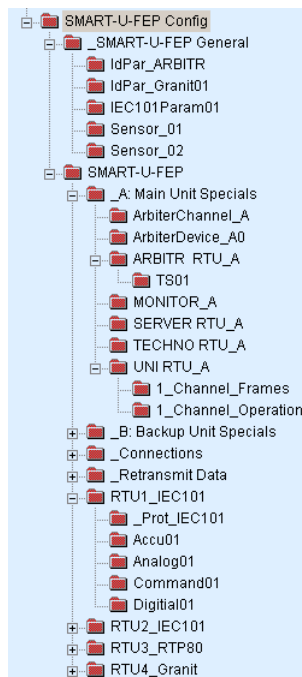


Figure 3-6.11: SMART-U-FEP: Instance Tree of Sample Setup

The exemplary tasks comprise:

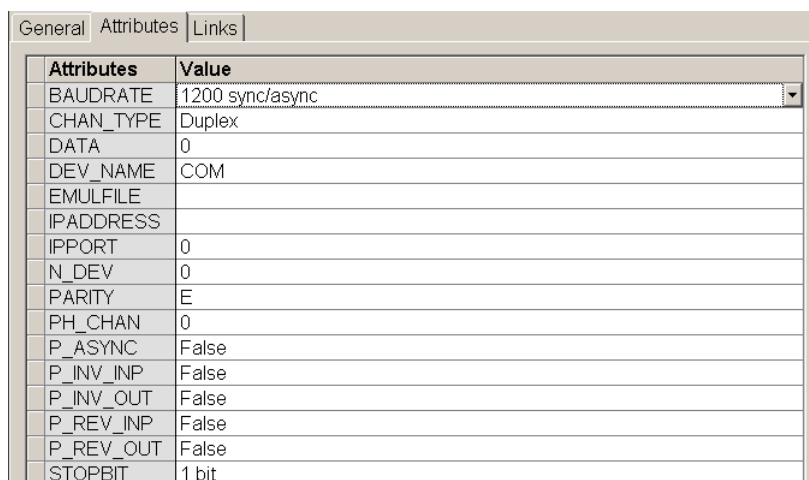
1. Setting-up a serial interface at SMART-U-FEP
2. Setting-up a (communication) channel for an RTU

3. Setting-up an IP-communication link to CFE
4. Setting-up an RTU (KP) for Protocol IEC101
5. Setting-up an RTU (KP) for Protocol Granit
6. Setting-up an RTU (KP) for Protocol RPT80
7. Setting-up data points
8. Setting-up retransmitted data points

1. Setting-up a Serial Interface (Device)

To accomplish this task, perform these steps:

1. Right-click instance '_Connections', select instance type *DEVICES* from 'Type Selection Box', click **Create Instance**. Name the new instance, e.g. "Device A4_COM".
2. Open tab **Attributes** and edit the values as appropriate, see Figure 3-6.12 and Table 3-6.7



Attributes	Value
BAUDRATE	1200 sync/async
CHAN_TYPE	Duplex
DATA	0
DEV_NAME	COM
EMULFILE	
IPADDRESS	
IPPORT	0
N_DEV	0
PARITY	E
PH_CHAN	0
P_ASYNC	False
P_INV_INP	False
P_INV_OUT	False
P_REV_INP	False
P_REV_OUT	False
STOPBIT	1 bit

Figure 3-6.12: Attributes of the Instance "Device"

If SMART-U-FEP is redundant, also create the Device for the backup unit: "Device B4_COM"

2. Setting-up a Serial Channel for an RTU

To accomplish this task, perform these steps:

1. Right-click instance '_Connections', select instance type *IDCHAN* from 'Type Selection Box', click **Create Instance**. Name the new instance, e.g. "Channel 4".
2. Open tab **Attributes** and edit the values as appropriate, see Table 3-6.9
3. Open tab **Links** and establish two system links (for "N_INP_DEV ..." and "N_OUT_DEV ...") to the associated device, here: to "Device B4" and also to the RTU that uses this channel, see Figure 3-6.10

3. Setting-up an IP-Connection to CFE

To accomplish this task, perform these steps:

1. Right-click instance '_Connections' (type *ConnectionDefinition*), select instance type *VALIDIP* from 'Type Selection Box', click **Create Instance**. Name the new instance, e.g. "IP-Connection 4".
2. Open tab **Attributes** and enter the IP address of the associated CFE LAN Interface. The other attributes can be left as per default, see Table 3-6.6.

4. Setting-up a Type 'IEC101' RTU

To accomplish this task, perform these steps:

1. Right-click instance SMART-U-FEP, select instance type *IDKP* from 'Type Selection Box', click **Create Instance**. Name the new instance, e.g. "RTU5".
2. Open tab **Attributes** and edit the values as appropriate, see Table 3-6.5 and Table 3-6.23. Make sure that the following attributes are properly set:
 - a. KP_TYPE: select "MASTER" for IEC101 type RTU
 - b. TM_TYPE: select "IEC101" for protocol IEC60870-5-101
3. Open tab **Links** and establish a system link to the instance "IEC101Param_n", located in the "SMART-U-FEP General" container.

Save the instance.

4. Right-click the new instance RTU5, select instance type *IEC101* from 'Type Selection Box', click **Create Instance**. Name the new instance, e.g. "Prot_IEC101"

5. Setting-up a Type 'Granit' RTU

To accomplish this task, perform these steps:

1. Right-click instance SMART-U-FEP, select instance type *IDKP* from 'Type Selection Box', click **Create Instance**. Name the new instance, e.g. "RTU6"
2. Open tab **Attributes** and edit the values as appropriate, see Table 3-6.5 and Table 3-6.23. Make sure that the following attributes are properly set:
 - a. KP_TYPE: select "NORMAL" for type Granit RTU
 - b. TM_TYPE: select "GRANIT" for this protocol
3. Open tab **Links** and establish a system link to the General Granit Protocol Parameter instance
Save the instance.
4. Right-click the new instance "RTU6", select instance type *GRANITAFB* from 'Type Selection Box', click **Create Instance** and name it.
5. Repeat this procedure for each group of 8 analogs as well as for each group of 8 digital values as follows:
 - a. Group 1 of 8 analog values, suggested name: "Granit_Ana-Gr1"
 - b. Group 2 of 8 analog values, suggested name: "Granit_Ana-Gr2"
 - c. Group n of 8 analog values, suggested name: "Granit_Ana-Gr n"
 - d. Group 1 of 8 digital values, suggested name: "Granit_Dig-Gr1"
 - e. Group 2 of 8 digital values, suggested name: "Granit_Dig-Gr2"
 - f. Group n of 8 digital values, suggested name: "Granit_Dig-Gr n"
6. Open tab **Links** and establish system links to the data points associated to the respective group, see Figure 3-6.13

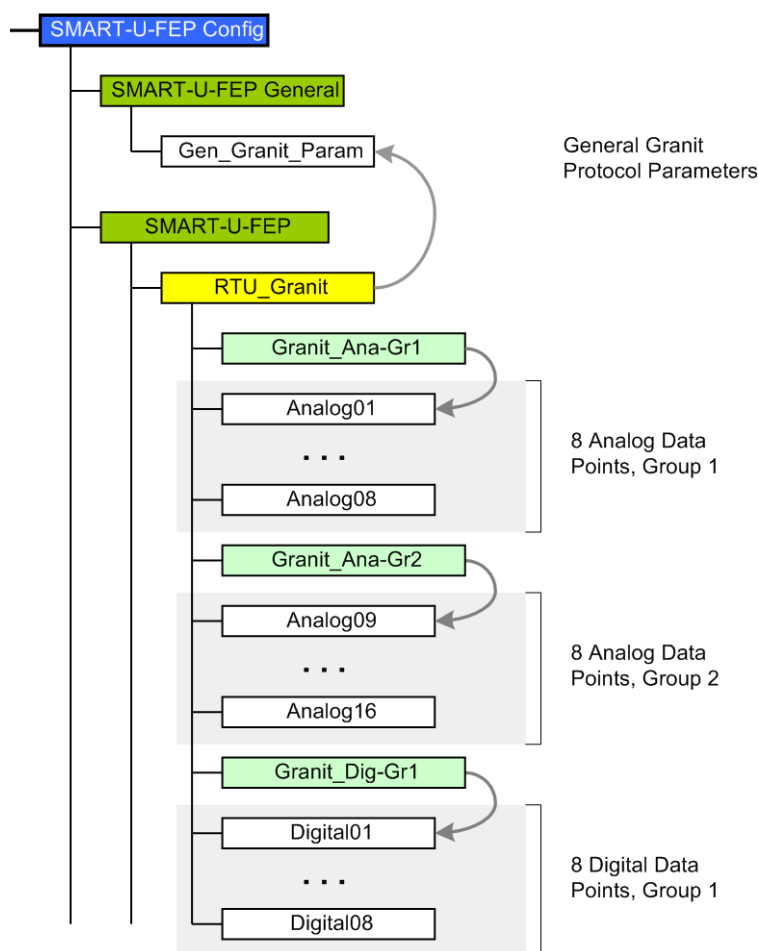


Figure 3-6.13: Protocol Granit: Data Points, Data Groups and System Links



Note

- Only one link is needed from protocol instance to any data point from the group
- If data points of a group are not yet available, you can postpone this step until these have been created

6. Setting-up a Type 'RPT80' RTU

To accomplish this task, perform these steps:

1. Right-click instance SMART-U-FEP, select instance type *IDKP* from 'Type Selection Box', click **Create Instance**. Name the new instance, e.g. "RTU7".
2. Open tab **Attributes** and edit the values as appropriate, see Table 3-6.5 and Table 3-6.23. Make sure that the following attributes are properly set:
 - a. KP_TYPE: select "NORMAL" for type RPT80 RTU
 - b. TM_TYPE: select "RPT80" for this protocol
3. Right-click the new instance "RTU7", select instance type *RPT80* from 'Type Selection Box', click **Create Instance** and name it.

The RPT80 protocol uses two types of data point allocation:

- a. For analog values, one-to-one allocation: Each RPT80 protocol instance is linked to just one analog data point
 - b. For digital values, group allocation: The protocol instance is linked to a group of up to 8 digital data points
4. Repeat this procedure for each analog data point and each group of 8 digital data points as appropriate. (Details on this issue can be found in the respective protocol literature)
 5. Open tab **Links** and establish system links from protocol instance to data points as illustrated by Figure 3-6.14

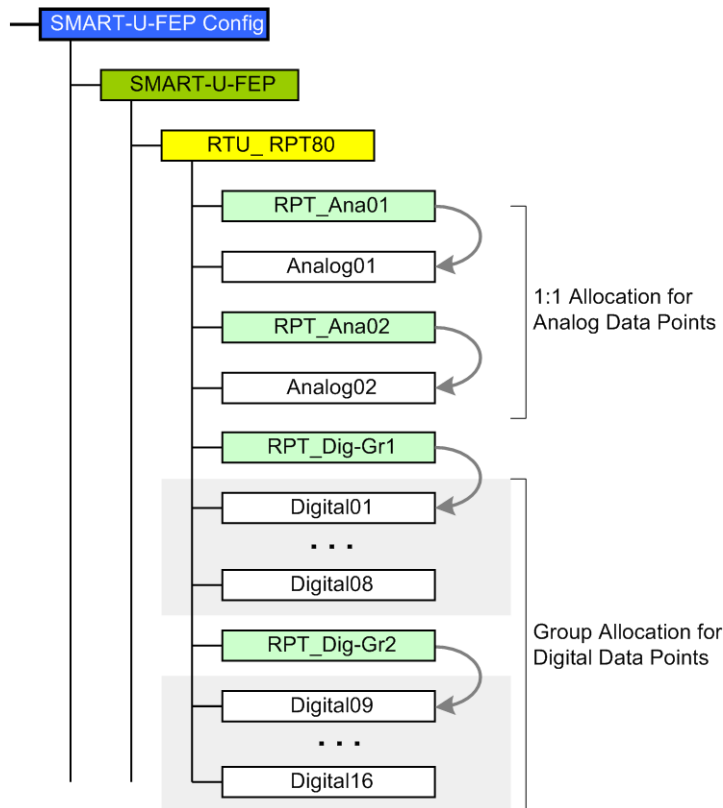


Figure 3-6.14: Protocol RPT80: Data Points, Data Groups and System Links



Note

- Only one link is needed from protocol instance to any data point from the group
- If data points of a group are not yet available, you can postpone this step until these have been created

7. Setting-up Data Points

To add data points to an RTU perform these steps:

1. Right-click instance RTU (type *IDKP*), select instance type
 - a. *IDTII* for an accumulator (counter value)
 - b. *IDTI* for an analog value
 - c. *IDTS* for a digital value
 - a. *IDTU* for a digital commandfrom 'Type Selection Box', click **Create Instance** and name it.
2. Open tab **Attributes** and edit the values as appropriate, see Attribute Tables (Table 3-6.18 through Table 3-6.21) for the respective data type. Save the instance.
3. In case of protocols Granit and RPT80: Check, whether at least one data point of a group has a link from protocol instance.

8. Setting-up Retransmitted Data Points

Retransmitted data points are those that are routed from one RTU to another RTU. Before configuring the routing instance, make sure that both, source data point and target data point have already been created and configured.

Then perform these steps:

1. Right-click instance 'SMART-U-FEP', select instance type *RetransmitData* from 'Type Selection Box', click **Create Instance**. Name the new container instance as appropriate
2. Right-click the (new) instance 'Retransmit Data' (type *RetransmitData*) and select either instance type from 'Type Selection Box' :
 - a. *IDRTI* for an analog data point
 - a. *IDRTS* for an digital data point... and click **Create Instance**
Repeat step 2 for every data point that is to be retransmitted
3. Establish for every data point for being retransmitted, two system links: one for source data point and one for target data point, see Figure 3-6.15

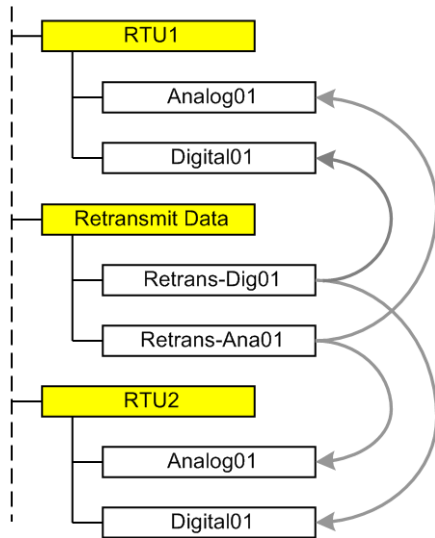


Figure 3-6.15: System Links for Retransmitted Data Points

3.6.3 Configuring CFE (SMART-U-FEP Part)

3.6.3.1 Overview

This section describes how SMART-U-FEP is to be configured as a standard component of the Process Interface.

Two IMM areas are involved:

- CFE: Container instance that comprises all CFE relevant components, such as CFE interfaces, CFE interface links (connections)
- CFE RTUs: Container instance that comprises RTUs, Data Concentrators, Sub-RTUs and data points

The following diagram shows a redundant CFE with two interfaces and two connections each and SMART-U-FEP as a child instance of 'CFE RTUs'. SMART-U-FEP has to be configured as a non-redundant Data Concentrator (DaCo) with Sub-RTUs, see Figure 3-6.16

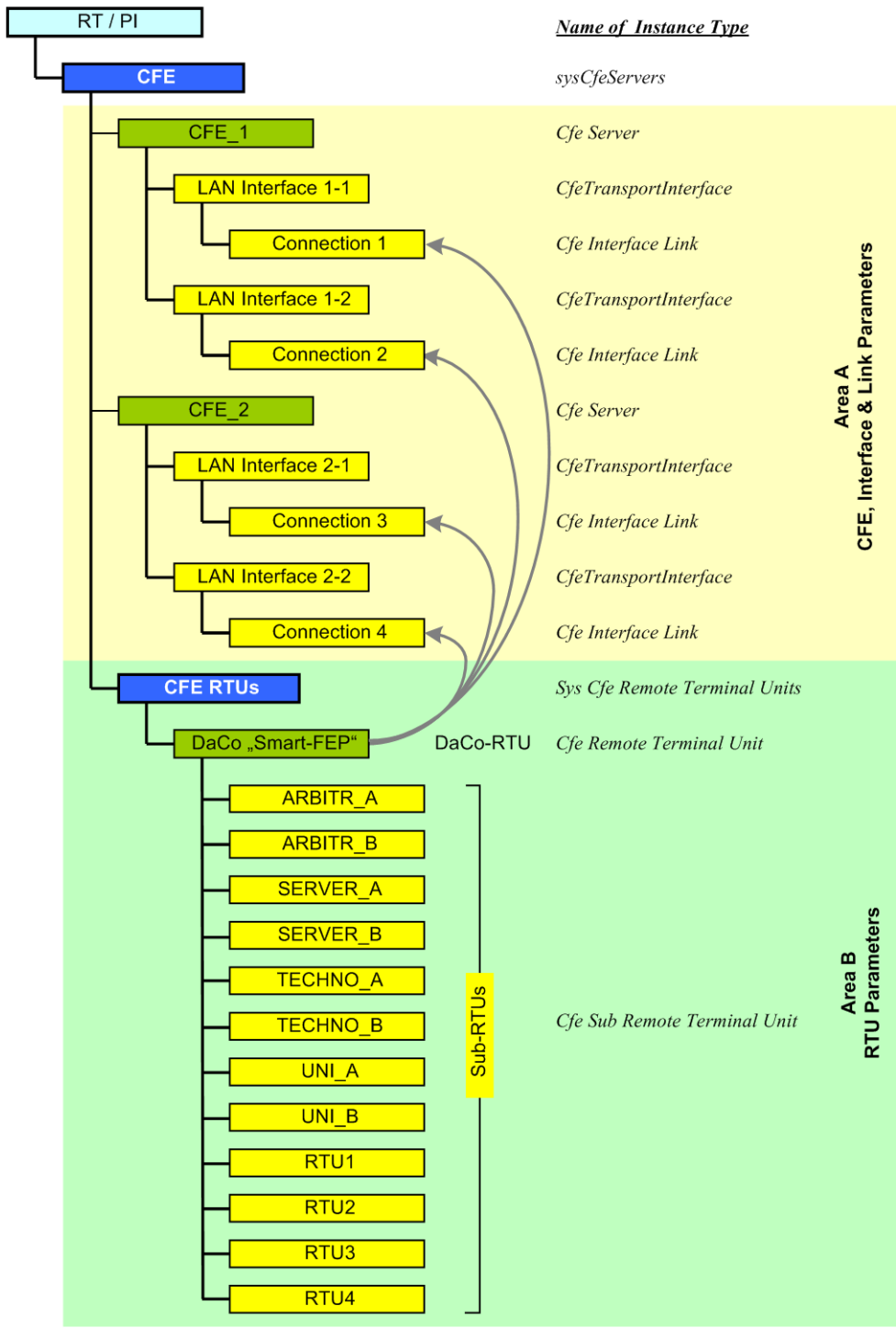


Figure 3-6.16: CFE Configuration Data for SMART-U-FEP and Sub-RTUs - Overview

When configuring SMART-U-FEP as a CFE component, the following specific characteristics are to be taken into account:

- Redundant and non-redundant "SMART-U-FEP" have the same non-redundant arrangement in IMM. Main Unit and Backup Unit are represented as one Data Concentrator

- Redundant "SMART-U-FEP" is connected over four IP-connections to CFE
- Non-redundant "SMART-U-FEP" has only two IP-connections to CFE
- Unlike standard, the instance CFE Interface Link (IP-connection) provides two extra status bits: "Unit Active" and "Unit Main"
- Virtual RTUs like ARBITER_A & B, SERVER_A & B, TECHNO A & B, UNI_A & B are treated as normal Sub-RTUs
- The way how Sub-RTUs are connected to SMART-U-FEP (redundant channel, non-redundant channel, communication protocol) is not reflected in IMM

3.6.3.2 Status Information

Next diagram shows details on CFE LAN interfaces and IP-connections as well as status information of these components.

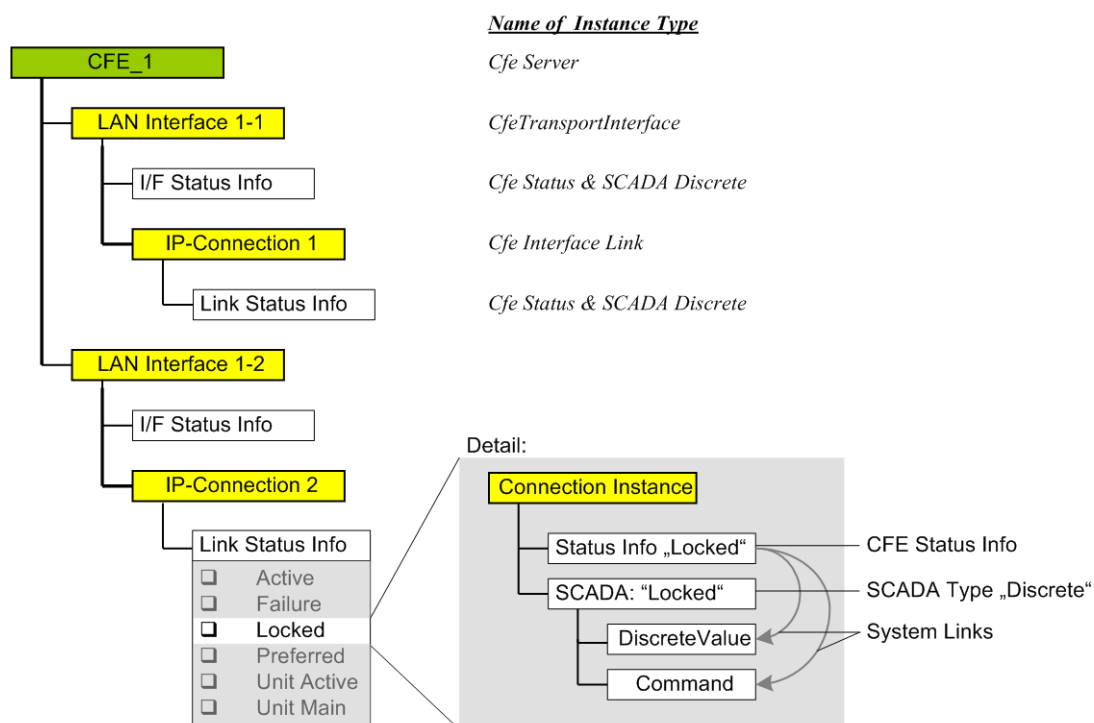
The instance 'LAN Interface' (type *CfeTransportInterface*) provides the following status information:

- Active (Value)
- Failure (Value)
- Locked (Value & Control)
- Preferred (Value & Control)

The instance 'IP-Connection' (type *Cfe Interface Link*) provides the following status information:

- Active (Value)
- Failure (Value)
- Locked (Value & Control)
- Preferred (Value & Control)
- Unit Active (Value & Control)
- Unit Main (Value)

For pure operation, there is no need to configure standard status information such as Active, Failure, Locked, Preferred. They exist as soon as the respective instance has been created. Status information is displayed in 'Runtime Explorer' / CFE-View without being configured in IMM. However, customized status information Unit Active and Unit Main has to be configured for proper operation. If status information has to be displayed in diagrams or put into logs or, if the status of a component has to be controlled, corresponding SCADA Measurement Values / Controls have to be created and linked to the Status Information, as illustrated by the next figure (Figure 3-6.17).



Note: the same applies for CFE_2, accordingly

Figure 3-6.17: Status Information for CFE - Smart FEP Interconnections

3.6.3.3 General Interrogation, Scan Groups

Depending on the type of Sub-RTU special settings for General Interrogation may be required, such as:

- Time controlled (Periodical) Interrogations
- Enabled / disabled Startup Interrogation
- Delayed Startup Interrogation

To cope different requirements, adequate "Scan Groups" are to be established and linked to the associated data points, see Figure 3-6.18

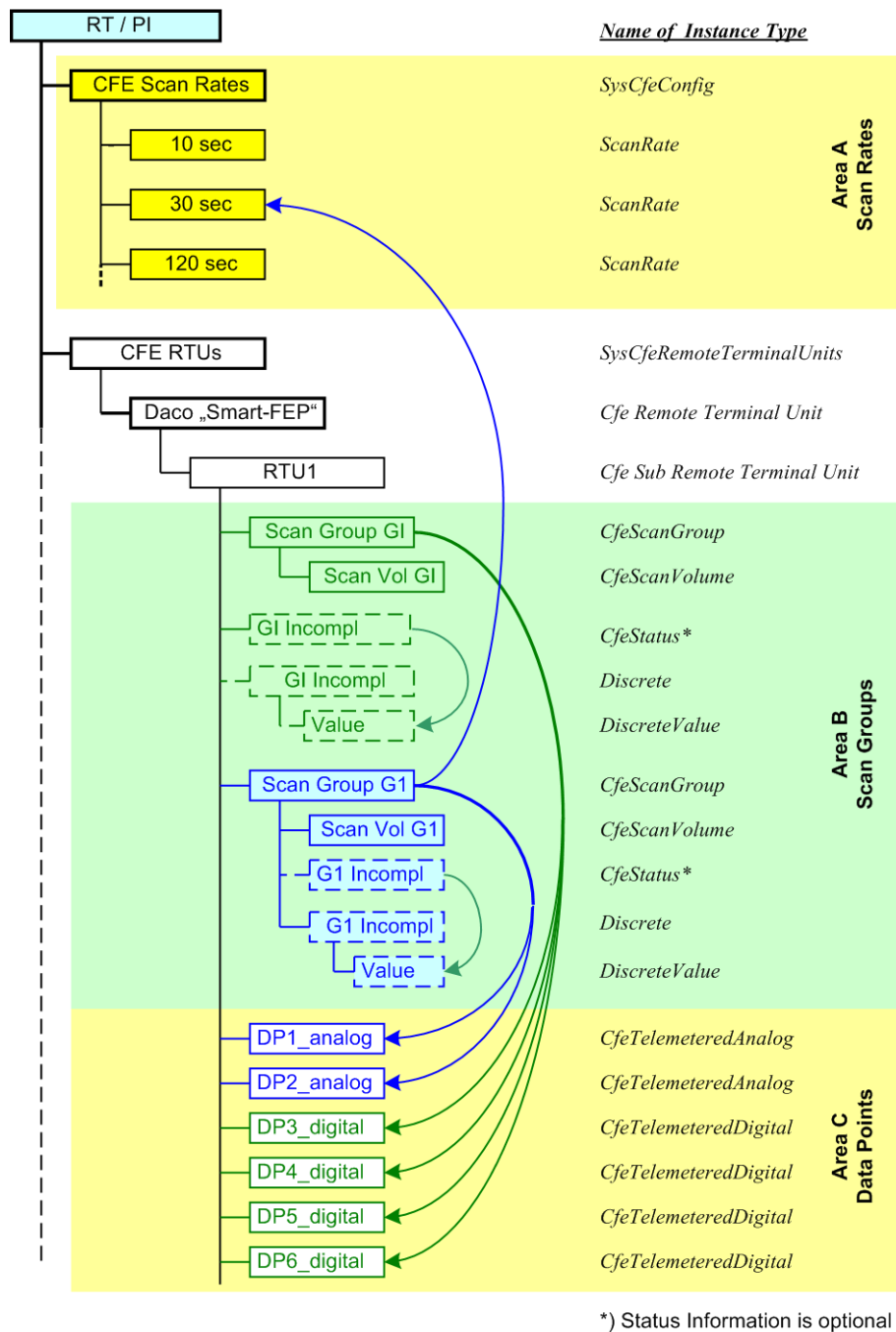


Figure 3-6.18: Interlinking Scan Groups and Data Points

3.6.3.4 Links from RTU Data Points to SCADA

Finally, all RTU data points are to be linked to SCADA "Measurements" in the IMM area "Net.../Substations .../Sub_XXX" as follows:

- For digital data points: The Link Association "Sends To (DiscreteValue)" is to be connected to a Network instance of the type *DiscreteValue*

- For analog data points: The Link Association "Sends To (AnalogValue)" is to be connected to a Network instance of the type *AnalogValue*
- For digital commands (if applicable): The Link Association "Receives From (Command)" is to be connected to a Network instance of the type *Command*
- For analog setpoints (if applicable): The Link Association "Receives From (Setpoint)" is to be connected to a Network instance of the type *Setpoint*

This is a standard procedure applicable for every data point, irrespective which type of RTU it might come from.

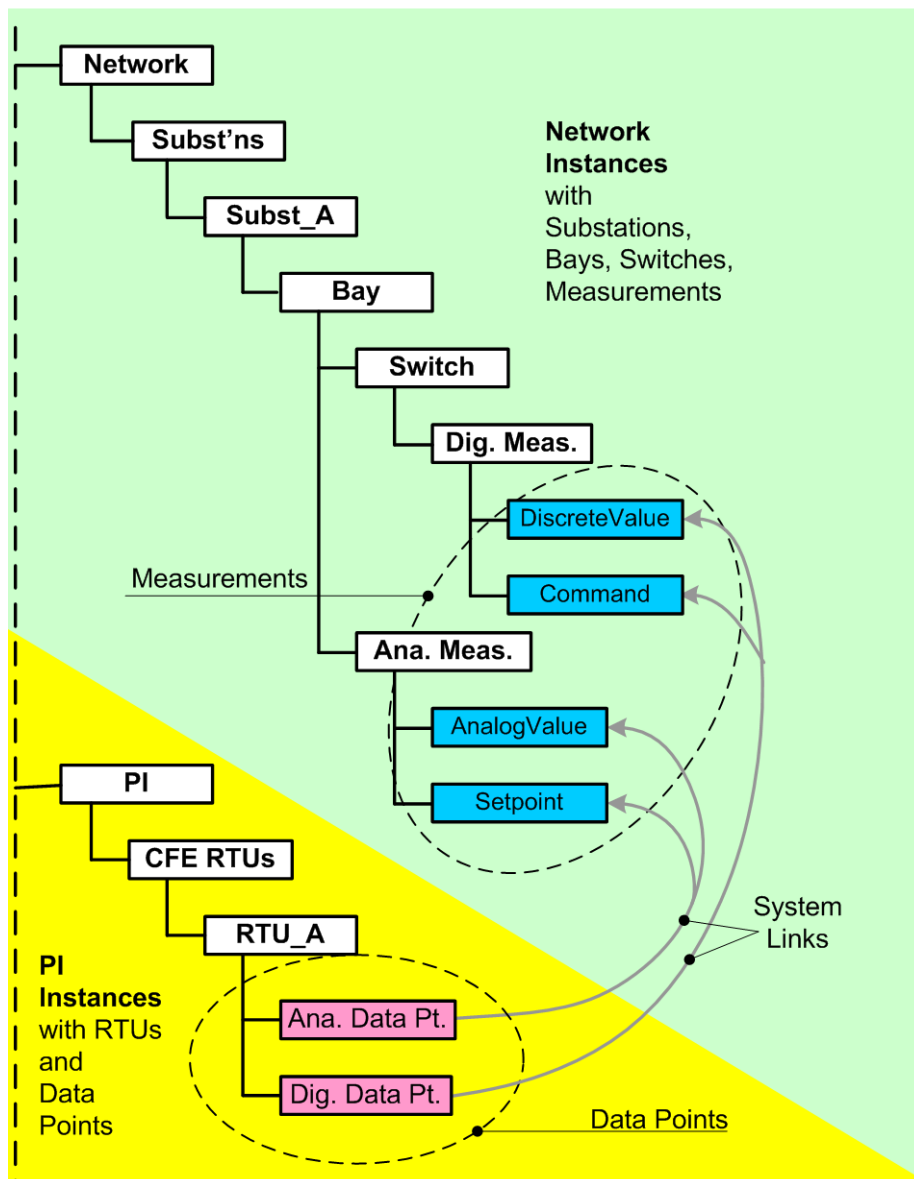


Figure 3-6.19: System Links Between RTU Data Points and Network Measurements

3.6.4 DBase Converters

Siemens PowerCC provides two types of data converters for SMART-U-FEP, see Figure 3-6.20.

1. **Import Converter:** For importing dBase files into IMM, the dBase-XML Import Converter is provided. Existing dBase files containing a valid Smart-U-FEP configuration can be imported for initial setup. As IMM does not support dBase file format, dBase files have to be converted to XML-files before they can be imported. The conversion can be executed using the tool 'SmartFEPInitialConfig'. The usage of this tool is described hereafter (Chapter 3.6.4.1 *Initial Data Import - 'SmartFEPInitialConfig' Usage*)
2. **Config Converter:** When a running SMART-U-FEP is to be updated via IMM, a new set of dBase files is to be generated for configuring the SMART-U-FEP. This is done by the 'Config Converter', a CFE process that runs automatically. The process including its options have to be setup in SMI before first usage. Details are given in Chapter 3.6.4.2 *Configuring 'SmartFEPConfigConverter' in SMI*.

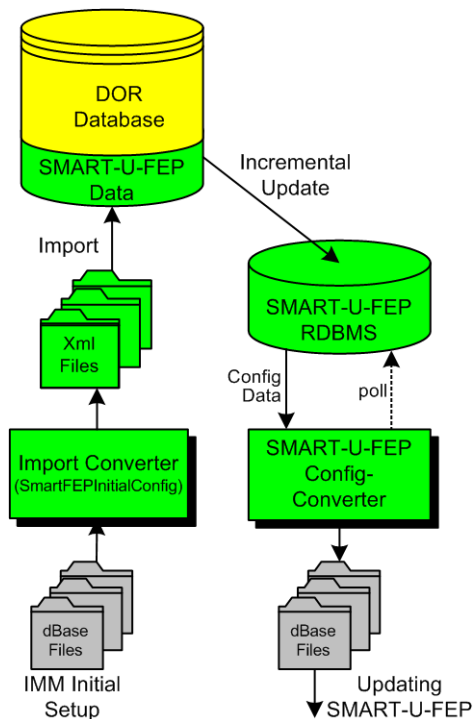


Figure 3-6.20: DBase-XML Converters Providing Import and Export Facility

3.6.4.1 Initial Data Import - 'SmartFEPInitialConfig' Usage

The SmartFEPInitialConfig tool is to be started from command line including appropriate command line options that control the behavior of the application. The command line parameters are divided into the following three groups:

1. Mandatory parameters describing the source directories (dBase) and the destination file (XML), options 1 through 3
2. Mandatory parameters describing the hierarchy under which this Smart FEP configuration shall be stored in the IMM after import, options 4 through 6
3. Optional parameters for controlling logging and text conversion, options 7 and 8

no.	Option	Description	Example (for 'SmarFEP01')
1	-MainUnitDir %directory%	Path of the directory where the dBase files of the main unit configuration are stored	-MainUnitDir \SmartFEP01\main
2	-BackupUnitDir %directory%	Path of the directory where the dBase files of the backup unit configuration are stored	-MainUnitDir \SmartFEP01\backup
3	-XMLOutputFile %filename%	Filename of the XML output file	-XMLOutputFile SmartFEP01.xml
4	-SmartFepName %name%	Name of the Smart Fep configuration instance in IMM (name of type <i>sysSmartFep</i> instance).	-SmartFepName SmartFEP01
5	-GeneralConfigName %name%	Name of the General Configuration instance of the Smart FEP (name of type <i>sysSmartFepGeneralConfig</i> instance)	-GeneralConfigName SmartFEP01_General
6	-ParentPath %parent%	Path and name of the IMM parent (name of the <i>sysSmartFepConfig</i> instance)	-ParentPath PI/SmartFEPConfig
7	-Logfile %filename%	Filename of the import logfile. Default: no logging.	-Logfile SmartFEPImport.log
8	-CodePage %codepage%	Codepage of the *.dbf files, used for text conversion from ANSI codepage to UTF-8. Default: Current system codepage	-CodePage 866



Note

Installation path of the SmartFEPInitialConfig.exe is %SR5_INSPATH%\Bin, which is by default C:\Siemens\PowerCC\Bin

Example

The usage of the tool is demonstrated by an example with typical settings. The following command string may be entered directly into the command line (cmd window) or - which is more convenient - stored in a batch file and executed.

```

C:\WINNT\system32\cmd.exe

Z:\>C:\Siemens\PowerCC\Bin\SmartFEPInitialConfig.exe -MainUnitDir \SmartFEP_Example\main -BackupUnitDir \SmartFEP_Example\backup -XMLOutputFile SmartFEP_Example.xml -SmartFepName SmartFEP_Example -GeneralConfigName SmartFEP_Example_General -ParentPath PI/SmartFEPConfig -Logfile SmartFEPImport.log -CodePage 866_

```

Figure 3-6.21: Starting the *SmartFEPInitialConfig* Tool from Command Line

The example shown in above figure will run the 'SmartFEPInitialConfig.exe' application, and will create a XML output file Z:\SmartFEP_Example.xml from the dBase files stored in the directories Z:\SmartFEP_Example\main and Z:\SmartFEP_Example\backup.

After import into IMM, the configuration data will be stored under PI/SmartFEPConfig/SmartFEP_Example_General (IDPAR, IDSENS and IEC101PARAM instances) and PI/SmartFEPConfig/SmartFEP_Example (all other data).

Errors during conversion are stored in the file Z:\SmartFEPImport.log.

For character conversion from ANSI to UTF-8, codepage 866 (DOS/Cyrillic) is used.

The result of the example should look like shown by next figure. If everything went fine, no errors should be reported in the output window, and the logfile (if configured) should be empty .

```

C:\WINNT\system32\cmd.exe
SmartFEP initial configuration converter

Main unit directory: \SmartFEP_Example\main
Backup unit directory: \SmartFEP_Example\backup
SmartFEP folder name: SmartFEP_Example
SmartFEP general config folder name: SmartFEP_Example_General
Parent: PI/SmartFEPConfig
XML output file: SmartFEP_Example.xml
Logfile: SmartFEPImport.log
Conversion finished. Press any key to exit.

```

Figure 3-6.22: Console Information After (Successful) Run of the *SmartFEPInitialConfig* Tool

The following figure shows the working directory with dBase folders and generated output files

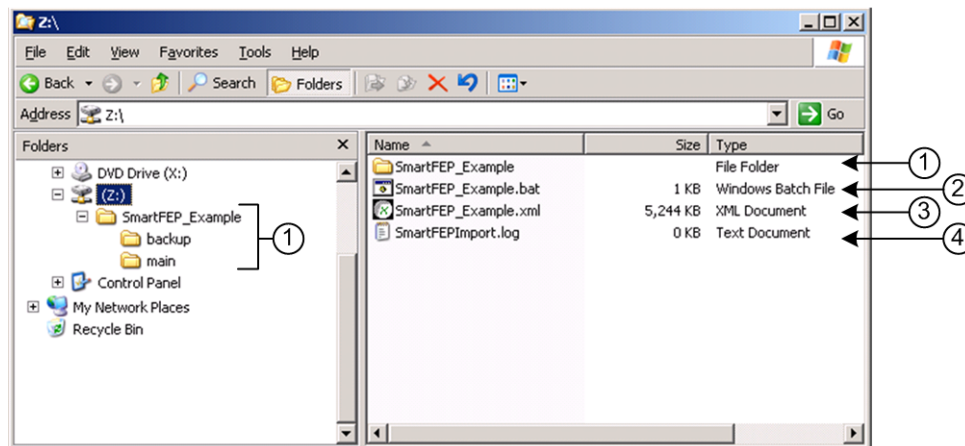


Figure 3-6.23: Example: Folders and Files after Running the Converter

Legend

- 1 Folder containing sub-folders "main" and "backup" with dBase files
- 2 Batch file with command and options
- 3 Generated XML file (after execution)
- 4 Log file (Size zero means no entries)



Note

Import of configurations consisting of one non-redundant single unit (there is only one directory with dBase configuration files) is also possible. In this case, the directory path must be given in both command line options `-MainUnitDir` and `-BackupUnitDir`. All other options must be specified as described above.

Importing XML File to IMM

After running the SmartFEPInitialConfig tool, the created XML file must be imported via IMM. First, create a new job, select the instance PI and create a new instance of type sysSmartFEPConfig. The name of the new 'sysSmartFEPConfig' instance must fit to the path given in the -ParentPath command line option when running the SmartFEPInitialConfig tool.

After the new sysSmartFEPConfig instance has been created, select option 'Import' from the 'File' menu. In the import window, select the option 'XML' and deselect all other options. Highlight the import XML file created by the SmartFEPInitialConfig tool and click the "Import" button.

The import may take some time depending on the size and complexity of the configuration. The import should finish without warnings or errors.

Compatibility Issues (SCADA 2003 vs. CFE)

Import of configurations compatible to SCADA 2003 will give slightly different results than import of configurations compatible to CFE. The difference between these configurations is that the virtual KPs (ARBITR, TECHNO, SERVER, and UNI) have identical SYS_KP number for main and backup unit in SCADA 2003 configuration, and different SYS_KP in CFE configuration.

After import of SCADA 2003 compatible configuration, only ARBITR will be stored separately for main and backup unit in the corresponding sysSmartFepSpecialConfig container due to different LOC_KP value, TECHNO, SERVER and UNI will be stored within the real KPs.

After import of CFE compatible configuration, ARBITR, TECHNO, SERVER and UNI KPs will be stored separately for main and backup unit in the corresponding sysSmartFepSpecialConfig container. In case of SCADA 2003 compatible configuration, the storage location of TECHNO, SERVER and UNI KPs may be changed by the operator after import and adapted to CFE compatible configuration.



Note

For a successful conversion from dBase to XML, successful import into IMM and successful recreation of dBase files, the original dBase files must contain a proper and valid configuration. The SmartFEPInitialConfig tool will report undefined attribute values (e.g. attribute TM_TYPE in type IDKP) during conversion, the IMM will report values that are out of the range of the data type (e.g. boolean values that are not [0, 1]) during import. On demand, the SmartFEPConfigConverter will perform post activation data validation (e.g. uniqueness of SYS_KP in type IDKP, uniqueness of data point addresses etc.) during conversion and report errors in the private process log.

Default Names for Created Instances

In the IMM, a unique name is required for each instance within its container. Unique default names for all instances are created during creation of the XML import file using the following rules. The names may be changed after import by the user.

Instance Type	Default Name(s)
MONITOR	As there is only one MONITOR instance per sysSmartFEPsSpecialConfig instance, the MONITOR instance is simply named "MONITOR"
IDSENS, IDPAR, IEC101PARAM	Instances are named with table name and a serial number starting with 1, SENS_(SerNum), e.g. "SENS_001", "PAR_003", "IEC101PARAM_002". The required number of digits is determined from the number of instances.

Instance Type	Default Name(s)
VALIDIP	Instances are named with table name and serial number starting with 1, VALIDIP_(SerNum), e.g. "VALIDIP_001". The required number of digits is determined from the number of instances.
IDKP	Instances are named with table name, the SYS_KP value and the KP_TYPE value, KP_(SYS_KP)_(KP_TYPE), e.g. "KP_025_NORMAL". The number of digits is fixed to three.
IDTI, IDTS, IDTII, IDTU	Instances are named with table name and the N_GR and N_CH values, TI_(N_GR)_(N_CH), e.g. "TI_001_004". The number of digits is fixed to three for N_GR and N_CH.
RPT80, IDMKT, UTM7INP, KIO	Instances are named with table name, data point type, and the N_GR and N_CH (only IDTI) values of the linked data point, RPT80_(type)_(N_GR)_(N_CH), e.g. "RPT80_TI_001_004" or "RPT80_TS_003". The number of digits is fixed to three for N_GR and N_CH.
GRANITAFB	Instances are named with table name, data point type, and the N_GR value of the linked data point, GRANITAFB_(type)_(N_GR), e.g. "GRANITAFB_TI_001". The number of digits is fixed to three for N_GR.
IEC101, IEC104, TM800	As there is only one instance allowed per KP, instances are named with table name only, e.g. "IEC101".
DEVICES	Instances are named with table name, the N_DEV value and the DEV_NAME value, DEVICE_(N_DEV)_(DEV_NAME), e.g. "DEVICE_018_COM". The number of digits is fixed to three.
IDCHAN	Instances are named with table name, the N_CHAN value and the TM_TYPE value, CHAN_(N_CHAN)_(TM_TYPE), e.g. "CHAN_018_IEC101". The number of digits is fixed to three.
KIO_UTM	Instances are named with table name and serial number starting with 1, KIO_UTM_(SerNum), e.g. "KIO_UTM_001". The required number of digits is determined from the number of instances.
IDRTI, IDRTS	Instances are named with table name and the N_KP, N_TI, N_KP_R and N_TI_R values, RTI_(N_KP)_(N_TI)_(N_KP_R)_(N_TI_R), e.g. "RTI_003_010_033_410". The number of digits is fixed to three for the N_KP and N_KP_R values, and determined from the maximum value for the N_TI and N_TI_R values.

3.6.4.2 Configuring 'SmartFEPConfigConverter' in SMI

The SmartFEPConfigConverter must be configured in PowerCC IMM node 'SMI' (System Management Information) to run properly in the system environment. A new software group "SmartFEPConfigConverter – Group HotStandby" is to be created in the appropriate parent instance. Inside this software group, a new Process instance "SmartFEPConfigConverter" must be created, see Figure 3-6.24.

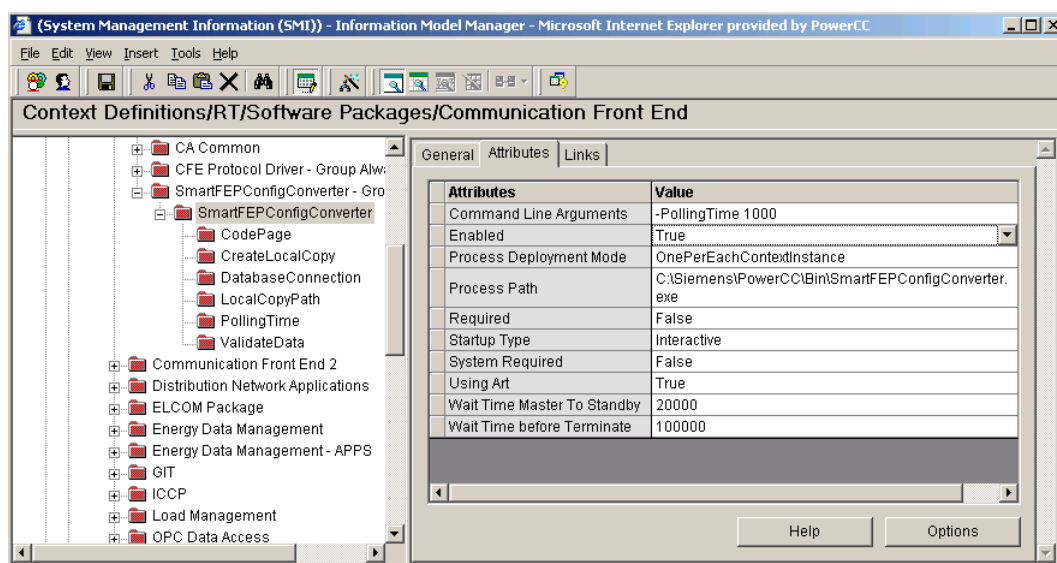


Figure 3-6.24: Configuring the Converter in IMM / SMI

All process options can be set either in SMI using child instances of the type *Process Attributes* or be specified as string value of the attribute 'Command Line Arguments' of the parent instance 'SmartFEPConfigConverter'. If 'Command Line Arguments' and Process Attributes specify different values for the same option, the 'Command Line' value overrides the Process Attribute value. 'Command Line Arguments' may be used for testing and debugging new systems. Preference is given to dedicated *Process Attributes*.

In the following tables all process options are described in detail.

PollingTime	
Description ->	The 'PollingTime' option sets the polling interval, i.e. the time between two database version checks by the SmartFEPConfigConverter. Polling time is given in milliseconds. Minimum polling time is 1000 milliseconds. If this option is not set, a warning message is created in the EventLog and the default value 10000 is used.
Using attribute 'Command Line Arguments'	Add "-PollingTime 5000" to the Command Line parameters. This sets the polling interval to 5 seconds.
Using instance type 'Process Attribute'	Add a child instance of the type <i>Process Attribute</i> and name it "PollingTime". Set 'Data Type' to 'LongInt'. Set 'Value' to 5000. This sets the polling interval to 5 seconds.

DatabaseConnection	
Description ->	The 'DatabaseConnection' option sets the database connection string, i.e., the TSN name of the SmartFEP RDBMS. If this option is not set a warning message is created in the EventLog and a default value is used (standard configuration for PowerCC).
Using attribute 'Command Line Arguments'	Add "DatabaseConnection <username>/<password>@<SID>" to the Command Line parameters
Using instance type 'Process Attribute'	Add a child instance of the type <i>Process Attribute</i> and name it "Database Connection". Set 'Data Type' to 'String'. Set 'Value' to <username>/<password>@<SID>

CreateLocalCopy	
Description ->	The "CreateLocalCopy" option determines whether the SmartFEPConfigConverter shall create local copies of the dBase files or not. If this option is not set, a warning message is created in the EventLog and the default value 'no' (not creating local copies) is used.
Using attribute 'Command Line Arguments'	Add "-CreateLocalCopy yes" to the Command Line parameters. This enables the creation of local copies. Allowed values are 'yes' and 'no'.
Using instance type 'Process Attribute'	Add a child instance of the type <i>Process Attribute</i> and name it "CreateLocalCopy". Set 'Data Type' to 'Bool'. Set 'Value' to 0 (do not create local copies) or 1 (create local copies).

LocalCopyPath	
Description ->	The 'LocalCopyPath' option determines the storage location of the local copies if these are desired (see 'CreateLocalCopy' option). If this option is not set, a warning message is created in the EventLog and the default value stored in the 'SR5_TEMP' environment variable (temporary directory of PowerCC) is used.
Using attribute 'Command Line Arguments'	Add "-LocalCopyPath c:\smartfepconfig" to the Command Line parameters. This sets the path where the local copies are stored to 'c:\smartfepconfig'.
Using instance type 'Process Attribute'	Add a child instance of the type <i>Process Attribute</i> and name it "LocalCopyPath". Set 'Data Type' to 'String'. Set 'Value' to "c:\smartfepconfig". This sets the path where the local copies are stored to "c:\smartfepconfig".

CodePage	
Description ->	The 'CodePage' option determines the ANSI code page that is used to convert UTF-16 strings (from the RDBMS) to ANSI strings (in the dBase files). If this option is not set, a warning message is created in the EventLog and the current system code page is used for conversion.
Using attribute 'Command Line Arguments'	Add "-Codepage 866" to the Command Line parameters. This sets the code page to 866 (DOS/Cyrillic).
Using instance type 'Process Attribute'	Add a child instance of the type <i>Process Attribute</i> and name it "CodePage". Set 'Data Type' to 'LongInt'. Set 'Value' to 1252. This sets the code page to 1252 (Windows/Cyrillic).

ValidateData	
Description ->	The 'ValidateData' option determines if the SmartFEPConfigConverter should perform validation of the SmartFEP configuration data. If validation is desired, detected errors are written in detail to the private process log (if enabled in system diagnosis) and a warning message is created in the event log. Default is 'no'.
Using attribute 'Command Line Arguments'	Add "-ValidateData yes" to the Command Line parameters. This enables data validation.
Using instance type 'Process Attribute'	Add a child instance of the type <i>Process Attribute</i> and name it "ValidateData". Set 'Data Type' to 'Bool'. Set 'Value' to '1' (for yes) or '0' (for no). This enables or disables data validation.

Volume 4 Test and Diagnosis

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4.1 Introduction

The Test and Diagnosis Volume provides information about the facilities and the usage of Test and Diagnostic Tools. This includes examples of setting input parameters and the description of output messages.

Content

Applying the Trace Functions to execute technological tracking of input information from RTU to the interface of the Realtime Processing System:

- Open trace
- Close trace
- Set Trace Filter
- Set Debug level

Applying Command Functions to get the following information about internal states, statistics and process images:

- Redundancy Status
- Communication Statistic
- Renewal CheckState
- GI Status
- Process Image
- Server Time

Intended Audience

- Service and Systems Engineers
- System Support Users

4.2 Test and Diagnosis Tools

Table of Content

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4.2.1 General

Purpose

Test and Diagnostic Tools are used to analyze and solve problems during configuration and operation of the Communications Front-End (especially during commissioning).

Test and Diagnosis Commands Provided

CFE provides a set of test and diagnostic utilities to get an overview of the current configuration of the devices and measurement relevant for the CFE. It is also possible to get information about communication statistics and to query internal status information. In order to achieve a high user acceptance the approach to Test and Diagnosis capability is completely done by means of a graphics user interface. The following figure depicts the different kinds of Test and Diagnosis commands.

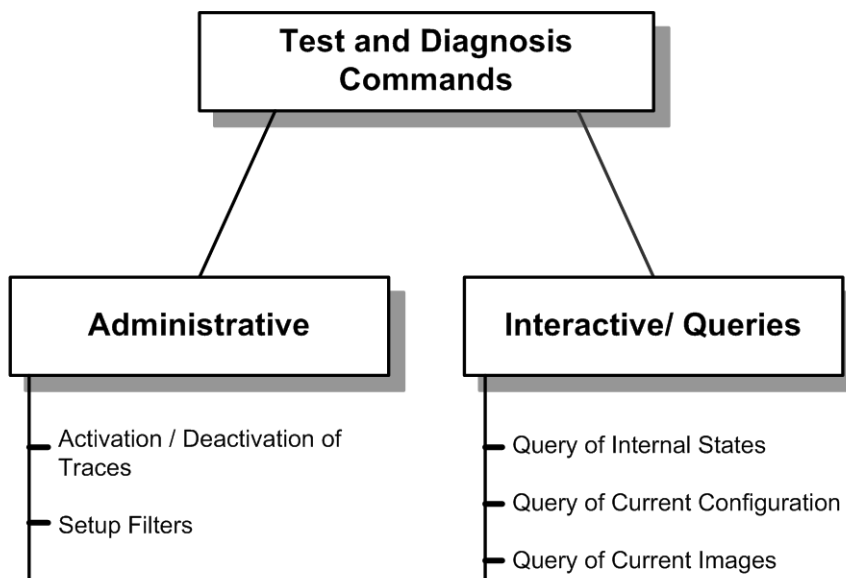


Figure 4-2.1: Test & Diagnosis Commands - Overview

Trace Commands

Trace Commands are used to control the trace output for a selected CfeInterface respectively its corresponding CFE process. In that context it is possible to activate the trace and setup filters. Chapter 4.2.3 *Using Test and Diagnosis Tools* describes this topic more detailed.

Query Commands

Query Commands provide a direct interaction between the user and the CFE application so it is possible to query internal states of devices (interfaces and links). It is also possible to read out the current status of the redundancy management and (if available) to query internal process images of a whole remote terminal unit or selected values. Furthermore Query Commands offer the facility to get information about pre-processing, e.g. the result of the last general interrogation (in order to find out what GI-relevant information objects have not been transmitted) or the status of the renewal control.

Operational Commands

Operational commands used for instance to lock or unlock a CfeInterface or CfeInterfaceLink or setting a preference are not part of Test and Diagnosis User Interface. Commands of that kind are integrated in the Runtime Explorer and not subject of this description.

4.2.2 Test and Diagnosis Facilities Overview

As mentioned above, there are two kinds of Test & Diagnostic Tools to help users to get knowledge about the internal status of devices and communication links and the data flow from the Remote Terminal Unit to the RTS processing. These are the trace facilities and the queries.

Trace Facilities

- Set individual trace level and filters adjustable per CfeInterface,
- Execute technological trace in order to track
 - a single information object (specified by the path name of a Measurement) or
 - the complete transferred information of a selected Remote Terminal Unit from the received protocol specific telegram up to the SCADA interface. The processing of that information through all layers of the CFE is visible.
- The amount of trace information is scalable due to adjustable debug levels.

Query of Internal States

- Getting redundancy information on:
 - the selected interface,
 - one or all links to the selected interface.
- Getting preprocessing state, i.e.
 - result of the last General Interrogation in order to identify missing information objects those raised a failed Completeness Check,
 - detailed result of last Renewal Check.

Query of Communication Statistics

Getting data throughput and Communication errors

Query Process Image

Reading the image of an individual substation or of selected substations of the same party line.

4.2.3 Using Test and Diagnosis Tools

4.2.3.1 Graphical User Interface

The Graphical User Interface of the Test and Diagnosis Tools provides an interaction between user and Test and Diagnosis application. The Test and Diagnosis User Interface receives diagnostic commands from the user and writes the output back to the Test and Diagnosis window. It also provides the facility to get remote access to Test and Diagnosis functions from any computer outside the PowerCC installation. The following features are provided by the user interface:

- The trace output is displayed directly in the User Interface. It is furthermore possible to access the relevant output file on disk for an evaluation at a later date, if necessary
- The format of output (both trace and response for commands) is human-readable
- The filter functions are applicable for all implemented protocols

4.2.3.2 Starting Test and Diagnosis User Interface

Normally the graphical user interface for Test & Diagnosis is not active. Therefore before being able to operate Test & Diagnosis Tools the user interface has to be started. This can be done by opening the executable file 'CfeTestAndDiagnosisUI.exe' by means of the windows explorer. The executable can be found on the system disk under the default path C:\Siemens\PowerCC\Apps\CFE\Bin (see Figure 4-2.2).

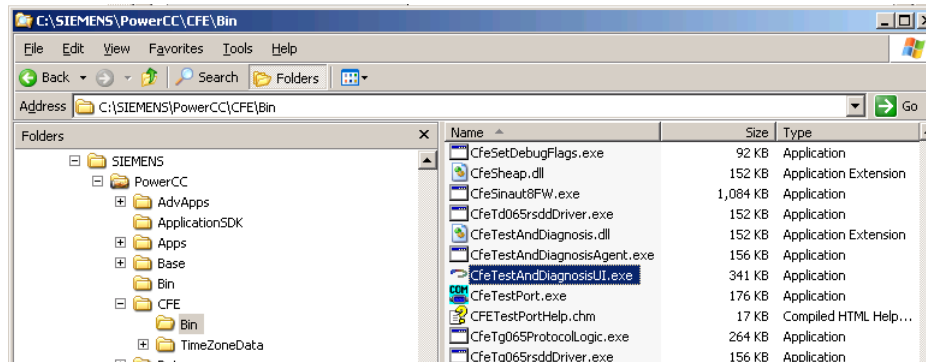


Figure 4-2.2: Starting Test & Diagnosis User Interface

At the end of the runup procedure, the Test and Diagnosis Tool displays the User Interface window, which consists of

- An overview pane, showing all CFEs connected to the PowerCC system
- Two Working Panes headed Command Output and Trace Output

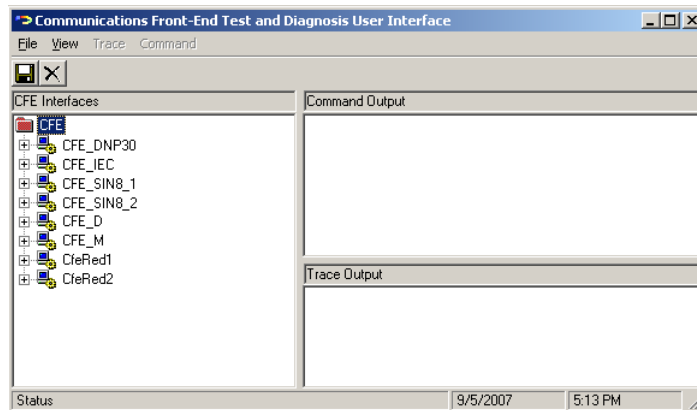


Figure 4-2.3: Test & Diagnosis User Interface Initial Display

The window toolbar shows the Windows standard buttons *File*, *View* and *Help* together with the Test and Diagnosis buttons **Trace** and **Command**, which are not yet active. To activate Test and Diagnosis facilities one of the CFEs' interfaces has to be selected. (Select a CFE by clicking on its symbol or name, then choose an interface (click on symbol or name). Now the menu items *Trace* and *Command* are highlighted, so that they can be used (see Figure 4-2.4).

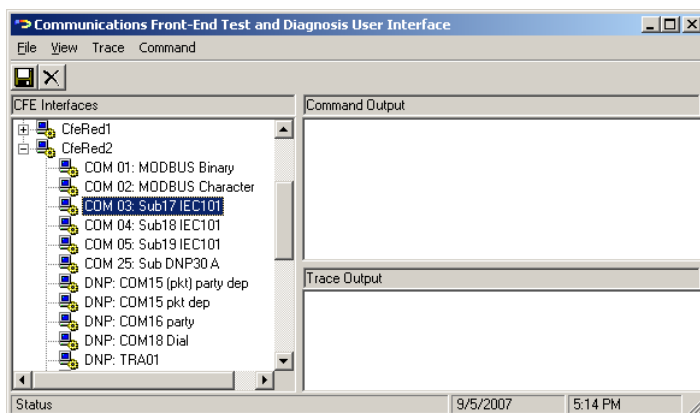


Figure 4-2.4: Test & Diagnosis: Selecting an Interface

4.2.3.3 Activation of Trace and Filters

Traces are primarily used to supervise the data flow inside the subcomponent CFE in order to find out why and on which level of data processing values get lost or changed. In order to get a human-readable diagnosis output it is necessary to limit its content by different filter functions.

Such filter functions are implemented inside the application as the trace output has an influence on the performance (and subsequently the data throughput). The adjustment of filters functions and the activation/deactivation of trace is done by means of the Trace menu items.

Trace Menu

The *Trace* Menu is evoked by clicking on the **Trace** button, where the following pull-down menu opens, which consists of two menu items for activation / deactivation and another three menu items for filter functions.

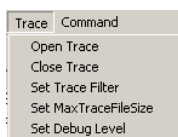


Figure 4-2.5: Trace Menu Options

Trace Filter

The possibilities to place trace outputs within subcomponent CFE to supervise the data flow inside are determined by the levels of data preprocessing. These levels from the incoming byte stream to the SCADA interface can be seen in Figure 4-2.6.

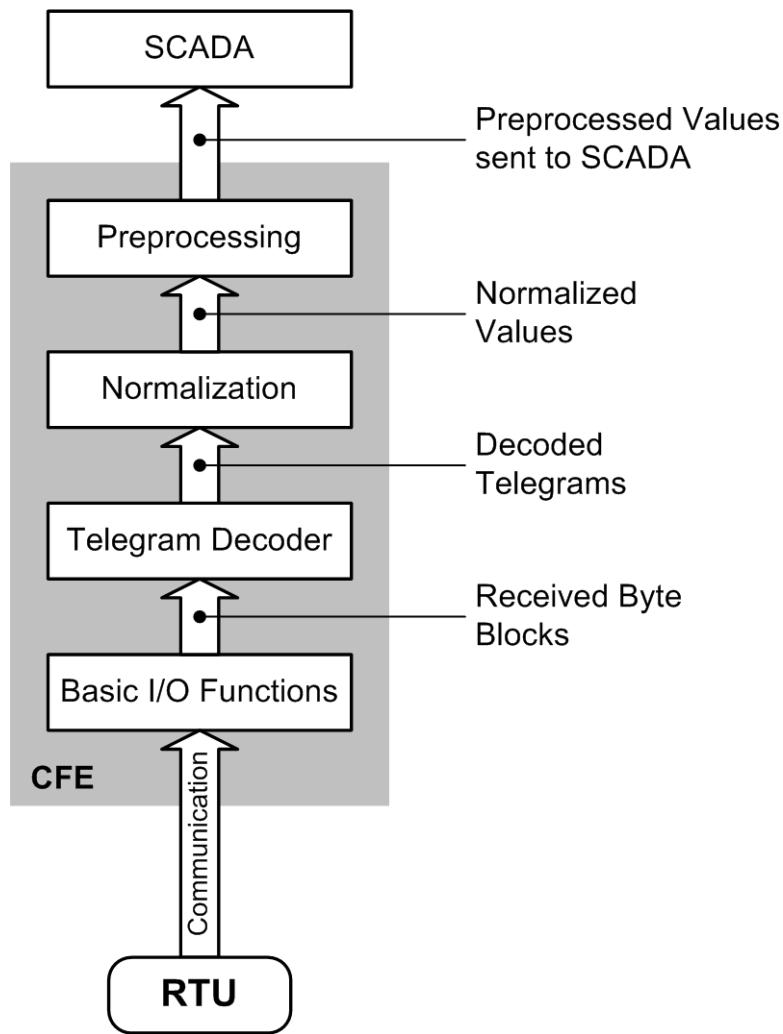


Figure 4-2.6: Data Flow and Possible Trace Points

Table 4-2.1 gives an overview of the different levels of data processing and trace capabilities for incoming RTU data and show how the output information can be filtered.

Table 4-2.1: Overview Processing Functions for Incoming RTU Data

Processing Function	Generated Data possible Output	Filter
Basic I/O Function	Incoming data as byte blocks.	Yes / No decision, no special filter possible.
Telegram Decoder	Incoming data in a protocol-specific format (telegram structure). It is possible to trace received information objects that do not match the CFE configuration (unknown station or object address).	- Faulty telegrams (wrong or not-interpretable structure) - Telegrams which cannot assigned to an parameterized RTU or data point - By substation or by data point single objects
Normalization	Incoming data in a protocol-independent format.	By substation or by data point single objects or address ranges possible

Processing Function	Generated Data possible Output	Filter
Preprocessing	Preprocessed data in a protocol-independent and for the SCADA-Interface prepared format. It is also possible to trace if a value will be suppressed in that level by old/new comparison or threshold processing. Such a trace makes it possible to find out what data will be passed to SCADA	By substation or by data point single objects)

Trace Filter for RCC Communication

The following figure shows the data flow and possible trace points for the communication to a Remote Control Center (RCC).

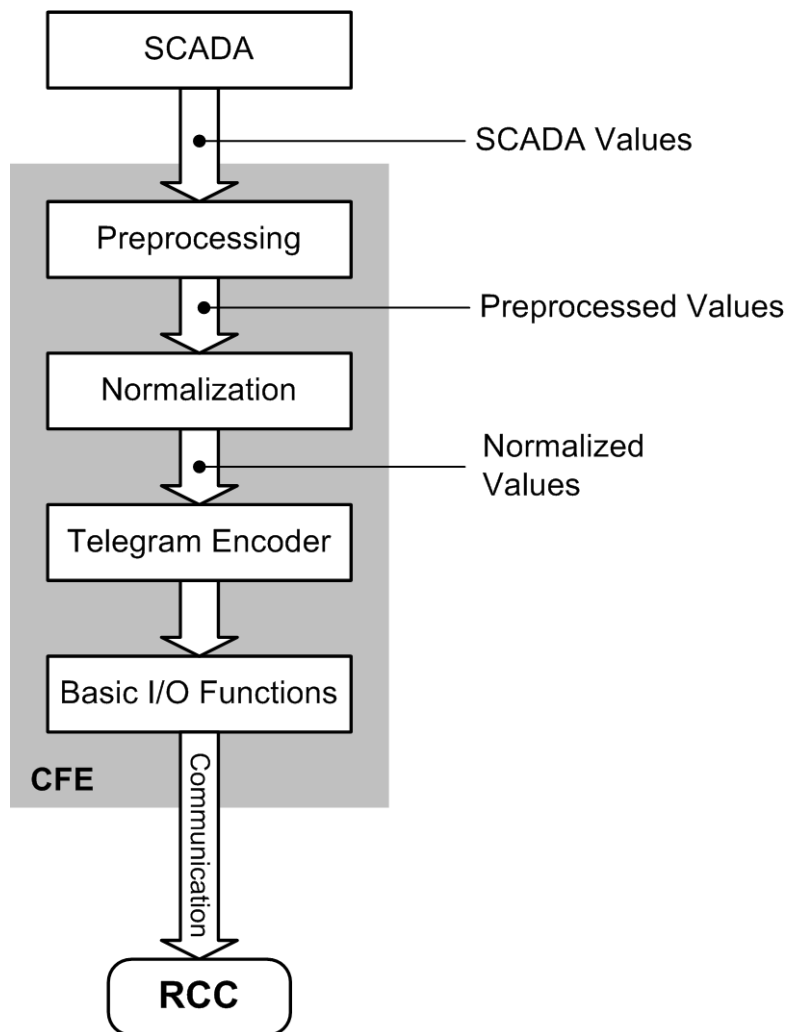


Figure 4-2.7: Data Flow and Trace Points for RCC Communication

Table 4-2.1 The following gives an overview of the different levels of data processing and trace capabilities and show how that output can be filtered.

Table 4-2.2: Overview Processing Functions for Outgoing RCC Data

Processing Function	Generated Data possible Output	Filter
(from SCADA)	Data in SCADA format	By substation or by data point
Preprocessing	scaled analog values	By substation or by data point
Normalization	Outgoing data in a protocol-dependent format	By substation or by data point or address ranges possible
Telegram Encoder	n/a	n/a

Open Trace

When evoking the Open Trace function by clicking on the appropriate menu item (see Figure 4-2.8) the Open Trace mask is displayed.

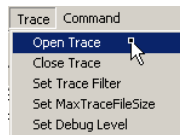


Figure 4-2.8: Menu Option 'Open Trace'

Figure 4-2.9 shows the **Open Trace** dialog window including the ability to adjust Trace Switches and Filters. The upper part provides Trace Switches, which directly limit the output of the trace functions to the specified levels of preprocessing. The lower part of the window headed "Filter By" restricts the output of the trace to CFE Data Points of specified Remote Terminal Units (in case of a party line interface) or to specified CFE Data Points of a single Remote Terminal Unit.

The example in Figure 4-2.9 shows the adjustment of the Trace Switches to enable technological tracking on the whole party-line consisting of four RTUs.

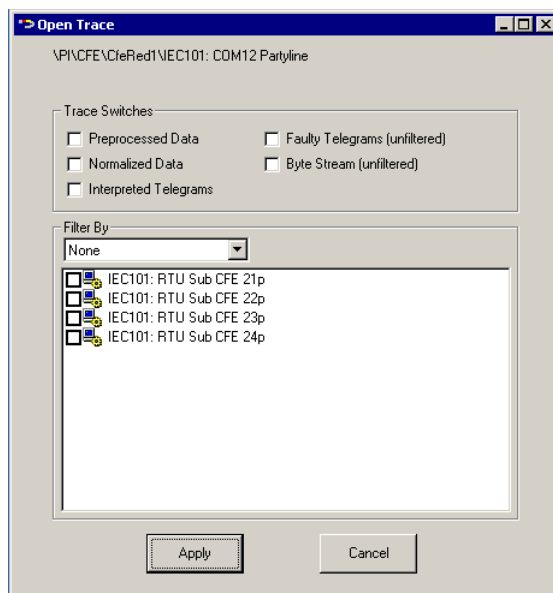


Figure 4-2.9: 'Open Trace' Mask

The Trace Switches can be selected by clicking on the preceding boxes. That will arise a hook within the concerning box. The next click on the box will deselect the Trace Switch again.

The Filter can be adjusted in two steps:

- Select a Filter function out of the pull-down menu from the 'Filter By' headline:

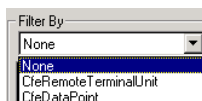


Figure 4-2.10: Selecting Trace Filter

None: no filtering by RTU or data points,

CfeRemoteTerminalUnit: filtering by selection of RTU,

CfeData Point: filtering by selection of data points.

- select one or more CFE RTUs and / or CfeTelemetered data points.

The filtering of RTU is just done by adjusting the filter function and then selecting RTUs by means of the offered check boxes (Figure 4-2.9).

The filtering of data points is done 1. by adjusting the filter function, then double clicking on the RTUs so that the names of the CFE data points get visible (Figure 4-2.9). Then the data points can be selected by means of check boxes (see Figure 4-2.11).

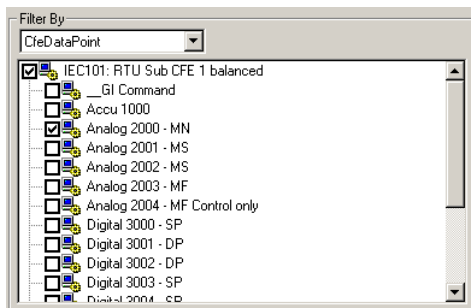


Figure 4-2.11: Filtering By Option 'CfeDataPoint'

The trace is activated by clicking the **Apply** button. This effects that the Open Command is sent to the CFE application together with Trace Switches and Filter information.

As a result, the **Open Trace** command and the response of the CFE application is displayed in Command Output pane. The trace output appears in the Trace Output pane.

Due to the filter setting, the Trace Output consists of the different types of entries containing the following parts (see Figure 4-2.12):

- Headline with the path of traced interface, protocol type
- Data type depending on traced data point (preprocessing level)
- Protocol and level specific information of the traced data point: e.g. Common Address Link Address, Object Address, Information type, Cause of Transmission, Value, Qualifier in case of IEC101 protocol and trace level interpreted telegrams

- Data point name used in CFE and network, RTU name, value type, value, qualifier, time stamp, additional Information

```

Command Output
12/20/2005 2:57:45 PM
\PI\CFE\CfeRed1\IEC101: COM13 balanced dependent
OpenTrace
OpenTrace

12/20/2005 2:58:08 PM
\PI\CFE\CfeRed1\IEC101: COM13 balanced dependent
Close Trace
CloseTrace

Trace Output
14:58:04.628 /PI/CFE/CfeRed1/IEC101: COM13 balanced dependent
Interpreted Telegram Received info object:
Common Address: 101
Link Address: 202
Object Address: 2000
Type: Measured value, normalized value (9)
Cause of Transmission: Periodic/Cyclic(1)
Structured Qualifier: false
Value : 15, Qualifiers:

14:58:04.628 /PI/CFE/CfeRed1/IEC101: COM13 balanced dependent
--> z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101read.cpp(2809)
void SetLocalTimeInT101Time(...)

14:58:04.628 /PI/CFE/CfeRed1/IEC101: COM13 balanced dependent
--> z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101dn.cpp(216)
void deviceMasterNormalizeContainer(...)

14:58:04.628 /PI/CFE/CfeRed1/IEC101: COM13 balanced dependent
Normalized Data : Analog 2000 - MN ---|--- /Network/Substation_CFE/IEC101: Sub
CFE 1 balanced/Analog.001 (typeid/address: 9/2000) on RTU IEC101: RTU Sub CFE
1 balanced
Normalized Float : 15.000000, Validity : Valid
Time Stamp : 12/20/2005 - 14:58:04,628 msec - Valid
Source : Telemetered, Normal Source : Telemetered, Normal Range : Normal
Additional Info : 0, Change of Value Counter : 0
  
```

Figure 4-2.12: Trace Output (Clipping) within Pane

The Trace Output pane is limited by height and width of the screen, but it is possible to scroll over the text by means of scroll bars at the bottom and right border of the pane. As the trace output is stored in a file, it is possible to open this file by means of Windows Accessories like Notepad or Word. The trace files can be found under the pre-configured path (default: Siemens\PowerCC\Logs\CFE\)

<name_of_CFE><name_of_COM>_trc.txt

An example (clipping) of a trace protocol can be seen in (Figure 4-2.13).

```

TnD_CfeRed1IEC101_COM13_balanced_dependent_trc.txt - Notepad
File Edit Format Help
2005-12-20 14:58:04.628 int findvalueEntry(...) p_objectAddress:2000 Link Address:202
Interpreted Telegram Received info object:
Common Address: 101
Link Address: 202
Object Address: 2000
Type: Measured value, normalized value (9)
Cause of Transmission: Periodic/cyclic(1)
Structured Qualifier: false
Value: 15, Qualifiers:
2005-12-20 14:58:04.628 --> Z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101read.cpp(2809)
void SetLocalTimeInT101Time(...)
2005-12-20 14:58:04.628 --> Z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101dn.cpp(216)
void deviceMasterNormalizeContainer(...)
2005-12-20 14:58:04.628 Normalized Data : Analog 2000 - MN --- /Network/Substation_CFE/IEC101: Sub CFE 1
balanced/Analog.001 (typeid/address: 9/2000) on RTU IEC101: RTU Sub CFE 1 balanced
Normalized Float : 15.000000, Validity : valid
Time Stamp : 12/20/2005 - 14:58:04,628 msec - valid
Source : Telemetered, Normal Source : Telemetered, Normal Range : Normal
Additional Info : 0, Change of Value Counter : 0
2005-12-20 14:58:04.628 --> Z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101read.cpp(2732)
void RestartGTimer(...) object address:2000 Link Address:202
2005-12-20 14:58:04.628 --> Z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101.cpp(4652)
void sendFixedSlave(devType *p_pdevType, char p_Function) p_pdevType:42440928 p_Function:0
2005-12-20 14:58:04.628 --> Z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101.cpp(4890)
void sendSlaveData(void)
2005-12-20 14:58:04.628 --> Z:\SR5\apps\CA\CFE\Common\interface\_src\Interface.cpp(277)
Interface::SetParameter()
2005-12-20 14:58:04.628 --> Z:\SR5\apps\CA\CFE\Common\interface\_src\ComPort.cpp(979)
CEVPccComPort::SetParameterPrivate(const char* pParameterName, void* pParameterValue)
PHONENUMBER:5241512 pParameterValue:42441808
2005-12-20 14:58:04.628 --> Z:\SR5\apps\CA\CFE\Common\interface\_src\Interface.cpp(198)
Interface::write() nNumberOfBytesToWrite:5
2005-12-20 14:58:04.628 --> Z:\SR5\apps\CA\CFE\Common\interface\_src\ComPort.cpp(438)
CEVPccComPort::write(synchronous) nNumberOfBytesToWrite:5 COM port:13
2005-12-20 14:58:04.675 --> Z:\SR5\apps\CA\CFE\Common\interface\_src\Interface.cpp(299)
Interface::getParameter()
2005-12-20 14:58:04.675 --> Z:\SR5\apps\CA\CFE\Common\interface\_src\ComPort.cpp(1219)
CEVPccComPort::getParameterPrivate() pParameterName: TXErr:5241132
2005-12-20 14:58:04.675 --> Z:\sr5\apps\ca\cfe\protocols\iec870-5-101\master\t101.cpp(4698)
sendFixedSlave: Exit

```

Figure 4-2.13: Protocol Output Displayed through Notepad

Close Trace

The *Close Trace* command terminates the currently running trace. This command is immediately sent to the application when clicking on the Close menu item with the effect, that the application stops tracing and the trace output is finished,

The **Close** command and the response of the CFE application are displayed in the Command Output pane, see Figure 4-2.12.

Set Trace Filter

The *Set Trace Filter* Function allows the user to change Trace Switches and Filter settings during a currently running trace. The Set Trace Filter function is evoked by clicking on the appropriate menu item (Figure 4-2.8) which effects that the Set Trace Filter mask is displayed.

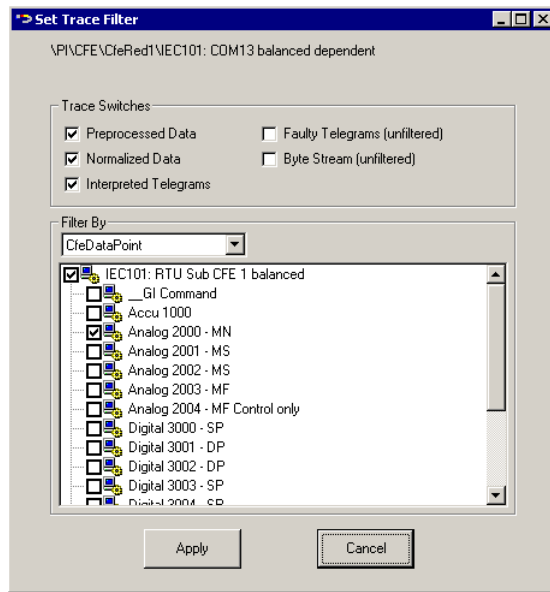


Figure 4-2.14: 'Set Trace Filter' Mask

Figure 4-2.14 shows the Set Trace Filter window which provides the same Filter facilities as the Open Trace window (Figure 4-2.9). The new filter settings are activated by clicking on the **Apply** button within the Set Trace Filter window. The command and the response of the CFE application are displayed in the Command Output pane, see Figure 4-2.15.

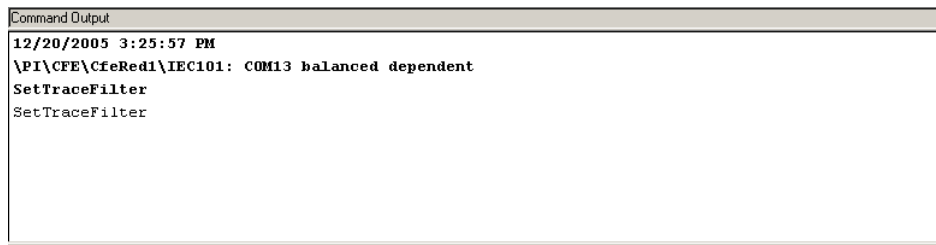


Figure 4-2.15: Command Output Pane with 'SetTraceFilter' Entry

Debug Level

The trace facilities described in previous paragraphs usually contain information for common use. In special cases of problem tracking it might be useful to get a more detailed output that contains additional information of the internal data flow and status information. Such a trace is primarily addressed to the developer due to support problem analysis. A diagnosis mask is provided to setup different debug levels. The meaning of debug level is protocol specific and will be explained later.

Set Debug Level

The *Set Debug Level* function is evoked by clicking on the appropriate menu item (Figure 4-2.8). This effects that the Debug Level mask is displayed. The mask allows the user to set one or more switches under the headline Master Function by means of check boxes to get more (internal) information on processing environment (see Figure 4-2.16).

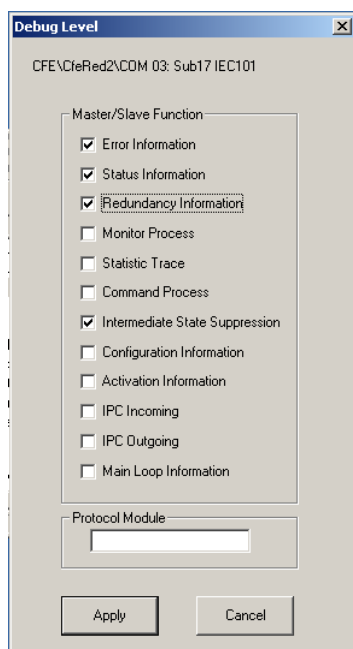


Figure 4-2.16: Setting Debug Level

Tagging the check boxes above has the following effects:

- **Error Information:** Errors detected by 'Device Master' such as faulty data on activation or errors in program flow (for experts only)
- **Status Information:** Generates additional information in Control output pane about interface, link and control states (blocked, preferred)
- **Intermediate State Suppression:** Effects, that information on Intermediate States is printed to the Control output pane
- **Command Process:** Provides information of command management, e.g., whether the command queue is empty or whether the command is to be buffered (for experts only)
- **Redundancy Information:** Generates information in Control Output pane about redundancy status if this changes during trace
- **Statistic Trace:** The counter of 'Information Messages Sent to SCADA system' within the last 20 seconds is printed cyclically to the protocol output
- **Activation Information:** Prints details on activation process to the output pane
- **Main Loop Information:** Provides details of internal program flow. This information might be useful for analysis and debugging (for experts only)
- **IPC Incoming:** Provides information about the incoming IPC message flow (CFE internal communication mechanism between Interfaces - CfeRti - Gateway (which is not traced here))
- **IPC Outgoing:** Provides information about the outgoing IPC message flow

The input box *Protocol Module* serves to type in a protocol specific debug level. The input box demands to enter a numerical (decimal or hexadecimal) number that depends on the applied protocol and the specific needed information. As the interpretation of the extended information is a matter of the developer due to support problem analysis, the implemented debug levels will not be explained within this documentation.

4.2.3.4 Query Functions

Test and Diagnosis Query functions provide a direct interaction between user and CFE application. With Test and Diagnosis Query functions it is possible:

- to read out the current status of the redundancy management
- to query internal process images of a whole remote terminal unit
- to get information about preprocessing, e.g. the result of the last general interrogation (in order to find out what GI-relevant information objects haven't been transmitted)
- to get the status of the renewal control.

By means of the Command Menu the following Query functions are available (see Figure 4-2.17):

- GetRedundancyStatus
- Communication Statistic
- GetRenewalCheckState
- Get GI Status
- Get Process Image
- Get Protocol Info - prints some information about available trace levels
- Server Time
- Print State (CfeRti - accessible through the Interface instance) - prints internal information about the interface and its RTUs and Sub-RTUs
- Print Memory (CfeRti - accessible through the Server instance) - prints the memory usage of the CfeRti internal classes
- Print Admin (CfeRti - accessible through the Server instance) - prints the IPC queue status and counts of sent messages

Command Menu

The Command Menu is evoked by clicking on the **Command** button, where the following pull-down menu opens:

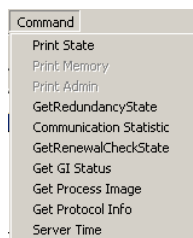


Figure 4-2.17: Command Menu Options

On clicking one of the Command Menu items, the command is either immediately executed, that means it is sent to the CFE application or a consecutive dialog window is displayed, when a further choice is possible.

If a consecutive window exists the desired command is activated by clicking on the **Apply** button.

Activation means that the command is sent to the CFE application together with the mask information.

As a result the command itself and the response of the CFE application are displayed in Command Output pane. The amount of information depends on the selected command and the additional options.

Query of Redundancy State

The GetRedundancyState command returns status information of the redundancy management (configured priorities, redundancy partners including their status information which are relevant for redundancy). On clicking on the **GetRedundancyState** item the 'Redundancy Status' mask is displayed. By means of a pull-down menu, it is possible, to optionally request the redundancy information of the selected interface, of a selected link or of all existing links of the selected interface, see Figure 4-2.18.

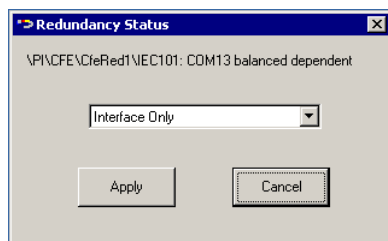


Figure 4-2.18: 'Redundancy Status' Mask

The command and the response of the CFE application are displayed in the Command Output pane.

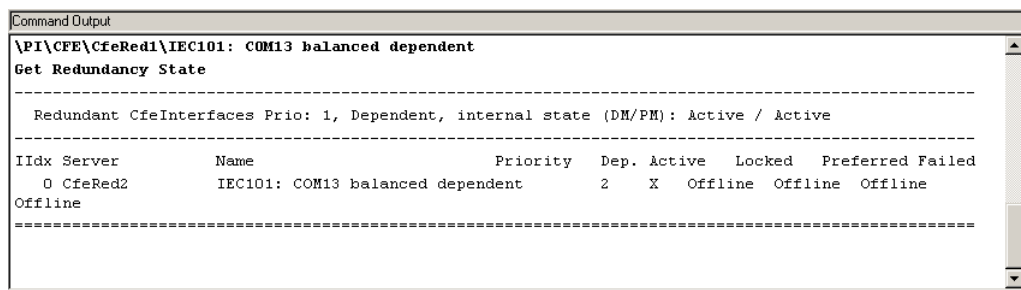


Figure 4-2.19: Command Output Pane Showing Redundancy Status

The Command Output consists of the following parts:

- Date and time, name of the examined interface, command name depending on traced data point (preprocessing level)
- Redundancy status of interface (and link if selected)

Communication Statistics

The Communication Statistic command returns statistical information, which is collected within the CFE application. Using the Test and Diagnosis user interface, you can call a Communication Statistics window providing its own interface. For details, refer to Chapter 4.2.3.5 *Communication Statistics* further down.

Query of the Renewal Check Result

The command option 'GetRenewalCheckState' returns the result of the last executed renewal check for the specified RTU and its renewal cycle 1 or 2. On clicking on the **GetRenewalCheck** item the GetRedundancyStatus mask is displayed.

By means of a pull-down menu it is possible to optionally request the Renewal Check Status information of one or of all RTUs connected to the concerning interface, see Figure 4-2.20.

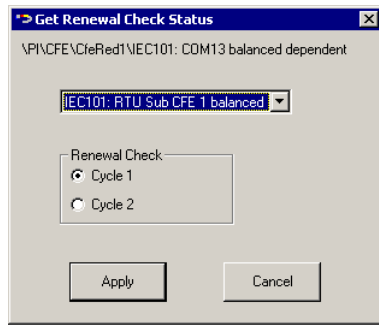


Figure 4-2.20: 'Get Renewal Check Status' Mask with Selected RTU

Clicking on the **Apply** button, the command is sent to the CFE application. Command and response will be displayed in the Command Output pane (see Figure 4-2.21).

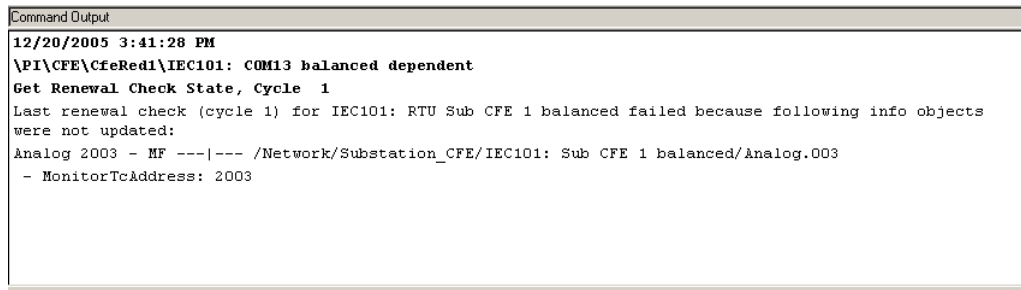


Figure 4-2.21: Command Output Pane with Renewal Check Response

Query of the GI Status

The command option 'Get GI Status' returns the result of the last general interrogation. On clicking the **Get GI Status** item the *Status of General Interrogation* window is displayed, see Figure 4-2.22.

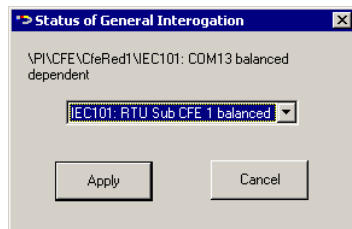


Figure 4-2.22: 'Get GI Status' Mask with Selected RTU

By means of a pull-down menu, it is possible, to optionally request the GI Status of one or of all RTUs connected to the selected interface. By clicking on the Apply button the command is activated and the result is displayed in the Command Output pane.

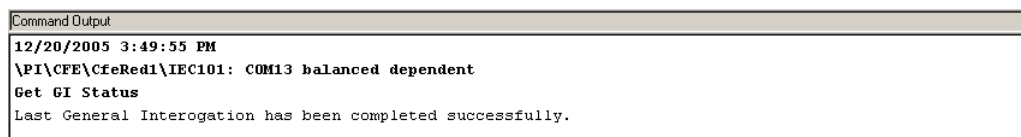


Figure 4-2.23: Command Output Pane Showing GI Status Information (GI Complete)

If the GI is incomplete, all the missing data points are listed in the window.

```

Command Output
12/20/2005 3:51:18 PM
\PI\CFE\CfeRed1\IEC101: COM13 balanced dependent
Get GI Status
Last General Interrogation was incomplete, following values are missing :
Digital 3007 - SP ---|--- /Network/Substation_CFE/IEC101: Sub CFE 1 balanced/Digit.2204
- MonitorTcAddress : 3007
Digital 3008 - SP ---|--- /Network/Substation_CFE/IEC101: Sub CFE 1 balanced/Digit.2205
- MonitorTcAddress : 3008

```

Figure 4-2.24: Command Output Pane Showing GI Status Information (GI Incomplete)

Query of the Process Image

The command option 'Get Process Image' returns the current process image on the CFE of a selected RTU or of selected CFE data points. When clicking on the **Get Process Image** item the *Get Process Image* window is displayed. It shows all RTUs connected to the Interface by means of a party-line. Each RTU is provided with a check box. Those RTUs, from which the Process Image shall be sent, have to be tagged accordingly.

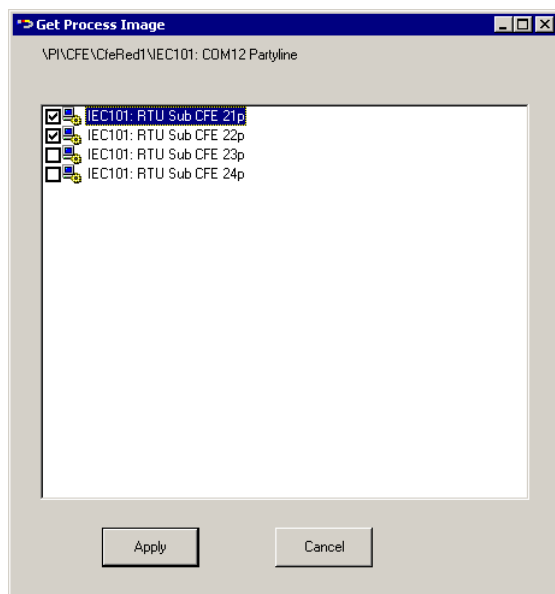


Figure 4-2.25: 'Get Process Image' Mask with Selected RTU

To select the information of single Data points, double-click on the desired RTU, so that the data points of this RTU get visible (see Figure 4-2.26).

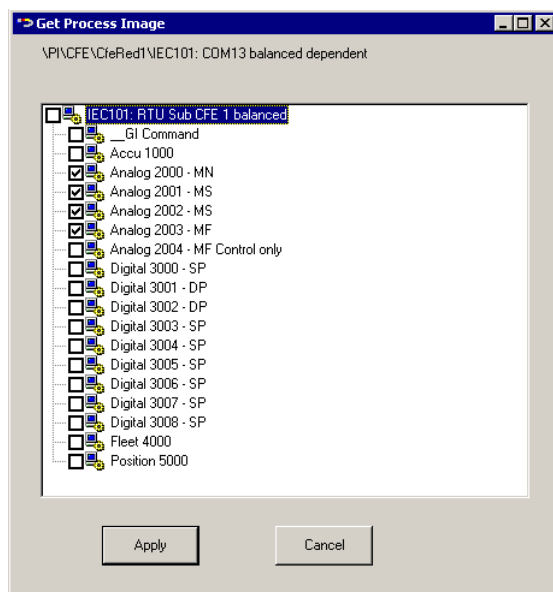


Figure 4-2.26: 'Get Process Image' Mask with Selected Datapoints

The command is activated by means of the **Apply** button, the result is displayed in the Command Output pane.

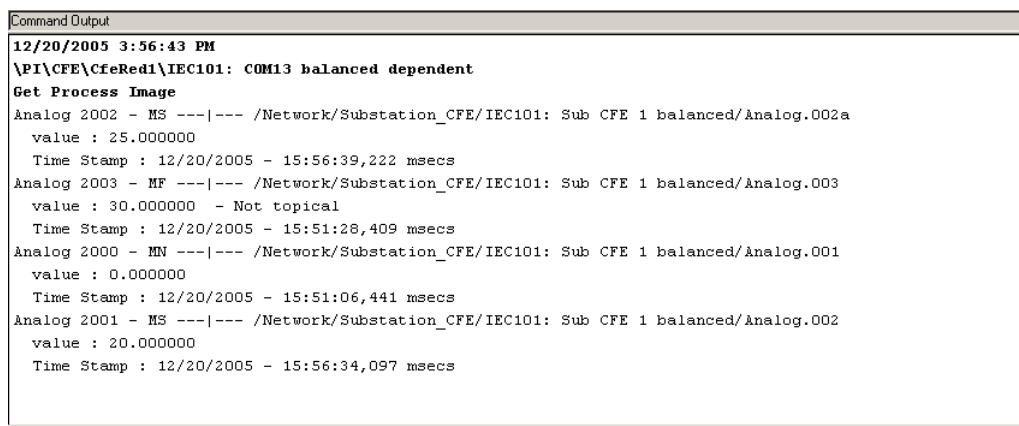


Figure 4-2.27: Command Output Pane Showing Process Image Data

The output consists of a single value in case a Single Data Point is requested comprising:

- Headlines with time and date, command
- Name and type of the requested value, network path, (type notion according to IEC101 protocol convention)
- The value itself and the appropriate qualifier
- The time stamp associated with the value

(See upper part of Figure 4-2.27.)

If more than one data point is requested, e.g. all data points of the RTU, then the output consists of a list of value items with the structure described above (see lower part of Figure 4-2.27).

Server Time

The command option 'Server Time' returns the current time of the selected CFE-Server. On clicking on the Server Time command item, the command is immediately sent to the CFE application.

The command and the response of the CFE application are displayed in the Command Output pane. The command provides the following information shown in the Output pane:

Local time, UTC-time, time zone, time difference between UTC time and local time and daylight saving time

```

Command Output
12/20/2005 3:57:22 PM
\PI\CFE\CfeRed1\IEC101: COM13 balanced dependent
Server Time
CFE Server Time information
Local Time: 2005/12/20 - 15:57:22,456
UTC      : 2005/12/20 - 14:57:22,456
Time Zone : W. Europe Standard Time
Difference, in minutes, between Coordinated Universal Time (UTC) and local time: -60
Difference, in minutes, between daylight saving time and standard: -60

```

Figure 4-2.28: Command Output Pane Showing Server Time

4.2.3.5 Communication Statistics

4.2.3.5.1 Overview

Messages Counted by Communication Statistics

The Communication Statistics enables you to measure the communication quality of ports and links related to connected RTUs. This is achieved by counting of messages each of which relates to a specific interface (port) and link. The following table shows the standard types of counter categories that are provided by the Communication Statistics function.



Note

For DNP3.0 the number of error counters has been extended, please refer to the document Communications Front-End, Technical Specification

Counter Category	Description
Messages Sent	Counts outgoing (sent) telegrams per interface and link
Received Messages	Counts incoming (received) telegrams per interface and link
Received Errors	Counts received errors per interface
Timeout	Counts timeouts per interface and link

Counter Accessing Methods

The resulting values of these counters can be accessed in:

- CFE: Via the CFE Test and Diagnosis interface you can call the Communication Statistics window from which you can retrieve the counters for the selected interface and RTU. With this method, no additional configuration is needed.

- SCADA: By selecting a telemetered accumulator measurement linked to a specific CFE system accumulator measurement. These data points and the link between both has to be defined manually. This method is used for displaying the communication quality of the ports and links via an object within a SCADA diagram and for further processing such as Energy Accounting and others.

4.2.3.5.2 CFE Communication Statistics Interface

Calling Communication Statistics Interface

The CFE Communication Statistics interface can be called from the CFE Test and Diagnosis user interface window described previously in Chapter 4.2.3.2 *Starting Test and Diagnosis User Interface*.

To call the CFE Communication Statistics interface:

1. In the CFE Test and Diagnosis window, select the Command option of the menu bar.

The Command menu is displayed.

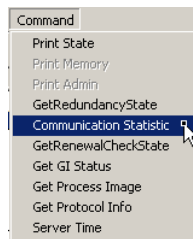


Figure 4-2.29: Command Menu Options

2. In the Command menu, select **Communication Statistic**.

The Communication Statistics user interface window is displayed.

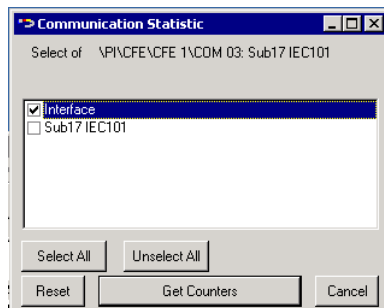


Figure 4-2.30: Communication Statistics Window

Communication Statistics Window

The above shown Communication Statistics user interface window consists of:

- Display of the instance path of the communication port from which the Test and Diagnosis window has been called (\PINC\FE\CFE_n\COM ...)
- List box for the selection of:
 - Interface: COM port (CFE Interface) for which Communication Statistics are desired
 - Link(s): link(s) appended to the selected interface for which Communication Statistics to be called up

- Selection buttons:
 - Select All: Selects the Interface option and all the available CFE Interface Link options in the list box.
 - Unselect All: Deselects all the selected options of the list box.
- Reset: Sets the currently selected counters to zero.
- Get Counters: Displays in the Command Output window the statistics for the currently selected items of the list box (i.e. of the selected interface and CFE Interface Links).
 - For the Interface, 4 counters are displayed.
 - For each of the CFE Interface Links, 3 counters are displayed.
- Close: Closes the Communication Statistics window.

Examples

Example 1:

In this example, both the COM port (Interface) and the CFE link appended to it was selected. After clicking **Get Counters** the Command Output pane displays the statistical information for both items (see the following Figure).

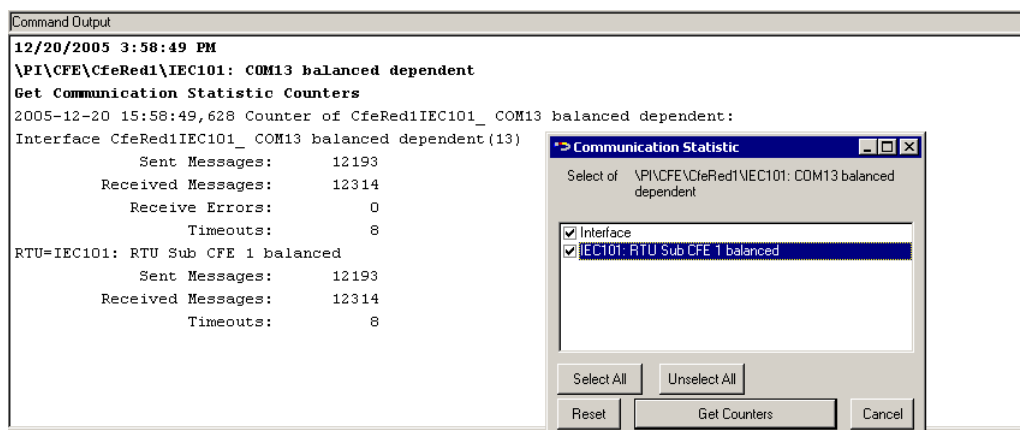


Figure 4-2.31: Communication Statistics Window and Command Output Pane with Statistic Values - Example 1

In the above example, we can observe that for the Interface and the interface link identical values are shown. This is typical for point-to-point connections. As such, it is sufficient, to select just the interface to get relevant information. This is demonstrated in the next example.

Example 2:

In this example, only the CFE Interface has been selected in the Communication Statistics dialog box. After clicking **Get Counters**, the Command Output pane displays the statistical information for the CFE Interface only (see the following figure).

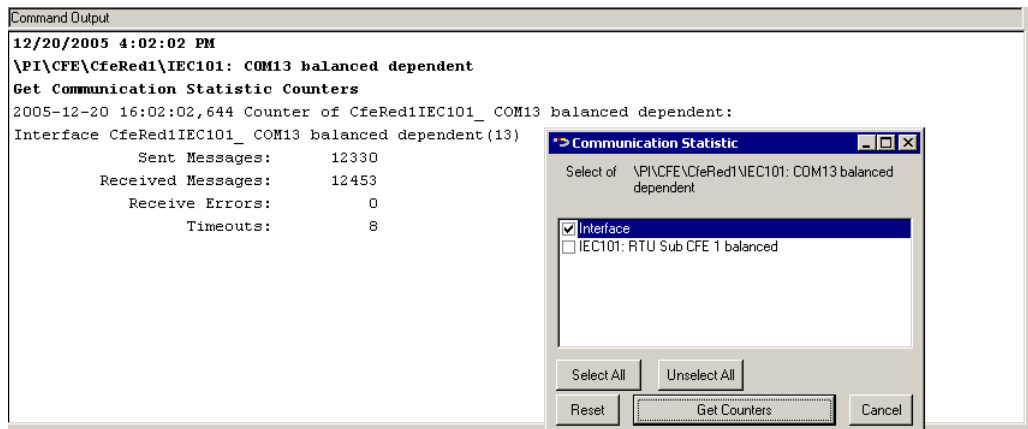


Figure 4-2.32: Communication Statistics Window and Command Output Pane with Statistic Values - Example 2

Example 3:

This example shows the extended communication statistics for DNP3.0, both for the link (Figure 4-2.33) and the CFE COM interface (Figure 4-2.34).

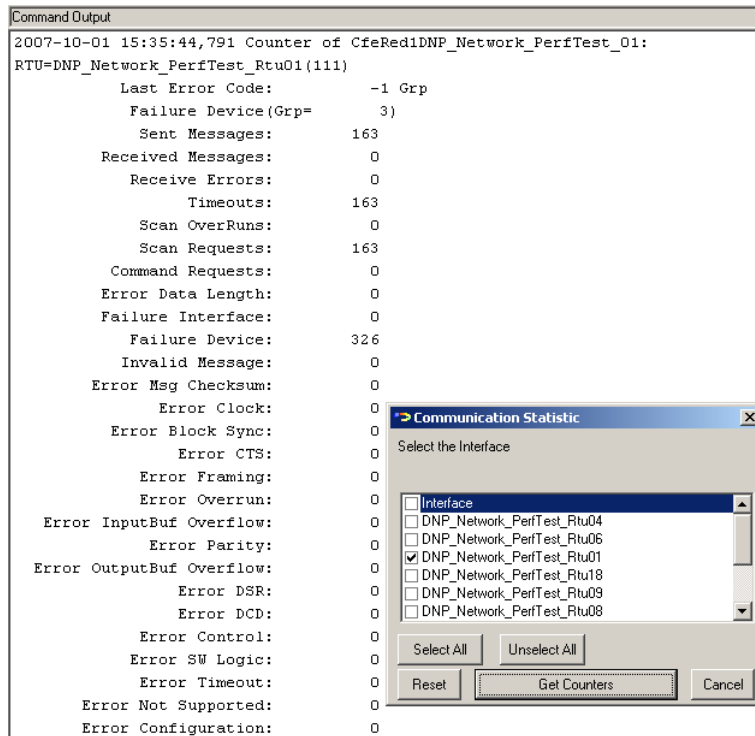


Figure 4-2.33: DNP3.0 Communication Statistics for the Link

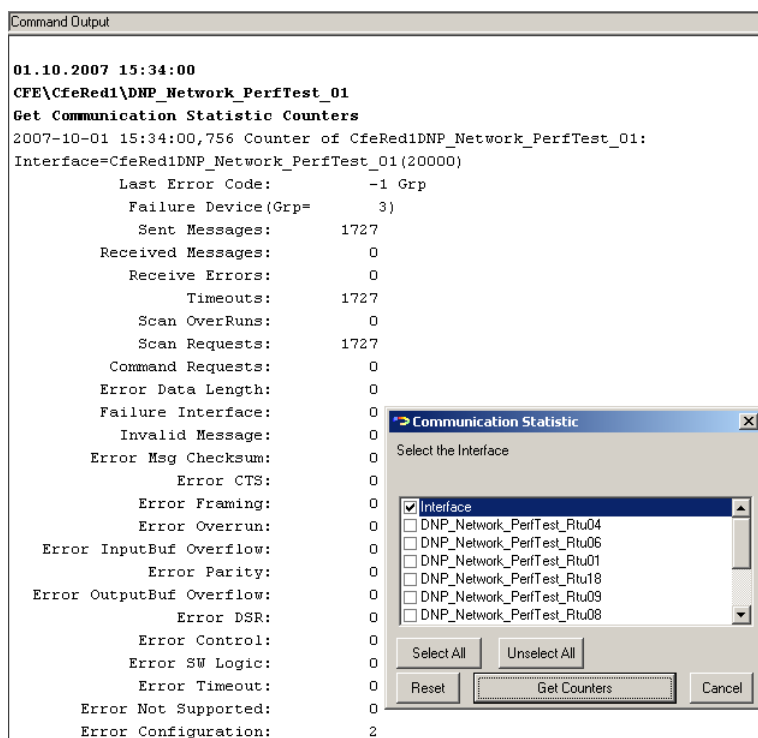


Figure 4-2.34: DNP3.0 Communication Statistics for the Interface

